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Moment map equations in Kähler geometry and gauge theory

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Introduction

In 1965 M.S. Narasimhan and C.S. Seshadri proved the so-called Narasimhan-Seshadri theorem [46]. In their original formulation, their result provided a relation between the notion of stability for holomorphic vector bundles over a compact Riemann surface, which had been introduced by D. Mumford in 1963 in the context of Geometric Invariant Theory, and representations of the fundamental group of the surface. Briefly, stability of vector bundles is defined by assigning a numerical invariant to each bundle, called its slope. A vector bundle over a Riemann surface is stable if its slope is greater than the slope of any of its proper holomorphic subbundles. In the 1970s and 1980s, the works by M. Atiyah, R. Bott, S. Kobayashi, S. Donaldson, N. Hitchin and others showed that the result by Narasimhan and Seshadri could be translated to the language of differential geometry, and more specifically, in terms of Yang-Mills connections, notions arising from aspects of differential geometry and theoretical physics that were being studied at the time. More precisely, Donadlson gave a new proof of the theorem by Narasimhan and Seshadri in 1983 [14]. He proved that an indecomposable holomorphic vector bundle over a compact Riemann surface $\mathcal{E} \to X$ is stable if and only if its hermitian fiber metric h, whose Chern connection ∇ satisfies the equation $\star F_\nabla = -\mathrm{i}\, c \cdot \mathrm{id}_\mathcal{E}$ for a fixed constant c that depends on the topology of the bundle and the surface. This is called the *constant central curvature* condition. If the condition of being decomposable is omitted, one gets that a hermitian holomorphic vector bundle is polystable if and only if it admits a hermitian fiber metric whose Chern connection satisfies the constant central curvature condition. The relation with Yang-Mills theory can be seen even more directly if one realizes that the theorem by Donaldson is equivalent to the following statement: a smooth complex vector bundle $E \to X$ over a compact Riemann surface endowed with a hermitian fiber metric has a *Hermite-Einstein-Yang-Mills connection*, i.e. a connection satisfying $\star F_\nabla = -\mathrm{i}\, c \cdot \mathrm{id}_E$ if and only if it admits a holomorphic structure making the bundle polystable.

In 1980, Kobayashi generalized the notion of stability to holomorphic vector bundles over higher-dimensional compact Kähler manifolds. This was a remarkable contribution, as it required the introduction of coherent sheaves. Kobayashi and Hitchin conjectured independently that the theorem of Narasimhan and Seshadri holds for compact Kähler surfaces (hence the name, *Hitchin-Kobayashi correspondence*). That is, a holomorphic vector bundle over a compact Kähler manifold is polystable if and only if it admits a hermitian fiber metric whose Chern connection satisfies the equation $\mathrm{i}\,\Lambda_\omega F_\nabla = c \cdot \mathrm{id}_E$ where Λ_ω is the dual Lefschetz operator. Such a fiber metric is called a Hermite-Einstein metric. The fact that the existence of the metric implies polystability is relatively easy to prove. The converse, however, was remarkably difficult, and it was proved by Donaldson and K. Uhlenbeck and S.T. Yau in successive works in the 1980s. This result is surprising because it provides a direct relation between a purely algebraic notion, stability, and a differential-geometric notion, the existence of a Hermite-Einstein metric. Again, this can be stated equivalently by saying that a complex vector bundle $(E, h) \to X$ has a unitary connection satisfying the *Hermite-Yang-Mills equations*

$$\begin{cases} F_\nabla^{(0,2)} &= 0 \\ \Lambda_\omega F_\nabla &= -\mathrm{i}\, c \cdot \mathrm{id}_E \end{cases}$$

if and only it has a holomorphic structure making it polystable.

This correspondence acts like a dictionary in the sense that it allows us to translate algebraic problems into differential-geometric problems and vice versa. The theorem above can be interpreted as fixing a hermitian fiber metric on a complex smooth vector bundle. Then, it can be interpreted in terms of moduli spaces: there is a homeomorphism between the moduli spaces of polystable holomorphic structures on the vector bundle and the moduli space of *Hermite-Yang-Mills connections* on E. We insist on the fact that this result is extraordinary, in the sense that it is a purely gauge-theoretic approach to an algebraic notion.

In 1975, the result by Narasimhan and Seshadri was generalized by Ramanathan for principal bundles over a compact Riemann surface with compact connected reductive complex Lie groups [48]. Among the contributions of Ramanathan to the discipline, we should highlight the fact that he gave a notion of stability. This was generalized for compact Kähler manifolds (which again required modifying the notion of stability) by Ramanathan and Subramanian [61]. From the beginning of the century onwards, several generalizations of the correspondence have been proven. Namely, I.

Mundet i Riera gave a generalization to Kähler fibrations [49], and O. García Prada and P.B. Gothen gave a generalization of the correspondence for Higgs bundles [18].

The remaining piece of the puzzle was given by M. Atiyah and R. Bott, who published a groundbreaking paper [4] in 1983 that set the bases of the moment map picture. Basically, they showed that the moduli space of Hermite-Einstein connections could be described by means of a symplectic reduction for an appropriate moment map on the set of connections, which is acted on by the gauge group of the bundle. Phrased that way, the Narasimahn-Seshadri theorem and, more generally, the Hitchin-Kobayashi correspondence, can be thought of as infinite-dimensional versions of the Kempf-Ness theorem.

The aim of this thesis is to present an account of the Hitchin-Kobayashi correspondence, explaining the moment map picture and the sense in which it can be seen as an infinite-dimensional version of the Kempf-Ness theorem, as well as the way in which the finite-dimensional version of the Kempf-Ness theorem motivates the correspondence. Detailed proofs of these results are beyond the scope of this thesis. However, for completeness, we will try to sketch them. We will first give a brief account of the vector bundle version of the correspondence both, to motivate the principal bundle, and so that we can compare it with the principal bundle version. We will, however, devote our efforts mostly to the principal bundle version of the correspondence. We have followed an exposition close to that of [49] and [18], but making the corresponding simplifications (which basically consist of ignoring any extra structure). When studying moduli spaces for these structures, rigorous constructions require working with Sobolev completions of several spaces, something that is often dismissed in the literature omitting the technical details. We have tried to specify, at least for the principal bundle version of the theorem, why Sobolev completions are needed to make everything rigorous and how to work with them. The second objetive of the thesis is to set the first steps towards a generalization of the correspondence for principal bundles over non-connected structure groups. Again, this requires both a new Hermite-Einstein-Yang-Mills equation and a new notion of stability, but more importantly, it relies on the work by García Prada, G. Barajas, Gothen and Mundet i Riera in [5] on twisted equivariant bundles over coverings. The key ideas and results concerning this generalization have been taken from a draft of a yet unpublished article that my supervisor kindly shared with me.

As a brief outline of the thesis, in chapter 1 we address the finite-dimensional version of symplectic reduction and the Kempf-Ness theorem. The former is generalized to Banach manifolds in chapter 4. Chapters 2 and 3 introduce connections in the context of vector and principal bundle, and they treat the relation between connections and complex structures; they so-called Chern or Chern-Singer correspondences. Finally, chapter 5 builds on all the previous work and introduces the Hitchin-Kobayashi correspondence.

I have tried to make this thesis as self-contained as possible, an ultimately impossible task, given the sheer amount on concepts that play a role in the Hitchin-Kobayashi correspondence. For the convenience of the reader, and for my own convenience when citing results, I have included 3 appendices that give short introductions to some of the prerequisites needed for the main body of the text. Appendix A is a short introduction to the theory of complex manifolds and vector bundles. It should be read before the main body of the text. Appendix B contains some notions of the theory of Lie groups which may not be covered in a first course on the subject, as well as some tools from [5] which set the basic framework needed to work with principal bundles with non-connected structure group. It should be read before chapter 5 although some of its results are used earlier in the text. Finally, appendix C gives a short introduction to Banach manifolds. It mostly contains relevant definitions, and statements that show that Banach manifolds are sufficiently well-behaved to still satisfy much of the properties of usual finite-dimensional smooth manifolds (with the exception, of course, of the existence of charts of the form $(x^1, \dots x^n)$). It should be read before chapter 4. With my deepest sorrow, and in an effort to keep the length of the thesis manageable, I have omitted a lot of proofs and only sketched others. However, whenever a proof has been omitted, I have tried to give a reference in the literature where it can be found. On the other hand, I have tried to pay special attention to detail in the proofs that I have included. This is a matter of personal taste, but I hope it will be helpful to future readers. I have also included an alphabetical index and an index of notations at the end for the convenience of the reader.

Chapter 1

Finite-dimensional Kempf-Ness theorem

The aim of this chapter is to provide a very brief account of the Kempf-Ness' theorem. The motivation behind including this chapter is the fact that the Hitchin-Kobayashi correspondence, treated in chapter 5 can be seen as an infinite-dimensional version of this theorem. It is important to remark that all of the results of chapter 5 can be stated and proved without ever mentioning moment maps. However, the framework of moment maps and symplectic reduction proves to be very useful, because, together with the results we are about to expose in this chapter, it motivates the correspondence, and hints at a lot of correct results.

In this chapter we will assume basic knowledge of symplectic geometry, Lie theory, and basic projective algebraic geometry. We will recall some basic concepts, but most results here will be stated without proof. Good references on this topics are [59], [24] and [23].

1.1 Symplectic geometry

Recall that, given a real vector space V, a *(linear) symplectic form* is a skew-aymmetric bilinear map $\Omega\colon V\times V\to\mathbb{R}$ such that the induced map $\Omega^\flat\colon V\to V^*$ given by

$$\begin{array}{rccc} \Omega^\flat\colon & V & \longrightarrow & V^* \\ & v & \mapsto & \Omega^\flat(v)\colon \end{array} \begin{array}{ccc} V & \longrightarrow & \mathbb{R} \\ u & \mapsto & \Omega(v,u) \end{array} \tag{1.1}$$

is a vector space isomorphism. We say that the pair (V,Ω) is a symplectic vector space. Observe that, for finite-dimensional spaces, is suffices to check that

$$\{0\}=\ker\Omega^\flat=\{v\in V:\Omega(v,u)=0\ \ \forall\ u\in V\}$$

Symplectic manifolds are smooth manifolds that are equipped with an extra structure: a symplectic form. A symplectic form is a 2-form such that, when restricted to the tangent space at each point, it defines a symplectic skew symmetric bilinear map. To make sense of this, we will start by briefly studying symplectic vector spaces and linear symplectic forms.

The definition of symplectic manifold then arises naturally by asking for each tangent space to be a symplectic vector space, and in a such a way that the corresponding symplectic forms vary smoothly.

Definition 1.1 (Symplectic manifold). A symplectic manifold is a pair (M,ω), where M is a smooth manifold, and $\omega\in\Omega^2(M)$ is a symplectic form. That is, a closed ($\mathrm{d}\omega=0$) 2-form such that

$$\omega_p\colon T_pM\times T_pM\to\mathbb{R}$$

is a linear symplectic form.

This definition is quite similar to the definition of a riemannian manifold (at least in spirit). Indeed, a symplectic manifold is a smooth manifold such that every tangent space is equipped with a skew-symmetric bilinear form that varies smoothly. This idea is quite similar to the definition of a riemannian manifold, where the role of ω is played by the metric. Nonetheless, the symplectic

structure turns out to be far more restrictive than the riemannian one, as we shall see. Observe that every symplectic vector space must have even-dimension (see [59, Th. 1.1]), hence every symplectic manifold is even-dimensional.

Example 1.2. $M = \mathbb{R}^{2n}$ endowed with the symplectic form given by

$$\omega_0 := \sum_{i=1}^{n} \mathrm{d}x^i \wedge \mathrm{d}y^i,$$

is a symplectic manifold. Here $(x^1, \dots x^n, y^1, \dots y^n)$ are the usual coordinates on $\mathbb{R}^{2n}$. In particular, $\mathbb{C}^n \equiv \mathbb{R}^{2n}$ is a symplectic manifold.

Definition 1.3 (Symplectomorphism). Let (M_1, ω_1) and (M_2, ω_2) be two symplectic manifolds. We say that a diffeomorphism (resp. local diffeomorphism) $\varphi\colon M_1 \to M_2$ is a *symplectomorphism* (resp. *local symplectomorphism*) if $\varphi^*\omega_2 = \omega_1$.

Example 1.2 may seem trivial, but it turns out that the symplectic structure is so rigid that every symplectic manifold looks locally like $\mathbb{R}^{2n}$ with its standard symplectic structure. To be more precise, every symplectic manifold is locally symplectomorphic to $\mathbb{R}^{2n}$ endowed with its standard symplectic structure. That is the content of Darboux's theorem, which basically states that given any symplectic manifold (M, ω) we may always find local coordinates in which the symplectic form ω is expressed as a wedge product of the exterior differentials of the coordinate functions.

Theorem 1.4 (Darboux). Let (M, ω) be a symplectic manifold. For any point $p \in M$, there are coordinates $(x^1, \dots x^n, y^1, \dots y^n)$ defined on an open neighborhood of p such that

$$\omega = \sum_{i=1}^{n} \mathrm{d}x^i \wedge \mathrm{d}y^i.$$

Proof. See [36, Prop. 22.13]. □

These coordinates are called *Darboux coordinates* or *canonical coordinates*.

Example 1.5. A non-trivial example of symplectic manifold is the complex projective space $\mathbb{CP}^n$. One way to endow it with a symplectic structure is theorem 1.10 bellow. Another option is to do it directly by defining a symplectic form. This is called the *Fubini-Study form* (see [35, Ex. 8.14]). In particular, for any embedded submanifold $X \subset \mathbb{CP}^n$, the pullback through the inclusion $\imath\colon X \hookrightarrow \mathbb{CP}^n$ induces a closed two-form $\imath^*\omega \in \Omega^2_{\mathbb{R}}(X)$ which turns out to be symplectic. The latter is not true in general: not every embedded submanifold of a symplectic manifold is symplectic, but it is true in this case because $\mathbb{CP}^n$ is actually Kähler (see appendix A). Consider the holomorphic coordinates of $\mathbb{CP}^n$ given by the affine open covering $\{U_i\}_{i=0}^n$, where $U_i = \{[z_0 : z_1 : \dots : z_n] : z_i \neq 0\}$. Recall that we have holomorphic coordinates

$$\varphi_i\colon U_i \to \mathbb{C}^n, [z_0 : \dots : z_i : \dots z_n] \mapsto \left(\frac{z_0}{z_i}, \dots \frac{\hat{z}_i}{z_i}, \dots \frac{z_n}{z_i}\right),$$

where the hat means that the corresponding term is missing. Writting w^j the projection of φ_i onto the j-th coordinate, the coordinate expression of the Fubini-Study form is

$$\omega_{\mathrm{FS}} = \sum_{j,k=1}^{n} \left(\frac{\delta_{jk} \cdot (1 + ||w||^2) - \bar{w}^j \cdot w^k}{(1 + ||w||^2)^2}\right) \cdot \mathrm{d}w^j \wedge \mathrm{d}\bar{w}^k. \tag{1.2}$$

1.1.1 Symplectic reduction

Symplectic reduction is a technique whose origin is found in the formalization of mechanics in the context of symplectic geometry. Symplectic geometry provides an ideal framework to study classical mechanics. Briefly, when studying systems with (holonomic) constrains, we describe them using the cotangent bundle of some smooth manifold, T^*M. It turns out that every cotangent bundle is canonically endowed with a symplectic structure (see [59, Ch. 2]). Sometimes, the system under

study has some kind of symmetry, which is modelled through the action of a Lie group G. Under some assumptions, this action can be used to build a quotient out of M whose dimension is lower, but which still encodes all the physical information of the system. This is called *symplectic reduction*. Of course, although much of the motivation arises from physics, the theory of reduction can still be presented in a completely abstract setting, which we do bellow.

The first ingredient we need is a way to take quotients in the category of smooth manifolds. Observe that quotients are always well-defined in the category of topological spaces, i.e. there is a canonical quotient topology. However, things are quite different in the category of smooth manifolds. When a group acts on a smooth manifold, the quotient space endowed with the quotient topology may not be a topological manifold (for example, it may fail to be Hausdorff), let alone a smooth manifold. However, if we ask some conditions to the action, we can overcome these limitations.

Let G be a Lie group, and M be a smooth manifold, and suppose that G acts on M through the left (right actions are handled analogously). We say that the action is *smooth* if the map $\phi\colon G\times M\to M, (g,p)\mapsto g\cdot p$ is smooth. Equivalently, the action is smooth if for every $g\in G$, the maps $\phi_g\colon M\to M, p\mapsto g\cdot p$ are smooth. Moreover, we say that the action is *proper* if the map $G\times M\to M\times M, (g,m)\mapsto (g\cdot m, m)$ is a proper map.

As we will work extensively with actions of Lie groups on smooth manifolds, let us fix some notation. When a Lie group acts on a smooth manifold, we will denote, unless otherwise stated, the action map by $\phi(g,p)$ or simply $g\cdot p$ for left actions (for right actions we will use $p\cdot g$). We will also denote $\phi_g\colon M\to M, p\mapsto \phi(g,p)$ for $g\in G$, and $\phi^{(p)}\colon G\to M, g\mapsto \phi(g,p)$ for $p\in P$. Both of them are smooth under the assumption that the action is smooth.

Proposition 1.6. Let G be a Lie group that acts smoothly and properly on a smooth manifold M. Then, the quotient map is an open map, and the quotient space is Hausdorff.

Proof. See [36, Lemma 21.2 and Prop. 21.4]. □

This allows to prove a key result for the theory of symplectic reduction: the quotient manifold theorem.

Theorem 1.7 (Quotient manifold theorem). Let G be a Lie group acting smoothly, freely and properly on a smooth manifold M. Then, the orbit space M/G is a topological manifold of dimension equal to $\dim M-\dim G$. Moreover, it admits a unique smooth structure such that the topological quotient map $q\colon M\to M/G$ is a surjective smooth submersion.

Proof. See [36, Th. 21.10]. □

The second ingredient needed to understand symplectic reduction are moment maps, but before diving into the definition, let us recall some facts about representations of Lie groups and Lie algebras. Let G be a Lie group with lie algebra $\mathfrak{g}$. Recall that the *adjoint representation* is the Lie group representation $G\to \mathrm{GL}(\mathfrak{g})$ given by $\mathrm{Ad}_g := (C_g)_*$, where $C_g\colon G\to G, h\mapsto ghg^{-1}$ is the conjugation map. That is, for each $\xi\in\mathfrak{g}$ $\mathrm{Ad}_g\xi\in\mathfrak{g}$ is the unique vector field C_g-related to ξ. If we see the Lie algebra as T_eG, then $\mathrm{Ad}_g=\mathrm{d}_eC_g$. On the other hand, let

$$\begin{array}{rccc}(\cdot,\cdot)\colon & \mathfrak{g}^*\times\mathfrak{g} & \longrightarrow & \mathbb{R}\\ & (\varphi,\xi) & \mapsto & \varphi(\xi)\end{array} \tag{1.3}$$

be the pairing induced by the dual (as a vector space) of the Lie algebra. We define the *coadjoint representation* $\mathrm{Ad}^*\colon G\to\mathrm{GL}(\mathfrak{g}^*), g\mapsto \mathrm{Ad}^*_g$, where Ad^*_g is given by

$$(\mathrm{Ad}^*_g(\varphi),\xi) := (\varphi, \mathrm{Ad}_{g^{-1}}(\xi)).$$

The definition using $\mathrm{Ad}_{g^{-1}}$ instead of Ad_g ensures that we actually get a representation.

Finally, let us introduce some notation:

1. Consider a Lie group G that acts smoothly on a smooth manifold M. Let $\xi \in \mathfrak{g}$. We define the *infinitesimal generator of the one parameter subgroup* $\exp(t\xi)$, also called *Killing vector field* generated by $\xi \in \mathfrak{g}$ is given by the vector field

$$X_{\xi,p} := \frac{\mathrm{d}}{\mathrm{d}t}\bigg|_{t=0} p \cdot \exp(t\xi) = \mathrm{d}_e \phi^{(p)}(\xi). \tag{1.4}$$

2. Let $\mu\colon M \to \mathfrak{g}^*$ be a smooth map. For each $\xi \in \mathfrak{g}$, we set $\mu^\xi\colon M \to \mathbb{R}, p \mapsto (\mu(p), \xi)$. We also say that the vector $X_{\xi,p}$ is the infinitesimal action of ξ on p.

3. When a Lie group G acts on a symplectic manifold (M,ω), we say that the action is symplectic, or that G *acts by symplectomorphisms*, if ϕ_g is a symplectimorphism for every $g \in G$. In that case, the action defines group homomorphism $G \to \mathrm{Diff}(M)$, whose image lies in $\mathrm{Sympl}(M,\omega)$ (the groups of diffeomorphisms and symplectomorphisms respectively).

Definition 1.8. Let (M,ω) be a symplectic manifold, and let G be a Lie group that acts smoothly, freely and properly by symplectmorphisms on M. The action is said to be *Hamiltonian* if there exists a smooth map, called *moment map* $\mu\colon M \to \mathfrak{g}^*$ satisfying:

1. $\mathrm{d}\mu^\xi = i_{X_\xi}\omega$ for all $X \in \mathfrak{g}$,
2. μ is equivariant with respect to the action and coadjoint representation. That is, for every $g \in G$, we have $\mu \circ \phi_g = \mathrm{Ad}^*_g \circ \mu$.

$$\begin{array}{ccc} M & \xrightarrow{\phi_g} & M \\ {\scriptstyle\mu}\downarrow & & \downarrow{\scriptstyle\mu} \\ \mathfrak{g}^* & \xrightarrow{\mathrm{Ad}^*_g} & \mathfrak{g}^* \end{array}$$

The quadruple (M,ω,G,μ) is called a *Hamiltonian space* or *Hamiltonian G-space*.

The first condition of the definition is equivalent to the following $(\mathrm{d}_p\mu(v),\xi) = \omega_p(X_{\xi,p}, v)$ for all $p \in M$, $v \in T_pM$ and $X \in \mathfrak{g}$. Moment maps may or may not exist for a symplectic action. However, if they exist, they are unique up to $\mathfrak{g}^*$ constant factor. revisar

Remark 1.9. The fact that μ is called a moment map is due to the fact that it can be seen as a generalization of both the linear and angular momentums. Perhaps more specifically, if we consider $\mathbb{R}^6$ as a symplectic manifold, thinking of it as describing the unconstrained movement of a point-particle in 3-dimensional euclidean space, then the first 3 coordinates should describe position, and the last 3 should describe linear momentum. After identifying $\mathbb{R}^3$ to its Lie algebra (and with its dual via the pairing given by the euclidean inner product), a suitable moment map is the projection to the last 3 coordinates, which represents linear momentum. Here, $\mathbb{R}^3$ is the Lie group acting on the $\mathbb{R}^3$ by translating the first coordinates. Similarly, if we let $\mathrm{O}(3)$ act on $\mathbb{R}^6$ by rotating the first and second triplets of coordinates on their own, a suitable moment map is $\mu(\boldsymbol{x},\boldsymbol{p}) = \boldsymbol{x} \times \boldsymbol{p}$, which is angular momentum.

We are finally ready to state the main result in the theory of finite-dimensional symplectic reduction.

Theorem 1.10 (Masden-meyer-Weinstein). Let (M,ω,G,μ) be a Hamiltonian G-space. Then, $M_{\mathrm{red}} := \mu^{-1}(0)/G$ is a symplectic manifold, whose symplectic form ω_{red} satisfies

$$i^*\omega = q^*\omega_{\mathrm{red}},$$

where $i\colon \mu^{-1}(0) \hookrightarrow M$ is the inclusion, and $q\colon \mu^{-1}(0) \to \mu^{-1}(0)/G$ the quotient map.

We will not give a detailed proof here because chapter 4 gives a proof of the theorem for Banach manifolds, which can be easily adapted (in fact, simplified) to the finite-dimensional case. A direct proof for the finite-dimensional case can be found in [59, Part. IX], however, it is worth to make some comments on the result. Firstly, $\mu^{-1}(0)$ is stable under the G-action. That is, if $p \in \mu^{-1}(0)$, then $G \cdot p \in \mu^{-1}(0)$. Indeed,

$$\mu(g \cdot p) = \mathrm{Ad}^*_{g^{-1}} \circ \mu(p) = \mathrm{Ad}^*_{g^{-1}}(0) = 0. \tag{1.5}$$

Moreover, $0 \in \mathfrak{g}^*$ is a regular value of the smooth map μ, which shows that $\mu^{-1}(0) \subset M$ is an embedded submanifold. It can be shown that $\dim \mu^{-1}(0) = \dim M - \dim G$, and $T_p\mu^{-1}(0) = \ker(\mathrm{d}_p\mu) \subset$

T_pM. What is more, $G \cdot p$ is an embedded submanifold of $\mu^{-1}(0)$ for all $p \in \mu^{-1}(0)$, and the tangent spaces of all the manifolds involved relate as follows:

$$T_{q(p)}M_{\text{red}} \cong T_p\mu^{-1}(0)/T_p(G \cdot p).$$

1.2 Geometric invariant theory

Kempf-Ness' theorem relates GIT (geometric invariant theory) to symplectic geometry; two braches of geometry that seem completely unrelated at first glance, but end up being intertwined. GIT is the theory developed by David Mumford in is book [42] which basically provides the framework needed to define quotients is algebraic geometry. We will not follow his approach, because he develops his theory in the language of schemes, and that is a level of generality beyond what we need. A good reference for GIT that requires less abstraction is [24]. We will follow this source, and specially [62]. Our objetive for this section is to give a brief account of projective GIT theory that requires minimal previous knowledge. We choose this style of exposition because the Kempf-Ness theorem, or rather, seeking an infinite-dimensional generalization, is what motivates the Hitchin-Kobayashi correspondence.

Defining an appropriate concept of quotient in algebraic geometry turns out not to be as straightforward as it is in other categories, such as topological spaces, where quotients always exist and are defined as set of orbits. The problem with algebraic varieties (we will focus on affine and projective varieties) is that group actions will produce orbit sets that cannot be given the structure of an algebraic variety in a reasonable way. To overcome this difficulty, we have to give up the idea that quotients should always be orbit spaces, and look at more abstract definitions. This is seen in the following example (this is worked out for affine varieties, but it will illustrate our point).

Example 1.11. Consider the action of $\mathbb{C}^*$ on $\mathbb{C}^2$ given by

$$\mathbb{C}^* \times \mathbb{C}^2 \longrightarrow \mathbb{C}^2, \quad ((t, t^{-1}), (x, y)) \mapsto (tx, t^{-1}y).$$

It is easy to see that the orbits of this action are:

1. Hyperbolas $XY = \alpha, \alpha \in \mathbb{C}^*$.
2. The punctured x and y axis, that is, the axis minus the origin.
3. The origin.

The punctured axis are not closed, since the origin is in their Zariski-closures. When taking quotients in any category, it is reasonable to ask the quotient map to be a morphism in the corresponding category. Hence, we would like to see $\mathbb{C}^2/\mathbb{C}^*$ as an affine algebraic variety (the zero set of a family of polynomials) in such a way that $q\colon \mathbb{C}^2 \to \mathbb{C}^2/\mathbb{C}^*$ was a morphism of affine varieties. In particular, continuous. However, points in algebraic varieties are always closed, so $\mathbb{C}^2/\mathbb{C}^*$ cannot have the structure of an algebraic variety. Let us more precise. We would like $\mathbb{C}^2/\mathbb{C}^* = \{[(0,0)], [(1,0)], [(0,1)], [(1,\alpha)] : \alpha \in \mathbb{C}^*\}$ to be an affine variety. If q was continuous, we would have

$$[(0,0)] = q\big((0,0)\big) \in q\left(\overline{x \text{ axis} \setminus \{(0,0)\}}\right) \subset \overline{q\left(x \text{ axis} \setminus \{(0,0)\}\right)} = \overline{[(1,0)]}.$$

Hence, $[(0,1)]$ is not a closed point, but points in affine varieties are closed.

The key idea that one should get from the previous example is that equivalence classes are not good quotients in algebraic geometry.

Recall from basic projective algebraic geometry that we define a projective variety as the zero set of a family of homogeneous polynomials, i.e. $X = V(F_1, \dots F_r) \subset \mathbb{CP}^n$ where $F_i \in \mathbb{C}[X_0, \dots X_n]$ is homogeneous. If we ask X to be irreducible, then $I(X) := \{F \in \mathbb{C}[X_0, \dots X_n] : F(x) = 0 \ \ \forall\, x \in X\}$ is prime. Then, we define the *homogeneous coordinate ring*

$$R(X) := \mathbb{C}[X_0, \dots X_n]/I(X),$$

which is a graded ring graded by degree. Every finitely generated graded C-algebra that can be generated by elements of degree one gives rise to a projective variety as follows: first, choose such a set

of generators $r_0, \dots r_n \in R$, and consider the $\mathbb{C}$-algebra homomorphism given by $\varphi\colon \mathbb{C}[X_0, \dots X_n] \to R, X_i \to r_i$. Let $I := \ker \varphi$. Then, by the first isomorphism theorem we have $R \cong \mathbb{C}[X_0, \dots X_n]/I$, and we define $\mathrm{Proj}(R) := V(I) = \{x \in \mathbb{CP}^n : F(x) = 0 \ \ \forall\, F \in I\}$. This is a projective variety. The choice of generators is irrelevant as different choices yield isomorphic varieties and, in fact, if we see these varieties as subsets of $\mathbb{CP}^n$ with the induced topology obtained from the usual quotient topology on complex projective space, different realizations of the same algebra are homeomorphic.

Consider now a complex reductive group G (we refer the reader to appendix B for the relevant notions), and suppose that G acts on a projective variety through a representation $\rho\colon G \to \mathrm{GL}(n+1, \mathbb{C})$. Formally, the action is given by $g \cdot [v] := [\rho(g)(v)]$ where $[\cdot]$ denotes the equivalence class in complex projective space. Equivalently, G acts through a representation in the affine cone associated to X. We call these actions *linear*. This induces an action of G on $\mathbb{C}[X_0, \dots X_n]$ given by

$$G \times \mathbb{C}[X_0, \dots X_n] \longrightarrow \mathbb{C}[X_0, \dots X_n], \ g \cdot (F(X)) \to F(g^{-1}(X)).$$

By a theorem of Nagata [45], the ring $R(X)^G := \{F \in \mathbb{C}[X_0, \dots X_n] : g \cdot F = F\}$ is finitely generated, and one may show that there is an algebra

$$\tilde{R}(X)^G = \bigoplus_{k \geq 0} R(X)^G_{dk}$$

for some $d \geq 0$ that is generated by elements of degree d. Changing the grading to make degree d the new degree one, $\tilde{R}(X)^G$ is generated by degree one elements. Here, $R(X)^G_k$ is the degree k piece defined by the grading.

Definition 1.12 (Projective GIT quotient). Let G be a complex reductive group that acts linearly on a projective variety $X \subset \mathbb{CP}^n$. The GIT quotient is the projective variety $X/\!\!/G := \mathrm{Proj}\left(\tilde{R}(X)^G\right)$.

The double bar notation is used to emphasize that this is generaly not a usual quotient in the sense of equivalence classes for an action. Right now it is quite unclear why we choose to name this a quotient, or what the geometric meaning of this set is, so we will try to answer those questions. To answer the first one, choose $F_0, \dots F_n$ degree one generators of $\tilde{R}(X)^G$, and define a "map"

$$\varphi\colon X \dashrightarrow \mathbb{CP}^n, \ x \mapsto [F_0(x) : \dots : F_n(x)]. \tag{1.6}$$

This is well-defined only points $x \in X$ that do not annihilate all generators, i.e. there is at least one element in $\tilde{R}(X)^G$ that does not vanish on x. The fact that it is well-defined follows from the fact that all the polynomials have the same degree. Observe that the image of φ is precisely $X/\!\!/G$. This follows from the definition of Proj and the fact that

$$\mathrm{Proj}\left(\tilde{R}(X)^G\right) = \{x \in \mathbb{CP}^n : Q(x) = 0 \ \ \forall\, Q \ \text{ such that } \ Q(F_0, \dots F_n) = 0\}.$$

Definition 1.13. Let G be a complex reductive group that acts linearly on a complex projective variety $X \subset \mathbb{CP}^n$.

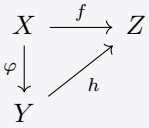
1. A point $x \in X$ is *semistable* if there is $F \in \tilde{R}(X)^G$ such that $F(x) \neq 0$.
2. A point $x \in X$ is *stable* if the orbit $G \cdot x$ is closed in X^{ss} and the stabilizer G_x if finite.
3. A point $x \in X$ is *polystable* if the orbit $G \cdot x$ is closed in X^{ss}.

Else, we say that x is *unstable*. The sets of stable, semistable and polystable points are denoted, respectively, X^s, X^{ss} and X^{ps}.

It is now appropriate to introduce the following definition.

Definition 1.14 (Categorical quotient). Let X be a projective variety, and G be a complex reductive group acting linearly on X. A *categorical quotient* for the action G is a pair (φ, Y), where Y is another variety, and $\varphi\colon X \to Y$ is a G-invariant morphism of varieties such that the following universal property is satisfied:

For every other G-invariant morphism $f\colon X \to Z$ there exist a unique morphism $h\colon Y \to Z$ such that the following diagram is commutative. Making some abuse of notation, it is common to say that φ is the quotient.



Then:

1. $\varphi\colon X^{ss} \to X/\!\!/G$ is a categorial quotient for the restricted action (in fact a good quotient, see [**24**, Prop. 4.8]). This justifies calling φ a quotient.
2. The quotient $X/\!\!/G$ parametrizes orbits for which there is at least one non-vanishing element of $\tilde{R}(X)^G$.
3. φ identifies orbits that cannot be distinguished by elements of $\tilde{R}(X)^G$. Observe in (1.6) that φ maps points in the same orbit to the same point, but it may map points in different orbits to the same orbit as well.
4. X^s and X^{ss} are open subsets of X.
5. The quotient $X/\!\!/G$ is the quotient (in the sense of equivalence classes) of X^{ss} by the equivalence relation that identifies x_1 and x_2 if and only if $\overline{G\cdot x_1} \cap \overline{G\cdot x_2} \cap X^{ss} \neq \varnothing$. More rigorously, the letter is equivalent to $\varphi(x_1) = \varphi(x_2)$, see [**24**, Th. 4.11].
6. The restriction of the action to X^s yields what is called a *geometric quotient*, i.e. a quotient that is indeed a set of equivalence classes.

The different definitions of stability are now clear from a geometric, but in practice it is quite difficult to decide whether a point is stable, semistable, polystable or unstable. To do so, we have the Hilbert-Mumford criterion. First, by a one parameter subgroup we mean a nontrivial group homomorphism $\mathbb{C}^* \to G$.

Let $\lambda\colon \mathbb{C}^* \to G$ be a one-parameter subgroup, where G is a complex reductive Lie group acting linearly on a projective manifold X. For any $x \in X$ the *weight* $w(x,\lambda)$ is defined as follows: first, let $y := \lim_{t\to 0} \lambda(t)\cdot x$. By the limit here we mean the equivalence class of the limit of any representative of the point, that is, the limit is taken in $\mathbb{C}^n$, and one can show that it always exists. Then, y is fixed by λ, so there exists an integer $w = w(x,\lambda)$ such that $\lambda(t)\cdot\hat{y} = t^{-w}\hat{y}$ where $\hat{Y}$ is any representative y (see [**24**, Sec. 5.2]).

Theorem 1.15 (The Hilbert-Mumford criterion). Let G be a complex reductive group that acts linearly on a projective variety $X \subset \mathbb{CP}^n$.

1. $x \in X^{ss}$ (resp. X^s) if and only if $w(x,\lambda) \geq 0$ (resp. $w(x,\lambda) > 0$) for all one-parameter subgroups.
2. $x \in X^{ps}$ if and only if $w(x,\lambda) > 0$ for all one-parameter subgroups λ for which $\lim_{t\to 0}\lambda(t)\cdot x \notin G\cdot x$.

Proof. See [**42**, Th. 2.1]. □

Finally, [**24**, Lemma 4.16] states that for every $x \in X^{ss}$, the set $\overline{G\cdot x}$ contains a unique polystable orbit. This result, together with point 5 above implies that the quotient $X/\!\!/G$ can be thought of as parametrizing polystable orbits. That is, there is a bijection $X/\!\!/G \leftrightarrow X^{ps}/G$, the latter being the usual set of equivalence classes. This uses the fact [**24**, Cor. 2.39] that the preimage of any point through φ contains a unique closed orbit.

1.2.1 The Kempf-Ness theorem

We refer the reader to appendix B for the basics on complex Lie groups and complexifications of real compact Lie groups. As we recall in the appendix, complex reductive Lie groups are characterized precisely by the fact that they are complexifications of compact real Lie groups.

We have defined a projective set $X \subset \mathbb{CP}^n$ as the zero set of a collection of homogeneous polynomials. Observe that $\mathbb{CP}^n$ is also a complex manifold (in the sense of complex differential geometry), and we may ask if projective sets can be equipped with a complex smooth structure such that they become embedded complex submanifolds of $\mathbb{CP}^n$. It turns out that for this to be true, we have to impose some extra conditions on the polynomials defining X. If $X = V(I)$, where $I = (F_1, \dots F_r)$ is a radical ideal, and

$$\mathrm{Jac}(F_1, \dots F_r)(z) = \begin{pmatrix} \frac{\partial F_1}{\partial X_0}(z) & \frac{\partial F_1}{\partial X_1}(z) & \dots & \frac{\partial F_1}{\partial X_n}(z) \\ \frac{\partial F_2}{\partial X_0}(z) & \frac{\partial F_2}{\partial X_1}(z) & \dots & \frac{\partial F_2}{\partial X_n}(z) \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial F_r}{\partial X_0}(z) & \frac{\partial F_r}{\partial X_1}(z) & \dots & \frac{\partial F_r}{\partial X_n}(z) \end{pmatrix} \tag{1.7}$$

has constant rank for every $z \in X$, then it is a complex manifold. Observe that this condition is well-defined and does not depend on the representative we choose for z. Indeed, choosing another representative λz yields a matrix obtained from (1.7) by multiplying each row by a power of λ, which does not change the rank. We may then show that if $s = \mathrm{rank}(\mathrm{Jac}(F_1, \dots F_r))$, and $d := n - s$, then X is a complex manifold of complex dimension $\dim_{\mathbb{C}}(X) = d$ (see [**44**, Prop. 5.42])

Suppose now that $\mathrm{U}(n+1)$ acts on $\mathbb{CP}^n$ through

$$\mathrm{U}(n+1) \times \mathbb{CP}^n \to \mathbb{CP}^m, (A, [z]) \mapsto [A \cdot z].$$

One can the show that, with respect to the Fubini-Study metric, the action is simplectic, and in fact Hamiltonian. The moment map is given by $\tilde{\mu} \colon \mathbb{CP}^n \to \mathfrak{u}(n+1)^*, [z] \mapsto \mu([z])$ and

$$\langle \tilde{\mu}([z]), A \rangle := \frac{\mathrm{Tr}(z^\dagger \cdot A \cdot z)}{2\mathrm{i} \cdot (z^\dagger \cdot z)} \qquad A \in \mathfrak{u}(n+1) \tag{1.8}$$

where $z^\dagger$ is the transpose conjugate [**24**, Ex. 8.10].

More generally, consider a reductive complex group $G = K^{\mathbb{C}}$ for some maximal compact subgroup. Suppose that G acts on a smooth projective set $X \subset \mathbb{CP}^n$ via a representation $\rho \colon G \to \mathrm{GL}(n+1, \mathbb{C})$. Restricting the action, there is also an action of K on $\mathbb{CP}^n$. We want to relate the GIT quotient with the symplectic quotient given by the Marsden-Weinstein-Meyer theorem. Thus, we take a smooth complex variety $X \subset \mathbb{CP}^n$, which is also a complex manifold, and therefore a symplectic manifold (the symplectic form is the pullback of the Fubini-Study form, see the comments in example 1.2). On the one hand, we may form the GIT quotient $X//G$, and on the other, if the restricted action of K was symplectic, we could consider the symplectic quotient. Using Weyl's unitary trick, we may choose coordinates appropriately, such that the representation is unitary $\rho \colon K \to \mathrm{U}(n+1)$ (see). In that case, the action of $\mathbb{C}$ on both X and $\mathbb{CP}^n$ is symplectic, and actually hamiltonian. A moment map may be given as follows: consider $\mathrm{d}_{e_K}\rho \colon \mathfrak{k} \to \mathfrak{u}(n+1)$. The dual map is $(\mathrm{d}_{e_k}\rho)^* \colon \mathfrak{u}(n+1)^* \to \mathfrak{k}^*$. Hence, we can define $\mu := (\mathrm{d}_{e_K}\rho)^* \circ \tilde{\mu}$, where $\tilde{\mu}$ is given in (1.8). Explicitly,

$$\langle \mu([z]), B \rangle := \frac{\mathrm{Tr}(z^\dagger \cdot \mathrm{d}_{e_K}\rho(B) \cdot z)}{2\mathrm{i}\ \ (z^\dagger \cdot z)}.$$

Now that both the symplectic and the GIT quotients make sense, their relation is given by the Kempf-Ness theorem.

Theorem 1.16 (Kempf-Ness). Let $G = K^{\mathbb{C}}$ be a complex reductive group acting linearly on a smooth complex projective variety, $X \subset \mathbb{CP}^n$. Suppose that its maximal compact subgroup $\mathbb{C}$ is connected and acts symplectically on X, and let $\mu\colon X \to \mathfrak{k}^*$ be the associated moment map. Then

1. $G\mu^{-1}(0) = X^{ps}$.
2. If $x \in X$ is polystable, then its orbit $G\cdot x$ meets $\mu^{-1}(0)$ in a single K-orbit, i.e. $K\cdot x \cap \mu^{-1}(0)$ is contained in a single G-orbit.
3. $x \in X$ is semistable if and only if its orbit's closure $\overline{G\cdot x}$ meets $\mu^{-1}(0)$.

Observe that point 1 means that polystable points contain a zero of the moment map on their orbit.

Corollary 1.17. In the above conditions, the inclusion $\mu^{-1}(0) \subset X^{ss}$ induces a homeomorphism

$$\mu^{-1}(0)/K \to X/\!\!/G.$$

Equivalently, we have a homeomorphism $\mu^{-1}(0)/K \cong X^{ps}/G$. We should remark that to speak about a homeomorphism, we see $X/\!\!/G$ endowed with the subspace topology it inherits from the usual topology of $\mathbb{CP}^n$.

Chapter 2

Connections on vector bundles

In this chapter, we develop the theory of connections over complex vector bundles. The notion of a connection itself is familiar from riemannian geometry, and the definition can be adapted to abstract vector bundles with minor modifications. The key idea behind connections is to provide a way of differenciating sections of vector bundles in a coordinate-independent way. The word connection is used with different meanings in the literature (all of them related of course), so we devote a part of the chapter to study these definitions. We also define the curvature of a connection from different points of view. The first definition we give should be familiar from riemannian geometry, but we will introduce a new way to think about curvatures as $\mathrm{End}(E)$-valued forms. Once these basic concepts are introduced, we give a proof of the existence of the Chern connection; a connection that is simultaneously compatible with a hermitian inner product and the complex structure. We also define the First Real Chern class and study some properties of the curvature of the Chern connection.

Definition 2.1. Let $\pi\colon E \to M$ be a smooth complex vector bundle of rank r over a smooth manifold M. A *connection* on E is a map

$$\begin{array}{rccc} \nabla\colon & \Gamma(T_{\mathbb{C}}M) \times \Gamma(E) & \longrightarrow & \Gamma(E) \\ & (X, \sigma) & \mapsto & \nabla_X \sigma \end{array}$$

satisfying the following properties:

1. ∇ is $\mathscr{C}^\infty(M, \mathbb{C})$-linear on its first component.
2. For every $f \in \mathscr{C}^\infty(M; \mathbb{C})$, and $\sigma \in \Gamma(E)$,

$$\nabla_X(f \cdot \sigma) = X(f) \cdot \sigma + f \cdot \nabla_X \sigma.$$

The expression $\nabla_X \sigma$ is usually called the *covariant derivative of σ in the direction of X*. One can show that $(\nabla_X \sigma)_p$ for $p \in M$ depends only on the value of X at p and the value of σ in an arbitrarily small neighborhood of p, so for any open subset $U \subset M$, restriction of the connection defines a connection on the restriction of the bundle $E|_U$ (which is a smooth manifold of its own). See [34, Prop. 4.3] for a proof for connections on the real tangent bundle of a manifold, which applies almost verbatim to this case.

Sometimes, one is given the following definition of a connection:

Definition 2.2. Let $\pi\colon E \to M$, a *connection* on E is a $\mathbb{C}$-linear map $\nabla\colon \Gamma(E) \to \Gamma(T^*_{\mathbb{C}}M \otimes E)$ such that for every $f \in \mathscr{C}^\infty(M; \mathbb{C})$, $\sigma \in \Gamma(E)$, we have

$$\nabla(f \cdot \sigma) = \mathrm{d}f \otimes \sigma + f \cdot \nabla\sigma \tag{2.1}$$

The last definition is sometimes called a *total covariant derivative*. Observe that a connection in the sense of definition 2.1 induces a total covariant derivative: for any $\sigma \in \Gamma(E)$, we define $\nabla\sigma \in \Gamma(T^*M \otimes E)$ to be the section such that $\nabla\sigma(X) = \nabla_X \sigma$ (we are using here the vector bundle isomorphism $T^*_{\mathbb{C}}M \otimes E \cong \mathrm{Hom}(T_{\mathbb{C}}M, E)$). Reciprocally, a total covariant derivative induces a connection by simply defining $(X, \sigma) \mapsto \nabla\sigma(X) =: \nabla_X \sigma$. By an abuse of notation, we will refer to both

constructions by the name connection. When we consider real vector bundles, we may still define connections analogously (we do not consider the complexified tangent and cotangent bundles in that case). Moreover, observe that $\Omega^1_{\mathbb{C}}(M;E) \cong \Gamma(\bigwedge^1 T^*_{\mathbb{C}}M \otimes E) = \Gamma(T^*_{\mathbb{C}}M \otimes E)$, hence a connection is a map $\nabla\colon \Omega^0_{\mathbb{C}}(M;E) \to \Omega^1_{\mathbb{C}}(M;E)$ satisfying the "Leibniz rule" of definition 2.2. This is the definition one can find in [26] and [28].

Remark 2.3 (Coordinate expression of connections). Consider a local frame $(s_1, \dots s_r)$ of a complex vector bundle E defined on some open set $U \subset M$. For each $1 \leq j \leq r$, $\nabla s_j \in \Omega^1_{\mathbb{C}}(E)$, and hence it may be written as

$$\nabla s_j = \omega^i_j \otimes s_i \tag{2.2}$$

for some $\omega^i_j \in \Omega^1_{\mathbb{C}}(M)$ that are called *connection one-forms*. If we consider a section $\xi = \xi^j \cdot s_j \in \Gamma(E)$, then the coordinate expression of its image is

$$\nabla\xi = \nabla(\xi^j s_j) = \mathrm{d}\xi^j \otimes s_j + \xi^j \cdot \nabla s_j = \left(\mathrm{d}\xi^k + \xi^j \cdot \omega^k_j\right) \otimes s_k.$$

Equivalently,

$$\nabla_X \xi = (X(\xi^k) + \xi^k \cdot \omega^k_j(X)) \cdot s_k.$$

A section ξ is said to be *parallel* if $\nabla\xi \equiv 0$. Observe that, with respect to the trivialization defined by the local frame $(s_1, \dots s_r)$ (proposition A.12), the section ξ can be identified with the smooth map $M \to \mathbb{C}^r, m \mapsto (\xi^1(m), \dots \xi^r(m))$. Then, $\Phi(\nabla\xi)$ (extending Φ to the tensor product $T^*_{\mathbb{C}}M \otimes E$) is given by $(\mathrm{d}\xi^1 + \xi^k \cdot \omega^1_k, \dots \mathrm{d}\xi^r + \xi^k \cdot \omega^r_k)$, which we may write as $\mathrm{d}\xi + \omega \cdot \xi$ for a matrix of one-forms $\omega = (\omega^j_k)$. This is sometimes stated by saying that, locally, every connection is $\mathrm{d} + A$ for some matrix of one-forms.

Moreover, if we choose another local frame $(\tilde{s}_1, \dots \tilde{s}_r)$ whose domain of definition overlaps with that of $(s_1, \dots s_r)$, proposition A.13 claims that there are smooth functions τ^ℓ_m such that $\tilde{s}_k = \tau^j_k \cdot s_j$. Then, a straightforward calculation (see [35, Page 195]) shows that the connection one-forms with respect to the new frame relate to the original ones by

$$\tilde{\omega} = (\tau^{-1})^p_j \cdot \mathrm{d}\tau^j_k + (\tau^{-1})^p_\ell \cdot \tau^j_k \cdot \omega^\ell_j. \tag{2.3}$$

Finally, observe that non-trivial connections always exist. A quick way of obtaining one is the following: choose an open cover of M by domains of local frames $\{U_\alpha\}_{\alpha\in A}$, and let $\{\varphi_\alpha\}_{\alpha\in A}$ be a partition of unity subordinate to the cover. On each U_α define a connection ∇^α by setting all connection one-forms to zero. Then,

$$\nabla := \sum_\alpha \varphi_\alpha \cdot \nabla^\alpha \tag{2.4}$$

is a non-zero connection on E.

2.1 Curvature of a connection

Every connection has an associated curvature form. One may find this object introduced in different equivalent ways in the literature. Some texts ([28] for instance) define the curvature by $\nabla \circ \nabla$. If we take this expression and definition 2.1 literally, it does not make sense. Indeed, ∇ is not an endomorphism; it can not be composed with itself. To give this definition properly, one has to extend ∇ to a family of operators $\mathrm{d}_\nabla\colon \Omega^p_{\mathbb{C}}(M;E) \to \Omega^{p+1}_{\mathbb{C}}(M,E)$. This is done in theorem 2.4 bellow. However, it is quite useful to see the curvature as an $\mathrm{End}(E)$-valued two-form, and that is not immediate from the definition $\mathrm{d}_\nabla \circ \mathrm{d}_\nabla$. To see the curvature that way, let us work with reference frames. Let $(s_1, \dots s_r)$ be a smooth local frame for E. Define the *curvature two forms* by

$$\Omega^k_j := \mathrm{d}\omega^k_j + \omega^k_\ell \wedge \omega^\ell_j. \tag{2.5}$$

Then, we define, locally, the curvature as

$$F_\nabla = \Omega^k_j \otimes s_k \otimes t^j. \tag{2.6}$$

, where $(t^1, \dots t^r)$ is the dual frame to $(s_1, \dots s_r)$. Since $E \otimes E^* \cong \mathrm{End}(E)$, we may see F_∇ as a (locally defined) $\mathrm{End}(E)$-valued two-form. However, if we choose another frame, and we use the transformation equation given by (2.3), a straightforward (but calculations) shows that $F_\nabla = \tilde{\Omega}^k_j \otimes$

$\tilde{s}_k \otimes \tilde{t}^j$. Hence, equation (2.6) defines F_∇ globally. We can now see how this relates to the operators d_∇.

Theorem 2.4 (The Exterior Covariant Derivative). Let M be a complex manifold, and $\pi\colon E \to M$ be a smooth complex vector bundle over M. Let ∇ be a connection on E. Then, there are unique operators called *exterior covariant derivatives* $\mathrm{d}_\nabla\colon \Omega^q_{\mathbb{C}}(M;E) \to \Omega^{q+1}_{\mathbb{C}}(M;E)$ for each $q \geq 0$, satisfying:

1. For every $\sigma \in \Gamma(E) = \Omega^0_{\mathbb{C}}(M;E)$, $\mathrm{d}_\nabla \sigma = \nabla\sigma$.
2. For every $\alpha \in \Omega^q_{\mathbb{C}}(M)$ and $\beta \in \Omega^p_{\mathbb{C}}(M;E)$,
$$\mathrm{d}_\nabla(\alpha \wedge \beta) = \mathrm{d}\alpha \wedge \beta + (-1)^q \alpha \wedge \mathrm{d}_\nabla \beta.$$
3. For every $\alpha \in \Omega^p_{\mathbb{C}}(M;E)$, $\mathrm{d}^2_\nabla \alpha = F_\nabla \wedge \alpha \in \Omega^{p+2}_{\mathbb{C}}(M;E)$.

Proof. First, observe that, with some abuse of notation, we denote all the exterior covariant derivatives the same (instead of something more reigorous like d^q_∇) to keep notation concise.

Now, define, for every $\sigma \in \Gamma(E)$, $\mathrm{d}_\nabla\sigma = \nabla\sigma$. For higher order E-valued forms, we are going to give an expression that depends on the chosen local frame, and then check that the expression is the same for any other local frame (this is similar to the way one constructs the exterior differential). Hence, let $(s_1, \dots s_r)$ be a local reference, and let $\alpha \in \Omega^q_{\mathbb{C}}(M;E)$ whose local expression is given by $\alpha = \alpha^j \otimes s_j$. We define $\mathrm{d}_\nabla\alpha := \mathrm{d}\alpha^j \otimes s_j + (-1)^q \alpha^j \wedge \mathrm{d}_\nabla s_j$. To check that this expression works for any local frame, let $(\tilde{s}_1, \dots \tilde{s}_r)$ be another local frame. In the intersection of the corresponding open sets, by proposition A.13 $\tilde{s}_k = \tau^j_k s_j$ for some smooth maps τ^j_k, such that $\tau = (\tau^j_k) \in \mathrm{GL}(n,\mathbb{C})$. Let $\tilde{\alpha}^j := (\tau^{-1})^j_i \cdot \alpha^i$. Then, we may write

$$s = \alpha^j \otimes s_j = \alpha^j \otimes \left((\tau^{-1})^k_j \cdot \tilde{s}_k\right) = \left((\tau^{-1})^k_j \cdot \alpha^j\right) \otimes \tilde{s}_k = \tilde{\alpha}^k \otimes \tilde{s}_k.$$

Thus,

$$\begin{aligned}
\mathrm{d}_\nabla\alpha &= \mathrm{d}\alpha^j \otimes s_j + (-1)^q \cdot \alpha^j \wedge \nabla s_j \\
&= \mathrm{d}(\tau^j_k \cdot \tilde{\alpha}^k) \otimes \left((\tau^{-1})^\ell_j \cdot \tilde{s}_\ell\right) + (-1)^q \cdot (\tau^j_k \cdot \tilde{\alpha}^k) \wedge \nabla((\tau^{-1})^\ell_j \cdot \tilde{s}_\ell) \\
&= (\mathrm{d}\tau^j_k \wedge \tilde{\alpha}^k) \otimes ((\tau^{-1})^\ell_j \cdot \tilde{s}_\ell) + (\tau^j_k \wedge \mathrm{d}\tilde{\alpha}^k) \otimes ((\tau^{-1})^\ell_j \cdot \tilde{s}_\ell) \\
&\quad + (-1)^q \cdot (\tau^j_k \cdot \tilde{\alpha}^k) \wedge \left[\mathrm{d}(\tau^{-1})^\ell_j \cdot \tilde{s}_\ell + (\tau^{-1})^\ell_j \cdot \nabla\tilde{s}_\ell\right] \\
&= (\mathrm{d}\tau^j_k \wedge \tilde{\alpha}^k) \otimes ((\tau^{-1})^\ell_j \cdot \tilde{s}_\ell) + \mathrm{d}\tilde{\alpha}^k \otimes \delta^\ell_k \cdot \tilde{s}_\ell + (-1)^q \cdot \left[(\tau^j_k \cdot \tilde{\alpha}^k) \wedge \mathrm{d}(\tau^{-1})^\ell_j \cdot \tilde{s}_\ell + \tilde{\alpha}^k \wedge \delta^\ell_k \cdot \nabla\tilde{s}_\ell\right] \\
&\overset{(2)}{=} \mathrm{d}\tilde{\alpha}^k \otimes \tilde{s}_k + (-1)^q \cdot \tilde{\alpha}^k \otimes \nabla\tilde{s}_k.
\end{aligned}$$

In (2), we used

$$\begin{aligned}
&(\mathrm{d}\tau^j_k \wedge \tilde{\alpha}^k) \otimes ((\tau^{-1})^\ell_j \cdot \tilde{s}_\ell) + (-1)^q \cdot (\tau^j_k \cdot \tilde{\alpha}^k) \wedge \mathrm{d}(\tau^{-1})^\ell_j \cdot \tilde{s}_\ell \\
&= \left[\mathrm{d}\tau^j_k \cdot (\tau^{-1})^\ell_j \wedge \tilde{\alpha}^k + (-1)^q \cdot \tau^j_k \cdot (\tilde{\alpha}^k \wedge \mathrm{d}(\tau^{-1})^\ell_j)\right] \otimes \tilde{s}_\ell \\
&= \left[\mathrm{d}\tau^j_k \cdot (\tau^{-1})^\ell_j + (-1)^{2q+1} \cdot \tau^j_k \cdot \mathrm{d}(\tau^{-1})^\ell_j\right] \wedge \tilde{\alpha}^k \otimes \tilde{s}_\ell = \mathrm{d}(\tau^j_k \cdot (\tau^{-1})^\ell_j) \wedge \tilde{\alpha}^k \otimes \tilde{s}_\ell \\
&= \mathrm{d}\left(\delta^\ell_k \wedge \tilde{\alpha}^k\right) \otimes \tilde{s}_\ell = 0.
\end{aligned}$$

Thus, d_∇ is well-defined, and by definition, 1 trivially is satisfied. To check 2, let $\beta \in \Omega^p_{\mathbb{C}}(M;E)$ whose expression in the domain of a local frame is $\beta = \beta^j \otimes s_j$, and let $\alpha \in \Omega^q_{\mathbb{C}}(M)$. Then,

$$\begin{aligned}
\mathrm{d}_\nabla(\alpha \wedge \beta) = \mathrm{d}_\nabla((\alpha \wedge \beta^j) \otimes s_j) &= \mathrm{d}(\alpha \wedge \beta^j) \otimes s_j + (-1)^{p+q} \cdot (\alpha \wedge \beta^j) \otimes \nabla s_j \\
&= \left[\mathrm{d}\alpha \wedge \beta^j + (-1)^q \cdot \alpha \wedge \mathrm{d}\beta^j\right] \otimes s_j + (-1)^{p+q} \cdot (\alpha \wedge \beta^j) \otimes \nabla s_j \\
&= \mathrm{d}\alpha \wedge (\beta^j \otimes s_j) + (-1)^q \cdot \alpha \wedge \left[\mathrm{d}\beta^j \otimes s_j + (-1)^p \cdot \beta^j \otimes \nabla s_j\right] \\
&= \mathrm{d}\alpha \wedge \beta + (-1)^q \cdot \alpha \wedge \mathrm{d}_\nabla\beta.
\end{aligned}$$

Finally, to check 3, let $\alpha \in \Omega^p_{\mathbb{C}}(M;E)$, and $\sigma \in \Gamma(E)$. By 2, we know that (observe that d_∇ s linear by definition)

$$\begin{aligned}\mathrm{d}^2_\nabla(\alpha\wedge\sigma) &= \mathrm{d}_\nabla\left(\mathrm{d}\alpha\wedge\sigma + (-1)^p\cdot\alpha\wedge\nabla\sigma\right) = \mathrm{d}_\nabla(\mathrm{d}\alpha\wedge\sigma) + (-1)^p\cdot\mathrm{d}_\nabla(\alpha\wedge\nabla\sigma) = \\ &= \mathrm{d}^2\alpha\wedge\sigma - (-1)^p\cdot\mathrm{d}\alpha\wedge\nabla\sigma + (-1)^p\cdot\left[\mathrm{d}\alpha\wedge\nabla\sigma + (-1)^p\cdot\alpha\wedge\mathrm{d}^2_\nabla\nabla\right] \\ &= \alpha\wedge\mathrm{d}^2_\nabla\sigma.\end{aligned}$$

There are two conclusions we may extract from the previous calculation. The first one is that the action of d^2_∇ on all the spaces $\Omega^p_{\mathbb{C}}(M;E)$ is determined by its action on $\Gamma(E)$ (every E-valued p-form is a sum of elements of the form $\alpha\otimes\sigma = \alpha\wedge\sigma$). The second one is that it suffices to check 3 for the elements of a local frame. Indeed, let $(s_1,\dots s_r)$ a local frame. Then, if we assume that 3 holds for the elements of the frame, for every $\alpha = \alpha^j\wedge s_j = \alpha^j\otimes s_j$, we get

$$\mathrm{d}^2_\nabla(\alpha) = \mathrm{d}^2_\nabla(\alpha^j\wedge s_j) = \alpha^j\wedge\mathrm{d}^2_\nabla s_j = \alpha^j\wedge(F_\nabla\wedge s_j) = F_\nabla\wedge\alpha,$$

where the last identity follows from checking that the coordinate expression of both sides is equal to $(\Omega^\ell_j\wedge\alpha^j)\otimes s_\ell$.

Finally, let us check that 3 holds for the sections of a local frame:

$$\begin{aligned}\mathrm{d}^2_\nabla(s_j) &= \mathrm{d}_\nabla(\nabla s_j) = \mathrm{d}_\nabla(\omega^k_j\otimes s_k) = \mathrm{d}\omega^k_j\otimes s_k + \omega^k_j\wedge\nabla s_k = (\mathrm{d}\omega^k_j + \omega^\ell_j\wedge\omega^k_\ell)\otimes s_k \\ &= \Omega^k_j\otimes s_k = \Omega^k_j\wedge s_k.\end{aligned}$$

Finally, from equation (2.6) we get

$$F_\nabla\wedge s_j = (\Omega^k_\ell\otimes s_k\otimes t^\ell)\wedge s_j = \delta^\ell_j\cdot\Omega^k_l\otimes s_k = \Omega^k_j\otimes s_k.$$

□

Some sources then define the curvature of the connection to be the map $\mathrm{d}^2_\nabla\colon\Gamma(E)\to\Omega^2_{\mathbb{C}}(M;E)$. It is common to find the notation $F_\nabla = \nabla\circ\nabla$, which is a slight abuse of notation that comes from using the symbol ∇ to refer to all maps d_∇. Observe that for sections $\xi\in\Gamma(M;E)$, our definition of the wedge product means that we can write $\mathrm{d}^2_\nabla(\xi) = F_\nabla(\xi)$.

2.2 Induced connections

Connections on vector bundles give rise to connections in other vector bundles, provided these are constructed using the ones we originally had. We will be concerned with 3 cases:

1. Let E and F be two vector bundles over a smoth manifold, endowed with connections ∇^E and ∇^F. Then, we may define a connection in $E\times F$ by extending by linearity the following expression
$$\left(\nabla^E\otimes\nabla^F\right)(\xi\otimes\eta) = \nabla^E\xi\otimes\eta + \xi\otimes\nabla^F\eta, \tag{2.7}$$
as a direct computation shows.

2. Let ∇ be a connection on E, then, it induces a connection on E^* as follows. There is a unique connection ∇^* on E^* such that for every $t\in\Gamma(E^*)$, and $\xi\in\Gamma(E)$,
$$\left[\nabla^*(t)\right](\xi) = \mathrm{d}(t(\xi)) - t(\nabla\xi), \tag{2.8}$$
where $t(\alpha\otimes s) := t(s)\cdot\alpha$. Combining this with the previous item, we get an induced connection in $\mathrm{Hom}(E,F)\cong E^*\otimes F$.

3. A connection ∇ on a complex vector bundle E induces a connection on $\bar{E}$ given by $\overline{\nabla}s := \overline{\nabla\bar{s}}$. The map $E\to\bar{E}, s\mapsto\bar{s}$ sends every element to itself, but it is antilinear: $\overline{\lambda * s} = \overline{\bar{\lambda}\cdot s} = \bar{\lambda}\cdot s$.

2.3 Hermitian inner products

Hermitian inner products can be seen as generalizations of Riemannian metrics to arbitrary vector bundles.

Definition 2.5 (Hermitian fiber metric). Let $\pi\colon E \to M$ a smooth complex vector bundle over a smooth manifold M. A *hermitian fiber metric* h is an assignment of a hermitian inner product h_x to each fiber E_x and such that for every pair of local smooth sections $\sigma, \tau \in \Gamma(U; E)$, the map $U \to \mathbb{C}, x \mapsto h_x(\sigma_x, \tau_x)$ is smooth. A vector bundle that is endowed with a hermitian fiber metric is called a *hermitian vector bundle*

That is, a hermitian fiber metric is just an assignment of a hermitian inner product to each fiber, with the condition that is varies smoothly. Under some abuse of language, we will often call h an *hermitian inner product*. We adopt the convention that hermitian inner products are antilinear with respect to their second entry, i.e. $h(v, \lambda \cdot u) = \bar{\lambda} \cdot h(v, u)$for $\lambda \in \mathbb{C}$. Of course, there is analogous idefinition for real vector bundles, where we assign an inner product to each fiber that varies smoothly. Riemannian metrics are examples of inner products in the bundle TM.

If E is endowed with a hermitian fiber metric h, we say that a connection is *compatible with the fiber metric*, *compatible with the hermitian inner product* or *unitary* if it satisfies the following relation ($U \subset M$ is some open set)

$$X\big(h(\xi,\eta)\big) = h(\nabla_X \xi, \eta) + h(\xi, \nabla_{\bar{X}} \eta) \qquad \text{for all} \qquad X \in \Gamma(U, T_{\mathbb{C}}M), \quad \xi, \eta \in \Gamma(U; E).$$

This is equivalent to the following condition

$$\mathrm{d}h(\xi,\eta) = h(\nabla\xi, \eta) + h(\xi, \nabla\eta). \tag{2.9}$$

Let us comment on the terms of equation (2.9) above. For any pair of smooth sections $\xi, \eta \in \Gamma(U; E)$, the definition of hermitian fiber metric implies that $h(\xi, \eta)$ is smooth, hence $\mathrm{d}h(\xi, \eta)$ is a smooth one form. On the right-hand side, we have $h(\nabla\xi, \eta)$ (and a symmetric term). Now, $\nabla\xi$ is a section of the bundle $T^*_{\mathbb{C}}M \otimes E$. The product of a section of $T^*M \otimes E$ and one of E through h is, by definition $h(\alpha \otimes \xi, \eta) := h(\xi, \eta) \cdot \alpha$, $h(\xi, \alpha \otimes \eta) := h(\xi, \eta) \cdot \bar{\alpha}$ (this is the conjugation of forms defined in section A.6), and it is extended by sesguilinearity to general sections of $T^*_{\mathbb{C}}M \otimes E$.

2.4 The First Real Chern Class

Definition 2.6 (First Real Chern Class). Let $\pi\colon E \to M$ be a complex vector bundle over a smooth manifold M, and let ∇ be a connection. We define the *First Chern form* of ∇ as

$$c_1(\nabla) := \frac{\mathrm{i}}{2\pi} \operatorname{tr} F_\nabla.$$

In the previous definition, the trace of an element $\alpha \otimes f \in \Omega^2_{\mathbb{C}}(M; \mathrm{End}(E))$ is defined as $\operatorname{tr}(\alpha \otimes f) := \operatorname{tr}(f) \cdot \alpha \in \Omega^2_{\mathbb{C}}(M)$.

Theorem 2.7. Let $\pi\colon E \to M$ be a complex vector bundle. Then, $c_1(\nabla)$ is closed, and the de Rham cohomology class of $c_1(\nabla)$ in $H^2_{\mathrm{dR}}(M; \mathbb{C})$ is independent of the choice of the connection.

Proof. Choose a connection ∇ (they always exist by equation (2.4)), a local frame $(s_1, \dots s_r)$ and its dual frame $(t^1, \dots t^r)$. Since $F_\nabla = \Omega^i_j \otimes s_i \otimes t^j$, we have $\operatorname{tr}(F_\nabla) = \Omega^i_j \cdot \delta^j_i = \Omega^j_j$ (sum implied). Thus,

$$c_1(\nabla) = \frac{\mathrm{i}}{2\pi} \cdot \Omega^j_j = \frac{\mathrm{i}}{2\pi} \cdot \left(\mathrm{d}\omega^j_j + \omega^j_\ell \wedge \omega^\ell_j\right) = \frac{\mathrm{i}}{2\pi} \cdot \mathrm{d}\omega^j_j,$$

where the second summand is zero because $\omega^j_\ell \wedge \omega^\ell_j = -\omega^\ell_j \wedge \omega^j_\ell = -\omega^j_\ell \wedge \omega^\ell_j$ (we renamed the dummy indexes in the last step). Hence, $c_1(\nabla)$ is locally exact, and therefore globally closed (recall that d acts locally). It need not be globally exact though, because the connection one forms are defined only locally. To check all connections define the same cohomology class, let ∇' be another connection.

Define the map $\mathscr{D}\colon \Gamma(T_{\mathbb{C}}M)\times\Gamma(E)\to\Gamma(E), (X,\sigma)\mapsto \nabla_X\sigma-\nabla'_X\sigma$. It is $\mathscr{C}^\infty(M;\mathbb{C})$-linear with respect to all of its components (see equation (2.10) below, for the X-componet it is obvious), so it defines an element of $\Omega^1_{\mathbb{C}}(M;\mathrm{End}(E))$.

$$\mathscr{D}_X(f\sigma)=\nabla_X(f\cdot\sigma)-\nabla'_X(f\cdot\sigma)=\mathrm{d}f\otimes\sigma+f\cdot\nabla_X\sigma-\mathrm{d}f\otimes\sigma-f\cdot\nabla'_X\sigma=f\cdot\mathscr{D}_X(\sigma) \tag{2.10}$$

For every local frame, we have $\mathscr{D}=(\omega_i^j-(\omega')_i^j)\otimes s_j\otimes t^i$, hence

$$c_1(\nabla)-c_1(\nabla')=\frac{\mathrm{i}}{2\pi}\left(\mathrm{d}\omega_j^j-\mathrm{d}(\omega')_j^j\right)=\mathrm{d}\left(\frac{\mathrm{i}}{2\pi}\cdot\mathrm{tr}\mathscr{D}\right),$$

because the coordinate expression we used is valid for every local frame. Therefore, $c_1(\nabla)$ and $c_1(\nabla')$ differ by an exact 2-form, as we wanted to see. □

The following proposition will be needed to define the first Chern class, and it is what explains the factor i in the definition of the Chern form.

Proposition 2.8. Let $\pi\colon E\to M$ be a complex vector bundle over a smooth manifold which is endowed with a hermitian fiber metric. If ∇ is a metric connection on E, then $c_1(\nabla)$ is a real 2-form.

Proof. Applying the Gram-Schmidt algorithm (see [36, Lemma 8.3]) to a frame around any point, we may always obtain an orthonormal frame, i.e. $h(s_i,s_j)=\delta_{ij}$. If ∇ is a metric connection, then $\omega_j^k=-\overline{\omega_k^j}$. Indeed, for every local vector field, the compatibility condition implies (no summation in the last expression)

$$0=X(h(s_i,s_j))=h(\nabla_X s_i,s_j)+h(s_i,\nabla_X s_j)=h(\omega_i^k(Z)s_k,s_j)+h(s_i,\omega_j^k(X)s_k)=\omega_i^j(X)+\overline{\omega_j^i(X)}.$$

We thus find $\omega_j^k+\overline{\omega_k^j}=0$ (no summation). Now, considering $j=k$ and taking the sum over j, $\omega_j^j+\overline{\omega_j^j}=0$. Hence, ω_j^j is purely imaginary, thus, so is Ω_j^j, and therefore, $c_1(\nabla)=\frac{\mathrm{i}}{2\pi}\cdot\Omega_j^j$ is real. □

Observe that, in general, $c_1(\nabla)$ defines an element of $H^2_{\mathrm{dR}}(M;\mathbb{C})$. However, the previous proposition shows that whenever ∇ is compatible with the metric, $c_1(\nabla)$ is real. Since we may always find a connection compatible with metric (for instance, the Chern connection, see theorem 2.12), we conclude that $c_1(\nabla)$ defines (for *any connection*) an element of the real De Rham cohomology group $H^2_{\mathrm{dR}}(M;\mathbb{R})$ (which we may see as the subspace of $H^2_{\mathrm{dR}}(M;\mathbb{C})$ of cohomology classes invariant under conjugation, see section A.6).

2.5 The Chern connection

Let $\pi\colon E\to M$ be a holomorphic vector bundle over a complex manifold M. Recall that in this case, there exists a Cauchy-Riemann operator $\overline{\partial}_E\colon\Omega^{(p,q)}_{\mathbb{C}}(M;E)\to\Omega^{(p,q+1)}_{\mathbb{C}}(M;E)$ (see proposition A.26). Since any connection ∇ can be seen as a map $\nabla\colon\Omega^0_{\mathbb{C}}(M;E)\to\Omega^1_{\mathbb{C}}(M;E)$, and there is a direct sum decomposition, $\Omega^1_{\mathbb{C}}(M;E)=\Omega^{(0,1)}_{\mathbb{C}}(M;E)\oplus\Omega^{(1,0)}_{\mathbb{C}}(M;E)$, we can split the connection into two components

$$\nabla^{(1,0)}\colon\Omega^0_{\mathbb{C}}(M;E)\to\Omega^{(1,0)}_{\mathbb{C}}(M;E)\qquad\qquad\nabla^{(0,1)}\colon\Omega^0_{\mathbb{C}}(M;E)\to\Omega^{(0,1)}_{\mathbb{C}}(M;E),$$

and therefore we may write $\nabla=\nabla^{(1,0)}+\nabla^{(0,1)}$. This is just the composition of $\Pi^{(1,0)}$ and $\Pi^{(0,1)}$ with ∇ (extending each projection to $T^*_{\mathbb{C}}M\otimes E$).

Remark 2.9. Since ∇ is a connection, it satisfies its "Leibniz rule", and so $\nabla(f\cdot s)=\mathrm{d}f\otimes s+f\cdot\nabla s$. Since $\mathrm{d}=\partial+\overline{\partial}$, after decomposition, we find, for all $f\in\mathscr{C}^\infty(M;\mathbb{C})$ and $s\in\Omega^0_{\mathbb{C}}(M;E)$

$$\nabla^{(0,1)}(f\cdot s)=\overline{\partial}f\otimes s+f\cdot\nabla^{(0,1)}s.$$

This is the same relation satisfied by $\overline{\partial}_E$ (see (A.21)). This motivates the following definition.

Definition 2.10 (Compatible connection). A connection ∇ on a holomorphic vector bundle $\pi\colon E \to M$ is *compatible with the holomorphic structure* if $\nabla^{(0,1)} = \overline{\partial}_E$

Our goal is to show that, for a holomorphic hermitian vector bundle, there is a special connection, the Chern connection, that is compatible with both, the holomorphic structure and the hermitian fiber metric. To do that, it is useful to have several equivalent characterizations of being compatible with the holomorphic structure. This is the content of the following lemma.

Lemma 2.11. Let $\pi\colon E \to M$ be a complex vector bundle over a complex manifold M, and let ∇ be a connection on E. The following are equivalent:

1. ∇ is compatible with the holomorphic structure ($\nabla^{(0,1)} = \overline{\partial}_E$).
2. If $U \subset M$ is open, $\xi \in \Gamma(U; E)$ is holomorphic, and $\bar{Z}$ is a smooth local section of $T^{(0,1)}M$, then $\nabla_{\bar{Z}}\xi = 0$.
3. For each holomorphic frame, the connection forms are of type $(1,0)$.

Proof. All the statements are local, so we may just work in the domain of a holomorphic local frame $(s_1, \dots s_r)$. Let ξ be a smooth local section defined on the same open subset U. Write $\xi = \xi^i s_i$. Then, from remark 2.3, we have $\nabla\xi = \mathrm{d}\xi^j \otimes s_j + \xi^j \cdot \omega_j^k \otimes s_k$. Projecting onto $\bigwedge_{\mathbb{C}}^{(0,1)} M \otimes E$, we find,

$$\Pi^{(1,0)}(\nabla\xi) = \nabla^{(0,1)}\xi = \overline{\partial}\xi^j \otimes s_j + \xi^j \cdot (\omega_j^k)^{(0,1)} \otimes s_k.$$

On the other hand, by definition of the Cauchy-Riemann operator for vector bundles, we have $\overline{\partial}_E\, \xi = \overline{\partial}\xi^j \otimes s_j$. Observe that $\overline{\partial}_E = \nabla^{(0,1)}$ if and only if $\xi^j \cdot (\omega_j^k)^{(0,1)} \otimes s_k = 0$ for every local section ξ, that is, if and only if $(\omega_j^k)^{(0,1)} = 0$ for all j, k. This shows that (1) and (3) are equivalent. Suppose now that (3) holds, and let $\xi = \xi^j \cdot s_j$ be a holomorphic local section of E (observe that the maps ξ^j are now holomorphic), and $Z \in \Gamma(U; T^{(0,1)}M)$. Then,

$$\nabla_Z\, \xi = Z(\xi^j) \cdot s_j + \xi^j \cdot \omega_j^k(Z) \cdot s_k = 0. \tag{2.11}$$

Where we used the comments in remarks A.22 and A.24. Finally, suppose that 2 holds, then, equation (2.11) implies that $\omega_j^k(Z) = 0$ for every $Z \in \Gamma(U, T^{(0,1)}M)$, hence by remark A.24, each ω_j^k is a $(1,0)$-form. □

Theorem 2.12 (Existence of the Chern connection). Let $\pi\colon E \to M$ be a complex vector bundle over a complex manifold M. There is a unique connection, called the *Chern connection* that is compatible with the metric and the holomorphic structure.

Proof. The strategy is to show uniqueness first, and use what we will find as a hint on how to define the connection. Let ∇ be a connection compatible with the holomorphic structure. By the previous lemma, the connection one-forms with respect to a holomorphic frame $(s_1, \dots s_r)$ are of type $(1,0)$. Let Z be a holomorphic section of $T^{(1,0)}M$. Then, $\bar{Z}$ is a smooth section of $T^{(0,1)}M$. Let $h_{jk} := h(s_j, s_k)$. Compatibility with the metric (3) and the connection (4) imply

$$Z(h_{jk}) = Z(h(s_j, s_k)) \overset{(3)}{=} h(\nabla_Z s_j, s_k) + h(s_j, \nabla_{\bar{Z}} s_k) \overset{(4)}{=} h(\omega_j^i(Z)s_i, s_k) = h_{ik} \cdot \omega_j^i(Z). \tag{2.12}$$

This is valid for every holomorphic section Z of $T^{(1,0)}M$. Observe that h_{jk} is a smooth function and $\mathrm{d}h_{jk}(Z) = Z(h_{jk})$. Moreover, since $\mathrm{d} = \partial + \overline{\partial}$, and $\overline{\partial}h_{jk}(Z) = 0$, we have $\partial h_{jk}(Z) = h_{ik} \cdot \omega_j^i(Z)$. Since h is a hermitian fiber metric, the matrix h_{ik} has an inverse, which we denote, as usual, by h^{ik}. Multiplying by that inverse on both sides, we get $h^{km}\partial h_{jk}(Z) = h^{km}h_{ik} \cdot \omega_j^i(Z) = \delta_i^m \cdot \omega_j^i(Z) = \omega_j^m(Z)$. Finally, since this is valid for every section Z of $T^{(1,0)}M$, we get

$$\omega_j^m = h^{km}\partial h_{jk}. \tag{2.13}$$

Observe that the right-hand side is a $(1,0)$-form, which was to expect after the result of the previous lemma. We have determined the connection one forms in terms of the coefficients h_{jk} for every

holomorphic local frame. Since a connection is uniquely given (at least locally) by its connection one forms, we have shown uniqueness.

Now, to show that it exists, the key idea is to use equation 2.13 to define a connection in the domain of each local holomorphic frame. Then, uniqueness ensures that the connection is well-defined where two such frames overlap, and thus, the connection is globally defined. Therefore, we can define ∇ in terms of each holomorphic local frame, and these definitions can be glued together to produce a globally defined connection. It remains to check that it is compatible with the metric and the holomorphic structure. Compatibility with the holomorphic structure follows from the previous characterization lemma, together with the fact each $\omega_j^m = h^{km}\partial h_{jk}$ is a $(1,0)$-form. Finally, compatibility with the metric can be deduced by direct computation as follows. Let $(s_1, \dots s_r)$ be an holomorphic local frame, and $\xi = \xi^i \cdot s_i$ and $\eta = \eta^j \cdot s_j$ smooth sections defined on the same open set. Let Z be a complex vector field, which may decomposed as $Z = X + Y$, where X is a section of $T^{(1,0)}M$, and Y is a section of $T^{(0,1)}M$. Equation 2.12 applies to X, and the analogous equation for Y is

$$Y(h_{ij}) = h(\nabla_Y s_i, s_j) + h(s_i, \nabla_{\bar{Y}} s_j) = h(s_i, \omega_j^k(Y)s_k) = \overline{\omega_j^k(Y)}s_k. \tag{2.14}$$

Now, on the one hand,

$$\begin{aligned} Z(h(\xi,\eta)) = Z(h_{ij}\xi^i\overline{\eta^j}) &= \xi^i\overline{\eta^j}Z(h_{ij}) + h_{ij}Z(\xi^i\overline{\eta^j}) = \xi^i\overline{\eta^j}\cdot(X(h_{ij}) + Y(h_{ij})) + h_{ij}Z(\xi^i\overline{\eta^j}) \\ &= \xi^i\overline{\eta^j}\cdot\left(\omega_j^k(X) + \overline{\omega_j^k(\bar{Y})}\right)\cdot h_{ik} + h_{ij}\cdot\left(Z(\xi^i)\overline{\eta^j} + \xi^i Z(\overline{\eta^j})\right). \end{aligned} \tag{2.15}$$

On the other hand,

$$\begin{aligned} h(\nabla_Z\xi,\eta) + h(\xi,\nabla_{\bar{Z}}\eta) &= h\left(Z(\xi^i)s_i + \xi^i\nabla_Z s_i, \eta\right) + h\left(\xi, \bar{Z}(\eta^j)s_j + \eta^j\nabla_{\bar{Z}}s_j\right) \\ &= Z(\xi^i)\overline{\eta^j}h_{ij} + \xi^i\overline{\eta^j}\cdot\omega_i^k(X)\cdot h_{ik} + \xi^i\cdot\overline{\bar{Z}(\eta^j)}\cdot h_{ij} + \xi^i\overline{\eta^j}\cdot\overline{\omega_j^k(\bar{Y})}\cdot h_{ik} \\ &\overset{(5)}{=} \left(Z(\xi^i)\overline{\eta^j} + \xi^i Z(\overline{\eta^j})\right)\cdot h_{ij} + \xi^i\overline{\eta^j}\cdot h_{ik}\cdot\left(\omega_j^k(X) + \overline{\omega_j^k(\bar{Y})}\right). \end{aligned} \tag{2.16}$$

In (5) we used that $\overline{\bar{Z}(\eta^j)} = Z(\overline{\eta^j})$. We thus get that the last expressions in equations (2.15) and (2.16) coincide, hence the connection is compatible with the hermitian metric. □

Observe that this proof is similar to the one given to construct the Levi-Civita connection; there, one writes the Christoffel symbols in terms on the coefficients of the riemannian metric on each local chart. When we work with several connections, we will denote by ∇_H the Chern connection.

As we anticipated before the previous theorem allows us to define the *First Real Chern class* of E as $c_1(E) := [c_1(\nabla)] \in H^2_{\mathrm{dR}}(M;\mathbb{R})$. This means that, associated to any holomorphic vector bundle, there is a real cohomology class that *does not depend* on the connection chosen, only on the vector bundle itself.

To close this chapter, we will show a very important property satisfied by the curvature of the Chern connection: it is of type $(1,1)$.

Proposition 2.13 (Curvature of the Chern connection). Let M be a complex manifold, and $\pi\colon E \to M$ be a complex vector bundle over M endowed with a hermitian fiber metric. Let ∇ be its Chern connection. Then, $F_\nabla \in \Omega_{\mathbb{C}}^{(1,1)}(M;\mathrm{End}(E))$.

Proof. From equations (2.5) and (2.6), we know that, with respect to a local frame, we may write

$$F_\nabla = (\partial\omega_j^k + \overline{\partial}\omega_j^k + \omega_\ell^k\wedge\omega_k^\ell)\otimes s_k\otimes t^j,$$

so we just need to check that, for the Chern connection, the curvature two-forms $\Omega_j^k = \partial\omega_j^k + \overline{\partial}\omega_j^k + \omega_\ell^k\wedge\omega_k^\ell$ is of type $(1,1)$. We now that the connection one forms of the Chern connection are of type $(1,0)$. This means that $\overline{\partial}\omega_j^k$ is of type $(1,1)$, and $\partial\omega_j^k + \omega_\ell^k\wedge\omega_k^\ell$ is of type $(2,0)$. Therefore, it suffices to show that the latter vanishes. This is a calculation using the expression of the connection one forms in terms of the metric (see equation (2.13)).

$$\begin{aligned} \partial\omega_j^k + \omega_\ell^k\wedge\omega_k^\ell &= \partial(h^{km}\partial h_{jk}) + (h^{pm}\partial h_{kp})\wedge(h^{qk}\partial h_{jq}) \\ &= \partial h^{km}\wedge\partial h_{jk} + h^{pm}h^{qk}\cdot\partial h_{kp}\wedge\partial h_{jq}. \end{aligned} \tag{2.17}$$

On the other hand,

$$0 = \partial\left(\delta_\ell^k\right) = \partial\left(h^{kp}h_{p\ell}\right) = \partial h^{kp}\cdot h_{p\ell} + h^{kp}\cdot\partial h_{p\ell},$$

therefore, $h_{p\ell}\cdot\partial h^{hp} = -h^{kp}\cdot\partial h_{p\ell}$. Multiplying on both sides by $h^{\ell m}$ and taking the sum yields,

$$h^{\ell m}h_{p\ell}\cdot\partial h^{kp} = \delta_p^m\partial h^{kp} \overset{(6)}{=} \partial h^{km} = -h^{\ell m}h^{kp}\cdot\partial h_{p\ell}.$$

Taking equality (6) and inserting it in equation (2.17), yields

$$\partial h^{km}\wedge\partial h_{jk} + h^{pm}h^{qk}\cdot\partial h_{kp}\wedge\partial h_{jq} = -h^{\ell m}h^{kp}\cdot\partial h_{p\ell}\wedge\partial h_{jk} + h^{pm}h^{qk}\cdot\partial h_{kp}\wedge\partial h_{jq} = 0.$$

□

2.6 The Chern correspondence

We devote this section to explaining, schematically, the Chern correspondence. Good references for this section are [28, Ch. 7] and [37, Ch. 1]. The main statement of the Chern correspondence is that there is a bijection between the set of connections on a complex vector bundle over a complex manifold and the set of Dolbeault operators (or semiconnections), which restricts to a bijection between connections of type $(1,1)$, and integrable Dolbeault operators.

Definition 2.14 (Dolbeault operator). Let $E\to M$ be a complex vector bundle over a complex manifold. A *Dolbeault operator* or *semiconnection of type* $(0,1)$ on E is a $\mathbb{C}$-linear map

$$\overline{\partial}_e\colon \Gamma(M;E)\longrightarrow\Omega_{\mathbb{C}}^{(0,1)}(M;E)$$

satisfying

$$\overline{\partial}_e(f\cdot\xi) = \overline{\partial}f\otimes\xi + f\cdot\overline{\partial}_e\xi. \tag{2.18}$$

We denote $\mathscr{D}(E)$ (or $\mathscr{D}$ when the bundle is clear from the context) the set of all Dolbeault operators.

Observe that every Dolbeault operator can be extended to higher order complex differential forms by linear extension of

$$\overline{\partial}_e(\alpha\otimes\xi) = \overline{\partial}\alpha\otimes\xi + (-1)^{p+q}\cdot\alpha\wedge\overline{\partial}_e\xi, \qquad \alpha\in\Omega_{\mathbb{C}}^{(p,q)}(M),\ \xi\in\Gamma(M;E). \tag{2.19}$$

Lemma 2.15. Let $E\to M$ be a complex vector bundle over a complex manifold. Then, $\mathscr{D}(E)$ is an affine space modelled on the vector space $\Omega_{\mathbb{C}}^{(0,1)}(\mathrm{End}(E))$.

Proof. First, fix $\overline{\partial}_e^0\in\mathscr{D}(E)$, and choose $\alpha\in\Omega_{\mathbb{C}}^{(0,1)}(\mathrm{End}(E))$. Let us see that $\overline{\partial}_e^0+\alpha\in\mathscr{D}(E)$. Observe that given $\xi\in\Gamma(M;E)$, $\alpha(\xi)\in\Omega_{\mathbb{C}}^{(0,1)}(M;E)$, so the sum of $\overline{\partial}_e^0$ and α makes sense as an operator. Moreover,

$$\begin{aligned}\left(\overline{\partial}_e^0+\alpha\right)(f\cdot\xi) &= \overline{\partial}_e^0(f\cdot\xi)+\alpha(f\cdot\xi) = \overline{\partial}f\otimes\xi + f\cdot\overline{\partial}_e^0\xi + f\cdot\alpha(\xi)\\ &= \overline{\partial}f\otimes\xi + f\cdot\left(\overline{\partial}_e^0+\alpha\right)(\xi).\end{aligned}$$

Reciprocally, choose another $\overline{\partial}_e\in\mathscr{D}(E)$. A direct computation shows that $\overline{\partial}_e^0-\overline{\partial}_e$ is $\mathscr{C}^\infty(M;\mathbb{C})$-linear, so $\overline{\partial}_e^0-\overline{\partial}_e\in\Omega_{\mathbb{C}}^{(0,1)}(M;\mathrm{End}(E))$. □

Let $\bar{\mathscr{A}}(E)$ be the set of all connections on a complex vector bundle over a complex manifold $E\to M$ (again, we will simply write $\bar{\mathscr{A}}$ if no confusion arises). Then, we have a map (see (2.9)):

$$\begin{array}{ccc}\bar{\mathscr{A}}(E) & \longrightarrow & \mathscr{D}(E)\\ \nabla & \mapsto & \nabla^{(0,1)}.\end{array}$$

Suppose now that E is endowed with a hermitian fiber metric h. This condition will allow us to give an inverse to the map we just defined. Let $\mathscr{A}(E)$ be the set of all the connections on E that are compatible with the fiber metric. By applying $\Pi^{(0,1)}$ to equation 2.9, we get

$$\overline{\partial} h(\xi,\eta) = h\left(\nabla^{(0,1)}\xi,\eta\right) + h\left(\xi,\nabla^{(1,0)}\eta\right).$$

This is key to define the inverse map. Indeed, we can define $\mathscr{D}(E) \to \mathscr{A}(E)$ as follows: given a Dolbeault operator $\overline{\partial}_e$, we define a connection by $\nabla := \nabla_e^{(1,0)} + \overline{\partial}_e$, where $\nabla_e^{(0,1)} \colon \Gamma(M;E) \to \Omega_{\mathbb{C}}^{(1,0)}(M;E)$ is uniquely determined by

$$\overline{\partial} h(\xi,\eta) = h\left(\overline{\partial}_e\xi,\eta\right) + h\left(\xi,\nabla_e^{(1,0)}\eta\right). \tag{2.20}$$

If the operator $\nabla_e^{(0,1)}$ exists, it is unique, because any other operator $\overline{\nabla}_e^{(1,0)}$ satisfying equation (2.20) should satisfy $h\left(\xi,\nabla_e^{(1,0)}\eta - \overline{\nabla}_e^{(1,0)}\eta\right) = 0$ for every $\xi,\eta \in \Gamma(M;E)$, thus $\nabla_e^{(1,0)} = \overline{\nabla}_e^{(1,0)}$. Moreover, equation (2.20) implies that

$$\nabla_e^{(1,0)}(f\cdot\xi) = \partial f \otimes \xi + f\cdot\nabla_e^{(1,0)}\xi. \tag{2.21}$$

Indeed, for any $\eta,\xi \in \Gamma(M;E)$ one has

$$\begin{aligned} h\left(\xi,\nabla_e^{(1,0)}(f\cdot\eta)\right) &= \overline{\partial} h(\xi,f\cdot\eta) - h\left(\nabla^{(0,1)}\xi,f\cdot\eta\right) = \overline{\partial}\left(\bar{f}\cdot h(\xi,\eta)\right) + \bar{f}\cdot h\left(\nabla^{(0,1)}\xi,\eta\right) \\ &= \overline{\partial}\bar{f}\cdot h(\xi,\eta) + \bar{f}\cdot\left(h(\xi,\eta) + h\left(\nabla^{(0,1)}\xi,\eta\right)\right) = \overline{\partial}\bar{f}\cdot h(\xi,\eta) + \bar{f}\cdot h\left(\xi,\nabla_e^{(1,0)}\eta\right) \\ &= h\left(\xi,\partial f\otimes\eta + f\cdot\nabla_e^{(1,0)}\eta\right). \end{aligned}$$

This implies that ∇ is a well-defined connection. To check that the so-defined connection ∇ is compatible with the metric, observe that equation (2.20) implies

$$\partial h(\xi,\eta) = \overline{\overline{\partial}\,\overline{h(\xi,\eta)}} = \overline{\overline{\partial} h(\eta,\xi)} = \overline{h\left(\overline{\partial}_e\eta,\xi\right) + h\left(\eta,\nabla_e^{(1,0)}\xi\right)} = h\left(\xi,\overline{\partial}_e\eta\right) + h\left(\nabla_e^{(1,0)}\xi,\eta\right).$$

Therefore,

$$\mathrm{d}h(\xi,\eta) = \partial h(\xi,\eta) + \overline{\partial} h(\xi,\eta) = h\left(\overline{\partial}_e\xi + \nabla_e^{(1,0)}\xi,\eta\right) + h\left(\xi,\overline{\partial}_e\eta + \nabla_e^{(1,0)}\eta\right).$$

To see that $\nabla_e^{(1,0)}$ actually exists, we follow an argument similar to the one given in the proof of theorem 2.12: we choose a local frame $(s_1,\dots s_r)$ of E and define the operator on the frame. There exist $(0,1)$-forms α_i^r such that $\overline{\partial}_e s_i = \alpha_i^r \otimes s_r$ (this is analogous to remark 2.3). Then, we can define $\nabla_e^{(1,0)}$ by its action on a local frame $\nabla_e^{(1,0)}(\xi^i\cdot s_i) = \left(\partial\xi^i + \xi^j\cdot\omega_j^k\right)\otimes s_k$, where

$$\omega_j^k := \overline{h^{mi}\cdot\left(\overline{\partial} h_{ij} - \alpha_i^r\cdot h_{rj}\right)}.$$

Finally, observe that $\nabla_e^{(0,1)} = \overline{\partial}_e$ by definition. We thus have a bijection $\mathscr{A}(E) \leftrightarrow \mathscr{D}(E)$.

It is appropriate now to state (without proof) the following result.

Theorem 2.16. Let $E \to M$ be a smooth complex vector bundle defined over a complex manifold X. Let $\overline{\partial}_e$ be a Dolbeault operator satisfying the *integrability condition* $\overline{\partial}_e \circ \overline{\partial}_e = 0$. Then, there is a unique holomorphic structure on E such that $\overline{\partial}_e = \overline{\partial}_E$.

Proof. See [28, I-Prop. 3.7]. Observe that by $\overline{\partial}_e^2 = 0$ we mean that the composition of $\overline{\partial}_e$ with its extension to one-forms (2.19) vanishes. □

Motivated by this result, let $\mathscr{H}(E) := \left\{\overline{\partial}_e \in \mathscr{D}(E) : \overline{\partial}_e^2 = 0\right\}$. Observe that each element on $\mathscr{H}(E)$ defines a holomorphic structure on E. We now ask ourselves which connections correspond to $\mathscr{H}(E)$ under the Chern correspondence. It turns out that this condition depends on the curvature. By (A.20), we have a decomposition $\Omega_{\mathbb{C}}^2(M;\mathrm{End}(E)) = \Omega_{\mathbb{C}}^{(2,0)}(M;\mathrm{End}(E)) \oplus \Omega_{\mathbb{C}}^{(1,1)}(M;\mathrm{End}(E)) \oplus \Omega_{\mathbb{C}}^{(0,2)}(M;\mathrm{End}(E))$, hence the curvature of any connection decomposes as $F_\nabla = F_\nabla^{(2,0)} + F_\nabla^{(1,1)} + F_\nabla^{(0,2)}$.

Lemma 2.17. Let $E \to M$ be a complex vector bundle, and let ∇ be a connection on E. Then, $\overline{\partial}_e := \nabla^{(0,1)}$ is an integrable Dolbeault operator if and only if $F_\nabla^{(2,0)} = 0$.

Proof. First, observe that the decomposition of the curvature induces a decomposition of $\mathrm{d}_\nabla^2 = \mathrm{d}_\nabla^{(2,0)} + \mathrm{d}_\nabla^{(1,1)} + \mathrm{d}_\nabla^{(0,2)}$, where $\mathrm{d}_\nabla^{(0,2)}(\alpha) = F_\nabla^{(0,2)} \wedge \alpha$ for any $\alpha \in \Omega_{\mathbb{C}}^{(p,q)}(M;E)$. The other operators are defined analogously. On the other hand, $\nabla = \nabla^{(1,0)} + \overline{\partial}_e$ means that, after extending these operators to act on higher degree forms as in (2.18) and (2.21), we have

$$\begin{aligned}\mathrm{d}_\nabla^2 &= (\nabla^{(1,0)} + \overline{\partial}_e) \circ (\nabla^{(1,0)} + \overline{\partial}_e) \\ &= \nabla^{(1,0)} \circ \nabla^{(1,0)} + \left(\nabla^{(1,0)} \circ \overline{\partial}_e + \overline{\partial}_e \circ \nabla^{(1,0)}\right) + \overline{\partial}_e \circ \overline{\partial}_e.\end{aligned}$$

Applying the projection $\Pi^{(0,2)}$ to both expressions for d_∇^2, we get $\mathrm{d}_\nabla^{(0,2)} = \overline{\partial}_e^2$, and the result follows. □

Observe that $F_\nabla^{(0,2)} = 0$ if and only if $F_\nabla^{(2,0)}$. This follows from the fact that $F_\nabla^{(2,0)}(\bar{X},\bar{Y}) = F_\nabla^{(0,2)}(X,Y)$ for all vector fields $X, Y \in \Gamma(M; T_{\mathbb{C}}M)$.

Motivated by the previous lemma, we set $\mathscr{A}^{(1,1)}(E) := \left\{\nabla \in \mathscr{A} : F_\nabla \in \Omega_{\mathbb{C}}^{(1,1)}(M;\mathrm{End}(E))\right\}$. We call its elements connections of type $(1,1)$ or *integrable connections*. By the previous lemma, the Chern correspondence restricts to a bijection $\mathscr{A}^{(1,1)}(E) \leftrightarrow \mathscr{H}(E)$. Observe that under this bijection, a connection becomes the Chern connection of the complex structure that its $(0,1)$ component induces on E.

Looking at theorem 2.16, we would like to identify $\mathscr{H}(E)$ with all possible holomorphic structures on E (whose underlying smooth complex vector bundle structure is that of E). Although theorem 2.16 does state that every element $\overline{\partial}_e \in \mathscr{H}(E)$ induces a holomorphic structure on E, it may happen that two different integrable Dolbeault operators induce biholomorphic complex structures, and it would not make sense to count those as different. To remedy this issue, let us define the following groups.

Definition 2.18. Let $E \to M$ be a complex vector bundle endowed with a hermitian fiber product h. The *complex gauge group* $\mathscr{G}^{\mathbb{C}} = \mathrm{GL}(E) \subset \Gamma(M;\mathrm{End}(E))$ is the group of complex smooth bundle automorphisms (equipped with fiberwise composition). The *gauge group* or *group of unitary automorphisms* $\mathscr{G} = \mathrm{U}(E,h) \subset \mathscr{G}^{\mathbb{C}}$ is the group of bundle isomorphisms of E satisfying $h_p(f_p(\xi), f_p(\eta)) = h_p(\xi,\eta)$, $\xi,\eta \in E_p$ for all $p \in M$.

The complex gauge group acts through the right on the space of Dolbeault operators as follows

$$\begin{array}{ccc} \mathscr{D}(E) \times \mathscr{G}^{\mathbb{C}} & \longrightarrow & \mathscr{D}(E) \\ (\overline{\partial}_e, f) & \mapsto & \overline{\partial}_e \cdot f \end{array}$$

where $\overline{\partial}_e \cdot f := \left(\mathrm{id}_{T_{\mathbb{C}}^*M} \otimes f^{-1}\right) \circ \overline{\partial}_e \circ f$. The identity component is usually omitted, and one writes $f^{-1} \circ \overline{\partial}_e \circ f$.

Proposition 2.19. Let $E \to M$ be a complex vector bundle over a smooth manifold. Let $f \in \mathscr{G}^{\mathbb{C}}$, and $\overline{\partial}_e, \overline{\partial}_e'$ two Dolbeault operators such that $\overline{\partial}_e' = \overline{\partial}_e \cdot f$. Let $\mathcal{E}$ and $\mathcal{E}'$ be the complex structures induced on E by theorem 2.16. Then, $f\colon \mathcal{E}' \to \mathcal{E}$ is an isomorphism between holomorphic vector bundles.

Proof. Since $f \in \mathscr{G}^{\mathbb{C}}$, it is an isomorphism of complex smooth vector bundles, so one needs only check that it is holomorphic. This follows from the fact that given two holomorphic vector bundles, holomorphicity of bundle homomorphism is determined by the condition $f \circ \overline{\partial}_e' = \overline{\partial}_e \circ f$. □

Thus, the true set of complex structures is given by the quotient of $\mathscr{H}(E)$ by the action of the complex gauge group:

$$\mathscr{C}(E) = \frac{\mathscr{H}(E)}{\mathscr{G}^{\mathbb{C}}},$$

$\mathscr{C}$ being the set of complex structures on E. Moreover, $\mathscr{G}^{\mathbb{C}}$ also acts on $\mathscr{A}(E)$ by $\nabla \cdot f := f^* \circ \nabla^{(1,0)} \circ (f^*)^{-1} + f^{-1} \circ \nabla^{(0,1)} \circ f$ (same abuse of notation as before), where f^* is the adjoint of f, i.e. the unique bundle homomorphism satisfying $h_p(\xi, f_p^*(\eta)) = h_p(f_p(\xi),\eta)$ for $\xi,\eta \in E_p$ and all $p \in M$. One

may show (see [28, Sec. 7.1]) that this action maps $\mathscr{A}^{(1,1)}(E)$ to itself, so it restricts to an action on $\mathscr{A}^{(1,1)}(E)$. We then get a bijection

$$\frac{\mathscr{A}^{(1,1)}(E)}{\mathscr{G}^{\mathbb{C}}} \leftrightarrow \frac{\mathscr{H}(E)}{\mathscr{G}^{\mathbb{C}}} = \mathscr{C}(E). \tag{2.22}$$

These spaces admit extra structures (i.e. topologies and even manifold structures). When endowed with those, we call them *Moduli spaces*.

Chapter 3

Principal bundles

The aim of this chapter in to introduce basic concepts regarding principal bundles. As we did for vector bundles, we will describe simultaneously principal bundles in the smooth and holomorphic category. Hence, we recover the notation $\mathcal{S}$ introduced in appendix A to mean either smooth or holomorphic. The expression *$\mathcal{S}$-Lie group* means a smooth or holomorphic Lie group, and an *$\mathcal{S}$-action* means either a smooth or holomorphic action (that is the map $G \times M \to M, (g, m) \mapsto g \cdot m$ is an $\mathcal{S}$-morphism). A good reference for principal G-bundles is [**51**, Ch. 1] (they work with smooth bundles, but some of the results and definitions can be generalized to the holomorphic setting).

We will also describe connections in principal bundles. Since principal bundles do not have linear fibers, connections will no longer be differential operators, but we will see that they can be seen from a very geometric viewpoint. There is also a concept of curvature, and this can be seen as a section of an associated vector bundle, which means that differential operators may be applied to the curvature of a connection. This is, as we will see in chapter 5, essential to write down the Hermite-Yang-Mills equations. We will also see how principal and vector bundles are related to one another (and how one goes from one to the other). The chapter finishes with a principal bundle version of the Chern correspondence which shows how reductions of the structure group (which are, in some sense, analogous to hermitian fiber metrics) parametrize complex structures on smooth principal bundles.

3.1 Definition and first properties

The notion of a vector bundle admits several generalizations. Among them, principal G-bundles will be very important for our purposes. The spirit of the definition is quite similar to that of vector bundles: we attach a Lie group to each point of the manifold in a way such that the resulting manifold becomes locally trivial. Before stating the definition, recall that an action ϕ of a group G on a set S is free if $\phi_g(s) = s$ for some $s \in S$ implies that $g = e$ (that is, only the group identity can have fixed points).

Definition 3.1 (Principal G-bundle). A *principal G-bundle*, or *principal bundle with structure group G* is a 4-tuple (P, G, M, π), where P and M are $\mathcal{S}$-manifolds, G is an $\mathcal{S}$-Lie group, and $\pi \colon P \to M$ is an $\mathcal{S}$-morphism satisfying

1. There is a free, $\mathcal{S}$, right-action of G on M.
2. $\pi^{-1}(\pi(p)) = \{p \cdot g : g \in G\} = p \cdot G$.
3. For every $m \in M$ there are an open neighborhood $U \subset M$ of m and an $\mathcal{S}$-isomorphism, called a *local trivialization* $T_U \colon \pi^{-1}(U) \to U \times G$, where $T_U(p) = (\pi(p), s_U(p))$, and $s_U \colon \pi^{-1}(U) \to G$ commutes with the G-action, that is $s_U(p \cdot g) = s_U(p) \cdot g$.

We say that M is the *base space*, and P is the *total space*. Observe that holomorphic principal bundles require G and M to be complex. They are also called *complex principal bundles*. If there is no room for confusion, we will denote a principal G by $P(M, G)$, or P_G if the base manifold is understood. Observe that the following diagram commutes

$$\begin{array}{ccc} P \supset \pi^{-1}(U) & \xrightarrow{T_U} & U \times G \\ & {\scriptstyle\pi}\searrow \quad \swarrow{\scriptstyle\pi_1} & \\ & M. & \end{array}$$

Note that this means that π must be surjective: for each $m \in M$, take a local trivialization (U, T_U) with $m \in U$. Then, $\pi = \pi_1 \circ T_U$, so m is in the image of the right-hand side. Moreover, by taking differentials, and applying the chain rule, we observe that π is a submersion. In particular, this implies that for every $m \in M$, $\pi^{-1}(m)$ is an embedded submanifold of P [**36**, Cor. 5.13]. Moreover, the second property of the definition of G-bundle implies that each fiber is a copy of G, and can be identified with G. In fact, each fiber is $\mathcal{S}$-isomorphic, as a manifold, to the structure group G. Moreover, from local triviality we may deduce that $\dim P(M, G) = \dim P + \dim M$.

Remark 3.2. If $P(M, G)$ is a principal G-bundle, then the action of G on P is proper, and $P/G \cong M$. Some references, such as [**30**] define principal G-bundles by replacing part 2 of definition 3.1 by the condition $P/G \cong M$. Here, we used the fact that the quotient manifold theorem (theorem 1.7) admits a generalization to free and proper holomorphic actions of complex Lie groups to complex manifolds (see [**56**, Lemma 3.2.6, page 334] for a proof). Recall that by a proper action, we mean that the map $P \times G \to P \times P, (g, p) \mapsto (g \cdot p, p)$ is a proper map.

Remark 3.3. Since principal bundles are equipped with a natural projection to the base manifold, we can define *global sections* (resp. *local sections*) as $\mathcal{S}$-morphisms $\sigma \colon M \to P$ (resp. $\sigma \colon U \to P$ where $U \subset M$ is open) satisfying $\pi \circ \sigma = \mathrm{id}_M$ (resp. $\pi \circ \sigma = \mathrm{id}_U$). They are sometimes called *global* and *local gauges* in the physics literature [**21**]. Just as happens with vector bundles, there is a one-to-one correspondence between local trivializations and local sections. Indeed, given a local section σ, we define a local trivialization by the inverse of $U \times G \mapsto \pi^{-1}(U), (m, g) \mapsto \sigma(m) \cdot g$. Explicitly, we define $T_U(p) = (\pi(p), s_U(p))$, where $s_U(p) \in G$ is the only element satisfying $p = \sigma(\pi(p)) \cdot s_U(p)$. Observe that this implies $s_U \circ \sigma \equiv e_G$. Reciprocally, given a local trivialization T_U, a local section is defined by $m \to T_U^{-1}(m, e_G)$. See [**51**, Prop. 1.1.6] for a proof in the smooth category, but observe that the claim holds in the holomorphic category.

Definition 3.4 (Principal bundle homomorphism). Let (P_1, M_1, G_1, π_1) and (P_2, M_2, G_2, π_2) be two $\mathcal{S}$-principal bundles with respective structure groups G_1 and G_2. A *principal bundle homomorphism* is a pair of mappings (ϑ, λ), such that $\vartheta \colon P_1 \to P_2$ is an $\mathcal{S}$-morphism, $\lambda \colon G_1 \to G_2$ is a $\mathcal{S}$-Lie group homomorphism, and $\vartheta(p \cdot g) = \vartheta(p) \cdot \lambda(g)$ for all $p \in P_1, g \in G_1$.

We will say that a principal bundle homomorphism is an *isomorphism* if ϑ is an $\mathcal{S}$-isomorphism, and λ is a Lie group isomorphism.

Remark 3.5.
In the situation described in the previous definition, ϑ induces a smooth map on the base manifolds M_1 and M_2. Indeed, ϑ must send fibers of π_1 to fibers of π_2, and therefore, we may define $\vartheta' \colon M_1 \to M_2$ by $\vartheta'(\pi_1(p)) := \pi_2(\vartheta(p))$ (which is an $\mathcal{S}$-morphism by local triviality). This makes the following diagram commutative.

$$\begin{array}{ccc} P_1 & \xrightarrow{\vartheta} & P_2 \\ {\scriptstyle\pi_1}\downarrow & & \downarrow{\scriptstyle\pi_2} \\ M_1 & \xrightarrow{\vartheta'} & M_2 \end{array}$$

Observe that $\vartheta'(\pi_1(p \cdot g)) = \pi_2(\vartheta(g \cdot p)) = \pi_2(\vartheta(p) \cdot \lambda(g)) = \pi_2(\vartheta(g))$ so its well-defined and independent of the representative chosen. It is smooth because π_1 is a surjective smooth submersion, and $\vartheta' \circ \pi_1 = \pi_2 \circ \vartheta$, the right-hand side being smooth. By the characteristic property of surjective smooth submersions, [**36**, Th. 4.29], we conclude that ϑ' is smooth.

We will generally work with different principal bundles over the same base manifold (although we may consider different structure groups). In that case, we will consider unless otherwise stated, morphisms where $\vartheta' = \mathrm{id}_M$ (these are sometimes called *vertical principal bundle morphisms*, or morphisms *covering the identity*). If the structure groups are the same, and $\lambda = \mathrm{id}_G$ then, ϑ is called a *G-morphism*.

In complete analogy to vector bundles (see (A.5) and the surrounding comments), given an $\mathcal{S}$-principal bundle $P(M, G)$, another $\mathcal{S}$-manifold, and an $\mathcal{S}$-morphism $\varphi \colon N \to M$, we can define a the

pullback principal bundle

$$\varphi^*P := \{(n,p) \in N \times P : \varphi(n) = \pi(p)\} \subset N \times P. \tag{3.1}$$

Endowed with the action given by $(n,p)\cdot g = (n, p\cdot g)$ and whose projection is the restriction of the projection $N \times P \to N$, it becomes a principal G-bundle [**51**, Remark 1.1.9].

Finally, sometimes vector bundles and fiber bundles are not enough; sometimes we need general manifolds to be the fibers of a bundle. This motivates the definition of a general fiber bundle, which generalizes vector and principal bundles.

Definition 3.6 (Fiber bundle). A *fiber bundle* is a 4-tuple (E, M, F, π), where M, E and F are $\mathcal{S}$-manifolds, $\pi\colon E \to M$ is a surjective $\mathcal{S}$ morphism, and for every $m \in M$ there are an open neighborhood $U \subset M$, and an $\mathcal{S}$-isomorphism $\Phi\colon \pi^{-1}(U) \to U \times F$ called a *local trivialization*, such that $\pi_1 \circ \Phi = \pi$.

Observe that we use the term *local trivialization* for all the types of bundles we have discussed. This shall lead to no confusion, as it will be clear from the context the type of bundle we are dealing with.

3.1.1 Frame bundles

This section is devoted to highlight the relation that exists between vector bundles and principal bundles. More precisely, given a rank r $\mathcal{S}$-vector bundle over the field $\mathbb{K} = \mathbb{R}$ or $\mathbb{C}$, and defined over an $\mathcal{S}$-manifold M, we are going to construct an $\mathcal{S}$-principal $\mathrm{GL}(r,\mathbb{K})$-bundle over the same manifold M. This principal bundle is called the *frame bundle of* E.

For each $m \in M$, let $L_m := \{\boldsymbol{u} := (u_1, \dots u_r) \text{ basis of } E_m\}$. Define (see (A.3))

$$L(E) := \bigsqcup_{m\in M} L_m. \tag{3.2}$$

Observe that there is a clear action of $\mathrm{GL}(k,\mathbb{K})$ on $L(E)$, which is given as follows

$$\begin{array}{ccc} L(E) \times \mathrm{GL}(r,\mathbb{K}) & \longrightarrow & L(E) \\ (\boldsymbol{u}, A = (a^i_j)) & \mapsto & \boldsymbol{u}A. \end{array} \tag{3.3}$$

Here, $\boldsymbol{u}A$ is the basis whose j-th element is given by $v_j := u_i \cdot a^i_j$. This is a free right-action because only the identity matrix can fix all the elements of a basis. Moreover, there is a canonical projection $\pi\colon L(E) \to M, (m,\boldsymbol{u}) \to m$. Since all the basis of one fiber are related by an invertible linear transformation, the orbit space for this action is M as a set. The only thing left is to endow $L(E)$ with an $\mathcal{S}$-manifold structure. To that end, choose a local trivialization of the vector bundle E, (Φ, U), and let $(s_1, \dots s_r)$ be the associated local frame (see proposition A.12). Then, for each $\boldsymbol{u} \in L_m$ there is a matrix $A(\boldsymbol{u}, \Phi) \in \mathrm{GL}(r,\mathbb{K})$ satisfying $u_j = a^i_j \cdot s_i$. Thus, we may define

$$\begin{array}{rccc} T_U\colon & \pi^{-1}(U) & \longrightarrow & U \times \mathrm{GL}(r,\mathbb{K}) \\ & (m,\boldsymbol{u}) & \mapsto & (m, A(\boldsymbol{u},\Phi)). \end{array}$$

The smooth structure on $L(E)$ is defined by requiring the maps T_U defined above to be diffeomorphisms (see [**21**, Th. 4.3.3]).

This construction can be refined when the vector bundle is equipped with some extra structure. For example, suppose that E is a smooth complex vector bundle equipped with a hermitian fiber metric. Then, we may define U_m the set of *unitary basis* with respect to the hermitian product h_m of the fiber E_m. Setting

$$U(E) := \bigsqcup_{m\in M} U_m, \tag{3.4}$$

we get a set endowed with a natural action of $\mathrm{U}(r)$ (recall that orthonormal basis are taken to orthonormal basis by $\mathrm{U}(r)$-transformations). The action is just as in (3.3). The principal bundle structure is given by choosing local trivializations associated to unitary frames. Thus, $U(E)$ has the structure of a principal $\mathrm{U}(r)$-bundle.

3.1.2 Associated bundles

Given an $\mathcal{S}$-principal G-bundle, and another $\mathcal{S}$-manifold F which is acted upon through the left by G by $\mathcal{S}$-morphisms, we may construct another fiber bundle called an *associated bundle* to the original principal G-bundle.

Firstly, observe that if $\phi\colon F \times G \to F$ is the left action originally defined, we may always define a right action as follows

$$\begin{array}{rccc} \tilde{\phi}\colon & G \times F & \longrightarrow & F \\ & (g,p) & \mapsto & \phi_{g^{-1}}(a). \end{array} \tag{3.5}$$

Thus, we can define a right action on the usual cartesian product $P \times F$ by

$$\begin{array}{ccc} (P \times F) \times G & \longrightarrow & P \times F \\ ((p,\xi),g) & \mapsto & (p \cdot g, g^{-1} \cdot \xi). \end{array}$$

If G acts freely and by $\mathcal{S}$-morphisms on F, then it does so on $P \times F$. Certainly, the action is component-wise free, therefore free. Moreover, by remark 3.2, the action of G on P is proper. Furthermore, the product of a proper action with any other action is again proper (see [50, Remark 6.3.9]). Having that in mind, we define $P \times_G F := (P \times F)/G$[1]. That is, we define $P \times_G F$ to be the set of equivalence classes for the equivalence relation $(p_1,\xi_1) \sim (p_2,\xi_2)$ if and only if $(p_1 \cdot g, g^{-1} \cdot \xi_1) = (p_2,\xi_2)$ for some $g \in G$. Let us denote by $[p,\xi]$ the equivalence class of a pair (p,ξ). We endow it with the quotient topology, and by the quotient manifold theorem (theorem 1.7 or theorem B.5 depending on the category we are workin in), $P \times_G F$ as a unique $\mathcal{S}$-manifold structure that makes the usual quotient map $q\colon P \times F \to P \times_G F$ into an $\mathcal{S}$-morphism and a submersion. Define the projection $\pi_G\colon P \times_G F \to M, [p,\xi] \mapsto \pi(p)$. It is well-defined because π is invariant by the action of G. Moreover, the following diagram commutes

$$\begin{array}{ccc} P \times F & & \\ {\scriptstyle q}\downarrow & \searrow^{\pi \circ \pi_1} & \\ P \times_G F & \xrightarrow[\pi_G]{} & M. \end{array}$$

Since $\pi_1 \circ \pi$ is an $\mathcal{S}$-morphism, the characteristic property of surjective smooth submersions implies that π_G is an $\mathcal{S}$-morphism (see [36, Th. 4.29] [35, Cor. 2.6] for the smooth and complex categories respectively).

The only thing left to check is that this structure admits local trivializations. Recall that the set of local trivializations is in bijection with the set of smooth local cross-sections (see remark 3.3), and let (U, T_U) be a local trivialization of P. Let $\sigma\colon U \to P$ be its associated cross-section. Define

$$\begin{array}{rccc} \tilde{\sigma}\colon & U \times F & \longrightarrow & \pi_G^{-1}(U) \\ & (m,f) & \mapsto & [\sigma(m), f]. \end{array} \qquad \begin{array}{rccc} \Phi\colon & \pi_G^{-1}(U) & \longrightarrow & U \times F \\ & [p,f] & \mapsto & (\pi_G([p,f]), s_U(p) \cdot f) \end{array} \tag{3.6}$$

These maps are $\mathcal{S}$-morphisms (this is again the characteristic property of surjective smooth submersions), and inverses of each other. *Phi* is a local trivialization. Observe that $\pi_1 \circ \Phi = \pi_G$ if and only if $\pi_1 = \pi_G \circ \Phi^{-1}$, and

$$\pi_G \circ \Phi^{-1}(m,f) = \pi_G[\sigma(m), f] = \pi(\sigma(m)) = \mathrm{id}_U(m) = m.$$

Remark 3.7. The natural projection $q\colon P \times F \to P \times_G F$ induces a natural $\mathcal{S}$-isomorphism between the fibers of π_G and F. Indeed, fix $p \in P$, and define the map $q_p\colon F \to P \times_G F, f \mapsto [p,f]$. The image is clearly contained in $\pi_G^{-1}(\pi(p))$, and q_p is equivariant:

$$q_{p\cdot g}(f) = [p \cdot g, f] = [p \cdot g, g^{-1} \cdot g \cdot f] = [p, g \cdot f] = q_p(g \cdot f),$$

thus $q_{p\cdot g} = q_p \circ \sigma_g$. If we choose an element in the fiber $\pi_G^{-1}(\pi(p))$, it will be $[p \cdot g, f] = q_p(g \cdot f)$, so each q_p is a bijection. In fact it is an $\mathcal{S}$-isomorphism.

[1] This is sometimes denoted $P \times_\phi F$ to emphasize the action.

Let us now discuss the case of the adjoint bundle. Let $P(M,G)$ be a principal G-bundle. As the manifold F, take the Lie algebra $\mathfrak{g}$. This is clearly a smooth manifold because it is a finite-dimensional vector space. Moreover, $\mathrm{Ad}\colon G \to \mathfrak{g}$ is a left action of G on $\mathfrak{g}$. We can thus define the *adjoint bundle* $\mathrm{ad}(P) := P \times_G \mathfrak{g}$. This case is special, in the sense that we actually obtain a vector bundle, as the following proposition shows.

Proposition 3.8. Let $P(M,G)$ be a principal G-bundle. Then, $\mathrm{ad}(P)$ is a smooth vector bundle.

Proof. One simply defines the vector space structure by $\lambda\cdot[p,\xi]+\mu\cdot[p,\zeta] := [p,\lambda\xi+\mu\zeta]$ for $p\in P, \xi,\zeta\in \mathfrak{g}$and checks that this independent of the chosen representatives. Actually, this does not use any particular feature of $\mathfrak{g}$, other than it is a vector space. This means that for every vector space V on which there is a representation of G, the bundle $P\times_G V$ is again a vector bundle. This works for complex vector bundles as well, and in that case $P\times_G V$ becomes a smooth vector bundle. If $P(M,G)$ is a holomorphic principal bundle, V is a complex vector space, and the representation $G\to \mathrm{GL}(V)$ is holomorphic, then the associated bundle is a holomorphic vector bundle. □

Example 3.9. Let us comment on another example of associated bundle that we will use: the *group adjoint bundle*. Let $P(M,G)$ be a principal G-bundle. Consider $F=G$ in the notation at the beginning of the section, where G acts on itself through the left by conjugation $g\cdot h := ghg^{-1}$. We then define $\mathrm{Ad}(G) := P\times_G G$. Observe that the fibers of $\mathrm{Ad}(P)$ have a natural group structure. The product is defined by $[p,g]\cdot[p,h] := [p, g\cdot h]$, and an argument analogous to that of proposition 3.8 shows that this is well-defined. In general, $\mathrm{Ad}(P)$ is a fiber bundle over M (it is usually not a vector bundle, as happens with $\mathrm{ad}(P)$).

Example 3.10. Consider a rank r complex vector bundle over a smooth manifold $E\to M$, and $L(E)$ its frame bundle as in (3.2). As we now, this construction can be used to obtain a principal bundle from a vector bundle. Now, consider the usual left action of $\mathrm{GL}(r,\mathbb{C})$ on $\mathbb{C}^r$ given by $(A,x)\mapsto A\cdot x$ (where we see $x\in\mathbb{C}^r$ as a column vector). The action is smooth and free, so we can consider the associated bundle $L(E)\times_{\mathrm{GL}(r,\mathbb{C})}\mathbb{C}^r$, which is a vector bundle. There is a vector bundle isomorphism $E\cong L(E)\times_{\mathrm{GL}(r,\mathbb{C})}\mathbb{C}^r$ given by

$$L(E)\times_{\mathrm{GL}(r,\mathbb{C})}\mathbb{C}^r \longrightarrow E, ([\boldsymbol{u},(x_1,\dots x_r)])\mapsto \sum_{i=1}^{r} x_i\cdot u_i.$$

Finally, there is a bijection between smooth sections of $P\times_G F$ and G-equivariant maps $P\to F$. Given a G-equivariant map $\tilde{s}\colon P\to F$, $\mathrm{id}_P\times\tilde{s}\colon P\to P\times F$ is G-equivariant, hence descends to the quotient and defines $s\colon P/G\cong M\to (P\times F)/G = P(F)$. Conversely, given a section $s\in\Gamma(M;P\times_G F)$, we define $\tilde{s}(m)=f$ if and only $s([p])=[p,f]$, where $[p]$ is the equivalence class of an element in $P/G\cong M$.

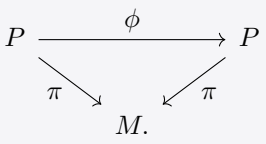
Remark 3.11. Of course, if F already had a right G-action, the same construction holds but there is no need to transform the action.

3.2 The gauge group of a principal bundle

Definition 3.12. Let $P(M,G)$ a principal G-bundle.

A *gauge transformation* of P is a G-equivariant diffeomorphism covering the identity on M. That is

$$\mathcal{G} = \{\phi\in\mathrm{Diff}(P) : \phi(p\cdot g)=\phi(p)\cdot g \quad\text{and}\quad \phi\circ\pi=\pi\}.$$



Clearly, the composition of two such automorphisms yields another one. Moreover, if $\phi\circ\pi=\pi$, since ϕ is a diffeomorphism, composition with ϕ^{-1} yields $\pi=\phi^{-1}\circ\pi$, and ϕ^{-1} is G-equivariant as well. Indeed, $\phi(\phi^{-1}(p)\cdot g)=\phi(\phi^{-1}(p))\cdot g = p\cdot g$ and after applying ϕ^{-1} on both sides, $\phi^{-1}(p)\cdot g=\phi^{-1}(p\cdot g)$. We thus conclude that $\mathcal{G}$ is a group. In fact, it is a subgroup of the group $\mathrm{Diff}(P)$. $\mathcal{G}$ is called *gauge group* or *group of gauge transformations*. Mathematics and physics literature differ in the terminology. Physicists often call gauge group what we have called structure group, and they call the group

$\mathcal{G}$ group of gauge transformations.

Al lot of sources claim that $\mathcal{G}$ is a Hilbert-Lie group (or a Banach-Lie group, it depends on the construction), and that is not very accurate. As we are going to see, the one one endows $\mathcal{G}$ with a Hilbert-Lie group structure is rather elaborate, and $\mathcal{G}$ itself *is not* a Hilbert-Lie group: we have to consider a group it is isomorphic to and complete it (in a certain sense we will precise in chapter 4) to get a Hilbert-Lie group. The construction is not straightforward at all, but we will give an outline in chapter 4.

We will, however, give now a proof of the following proposition, which provides a way of looking at the gauge group from different perspectives.

Proposition 3.13. Let M be a smooth manifold, and $P(M,G)$ a principal G-bundle. Then, $\mathcal{G} \cong \mathrm{Gau}(P) \cong \Gamma(M;\mathrm{Ad(P)})$ as groups, where

$$\mathrm{Gau}(P) := \left\{\tau \in \mathscr{C}^\infty(P,G) : \tau(p\cdot g) = g^{-1}\tau(p)g\right\},$$

and $\Gamma(M;\mathrm{Ad}(P))$ and $\mathrm{Gau}(P)$ are given a group structure through pointwise multiplication.

Proof. Let us establish first the isomorphism $\mathcal{G} \cong \mathrm{Gau}(P)$. Let $\phi \in \mathcal{G}$, since it covers the identity, it must preserve fibers, that is, $\pi(\phi(p)) = \pi(p)$ for every $p \in M$. Since fibers are the orbits of the G-actions on P, and the action is free, for every $p \in P$, there exists a unique $\tau(p) \in G$ such that $\phi(p) = p \cdot \tau(p)$ (see definition 3.1). To see that τ is smooth, it suffices to show it locally. Let $T_U : \pi^{-1}(U) \to U \times G$ be a local trivialization. The condition $\pi \circ \phi = \pi$ means that $\phi(\pi^{-1}(U)) = \pi^{-1}(U)$, so we can consider $T_U \circ \phi$ defined on $\pi^{-1}(U)$. On the one hand, $T_U(\phi(p)) = (\pi(p), s_U(\phi(p)))$. On the other hand, $\phi(p) = p \cdot \tau(p)$, so $T_U(p\cdot\tau(p)) = (\pi(p), s_U(p\cdot\tau(p))) = (\pi(p), s_U(p)\cdot\tau(p))$. In particular, $s_U(\phi(p)) = s_U(p)\cdot\tau(p)$, hence $\tau|_{\pi^{-1}(U)} = s_U^{-1}\cdot(s_U\circ\phi)$ is a composition of smooth maps, and therefore, it is smooth.

Moreover,

$$\phi(p\cdot g) = (p\cdot g)\tau(p\cdot g) = \phi(p)\cdot g = (p\cdot\tau(p))\cdot g,$$

and therefore $p\cdot(g\,\tau(p\cdot g)) = p\cdot(\tau(p)\cdot g)$. Since G acts transitively on P, it follows that $\tau(p\cdot g) = g^{-1}\tau(p)g$. Thus, we can define a map $\mathcal{G} \to \mathrm{Gau}(P)\colon \phi \mapsto \tau$. Let us check that it is a group homomorphism. Let τ_{12} be the image of $\phi_1 \circ \phi_2 \in \mathcal{G}$. Then:

$$p\cdot\tau_{12}(p) = (\phi_1\circ\phi_2)(p) = \phi_1(p\cdot\tau_2(p)) = (p\cdot\tau_2(p))\cdot\tau_1(p\cdot\tau_2(p)) = p\cdot\tau_2(p)\tau_2(p)^{-1}\tau_1(p)\tau_2(p) = p\cdot\tau_1(p)\tau_2(p),$$

and using again that the action is free, $\tau_{12}(p) = \tau_1(p)\tau_2(p)$. Finally, if we are given $\tau \in \mathscr{C}^\infty(P,M)$, we can define $\phi(p) = p\cdot\tau(p)$. This defines an element of $\mathcal{G}$. Smoothness is and the fact that it covers the identity are obvious. Moreover, $\phi(p\cdot g) = p\cdot g\cdot\tau(p\cdot g) = p\cdot gg^{-1}\tau(p)g = \phi(p)\cdot g$. The fact $\mathrm{Gau}(P)\to\mathcal{G}, \tau\mapsto\phi$ so defined is the inverse of the homomorphism above is immediate.

Let us consider now the isomorphism $\mathrm{Gau}(P) \cong \Gamma(M;\mathrm{Ad}(P))$. Let $\tau \in \mathrm{Gau}(P)$. Consider the map

$$\tilde{\tau}\colon M \longrightarrow \mathrm{Ad}(P), \pi(p) \mapsto [p,\tau(p)].$$

Observe that $\tilde{\tau}$ is well-defined; if we pick another element of the fiber $p\cdot G$, then $\tau(p\cdot g) = g\tau(p)g^{-1}$ by definition, and $[p,\tau(p)] = [p\cdot g, g\tau(p)g^{-1}] = [p\cdot g, \tau(p\cdot g)]$. The map $\tilde{\tau}$ is smooth. Indeed, π is a surjective smooth submersion, and $\tilde{\tau}\circ\pi = q\circ(\mathrm{id}\times\tau)$, where $\pi\colon P\to M$ is the principal G-bundle projection, and $q\colon P\times G\to\mathrm{Ad}(P)$ is the quotient map, so we can use the characteristic property of surjective smooth submersions [**36**, Th. 4.29]. Since the group operation is defined for both groups as pointwise multiplication, the assignment $\mathrm{Gau}(P)\to\mathrm{Ad}(P), \tau\to\tilde{\tau}$ is a group homomorphism.

To define an inverse, let $\tilde{\tau}\in\Gamma(M;\mathrm{Aut}(P))$. For each $p\in P$, there is a unique $\tau(p)\in G$ such that $\tilde{\tau}(\pi(p)) = [p,\tau(p)]$ (there is only one representative of the equivalence class whose first entry is p). Once again, τ is smooth, and $\tau(p\cdot g) = g\tau(p)g^{-1}$ because $[p,\tau(p)] = [p\cdot g,\tau(p\cdot g)] = [p\cdot g, g\tau(p)g^{-1}]$. To check smoothness we can use local trivializations again. Let $\Phi_U\colon \pi_G^{-1}(U)\to U\times G$ be a local trivialization (see equation 3.6). In this trivialization, $\tau|_{\pi_G^{-1}(U)} = s_U^{-1}\cdot(\mathrm{pr}_G\circ\Phi_U\circ\tilde{\tau}\circ\pi)$ is a composition of smooth maps. Similar arguments show that this is again a group homomorphism, whose inverse is the one defined above. □

3.3 Theory of connections

For this section, we will work in the smooth category. If a connection is mentioned in the context of a holomorphic principal bundle, we will always mean a connection on the underlying smooth principal. The last section of the chapter dives into the realtion of connections and complex structures on smooth vector bundles. G-bundle. As we will see, there are different equivalent definitions of a connection on a principal G-bundle. The geometric intuition behind the idea of a connection is best seen when one uses it to define a parallel transport, which is a way to identify (or connect) fibers over nearby points. However, we will not treat viewpoint further, but we refer the reader to [51, Sec. 1.7] for details.

Since every principal bundle has a natural Lie group action, every $\xi \in \mathfrak{g}$ defines the *Killing vector field* $X_\xi \in \mathfrak{X}(P)$ through

$$X_{\xi,p} := \left.\frac{\mathrm{d}}{\mathrm{d}t}\right|_{t=0} p \cdot \exp(t\xi) = \left.\frac{\mathrm{d}}{\mathrm{d}t}\right|_{t=0} \phi^{(p)}(\exp(t\xi)) = \left(\phi^{(p)} \circ \gamma\right)'(0) = \mathrm{d}_e\phi^{(p)}(\xi). \tag{3.7}$$

Where γ is the curve given by $\gamma(t) = \exp(t\cdot\xi)$, and we used [36, Cor. 3.25]. Observe that these vector fields are always tangent to the orbits of the action (because the curves $\exp(tX)\cdot p$ are contained in said orbits). The fact that the assignment $p \mapsto X_{\xi,p}$ defines a smooth vector is readily checked (see, for example [66, Prop. 27.11]). The natural projection $\pi\colon P \to M$ defines, for each $p \in P$ a *vertical subspace* given by $V_pP := \ker \mathrm{d}_p\pi$. Its elements are called *vertical vectors*, and a vector field that is always vertical is called a *vertical vector field*. Observe that Killing vector fields are always vertical. Indeed,

$$\mathrm{d}_p\pi(X_{\xi,p}) = \mathrm{d}_e\left(\pi \circ \phi^{(p)}\right)(\xi) = 0$$

because $\pi \circ \phi^{(p)}$ is constant. In fact, $\mathrm{d}_e\phi^{(p)}\colon \mathfrak{g} \to V_pP$ is a vector space isomorphism (see [51, Lemma 1.3.1] or [66, Prop. 27.18]). Moreover, $VP \cong P \times \mathfrak{g}$, so it is trivial, and the isomorphism is given by $(p,\xi) \mapsto X_{\xi,p}$.

Definition 3.14 (Connection on a principal G-bundle). Let $P(M,G)$ be a principal G-bundle. A *connection* on P is a smooth distribution HP such that

1. $T_pP = V_pP \oplus H_pP$ for every $p \in P$
2. $H_{\phi_g(p)}P = \mathrm{d}_p\phi_g(H_pP)$ for every $p \in P, g \in G$. We say that the distribution is G-invariant.

We call H_pP the *horizontal subspace* at p.

The direct sum decomposition $T_pP = H_pP \oplus V_pP$ on each tangent space implies that every vector $v \in T_pP$ can be uniquely decomposed as into a vertical and a horizontal vector $w = w^H + w^V$. In fact, given a vector field $X \in \mathfrak{X}(P)$, the fiberwise decomposition defines to smooth vector fields $X^V \in \Gamma(P;VP)$ and $X^H \in \Gamma(P;HP)$ which are called the *vertical* and *horizontal components* of X respectively [66, Prop. 28.3]. Observe that both components depend on the connection. Of course, we have the same decomposition for vectors of each tangent space.

Remark 3.15.

1. The projection $\mathrm{d}_p\pi$ defines a vector space isomorphism between H_pP and $T_{\pi(p)}P$. This follows from the fact that $\mathrm{d}_p\pi$ is surjective and $\ker \mathrm{d}_p\pi = V_pP$, so by the first isomorphism theorem for vector spaces $T_pP/V_pP \cong H_pP \cong T_{\pi(p)}M$.
2. The vertical-horizontal decomposition is G-invariant. More precisely, if $w = w^H + w^V \in T_pP$, then the vertical and horizontal parts of $\mathrm{d}_p\phi_g(w) \in T_{p\cdot g}P$ are $\mathrm{d}_p\phi_g\left(w^V\right)$ and $\mathrm{d}_p\phi_g\left(w^H\right)$ respectively. Indeed, $\mathrm{d}_p\phi_g(w) = \mathrm{d}_p\phi_g\left(w^H\right) + \mathrm{d}_p\phi_g\left(w^V\right)$. The latter is vertical because $\mathrm{d}_{g\cdot p}\pi\left(\mathrm{d}_p\phi_g\left(w^V\right)\right) = \mathrm{d}_p\pi(w^V) = 0$, and the former is horizontal because the distribution is G-invariant. The result follows from the uniqueness of the decomposition.

The main idea of the definition, is that a connection is a *choice* of a complementary subspace to the vertical subspace V_pP. Observe that, in general, there is no canonical way of choosing such a complement. This definition is quite intuitive, in the sense that the geometric meaning of what is

being done; choosing "smoothly" a complementary to the vertical subspace, is quite clear. There is another equivalent way of seeing this as a splitting of a short exact sequence. Observe that, at the level of vector spaces we have the following exact sequence

$$0 \to V_pP \xrightarrow{i} T_pP \xrightarrow{\mathrm{d}_p\pi} T_{\pi(p)}M \to 0.$$

This sequence can be extended to a sequence of vector bundles over P

$$0 \to VP \xrightarrow{i} TP \xrightarrow{\mathrm{d}\pi'} \pi^*TM \to 0, \tag{3.8}$$

where $\mathrm{d}_p\pi' \colon T_pP \to (\pi^*TM)_p, v \mapsto (p, \mathrm{d}_p\pi(v))$. Given a short exact sequence of vector bundles over a manifold, $0 \to A \xrightarrow{i} B \xrightarrow{j} C \to 0$ a *right splitting* is a bundle homomorphism $k \colon C \to B$ such that $j \circ k = \mathrm{id}_C$, and a *left splitting* is a bundle homomorphism $\ell \colon B \to A$ such that $\ell \circ i = \mathrm{id}_A$. There is a right splitting if and only if there is a left splitting. If there is such a splitting, we say that the short exact sequence splits. In our case, every horizontal subbundle induces a G-equivariant left splitting of (3.8) through the vertical projection $\mathsf{pr} \colon TP \to VP$ which maps each vector to its vertical component. G-equivariance in this context means that $\mathrm{d}\phi_g \circ \mathsf{pr} = \mathsf{pr} \circ \mathrm{d}\phi_g$. Reciprocally, a G-equivariant right splitting ℓ of (3.8) defines a horizontal vector bundle by $HP := \ker \ell$ (see [66, Prop. 27.20] for a proof of the equivalence, it is done using left splittings there).

Moreover, connections allow us to speak about horizontal lifts. Let $P(M, G)$ be a principal bundle endowed with a connection, which we see as a horizontal subbundle. Let $Y \in \mathfrak{X}(M)$. A *horizontal lift* is another vector field Z that is horizontal such that $\mathrm{d}_p\pi(Z_p) = Y_{\pi(p)}$ for each $p \in P$. As we see in the following lemma, horizontal lifts always exist.

Lemma 3.16 (Existence of horizontal lifts). Let $P(M, G)$ be a principal bundle, and let $Y \in \mathfrak{X}(M)$. Then, Y admits a unique horizontal lift. Moreover, if $Y^{\mathbf{HL}}$ is the lift, it satisfies $(\phi_g)_* Y^{\mathbf{HL}} = Y^{\mathbf{HL}}$.

Proof. By remark 3.15, the only option is to define $Y^{\mathbf{HL}}_p := (\mathrm{d}_p\pi|_{H_pP})^{-1}(Y_{\pi(p)})$. Since smoothness is a local issue, up to choosing a trivialization, we can assume that P is trivial. Hence, $TP \cong TM \oplus TG$. Then, we choose any smooth vector field $\hat{Y} \in \mathfrak{X}(P)$ satisfying $\mathrm{d}_{p,g}\pi(\hat{Y}_{p,g}) = Y_p$, for example, $(p, g) \mapsto (Y_p, 0) = \hat{Y}_p$. By uniqueness of the vertical-horizontal decomposition, $\hat{Y}^H = Y^{\mathbf{HL}}$, and it follows that the horizontal lift is smooth. Invariance under the action is clear from the fact that

$$\mathrm{d}_p\phi_g|_{H_pP} \circ \left(\mathrm{d}_p\pi|_{H_pP}\right)^{-1} = \left(\mathrm{d}_{p\cdot g}\pi|_{H_{p\cdot g}P}\right)^{-1},$$

so $\mathrm{d}_p\phi_g\left(Y^{\mathbf{HL}}_p\right) = \mathrm{d}_p\phi_g\left(\left(\mathrm{d}_p\pi|_{H_pP}\right)^{-1}(Y_p)\right) = \left(\mathrm{d}_{p\cdot g}\pi|_{H_{p\cdot g}P}\right)^{-1}(Y_p) = Y^{\mathbf{HL}}_{p\cdot g}$. □

We will write $Y^{\mathbf{HL}}$ for the horizontal lift of a vector field Y. It is not explicit in the notation, but it clearly depends on the connection.

The definitions we have for connections are geometrically clear, but not very practical form a computational point of view. In our seek for a better definition, we will need the following lemma.

Lemma 3.17. Let G be a Lie group that acts smoothly and freely from the right on a smooth manifold M, and let $\phi \colon M \times G \to M$ denote the action. Then,

$$(\phi_g)_*(X_\xi) = X_{\mathrm{Ad}_{g^{-1}}(\xi)}.$$

Proof. Pointwise, we may show that $\mathrm{d}_p\phi_g(X_{\xi,p}) = \left(X_{\mathrm{Ad}_{g^{-1}}(\xi)}\right)_{p\cdot g}$ for all $p \in P, g \in G$. Let $h \in G$ and observe that

$$\phi_g \circ \phi^{(p)}(h) = \phi_g(p \cdot h) = p \cdot hg = (p \cdot g) \cdot g^{-1}hg = \phi^{(p\cdot g)} \circ C_{g^{-1}}(h).$$

Hence,

$$\begin{aligned}\left[(\phi_g)_*(X_\xi)\right]_{p\cdot g} &= \mathrm{d}_p\phi_g\left(\mathrm{d}_e\phi^{(p)}(\xi)\right) = \mathrm{d}_e(\phi^{(p\cdot g)} \circ C_{g^{-1}})(\xi) = \mathrm{d}_e\phi^{(p\cdot g)}\left(\mathrm{d}_eC_{g^{-1}}(\xi)\right) \\ &= \mathrm{d}_e\phi^{(p\cdot g)}\left(\mathrm{Ad}_{g^{-1}}(\xi)\right) = \left(X_{\mathrm{Ad}_{g^{-1}}(\xi)}\right)_{p\cdot g}.\end{aligned}$$

□

Let V be a vector space. By a V-valued p-form we mean a p-form with values in the trivial vector bundle $M \times V$, that is, a section of the vector bundle $(M \times V) \otimes \bigwedge^p T^*M$ (see section A.7). Observe that a general (real) p-form may be written as a sum

$$\alpha = v_1 \otimes \alpha_1 + v_2 \otimes \alpha_2 + \dots v_r \otimes \alpha_r \tag{3.9}$$

where $v_i \in V$, and $\omega_j \in \Omega^p_{\mathbb{R}}(M)$. If V is a complex vector space and the forms are taken to be complex everything works the same. We will write $\Omega^p_{\mathbb{R}}(M;V)$ for the set of real V-valued p-forms.

Remark 3.18. Let us consider the particular case of a V-valued one-form, and analyze it more carefully. Suppose $\omega \in \Omega^1(M;V)$ can be written as in equation (3.9). Fix a point p. Then, $\omega_p = \sum_{i=1}^r v_i \otimes \omega_{i,p}$ is an element of $V \otimes T^*_pM$ (we may omit the manifold). When we evaluate ω_p at a vector $w \in T_pM$ we simply get a vector

$$\alpha_p(w) = \sum_{i=1}^r v_i \otimes \alpha_{i,p}(w) = \sum_{i=1}^r \alpha_{i,p}(w) \cdot v_i,$$

where we used that $\alpha_{i,p}(w)$ is a real number for every $1 \leq i \leq r$. Furthermore, since each $\alpha_{i,p}$ is linear, α_p is a linear map from T_pM to V. Taking this into account, a V-valued one-forms can be thought of as a family of linear maps $\alpha_p \colon T_pM \to V$ that vary smoothly with the point p. This idea of "varying smoothly" simply means that, whenever $X \in \mathfrak{X}(M)$, we may write $\alpha(X)$ as a linear combination of vectors where the coefficients are smooth functions (in this case, these smooth functions are simply $\alpha_i(X)$).

Definition 3.19 (Connection 1-form). Let $P(M,G)$ be a principal G-bundle, and let HP be a connection. Define the *connection 1-form* by

$$A_p(w) = \left(\mathrm{d}_e\phi^{(p)}\right)^{-1}\left(w^V\right) \qquad w \in T_pP.$$

Observe that we are using A instead of a greek letter to denote the one-form. This notation is not universal. The following theorem show that ω is smooth, and some properties that characterize it.

Theorem 3.20. Let $P(M,G)$ be a principal G-bundle, and consider a connection A. Let ω be its connection one-form. Then, A is a smooth one-form, and

1. $A(X_\xi) = \xi$ for every $\xi \in \mathfrak{g}$
2. $(\phi_g)^*A = \mathrm{Ad}_{g^{-1}} \circ A$. We say that A is *G-equivariant*.

Moreover, if $A \in \Omega^1(P;\mathfrak{g})$ satisfies the previous conditions, then there is a unique connection in P whose connection 1-form is A.

Proof. The fact that $A \colon TP \to \mathfrak{g}$ is smooth follows from the fact that it can be written as the following composition of smooth maps

$$TP \xrightarrow{\mathrm{pr}} VP \cong P \times \mathfrak{g} \xrightarrow{\mathrm{pr}_{\mathfrak{g}}} \mathfrak{g}.$$

Property 1 follows immediately from $\mathrm{d}_e\phi^{(p)}(\xi) = X_{\xi,p}$ and the fact that $X_{\xi,p}$ is vertical. For property 2, since both sides are linear, it suffices to check the equality independently for horizontal and vertical vectors. Observe first that for horizontal vector fields both sides yield 0, so the equality holds. Indeed, if $w \in T_pP$ is horizontal, by remark 3.15, so is $\mathrm{d}_p\phi_g(w)$. Now, to check it for vertical vector fields, we may use linearity to check it just for Killing vector fields. Thus,

$$(\phi_g)^*A(X_\xi) = A((\phi_g)_*(X_\xi)) = A(X_{\mathrm{Ad}_{g^{-1}}(\xi)}) = \mathrm{Ad}_{g^{-1}}(\xi) = \mathrm{Ad}_{g^{-1}} \circ A(X_\xi), \tag{3.10}$$

where we used lemma 3.17.

Now, let $A \in \Omega^1(P;\mathfrak{g})$ satisfy properties 1 and 2. A connection is given by setting $H_pP := \ker A_p$. Observe that we have a pointwise vector space short exact sequence given by $0 \to H_pP \to T_pP \xrightarrow{A_p}$

$\mathfrak{g} \to 0$, and the map $\mathrm{d}_e\phi^{(p)} \colon \mathfrak{g} \to V_pP \subset T_pP$ is a left splitting. The existence of the splitting implies (see [66, Prop. 27.20]) that there is an isomorphism $T_pP \cong \mathfrak{g} \oplus H_pP$, and since $\mathfrak{g} \cong V_pP$, we get $T_pP \cong V_pP \oplus H_pP$ thus satisfying condition 1 of definition 3.14. To check condition 2, choose $w \in H_pP$, and let us see that $\mathrm{d}_e\phi_g(w) \in H_{p\cdot g}P = \ker A_{g\cdot p}$.

$$A_{g\cdot p}(\mathrm{d}_p\phi_g(w)) = \left((\phi_g)^*A\right)_p(w) = \mathrm{Ad}_{g^{-1}}(A_p(w)) = \mathrm{Ad}_{g^{-1}}(0) = 0.$$

The only thing left to check if that HP is a smooth subbundle. This is done by showing that HP is locally spanned by smooth sections. It is a somewhat lengthy calculation, so we do not show it here. A complete proof can be found in [66, Prop. 28.5] □

Moreover, one may show that every principal bundle admits a connection. This is done through a partition of unity argument [51, Prop. 1.3.7]. The horizontal subbundle determined by a connection whose one-form is A will be denoted H^AP.

3.4 Curvature of a connection

First and foremost, when we deal with vector-valued p-forms, we can extend the exterior derivative as follows. If V is a real vector space, and $\beta \in \Omega^p_{\mathbb{R}}(M;V)$, we have seen that we may express β in the following manner (we may suppose that $\{v_1, \dots v_r\}$ is a basis of V) $\beta = \omega^j \otimes v_j$, where $\omega_j \in \Omega^p_{\mathbb{R}}(M)$. We can simply define

$$\mathrm{d}\beta := \mathrm{d}\beta_j \otimes v_j \in \Omega^{p+1}_{\mathbb{R}}(M;V).$$

Remark 3.21. The differential is well-defined, regardless of the basis we use to express β. The justification is quite similar to the proof of theorem 2.4. Indeed, choose two basis $\{v_1, \dots v_n\}$ and $\{\tilde{v}_1, \dots \tilde{v}_n\}$ of V, and suppose that $\beta = \beta^j \otimes v_j = \tilde{\beta}^k \otimes \tilde{v}_k$. Since any two basis are related by a unique linear isomorphism, there is a non-singular matrix $A = (a^i_j) \in \mathrm{GL}(r, \mathbb{R})$ such that $\tilde{v}_k = a^i_k \cdot v_i$. Then, $a^k_i \cdot \tilde{\beta}^i = \beta^k$. Thus,

$$\mathrm{d}\beta = \mathrm{d}\beta^j \otimes v_j = \mathrm{d}\left(a^j_k \cdot \tilde{\beta}^k\right) \otimes v_j = \mathrm{d}\tilde{\beta}^k \otimes (a^j_k \cdot v_j) = \mathrm{d}\tilde{\beta}^k \otimes \tilde{v}_k,$$

using the fact that the exterior differential is linear. Observe as well that $\mathrm{d}_\nabla = \mathrm{d}$ defines a connection on trivial vector bundles.

Definition 3.22 (Covariant exterior derivative). Let $P(M,G)$ be a principal G-bundle, and V be a finite dimensional vector space. Given $\alpha \in \Omega^p_{\mathbb{R}}(M;V)$, we define its *covariant exterior derivative with respect to a connection* A as the V-valued $p+1$ form given by

$$\mathrm{D}_A\alpha(X_0, X_1, \dots X_p) := \mathrm{d}\alpha\left(X_0^H, X_1^H, \dots X_p^H\right) \qquad X_1, \dots X_p \in \mathfrak{X}(P)$$

Given $\alpha \in \Omega^p_{\mathbb{R}}(M;V)$, it is common to denote α^H the V-valued p-form

$$(\alpha)^H(X_1, \dots X_p) := \alpha\left(X_1^H, \dots X_p^H\right).$$

Therefore, we may also write $\mathrm{D}_A\alpha = (\mathrm{d}\alpha)^H$. The symbol A in the notation is meant to make a reference to the connection one-form. This is because the covariant derivative depends on the connection (because the horizontal components of the vectors depend on the connection). We are now ready to define the curvature of a connection.

Definition 3.23 (Curvature of a connection). Let $P(M,G)$ be a principal G-bundle endowed with a connection with connection form A. We define the curvature of this connection as $\mathrm{D}_A A$.

In particular, the curvature of the connection is a $\mathfrak{g}$-valued two-form. Again, a geometric interpretation of the curvature (and a justification of the fact that we call *this* curvature) is better seen in terms of parallel transport.

If, given $\alpha \in \Omega^k_{\mathbb{R}}(M;\mathfrak{g})$ and $\beta \in \Omega^\ell_{\mathbb{R}}(M;\mathfrak{g})$, we define $[\alpha,\beta] \in \Omega^{k+\ell}_{\mathbb{R}}(M;\mathfrak{g})$ by

$$[\beta \wedge \alpha](X_1,\dots X_{k+\ell}) := \frac{1}{k!\cdot \ell!}\sum_{\sigma\in S_{k+\ell}} \operatorname{sgn}(\sigma)\cdot[\beta(X_{\sigma(1)},\dots X_{\sigma(\ell)}),\alpha(X_{\sigma(\ell+1)},\dots X_{\sigma(\ell+k)})], \tag{3.11}$$

then we have the following theorem, that gives an expression for the curvature in terms of the connection one-form. Equivalently, we may define $[\cdot\wedge\cdot]$ by linear extension of $[(\beta\otimes\xi)\wedge(\alpha\otimes\zeta)] := (\beta\wedge\alpha)\otimes[\xi,\zeta]$ for α,β usual differential forms and $\xi,\zeta\in\mathfrak{g}$ (see [21, Sec. 5.2]).

Theorem 3.24 (Structure equation). Let $P(M,G)$ be a principal G-bundle endowed with a connection, and let A be the corresponding connection one-form. Then,

$$\mathrm{D}_A A = \mathrm{d}A + \frac{1}{2}\cdot[A\wedge A].$$

Proof. The proof is rather lengthy, see [30, Th. 5.2] or [60, Th. 11.2]. The strategy is to consider separately the cases where $D_A A$ acts on two vertical vectors, two horizontal vectors, and a horizontal and a vertical vector. Then, one concludes using the fact that both sides are linear. □

We devote the rest of the section to show that the curvature of a connection on a principal G-bundle can be viewed in a canonical way as a $\mathrm{ad}(P)$-valued two-form, that is, a section of a vector bundle. Let us start by a definition. For now, we keep the discussion general, and we define horizontal differential forms any finite-dimensional representation of the Lie group.

Definition 3.25. Let $P(M,G)$ be a principal G-bundle with right action $\phi\colon P\times G\to G$, and consider a finite-dimensional representation $\sigma\colon G\to \mathrm{Aut}(V)$ for some finite-dimensional vector space V. Given $\alpha\in\Omega^k_{\mathbb{R}}(P;V)$ is called *horizontal (of type σ)* if it satisfies:

1. For every vertical vector field X, $i_X\alpha = 0$
2. $(\phi_g)^*\alpha = \sigma(g^{-1})\circ\alpha$ for every $g\in G$.

We denote the space of all horizontal k-forms with respect to σ by $\Omega^k_{\sigma,\mathrm{hor}}(P;V)$.

Remark 3.26. The space of connections (more precisely, connection one-forms) on a principal bundle, which we denote $\mathscr{A}(P_G)$ is an affine space modelled on the space of horizontal one-forms with respect to the adjoint representation, i.e. $\Omega^1_{\mathrm{Ad},\mathrm{hor}}(P;\mathfrak{g})$. Observe the clear parallelism between this claim and lemma 2.15. To see the claim, let A_1 and A_2 be the connection one-forms of two connections on a principal G-bundle, and let $A := A_1 - A_2$. Being the differential of two $\mathfrak{g}$-valued one-forms, it is a one-form, so we need only check that it is horizontal. The fact that $(\phi_g)^*A = \mathrm{Ad}_{g^{-1}}\circ A$ follows from the second statement of theorem 3.20. Now, the vertical bundle is spanned by vector fields X_ξ for $\xi\in\mathfrak{g}$, therefore $i_{X_\xi}A = A(X_\xi) - A_2(X_\xi) = \xi - \xi = 0$. Reciprocally, given a connection $A\in\Omega^1(P;\mathfrak{g})$, and a horizontal one-form $\alpha\in\Omega^1_{\mathrm{Ad},\mathrm{hor}}(P;\mathfrak{g})$, the sum $A+\alpha$ is a connection one-form. By property 2 of definition 2.15, $A+\alpha$ is G-equivariant. Moreover, given a vertical vector field X_ξ, we have $(A+\alpha)(X_\xi) = A(X_\xi)+\alpha(X_\xi) = \xi + 0$.

This discussion implies that if we fix a connection one-form A, then any other connection may be obtained as $A+\alpha$ for $\alpha\in\Omega^1_{\mathrm{Ad},\ \mathrm{hor}}(P;\mathfrak{g})$. It is then natural to ask how the curvature of $A+\alpha$ relates to that of $A+\alpha$. From the structure equation (theorem 3.24) it follows that

$$\begin{aligned}\mathrm{D}_{A+\alpha}(A+\alpha) &= \mathrm{d}(A+\alpha) + \frac{1}{2}\cdot[A+\alpha, A+\alpha]\\ &= \mathrm{d}A + \frac{1}{2}\cdot[A,A] + \mathrm{d}\alpha + \frac{1}{2}\cdot[\alpha,\alpha] + \frac{1}{2}\cdot[A\wedge\alpha] + \frac{1}{2}\cdot[\alpha\wedge A]\\ &= \mathrm{D}_A A + \mathrm{d}\alpha + [A\wedge\alpha] + \frac{1}{2}\cdot[\alpha,\alpha].\end{aligned}$$

Moreover, for $\alpha\in\Omega^p_{\mathrm{Ad},\ \mathrm{hor}}(P;\mathfrak{g})$, the following relation holds $\mathrm{D}_A\alpha = \mathrm{d}\alpha + [A\wedge\alpha]$ (see [51, Remark 1.4.12]). Taking this identity into account, the previous equation reads

$$\mathrm{D}_{A+\alpha}(A+\alpha) = \mathrm{D}_A A + \mathrm{D}_A\alpha + \frac{1}{2}\cdot[\alpha\wedge\alpha]. \tag{3.12}$$

As the following lemma shows, the curvature two-form is horizontal.

Lemma 3.27. Let $P(M,G)$ be a principal G-bundle and let $A \in \mathscr{A}(P(M,G))$ be a connection. Then, its curvature is horizontal, i.e. $\mathrm{D}_A A \in \Omega^2_{\mathrm{Ad,hor}}(P;\mathfrak{g})$.

Proof. The fact that $i_X(\mathrm{D}_A A) = 0$ for vertical vector fields is obvious. For the second condition, one uses the structure equation. See [51, Page 40]. □

We are going to show now that there is an isomorphism $\Omega^k_{\sigma,\mathrm{hor}}(P;V) \cong \Omega^k(M; P\times_G V)$. In particular, this means that there is an isomorphism $\Omega^2_{\mathrm{Ad,hor}}(P;\mathfrak{g}) \cong \Omega^2(M;\mathrm{ad}(P))$. By means of this isomorphism, we can view the curvature of the connection $\mathrm{D}_A A$ as a $\mathrm{ad}(P)$-valued 2-form.

Theorem 3.28. Let $P(M,G)$ be a principal G-bundle, and $\sigma\colon G \to \mathrm{Aut}(V)$ a finite-dimensional representation.

To every $\alpha \in \Omega^k_{\sigma,\mathrm{hor}}(P;V), k \geq 1$ corresponds a unique element $\Upsilon(\alpha) \in \Omega^k_{\mathbb{R}}(M; P\times_G V)$ such that the following diagram commutes

$$\begin{array}{ccc} \bigwedge^k TP & \xrightarrow{\mathrm{pr}\times\alpha} & P\times V \\ {\scriptstyle \Lambda^k \mathrm{d}\pi}\downarrow & & \downarrow{\scriptstyle q} \\ \bigwedge^k TM & \xrightarrow{\Upsilon(\alpha)} & P\times_G V \end{array}$$

Moreover, the assignment $\Upsilon\colon \Omega^k_{\sigma,\mathrm{hor}}(P;V) \to \Omega^k_{\mathbb{R}}(M; P\times_G V), \alpha \mapsto \Upsilon(\alpha)$ is a vector space isomorphism.

Proof. Recall that $P\times_G V = P\times V/\sim$ where $\sim$ is the equivalence relation $(p,v)\sim(p',v')$ if and only if $p' = p\cdot g$ and $v' = \sigma_{g^{-1}}(v)$. It is a vector bundle. To fix notation, set $q\colon P\times V \to P\times_G V$ the quotient map, $\mathrm{pr}\colon \bigwedge^k TP \to P$ the usual projection, and $\Lambda^k\mathrm{d}\pi\colon \bigwedge^k TP \to \bigwedge^k TM$ the bundle homomorphism uniquely determined by

$$\Lambda^k\mathrm{d}\pi(v_1\wedge v_2\ldots\wedge v_k) := (\mathrm{d}_p\pi(v_1))\wedge(\mathrm{d}_p\pi(v_2))\wedge\ldots\wedge(\mathrm{d}_p\pi(v_k)) \quad \text{if} \quad v_1,\ldots v_k \in T_pP.$$

$\Lambda^k\mathrm{d}\pi$ is a surjective smooth submersion. Define as well $q_p\colon V \to P\times_G V$ by $q_p(v) := [p,v]$. Having said that, let $m\in M$, $v_i \in T_mM$, $1\leq i\leq k$, and choose $p\in P$ that projects onto m (that is $\pi(p)=m$). Choose as well $w_i \in T_pP$ such that $\mathrm{d}_p\pi(w_i) = v_i$ for $1\leq i\leq k$. We define

$$(\Upsilon\alpha)_m(v_1,\ldots v_k) := (q_p\circ\alpha_p)(w_1,\ldots w_k). \tag{3.13}$$

Once we see that this is independent of the choices of p and vectors $w_1,\ldots w_k$, α will be well-defined as a map, and commutativity of the diagram will be obvious. Recall that

$$\Omega^k_{\mathbb{R}}(M; P\times_G V) = \Gamma\left(M; \left(\bigwedge^k TM\right)^* \otimes P\times_G V\right)$$

and $\left(\bigwedge^k TM\right)^* \otimes P\times_G V$ is identified with the bundle whose fibers are alternating multilinear forms from $T_mM\times\overset{k}{\cdots}\times T_mM \to (P\times_G V)_m$. Hence, $\Upsilon(\alpha)\circ\Lambda^k\mathrm{d}\pi(v_1\wedge\ldots\wedge v_k)$ is just the form $\Upsilon(\alpha)$ acting on $v_1,\ldots v_r$. As usual, smoothness follows from the characteristic property of surjective smooth submersions [36, Th. 4.29] applied to $\Lambda^k\mathrm{d}\pi$.

To see that (3.13) does not depend on our choices, choose another point p' projecting to m. Thus, we will have $p' = p\cdot g$ for some $g\in G$. Let $w_i' \in T_{p'}P$ such that $\mathrm{d}_{p'}\pi(w_i') = v_i$ for $1\leq i\leq k$. Then, since $\mathrm{d}_{p'}\pi(w_i' - \mathrm{d}_p\phi_g(w_i)) = v_i - \mathrm{d}_p(\pi\circ\phi_g)(w_i) = v_i - v_i = 0$, there are vertical vectors $u_i \in V_{p'}P$ satisfying $w_i' = \mathrm{d}_p\phi_g(w_i) + u_i$. Thus, we get

$$\begin{aligned} q_{p'}\circ\alpha_{p'}(w_1',\ldots w_k') &= q_{p\cdot g}\circ\alpha_{p\cdot g}(\mathrm{d}_p\phi_g(w_1)+u_1,\ldots,\mathrm{d}_p\phi_g(w_k)+u_k) \\ &\overset{(3)}{=} q_{p\cdot g}\circ\alpha_{p\cdot g}(\mathrm{d}_p\phi_g(w_1),\ldots,\mathrm{d}_p\phi_g(w_k)) \\ &\overset{(4)}{=} q_p\circ\sigma(g)\circ\left((\phi_g)^*\alpha\right)(w_1,\ldots,w_k) \overset{(5)}{=} q_p\circ\tilde{\alpha}_p(w_1,\ldots,w_k). \end{aligned}$$

In (3), we used the fact that, if α is horizontal, then it vanishes when one of its entries is a vertical vector field. (4) uses the fact that $q_{p\cdot g} = q_p\circ\sigma_g$ (see remark 3.7), and (5) follows from the identity

$\sigma(g^{-1})\circ\alpha = (\phi_g)^*\alpha$ for horizontal forms. Observe that the map Υ is clearly linear. For the converse, given $\beta\in\Omega^k_{\mathbb{R}}(M;P\times_G V)$, we define $\hat{\Upsilon}(\beta)$ by $(\hat{\Upsilon}(\beta))_p = q_p^{-1}\circ(\pi^*\beta)_p$. Observe that, looking at (3.13) this is the only reasonable choice. $\hat{\Upsilon}(\beta)$ is horizontal. Indeed, if w^V is vertical, $i_{w^V}\hat{\Upsilon}(\beta)=0$ because $\mathrm{d}_p\pi\left(w^V\right)=0$. Moreover,

$$\begin{aligned}\left((\phi_g)^*\hat{\Upsilon}(\beta)\right)_p(w_1,\dots w_k) &= \Big(\hat{\Upsilon}(\beta)\Big)_{p\cdot g}\left(\mathrm{d}_p\phi_g(w_1),\dots \mathrm{d}_p\phi_g(w_k)\right)\\ &= q_{p\cdot g}^{-1}\circ\beta_{\pi(p)}\left(\mathrm{d}_{p\cdot g}\pi(\mathrm{d}_p\phi_g(w_1)),\dots \mathrm{d}_{p\cdot g}\pi(\mathrm{d}_p\phi_g(w_k))\right)\\ &= q_{p\cdot g}^{-1}\circ\beta_{\pi(p)}(\mathrm{d}_p\pi(w_1),\dots \mathrm{d}_p\pi(w_k)) = \sigma(g^{-1})\circ q_p^{-1}\circ\beta_{\pi(p)}(\mathrm{d}_p\pi(w_1),\dots \mathrm{d}_p\pi(w_k)).\end{aligned}$$

See (4) for the last equality. The map $\hat{\Upsilon}\colon\Omega^k_{\mathbb{R}}(M;P\times_G V)\to\Omega^k_{\sigma,\mathrm{hor}}(P;V)$ is also linear. One readily verifies that the two compositions of Υ and $\hat{\Upsilon}$ yield the corresponding identities. □

Remark 3.29.

1. From remark 3.26 and theorem 3.28, it follows that $\mathscr{A}(P)$ can be seen as an affine space modelled on $\Omega^1_{\mathbb{R}}(M;\mathrm{ad}(P))$.

2. When we see the curvature of a connection as an element of $\Omega^2_{\mathbb{R}}(M;P\times_G\mathfrak{g})\cong\Omega^2_{\mathrm{Ad},\ \mathrm{hor}}(M;\mathrm{ad}(P))$, we will write F_A. More precisely, we define $F_A := \Upsilon(\mathrm{D}_A A)$.

3.5 Connections on associated vector bundles

Let $P(M,G)$ be a principal bundle over a smooth manifold M, and let $\sigma\colon G\to\mathrm{Aut}(V)$ be a finite-dimensional representation. Consider $P(V) := P\times_G V$ the associated vector bundle defined as in proposition 3.8. Our objective for this section is to show that a connection A on P induces a connection (in the sense of definition 2.1) on the associated vector bundle $P(V)$. Before giving the definition, recall the isomorphism $\Upsilon\colon\Omega^k_{\sigma,\mathrm{hor}}(P;V)\to\Omega^k(M;P\times_G V)$ given in theorem 3.28.

Definition 3.30. Under the previous hypothesis, let $\alpha\in\Omega^k_{\mathbb{R}}(M;P(V))$, and suppose that the principal G-bundle $P(M,G)$ is endowed with a connection A. The covariant exterior derivative of α induced by A is defined by

$$\mathrm{d}_A\alpha := \Upsilon\left(\mathrm{D}_A(\Upsilon^{-1}(\alpha))\right)$$

According to the proof given in theorem 3.28, $\mathrm{d}_A\alpha\in\Omega^k_{\mathbb{R}}(M;P(V))$ satisfies

$$(\mathrm{d}_A\alpha)_{\pi(p)}(\mathrm{d}_p\pi(w_0),\dots\mathrm{d}_p\pi(w_k)) = q_p\circ(\mathrm{D}_A\Upsilon^{-1}(\alpha))_p(w_0,\dots w_k) \tag{3.14}$$

In particular, this construction defines a family of maps

$$\mathrm{d}_A\colon\Omega^k_{\mathbb{R}}(M;P(V))\longrightarrow\Omega^{k+1}_{\mathbb{R}}(M;P(V)).$$

These maps are covariant exterior derivatives in the sense of theorem 2.4. Observe that, for now, we work with real representations, but most of the topics we covered in chapter 2 apply to real vector bundles if we do not complexify the tangent bundle. In particular, d_A defines a connection ∇^A in the sense of definition 2.1, that is, $\nabla^A=\mathrm{d}_A$ acting on sections (i.e. zero-forms) satisfies Leibniz's rule $\nabla^A(f\cdot s) = \mathrm{d}f\otimes s + f\cdot\nabla^A s$ [**51**, Prop. 1.5.4]. To make sense of this, we must observe first that theorem 3.28 does not cover the case $k=0$. In that case, we can still relate 0-forms (sections). Indeed, there is a vector space isomorphism [**51**, Prop. 1.2.6]

$$\left\{\begin{array}{c}G\text{-equivariant maps}\\ f\colon P\to V\end{array}\right\}\longleftrightarrow\Gamma(M;P\times_G V), \tag{3.15}$$

which assigns to a G-equivariant map $\hat{\xi}$ the section $\boldsymbol{\xi} := \Upsilon(\hat{\xi})$ given by $\boldsymbol{\xi}(m) = [p,\hat{\xi}(p)], m=\pi(p)$ (that is, $\sigma\circ\pi(p) = q_p(f(p))$). If we call the isomorphism Υ, we can extend d_A to act on sections, and it is precisely acting on sections where we have $\mathrm{d}_A=\nabla^A$.

Remark 3.31. Observe that for every $\alpha \in \Omega^k_{\sigma,\mathrm{hor}}(P;V)$, $\mathrm{D}_A\alpha$ is horizontal. Indeed, $i_X(\mathrm{D}_A\alpha) = 0$ for vertical vector fields is obvious, and one readily shows that $\phi_g^*(D_A\alpha) = \mathrm{D}_A(\phi_g^*\alpha) = \mathrm{D}_A(\sigma(g^{-1}) \circ \alpha) = \sigma(g^{-1}) \circ \mathrm{D}_A\alpha$. The first equality follows from remark 3.15. Indeed,

$$\begin{aligned}(\phi_g)^*\mathrm{D}_A A(X,Y) &= \mathrm{d}A\left(\mathrm{d}\phi_g(X)^H, \mathrm{d}\phi_g(Y)^H\right) = \mathrm{d}A(\mathrm{d}\phi_g\left(X^H\right), \mathrm{d}\phi_g\left(Y^H\right)) \\ &= \left((\phi_g)^*\mathrm{d}A\right)^H(X,Y) = \mathrm{d}((\phi_g)^*A)^H(X,Y) = \mathrm{D}_A((\phi_g)^*A)(X,Y)\end{aligned}$$

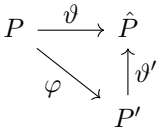
3.6 Reductions of the structure group

Definition 3.32 (Reduction of principal bundles). Let $P_1(M,G_1)$ and $P_2(M,G_2)$ two $\mathcal{S}$-principal bundles over the same manifold. If $\lambda\colon G_1 \to G_2$ is an embedding, and $\vartheta' = \mathrm{id}_M$, then we say that P_1 is a G_1-*reduction* or *reduction of the structure group* from G_2 to G_1.

Under the conditions of the previous definition, P_1 may be identified with its image in P_2 through ϑ. We say that P_1 is an *embedded subbundle*. The definition may be relaxed so that we consider λ an immersion. First, observe that ϑ is injective. Indeed, if $\vartheta(p) = \vartheta(p')$, the fact that $\vartheta' = \mathrm{id}_M$ implies that $\pi_1(p) = \pi_1(p')$, thus $p' = p \cdot g$, so $\vartheta(p) = \vartheta(p') = \vartheta(p) \cdot \lambda(g)$ which means $\lambda(g) = e_{G_2}$ because the action is free. Since λ is an embedding, it follows that $g = e_{G_1}$. The fact that ϑ is an immersion follows from local triviality. Schematically, choose $\sigma\colon M \to P_1$ a section, and observe that $\vartheta \circ \sigma\colon M \to P_2$ is a section as well. According to remark 3.3, these induce local trivializations of P_1 and P_2 over the same open set U, say T_1 and T_2. Then, a straightforward calculation shows that $T_2 \circ \vartheta \circ T_1^{-1} = \mathrm{id}_M \times \lambda$. Since the right-hand side is an embedding, it follows that ϑ is an immersion. Moreover, this means that ϑ is a local homeomorphism onto its image. Since it is injective, it is an embedding. This works in the smooth and holomophic categories.

When one identifies $P_1 \equiv \vartheta(P_1) \subset P_2$, P_1 is a principal G_1-bundle whose G_1 action is given by restriction of the G_2 action on P_2. Moreover, its projection simply is the restriction of the projection of P_2. This means that $VP_1 \subset VP_2$, and for each $p \in P_1$, $\mathrm{d}_e\phi^{(p)}|_{\mathfrak{g}_1}\colon \mathfrak{g}_1 \to V_pP_1$ defines a vector space isomorphism.

Two reductions $P(M,H)$ and $P'(M,H)$ of the structure group of a principal G-bundle $\hat{P}(M,G)$ are *equivalent* if there is an isomorphism of principal bundles $\varphi\colon P \to P'$ covering the identity such that the diagram is commutative.



Consider now $P(M,G)$ a principal G-bundle, and let $H \subset G$ be a closed subgroup. In that case:

1. The set $G/H := \{gH : g \in H\}$ has a unique manifold structure such that the projection $G \to G/H$ is a smooth embedding. This is a consequence of the quotient manifold theorem. See [**36**, Th. 21.17] for a detailed proof. Moreover, the projection $G \to G/H$ admits local smooth sections.

2. G acts on G/H through $G \times G/H \to G/H, (g, \tilde{g}H) \mapsto g\tilde{g}H$, and this action is smooth with respect to the previous smooth structure (see, for example, the proof of [**36**, Th. 21.17]).

Taking this into account, we may consider the associated bundle $P \times_G (G/H)$, where G acts on G/H by right multiplication as we just defined. This is a fiber bundle over M with fiber G/H. We will write $P(G/H) := P \times_G (G/H)$. Moreover, there is a diffeomorphism

$$\begin{array}{ccc} P \times_G (G/H) & \longrightarrow & P/H \\ [p, gH] & \mapsto & [p \cdot g]. \end{array}$$

The interest of this construction lies in the fact that sections of the bundle $P(G/H)$ parametrize reductions of the structure group of P to H.

Remark 3.33. Whenever P is a principal G-bundle and $H \subset G$ is a closed Lie subgroup, the projection $P \to P/H$ has the structure of a principal H-bundle. The free and proper action of G restricts to a free and proper action of H (this means that P/H is a well-defined manifold), and the condition of remark 3.2 holds trivially. Local triviality is not obvious, and it follows from the fact that $G \to G/H$ is a principal H-bundle as well. See [**30**, I Prop. 5.5].

Proposition 3.34. Let $P(M,G)$ be a principal bundle, and let $H \subset G$ be a closed subgroup. Then, there is a reduction of the structure group to H if and only if the bundle $P(G/H)$ has a global section. In fact, there is a bijection

$$\Gamma(M; P(G/H)) \longleftrightarrow \{\text{reductions of the structure group of } P(M,G) \text{ to } H\} / \sim$$

where $\sim$ stands for the relation of being equivalent.

Proof. We refer the reader to [30, I Prop. 5.6] for the details. We shall only sketch how the bijection goes. Let $\pi_{P/H}\colon P \to P/H$, $\pi\colon P \to M$ and $\hat{\pi}\colon P \times_G G/H \cong P/H \to M$ be the projections corresponding to all the principal bundles involved. Given a section $\sigma\colon M \to P(G/H) \cong P/H$, the reduction is given by

$$P_\sigma = \sigma^* P = \left\{(m,p) \in M \times P : \sigma(m) = \pi_{P/H}(p)\right\}.$$

Since $P \to P/H$ is a principal H-bundle, this endows P_σ with the structure of a principal H-bundle over M. Observe that if $(m,p) \in P_\sigma$, then $\pi(p) = m$. Certainly, form the definition of pullback $\pi_{P/H}(p) = \sigma(m)$. Applying $\hat{\pi}$ on both sides we get $\pi(p) = \hat{\pi}(\pi_{P/H}(p)) = \hat{\pi}(\sigma(m)) = m$. Finally, we consider $\vartheta\colon P_\sigma \to P, (m,p) \mapsto P$. Hence, we may identify P_σ with the set $P' := \left\{p \in P : \pi_{P/H}(p) = \sigma(\pi(p))\right\} \subset P$. Reciprocally, let $\pi'\colon P' \to M$ be a reduction of the structure group to H, and let $\vartheta\colon P' \to P$ the corresponding embedding. Then, the section is given by $\sigma(m) = \pi_{P/H}(\vartheta(p'))$ for and p' such that $\pi'(p') = m$. □

Remark 3.35. There is a natural identification $P_\sigma \times_H G \cong P$ where H acts on G by left multiplication. Just to be clear, recall that

$$P_\sigma \times_H G := \left\{(m,p,g) \in M \times P \times G : \pi_{P/H}(p) = \sigma(m)\right\} / \sim$$

where $(m,p,g) \sim (m, p \cdot h, h^{-1} \cdot g)$. This has the structure of a principal G-bundle over M, which is isomorphic to P though the map $P_\sigma \times_H G \to P, [m,p,g] \mapsto p \cdot g$. referencia. Observe that then we may identify $P_\sigma \equiv P_\sigma \times \{e_G\} \subset P_\sigma \times_H G \cong P$, and see it as an embedded submanifold of P. Note that by $P_\sigma \times \{e_G\}$ we mean the set of equivalence classes, that is $[m,p,e_G]$ such that $\pi_{P/H}(p) = \sigma(m)$. The G-action that turns $P \times_H G$ into a principal G bundle (equivalently the G-action on P seen under the isomorphism) is given by $[m,p,g] \cdot g' = [m,p,g \cdot g']$. This is clearly independent of the representative. Moreover, P_σ is invariant under the restriction of the action to H. Indeed, $(P_\sigma \times \{e_G\}) \cdot h = P_\sigma \times \{h\} = P_\sigma \cdot h^{-1} \times \{e_G\} = P_\sigma \times \{e_G\}$. Finally, observe that the restriction of the G-action on P to an action of H on P_σ is precisely the H-action that P_σ has as a principal H-bundle over M.

By the discussion above, we will speak about sections of $P(G/H)$ and reductions of the structure group of P to H intercheangeably. Let us see how connections on reductions of bundles induce connections on the original bundle. The statement is actually more general, but what we will need when giving a proof of the Chern-Singer correspondence is the particular case of reductions of the structure group.

Proposition 3.36. Let $(\vartheta, \lambda)\colon P_1(M_1, G_1) \to P_2(M_2, G_2)$ be a principal bundle homomorphism between smooth principal bundles. Suppose that the induced map on the base spaces $\vartheta'\colon M_1 \to M_2$ is a diffeomorphism. Let A be a connection on P_1. Then, there is a unique connection $A' = \vartheta(A)$ on P_2 such that $\mathrm{d}\vartheta(H^A P_1) \subset H^{\varphi(A)} P_2$. Moreover, the connection one-form and curvature two-form are related as follows

$$\vartheta^*(A') = \mathrm{d}\lambda(A) \qquad \vartheta^*(\mathrm{D}_{A'} A') = \mathrm{d}\lambda(\mathrm{D}_A A).$$

Proof. See [30, Chap. II, Prop. 6.1]. The connection $\vartheta(A)$ is defined as follows (we define it in terms of the horizontal subbundle). For each $p_2 \in P_2$ with $m_2 := \pi_2(p_2)$ choose $p_1 \in \pi_1^{-1}\left((\vartheta')^{-1}(p_2)\right) \subset P_1$. Then, $p_2 = \vartheta(p_1) \cdot g$ for some $g \in G$. We set

$$H^{\vartheta(A)}_{p_2} P_2 := \mathrm{d}_{\vartheta(p_2)}\phi_g \circ \mathrm{d}_{p_1}\vartheta \ \left(H^A_{p_1} P_1\right).$$

One then shows that this satisfies the conditions of definition 2.1. □

Remark 3.37. Consider now the particular case of a reduction of the structure group. With respect to the notation in the theorem above, this means $M = M_1 = M_2$, and $G_1 \subset G_2$ is a Lie subgroup. The map $\lambda = \imath$ is the inclusion, and so is ϑ when we see P_1 as an embedded subbundle of P_2.

3.7 Metrics and principal bundles

The aim of this section is to show, succinctly, that reductions of the structure group are analogous, at least in spirit, to fiber metrics on vector bundles. Thus, consider a smooth complex vector bundle of rank r over a smooth manifold $E \to M$. Let $L(E)$ be its frame bundle defined as in section 3.1.1. We claim that there is a one-to-one correspondence between the set of hermitian fiber metrics on E, and the set of reductions of the structure group to $\mathrm{U}(r)$ up to equivalence.

The bijection is given as follows: if h in a hermitian fiber metric on E, we define the bundle $U(E)$ of unitary frames as in (3.4). This is naturally a subset of $L(E)$, and it is a reduction of the structure group to $\mathrm{U}(r)$. Reciprocally, consider $\vartheta\colon P(M,\mathrm{U}(r)) \to L(E)$ a reduction of the structure group to $\mathrm{U}(r)$ in the sense of definition 3.32, where we choose $\lambda = \imath\colon \mathrm{U}(r) \hookrightarrow \mathrm{GL}(r,\mathbb{C})$ the inclusion. For every $m \in M$, and $p \in \pi^{-1}(m)$, we define h_m the metric on E_m whose basis with respect to $\vartheta(p)$ is I_r. This does not depend on the choice of p, as any other element in the fiber will be $p \cdot A$ for some $A \in \mathrm{U(r)}$, and $\vartheta(p \cdot A) = \vartheta(p) \cdot A$ (observe that $\vartheta(p)$ is a basis of E_m because it is on the fiber over m of $L(E)$). Hence, any two bases of E_m defined that way are transformed into each other through an element of $\mathrm{U}(r)$, and the matrix of h with respect to any of them is I_r. Hence, $m \mapsto h_m$ is a well-defined map. To check that it is smooth, recall that $L(E) \times_{\mathrm{GL}(n,\mathbb{C})} \mathbb{C}^r \cong E$ (example 3.10). Now, consider two local sections σ_1, σ_2 of E, that under the isomorphism we see as $\sigma_i(m) = [\vartheta(p), \mathbf{x}_i(m)]$ for some smooth maps (that we see as rows) $\mathbf{x}_i\colon U \to \mathbb{C}^r$. Then, $h(\sigma_1,\sigma_2) = \mathbf{x}_1 \cdot \mathbf{x}_2^\dagger$.

Remark 3.38. From proposition 3.34, it follows that the set of hermitian fiber metrics is parametrized by sections of the bundle $L(E)/\mathrm{U}(r)$.

This is of course specific to the case of the frame bundle, but it justifies thinking about reduction of the structure group in general fiber bundles as "metrics" on principal fiber bundles.

3.8 Complex principal bundles

Even though we have given the definition of a principal bundle for the general case of an $\mathcal{S}$-Lie group acting through $\mathcal{S}$-morphisms on an $\mathcal{S}$-manifold, most of the discussion of the preceding sections was done for smooth principal bundles. Sure, every complex principal bundle is in particular a smooth principal fiber bundle, but it is worth expending some time studying complex principal bundles on their own. We will adopt the following notational convention: whenever we have a complex principal G-bundle over a complex manifold M, we will use `mathcal` symbols (for example, $\mathcal{P}$). Then, the corresponding plain letter P will refer to the underlying smooth principal G-bundle. Moreover, whenever when we say "complex principal bundle" we will always assume that the structure group is a complex Lie group, and that the base space is a complex manifold (the definition does not make sense otherwise).

The case of connections deserves some special comments. Let $\mathcal{P}(M,G)$ be a holomorphic principal G-bundle, and let A be a connection on the underlying smooth manifold, which we specify by giving its associated one-form. From appendix B, we know that the Lie algebra of every complex Lie group is naturally a complex Lie algebra, but when seeing the connection on the underlying smooth principal bundle, we see it as $A \in \Omega^1_{\mathbb{R}}(P;\mathfrak{g}^{\mathbb{R}})$. Here, $\mathfrak{g}^{\mathbb{R}}$ denotes the underlying real Lie algebra. However, since $\mathfrak{g}^{\mathbb{R}} = \mathfrak{g}$ as sets, we can extend the connection one form to $T_{\mathbb{C}}P$ by complex bilinearity, i.e. $A(X \otimes \lambda) := \lambda \cdot A(X)$ for $X \in \Gamma(P; T_{\mathbb{C}}P)$. Under some abuse of notation, we will keep the same letter A to denote the complexified connection. Then, we can simply write $A \in \Omega_{\mathbb{C}}(\mathcal{P};\mathfrak{g})$. Observe that the conditions 1 and 2 on theorem 3.20 still apply yo the complexified form. The first property does not change because for each $\xi \in \mathfrak{g}$, the vector field X_ξ is real (this follows from the fact that the exponential as a complex Lie group is the same as the exponential as a real Lie group). As for the second one, we have to extend ϕ_g^* to act on complex differential forms by complex linearity, and since $\mathrm{Ad}_{g^{-1}}$ is complex-linear (see remark B.1), the right-hand side is holomorphic as well, and the equality still holds.

Definition 3.39 (Almost complex G-structure). Let $P(M,G)$ be a smooth principal G-bundle, where G is a complex Lie group, and M is a complex manifold. An *almost complex G-structure* is an almost complex structure on the smooth manifold P such that

1. $\mathrm{d}\pi \circ J = J_M \circ \mathrm{d}\pi$.
2. $J_{p\cdot g}(\mathrm{d}_{(p,g)}\phi(w,X)) = \mathrm{d}_{(p,g)}\phi(J_p(w),(J_G)_g(X))$, for each $p \in P, g \in G$ and $X \in T_gG, B \in T_pP$.

J_M and J_G are the canonical almost complex structures on M and G.

We say that an almost complex G-structure is integrable if it is as a usual almost complex structure in the sense of Newlander-Niremberg.

Remark 3.40.

1. The previous definition has a very clear meaning: if J is an integrable almost complex G-structure, then, by the Newlander-Niremberg theorem P admits a complex structure turning it into a complex principal bundle. In that case, since M and G are complex manifolds, theorem A.23 states that the projection and the action of G are holomorphic maps. In that case, one may show that P admits holomorphic local trivializations (see [53, Prop. 2.5.3]).
2. The second condition is equivalent to ask J to satisfy the following

$$J_{p\cdot g} \circ \mathrm{d}_p\phi_g = \mathrm{d}_p\phi_g \circ J_p \tag{3.16}$$
$$J_{p\cdot g} \circ \mathrm{d}_g\phi^{(p)} = \mathrm{d}_g\phi^{(p)} \circ (J_G)_g \tag{3.17}$$

The proof is just a straightforward application of the Leibniz rule [30, I Prop. 1.4]

$$\mathrm{d}_p\phi_g(J_p(w)) + \mathrm{d}_g\phi^{(p)}((J_G)_g(X)) = J_{p\cdot g}(\mathrm{d}_p\phi_g(w)) + J_{p\cdot g}(\mathrm{d}_g\phi^{(p)}(X))$$

and setting X or w equal to 0. Moreover, (3.17) is equivalent to

$$J_p(X_{p,\zeta}) = X_{p,\mathrm{i}\,\zeta}, \qquad \zeta \in \mathfrak{g}. \tag{3.18}$$

Indeed, (3.17) implies (3.18) setting $g = e$, and the converse is a direct calculation using the fact $\mathrm{d}_g\phi^{(p)}(X) = X_{\theta(X),p}$, θ being the Maurer-Cartan form, which commutes with J.

Let us denote $\mathscr{J}(P)$ the set of integrable almost complex G-structures, and $\bar{\mathscr{J}}(P)$ the set of almost complex G-structures.

Remark 3.41. Let $\mathcal{P}(M,G)$ be a complex principal bundle. If we consider the curvature $\mathrm{D}_A A$ of the underlying real smooth principal bundle, we may see it as an element of $\Omega^2_{\mathbb{R}}(P;\mathfrak{g}^{\mathbb{R}})$. We now, however, that $\mathfrak{g}$ has the structure of a complex Lie algebra. Hence, we may extend $\mathrm{D}_A A$ to act on $T_{\mathbb{C}}P$ by setting $\mathrm{D}_A A(\lambda\otimes X, \mu\otimes Y) := \lambda\cdot\mu\cdot \mathrm{D}_A A(X,Y)$. We may then see $\mathrm{D}_A A \in \Omega^2_{\mathbb{C}}(P;\mathfrak{g})$. Then the Whitney sum decomposition (A.20) induces a decomposition $\mathrm{D}_A A = \mathrm{D}_A A^{(2,0)} + \mathrm{D}_A A^{(1,1)} + \mathrm{D}_A A^{(0,2)}$. Observe that $\mathrm{D}_A A^{(0,2)}(X,Y) = \mathrm{D}_A A\left(X^{(0,1)}, Y^{(0,1)}\right)$, $\mathrm{D}_A A^{(1,1)}(X,Y) = \mathrm{D}_A A\left(X_J^{(1,0)}, Y_J^{(0,1)}\right) + \mathrm{D}_A A\left(X_J^{(0,1)}, Y_J^{(1,0)}\right)$ and similarly for $\mathrm{D}_A A^{(0,2)}$. Observe that $\mathfrak{g}$ and $(\mathfrak{g}^{\mathbb{R}})^{\mathbb{C}}$ are *not the same Lie algebra*. To see the complexify the curvature and the connection, we use the former. If we see the curvature as F_A, we may consider its complexification by seeing it as a form on the complex vector bundle $\mathrm{ad}(P)$ (see proposition 3.8).

3.9 The Chern-Singer correspodence

The Chern-Singer correspondence can be seen as a generalization of the Chern correspondence we studied in section 2.6. Recall that there, we saw that there was a bijection between the set of connections of type $(1,1)$ on a complex vector bundle and the set of integrable Dolbeault operators. Here, we shall see that, for principal bundles over reductive complex Lie groups, we can find a bijection between the set of integrable almost complex G-structures, $\mathscr{J}(P)$ and the set of connections on a reduction of the principal bundle to a compact subgroup K such that $G = K^{\mathbb{C}}$. This is section is mostly based on the thorough exposition given in [53].

Having said that, let $P_G := P(M, G)$ be a principal G-bundle over a complex manifold, and let its structure group G be a complex reductive Lie group. From appendix B, we know that G is the complexification of a compact real Lie group K, i.e. $G = K^{\mathbb{C}}$. Fix a reduction P_K of the structure group to K. We fix that data for the remainder of the section. Recall that we can see the reduction P_K as an embedded submanifold of P_G. Hence, TP_K is a subbundle of TP_G [**36**, Example 10.33-(c)].

Theorem 3.42 (Chern-Singer correspondence I). The map $\mathrm{CS}\colon \bar{\mathscr{J}}(P_G) \to \mathscr{A}(P_K), J \mapsto A_J$, where A_J denotes the connection corresponding to the distribution $H^J_p P_K := T_p P_K \cap J_p(T_p P_K)$, $p \in P_K$ is a bijection.

Proof. Let $J \in \bar{\mathscr{J}}(P_G)$, and let us show first that the assignment $p \mapsto H^J_p P_K$ is a smooth distribution, and that it satisfies properties 1 and 2 of definition 3.14. Note that $H^J_p P_K \cap V_p P_K = \{0\}$. Indeed, an element of the intersection is on the vertical subbundle, hence it is X_ζ for some $\zeta \in \mathfrak{k}$. Then, by (3.17)

$$J_p(X_{p,\zeta}) = J_{p\cdot e}\left(\mathrm{d}_e\phi^{(g)}(\zeta)\right) = \mathrm{d}_e\phi^{(p)}(J_{\mathfrak{g}}(\zeta)) = \mathrm{d}_e\phi^{(g)}(\mathrm{i}\,\zeta) = X_{p,\mathrm{i}\,\zeta}.$$

Moreover, since $X_{p,\zeta} \in H^J_p P_K$, it follows that $X_{p,\zeta} \in J_p(T_p P_K)$, therefore, $J(X_{p,\zeta}) \in J_p^2(T_p P_K) = T_p P_K$. Using the fact that $\mathrm{d}_e\phi^{(p)}\colon \mathfrak{g} \to V_p P_G$ is an isomorphism which restricts to an isomorphism $\mathfrak{k} \to V_p P_K$, it follows that $\zeta \in \mathfrak{k} \cap \mathrm{i}\,\mathfrak{k} = \{0\}$. Finally, since $T_p P_K + J_u(T_p P_K) = T_p P_G$, ($\dim_{\mathbb{R}} M$ is the dimension of M as a real manifold, and we used the fact that $\dim_{\mathbb{R}} G = 2 \cdot \dim K$, see appendix B)

$$\begin{aligned}\dim(H^J_p P_K) &= \dim(T_p P_K \cap J_p(T_u P_K)) = \dim(T_p P_K) + \dim(J_p(T_p P_K)) - \dim(T_p P_K + J_u(T_p P_K)) \\ &= \dim_{\mathbb{R}} M + \dim K + \dim_{\mathbb{R}} M + \dim K - \dim G + \dim_{\mathbb{R}} M = \dim_{\mathbb{R}} M\end{aligned} \tag{3.19}$$

so $\dim(H^J_p P_K \oplus V_p P_K) = \dim H^J_p P_K + \dim G = \dim T_p P_K$ and the sum covers the whole tangent bundle. Finally, $\mathrm{d}_p\phi_k(H^J_p P_K) \subset H^J_{p\cdot k} P_K$ (we need only check it for $k \in K$) and since the dimensions are equal, we have the equality. Indeed, choose, $w \in T_p P_K \cap J_p(T_p K)$. We may write $w = J_p(w')$ for some $w' \in T_p P_K$. Then, by (3.16), we have

$$\mathrm{d}_p\phi_k(J_p(w')) = J_{p\cdot k}(\mathrm{d}_p\phi_k(w')) \in J_{p\cdot k}(T_{p\cdot k} P_K) \cap T_{p\cdot k} P_K$$

where we used that $\mathrm{d}_p\phi_k(w') \in T_{p\cdot k} P_k$ because the action of G restricts to the action of K on P_K. Smoothness is somewhat tedious to proof, the key idea is to use a local frame of TP_K to obtain one of $H^J P_K$ and then apply the local frame criterion for subbundles (lemma A.18). We refer the reader to [**53**, Th. 3.1.1] for the details.

Let us now construct the inverse image. From proposition 3.36 and remark 3.37, we know that every connection $A \in \mathscr{A}(P_K)$ induces a connection on the original bundle P_G. Let us still denote by A de induced connection. We define an almost complex G-structure J^A on P_G as follows. For $p \in P$ and $w \in T_p P$, the connection A induces a decomposition $w = w^V + w^H$. Then, for some $\zeta \in \mathfrak{g}$, we have $w^V = X_{p,\zeta}$. Define

$$J^A_p(w) := X_{p,\mathrm{i}\,\zeta} + \left((J_M)_{\pi(p)}(\mathrm{d}_p\pi(w))\right)^{\mathbf{HL}}_p. \tag{3.20}$$

Observe that J_A is an almost complex G-structure. The fact that $(J^A)^2 = -\mathrm{id}$ is clear from the fact that (3.20) already gives us the decomposition of $J^A(w)$ as a sum of a vertical and horizontal vector. Hence,

$$\begin{aligned}J^A_p(J^A_p(w)) &= X_{-\zeta,p} + \left((J_M)_{\pi(p)}\left(\mathrm{d}_p\pi\left(\left((J_M)_{\pi(p)}(\mathrm{d}_p\pi(w))\right)^{\mathbf{HL}}_p\right)\right)\right)^{\mathbf{HL}}_p \\ &= -X_{\zeta,p} - \mathrm{d}_p\pi(w)^{\mathbf{HL}}_p = w^V + w^H = w.\end{aligned}$$

Now, $\mathrm{d}\pi \circ J = J_M \circ \mathrm{d}\pi$ and (3.18) are obvious. To check (3.16), let $w \in T_p P$. Let $w^V = X_{\zeta,p}$, and observe that $\mathrm{d}_p\phi(w)^V = \mathrm{d}_p\phi(w^V + w^H)^V = \mathrm{d}_p\phi_g(w^V)^V$ (horizontal vectors are mapped to horizontal vectors by $\mathrm{d}_p\phi_g$ by definition). Moreover, by lemma 3.17

$$\mathrm{d}_p\phi(X_{\zeta,p}) = \mathrm{d}_p\phi(w^V) = \mathrm{d}_p\phi_g\left(X_{\zeta,p}\right) = X_{\mathrm{Ad}_{g^{-1}}(\zeta),p\cdot g}, \tag{3.21}$$

which is again vertical. Thus,

$$\begin{aligned} J_{p\cdot g}(\mathrm{d}_p\phi_g(w)) &= X_{\mathrm{i}\,\mathrm{Ad}_{g^{-1}}(\zeta),p} + \left((J_M)_{\pi(p)}(\mathrm{d}_p\pi(w))\right)^{\mathbf{HL}}_{p\cdot g} \\ &\overset{(6)}{=} \mathrm{d}_p\phi_g(X_{\mathrm{i}\,\zeta,p}) + \mathrm{d}_p\phi_g\left(\left((J_M)_{\pi(p)}(\mathrm{d}_p\pi(w))\right)^{\mathbf{HL}}_{p}\right) \\ &= \mathrm{d}_p\phi_g(J_p(w)), \end{aligned}$$

where we used in (6) equation (3.21) but with $\mathrm{i}\,\zeta$ and lemma 3.16.

Finally, observe that these constructions are inverses of each other. Let us see first that $A = A_{J^A}$. This amounts to checking whether $H^A_pP_K$ and $T_pP_K \cap J^A_p(T_pP_K) = H^{J^A}_pP_K$ coincide for each $p \in P_K$. Choose $X \in H^A_pP_K \subset T_pP_K$, then $X^V = 0$, and

$$X = J^A_p\left(-\left((J_M)_{\pi(p)}(\mathrm{d}_p\pi(X))\right)^{\mathbf{HL}}\right) \in J^A_p(H^A_pP_K) \subset J^A_p(T_pP_K).$$

Hence, $X \in T_pP_K \cap J^A_p(T_pP_K)$. Thus, $H^A_pP_K \subset H^{J^A}_pP_K$ and since they have the same dimension (the calculation of equation (3.19) applies to both sides), they are equal.

Reciprocally, consider $J \in \bar{\mathscr{J}}(P_G)$. Let $A_J := \mathsf{CS}(J)$. Let us see that $J^{A_J} = J$. Both are linear and coincide on vertical vectors ($J_p(X_{p,\zeta}) = X_{p,\mathrm{i}\,\zeta}$ follows from applying (3.17) to ζ). Since both are linear as well, it suffices to check that they coincide when applied to horizontal vectors (with respect to A_J). Hence, if $w \in H^A_pP_K$, we have

$$\mathrm{d}_p\pi(J_p(w)) = (J_M)_{\pi(p)}(\mathrm{d}_p\pi(w)). \tag{3.22}$$

Moreover, $J_p(w)$ is horizontal (observe that $H^J_pP_K = T_pP_K \cap J_p(T_pP_K)$), so considering the horizontal lift of (3.17) we get

$$J^{A_J}(w) = \left((J_M)_{\pi(p)}(\mathrm{d}_p\pi(w))\right)^{\mathbf{HL}} = J_p(w).$$

□

Just as we found in the vector bundle case, it turns out that $(1,1)$-connections are precisely those that induce integrable almost complex G-structures.

Theorem 3.43 (Chern-Singer correspondence, II). The map CS restricts to a bijection $\mathsf{CS}|_{\mathscr{J}(P_G)}\colon \mathscr{J}(P_G) \to \mathscr{A}^{(1,1)}(P_K)$.

For the proof of this result we will diverge from [53]. We will instead make use of the following lemma, which is taken from [64].

Lemma 3.44. Let $A \in \mathscr{A}(P_K)$ be a connection, and let $J = \mathsf{CS}^{-1}(A) \in \bar{\mathscr{J}}(P_G)$. Then, J is integrable (that is, $J \in \mathscr{J}(P_G)$) if and only if the two form

$$\begin{array}{rccc} \mathfrak{f}_J\colon & \Gamma(P_G;T_{\mathbb{C}}P_G)\times\Gamma(P_G;T_{\mathbb{C}}P_G) & \longrightarrow & \mathscr{C}^\infty(P_G;\mathfrak{g}) \\ & (X,Y) & \mapsto & -A\left(\left[X^{(0,1)}_J, Y^{(0,1)}_J\right]\right) \end{array} \tag{3.23}$$

vanishes identically.

The notation $X^{(0,1)}_J$ means the $(0,1)$-component with respect to the Whitney sum decomposition that J induces on $T_{\mathbb{C}}P_G$ (see (A.8), even though there it is stated for complex manifolds the decomposition only requires an almost complex structure [35, Lemma 1.60]). Moreover, the connection one-form A in (3.23) should be unsderstood as the extension of A to P_G (as in referencia), which we still denote A.

Proof. This is a consequence of the classical Newlander-Niremberg theorem. Recall from appendix A.5 that the Newlander-Niremberg theorem implies that the integrability condition is equivalent to the condition $\left[T^{(0,1)}P_G, T^{(0,1)}P_G\right] \subset T^{(0,1)}P_G$. Thus, we are led to check that the condition $\mathfrak{f}_J$ is equivalent to the condition that for every pair of complex vector fields X, Y, we have $\left[X^{(0,1)}_J, Y^{(0,1)}_J\right] \in T^{(0,1)}P_G$. It will follow immediately from showing $T^{(0,1)}P_G = \ker A$. □

Lemma 3.45. Let $P(M,G)$ be a principal bundle. Then, for any pair of horizontal vector fields, $X,Y\in\Gamma(P;T_{\mathbb{C}}P)$, we have

$$\mathrm{D}_A A(X,Y) = -A([X,Y]).$$

Proof. First, for a $\mathfrak{g}$-valued one form and a vector field, we define $X(\omega\otimes\xi) := X(\omega)\cdot\xi$. Now, (/) follows from the corresponding identity for usual one-forms (see [36, Prop. 14.29]), and (8) from the fact that $A(X)=0$ for horizontal vector fields:

$$\mathrm{D}_A A(X,Y) = (\mathrm{d}A)^H(X,Y) = \mathrm{d}A(X,Y) \overset{(7)}{=} X(A(Y)) - Y(A(X)) - A([X,Y]) \overset{(8)}{=} -A([X,Y]).$$

□

Observe that, in the context of lemma 3.44, J preserves horizontal vector fields. More precisely, the complexification of $TP_G = H^A P_G \oplus VP_G$ yields a decomposition $T_{\mathbb{C}}P_G = H^A_{\mathbb{C}}P_G \oplus V_{\mathbb{C}}P_G$ which allows us to extend the notion of horizontal and vertical vector fields to the corresponding complexification. Moreover, the isomorphism $P\times\mathfrak{g}\cong VP$ extends to $P_G\times\mathfrak{g}^{\mathbb{C}}\cong V_{\mathbb{C}}P_G$. Now, if $Y\in\Gamma(P_G;H^A_{\mathbb{C}}P_G)$ i.e. Y is horizontal, then so is JY (this follows from the definition of J, equation (3.20) and the fact that we see now J as the complexified map). We can now proof the desired result. Moreover, observe that, from the froom of the previous lemma, it follows that $T^{(0,1)}P_G$ is G-invariant. Indeed, for any $Y\in T^{(0,1)}P_G$, $A(\mathrm{d}\phi_g(Y)) = \mathrm{Ad}_{g^{-1}}(A(Y)) = 0$.

Proof of theorem 3.43. By the previous lemma, we have to check that $\mathfrak{f}_J\equiv 0$. If either X or Y are vertical vector fields, then $\mathfrak{f}_J(X,Y)=0$, so we may restrict to the case where they are both horizontal. Indeed, for any $\zeta\in\mathfrak{g}^{\mathbb{C}}$ and any vector field Y, the vector field $\left[X_\zeta, Y_J^{(0,1)}\right]$ is of type $(0,1)$. This follows from the fact that $T^{(0,1)}P_G\subset T_{\mathbb{C}}P_G$ is G-invariant, and $\left[X_\zeta,Y_J^{(0,1)}\right] = \mathcal{L}_{X_\zeta}\left(Y_J^{(1,0)}\right)$. Invariance under the G-action means that the latter is of type $(0,1)$. Finally, since A is a form of type $(1,0)$, $\mathfrak{f}_J(X,Y)$ vanishes if either X or Y are vertical. revisar

Once we show that $\mathfrak{f}_J$ restricted to horizontal vector fields coincides with $F_A^{(0,2)}$ we will be finished (recall from lemma 3.27 that the curvature is a horizontal two-form). Hence, from proposition A.19 it follows that for all $Z,Y\in\Gamma(P_G;T_{\mathbb{C}}P_G)$

$$\begin{aligned} -A\left(\left[Z_J^{(0,1)},Y_J^{(0,1)}\right]\right) &= -\frac{1}{4}\cdot A\left([Z+\mathrm{i}\,JZ, Y+\mathrm{i}\,JY]\right) \\ &= -\frac{1}{4}\cdot A([Z,Y]) + \frac{1}{4}\cdot A([Z,Y]) - \frac{\mathrm{i}}{4}\cdot A([Z,JY]) - \frac{\mathrm{i}}{4}\cdot A([JZ,Y]). \end{aligned} \tag{3.24}$$

Similarly, for all $Z,Y\in\Gamma(P_G;T_{\mathbb{C}}P_G)$

$$\begin{aligned} \mathrm{D}_A^{(0,2)}(Z,Y) = \mathrm{D}_A\left(Z_J^{(0,1)},Y_J^{(0,1)}\right) &= \frac{1}{4}\cdot\mathrm{D}_A(Z+\mathrm{i}\,JZ, Y+\mathrm{i}\,JY) \\ &= \frac{1}{4}\cdot\mathrm{D}_A(Z,Y) - \frac{1}{4}\cdot\mathrm{D}_A(JZ,JY) + \frac{\mathrm{i}}{4}\cdot\mathrm{D}_A(Z,JY) + \frac{\mathrm{i}}{4}\cdot\mathrm{D}_A(JZ,Y). \end{aligned} \tag{3.25}$$

Finally, by lemma 3.45 for horizontal vector fields, $\mathrm{D}_A([X,Y]) = -A([X,Y])$, which applied to each component of (3.24) and (3.25) when the vector fields are horizontal yields the desired result. We have used thet fact that $\mathrm{D}_A A^{(0,2)}=0$ if and only if $F_A^{(0,2)}=0$. □

This proposition (together with the background we have from vector bundle theory) motivate the following definition: given an integrable almost complex G-structure on P_G, the connection induced on P_G after extending $\mathsf{CS}^{-1}(J)$ to P_G is called the *Chern connection* associated to J.

Finally, we will comment without proof how quotients by the gauge group play a role in the correspondence. That is, we briefly present a generalization to the bijection (2.22). We refer to [53] and references therein for further details. Again, we do not intend to give these spaces further structure for now (that is, we do not study moduli spaces), so we will not complete any space with respect to any Sobolev norm. Let $\mathscr{G}^{\mathbb{C}} = \Gamma(M;P_G\times_G G)$ and $\mathscr{G} = \Gamma(M;P_K\times_K K)$ the gauge groups associated to both bundles (see example 3.9 and theorem 3.13). Observe that, the fibers of $P_G\times_G G$ are isomorphic to $G=K^{\mathbb{C}}$, it seems reasonable to say that $\Gamma(M;P_G\times_G G)$ is a complexification of

$\Gamma(P_K \times_K K)$, hence the notation.

We may see $\mathscr{G}$ as a subgroup of $\mathscr{G}^{\mathbb{C}}$. Moreover, there is a right action of $\mathscr{G}$ on $\mathscr{A}(P_K)$ given by pullback $\mathscr{G} \times \mathscr{A}(P_K) \to \mathscr{A}(P_K), (A, \phi) \mapsto (\phi^{-1})^* A$. A straightforward computation shows that the pullback of a connection one-form is a connection one-form [7, Th. 3.2.5]. For an explicit expression, suppose that $\phi(p) = p \cdot \tau(p)$ (c.f. 3.13). Then, (see [7, Th. 3.2.14])

$$((\phi^{-1})^* A)_p = \mathrm{Ad}_{\tau(p)} \circ A_p + \mathrm{d}_{\tau(p)^{-1}} L_{\tau(p)} \circ \mathrm{d}_p \tau^{-1}. \tag{3.26}$$

The previous expression is sometimes written as $\phi^* A = \mathrm{Ad}_{\tau^{-1}} \circ A + \mathrm{d}\tau^{-1}\tau$. The second summand can also be written as $(\tau^{-1})^*\theta$, where θ is the *Maurer-Cartan* form. Hence, the action is equivalently written as

$$(\phi^{-1})^* A = \mathrm{Ad}_{\tau^{-1}}(A) + (\tau^{-1})^*\theta. \tag{3.27}$$

This action leaves $\mathscr{A}^{(1,1)}(P_K)$ invariant. On the other hand, there is a right action of $\mathscr{G}^{\mathbb{C}}$ on $\bar{\mathscr{J}}(P_G)$ given by $\bar{\mathscr{J}}(P_G) \times \mathscr{G}^{\mathbb{C}} \to \bar{\mathscr{J}}(P_G), (J, \varphi) \mapsto \varphi(J)$, where

$$\varphi(J) := (\mathrm{d}\varphi)^{-1} \circ J \circ \mathrm{d}\varphi.$$

This action leaves $\mathscr{J}(P_G)$ invariant. Moreover, via the Chern-Singer correspondence (version I, theorem 3.42), we get an action of $\mathscr{G}^{\mathbb{C}}$ on $\mathscr{A}(P_K)$ given by $(A, \varphi) \mapsto A_{\varphi(J_A)}$, which extends the original $\mathscr{G}$-action.

Now, to almost complex integrable G-structures in the same $\mathscr{G}^{\mathbb{C}}$-orbit induce isomorphic complex principal bundle structures on P_G. Thus, the quotient $\mathscr{J}(P_G)/\mathscr{G}^{\mathbb{C}}$ can be seen as the set of essentially different holomorphic structures on the bundle P_G. Now, the Chern-Singer correspondence (version II, theorem 3.43) establishes that there is a bijection

$$\frac{\mathscr{J}(P_G)}{\mathscr{G}^{\mathbb{C}}} \leftrightarrow \frac{\mathscr{A}^{(1,1)}(P_K)}{\mathscr{G}^{\mathbb{C}}}, \tag{3.28}$$

the right-hand side being the set of complex principal bundle structures on P_G.

Chapter 4

Infinite-dimensional symplectic reduction

The aim of this chapter is to give a Banach-manifold version of symplectic reduction (theorem 1.10), for the action of a Banach Lie group on a Banach manifold. We refer the reader to appendix C for the relevant notions concerning infinite-dimensional Banach manifolds. The chapter starts with the (straighforward) generalization of symplectic manifolds to infinite-dimensiones. Once this is done, we can give the definition of moment map, and state the reduction theorem for Banach manifolds. As we shall see, the proof does not really change much from the classical theorem, aside from the fact that we no longer have a quotient manifold theorem, and that has to be addressed by requiring additional properties to moment maps. However, if we assume that we can define quotients, and they are Banach manifolds, the proof is quite similar to the finite-dimensional case. In fact, the original article by Marsden and Weinstein [38] gives a proof of theorem 1.10 in the context of Banach manifolds. A very thorough exposition of the theorem can be found in [20]. This results will be used in chapter 5 to justify the fact that the Hitchin-Kobayashi may be seen as an infinite-dimensional version of the Kempf-Ness theorem introduced in chapter 1. To that end, we devote the second half of the chapter to justifying that the gauge group introduced in section 3.2 can be seen (with some subtelties) as a Hilbert Lie group.

4.1 Symplectic reduction theorem for Banach-manifolds

The definition of an infinite-dimensional symplectic manifold is quite similar to the finite-dimensional one except for the fact that we no longer ask $\Omega^\flat$ to be an isomorphism between $T_m\mathcal{M} \cong T_m\mathcal{M}^*$, but to be injective. Observe that in the finite-dimensional setting this already implies that $\Omega^\flat$ is an isomorphism because $T_m\mathcal{M}$ and its dual are vector spaces of the same fonote dimension.

Definition 4.1 (Banach symplectic manifold). Let $\mathcal{M}$ be a Banach manifold, we say that a pair $(\mathcal{M}, \Omega)$ is a *Banach symplectic manifold* if Ω is a closed two-form satisfying:

1. For each $m \in \mathcal{M}$, $\Omega_m \colon T_m\mathcal{M} \times T_m\mathcal{M} \to \mathbb{R}$ is continuous.
2. For each $m \in \mathcal{M}$, Ω_m is non-degenerate, that is, if $u \in T_m\mathcal{M}$ and $\Omega_m(u, v) = 0$ for all $v \in T_xX$, then $u = 0$.

We say that Ω is a symplectic form.

The notions of symplectomorphism, hamiltonian action of a Banach Lie group on a Banach manifold and moment map are defined just as in the finite-dimensional case (see section 1.1.1). Observe that given a Banach Lie group G and $\xi \in \mathrm{Lie}(G)$, the infinitesimal action, i.e. the vector field X_ξ can be defined just as in (1.4) because Banach Lie groups have an exponential map. The first major difference with respect to the finite setting is the fact that we do not have a direct generalization of the quotient manifold theorem. What ends up working is to ask more conditions to the moment map. Before dwelling in the problem of the lack of a quotient manifold theorem, let us consider the quotient $\mu^{-1}(0)/G$. The fact that μ is G-equivariant means that $\mu^{-1}(0)$ is invariant under the G-action (same justification as (1.5)). Hence, G acts on $\mu^{-1}(0)$, and the *reduced space* $\mathcal{M}_{\mathrm{red}} := \mu^{-1}(0)/G$ is well-defined as a topological space. In the classical theorem, one could show that 0 was always a regular value of μ, and this implied that $\mu^{-1}(0)$ was an embedded submanifold of the original manifold. Since we may not have that condition, we have to impose it on μ. This motivates the following definition.

Definition 4.2. Let G be a Banach-Lie group acting symplectically on a Banach-manifold $\mathcal{M}$, and let $\mu\colon \mathcal{M} \to \mathfrak{g}^*$ be a moment map for the action.

1. We say that $0 \in \mathrm{Lie}(G)^*$ is a *weakly regular value* of μ if $\mu^{-1}(0)$ is an embedded submanifold of $\mathcal{M}$ and $T_x(\mu^{-1}(0)) = \ker \mathrm{d}_x\mu$.
2. We say that μ *satisfies the slice condition* if the action of G on $\mathcal{M}$ is free, and for every $x \in \mu^{-1}(0)$ there is a submanifold $S_x \subset \mu^{-1}(0)$, called a *slice* such that

$$T_x(\mu^{-1}(0)) = T_xS_x \oplus T_x(G \cdot x).$$

Aside from the motivation we just gave for the definition of weakly regular value, the condition $T_x(\mu^{-1}(0)) = \ker \mathrm{d}_x\mu$ is always satisfied for a moment map in the classical sense. Moreover, if μ is a submersion at 0, then 0 is a weakly regular value (see [1, Th. 3.5.4]), but not every weakly regular value is also regular. It remains to somehow motivate the slice condition. That is precisely what will be the substitute of the quotient manifold theorem. Indeed, [1, Sec. 3.5] gives a detailed exposition of the theory of quotient Banach-manifolds. Under the hypothesis of the existence of a slice, we may use it to get local coordinates around every point (see the proof of [1, Th. 3.5.25 and Lemma 3.5.26]).

Moreover, observe that if $\phi\colon G\times\mathcal{M} \to \mathcal{M}$ is the action, then $\phi^{(x)}\colon G \to \mathcal{M}$ is an embedding for each $x \in \mathcal{M}$, and if $x \in \mu^{-1}(0)$, then $G \cdot x \subset \mu^{-1}(0)$ is an embedded submanifold. If the slice condition is satisfied, we have $T_x(\mu^{-1}(0)) = T_xS_x \oplus T_x(G\cdot x)$, hence $T_x(\mu^{-1}(0))/T_x(G\cdot x) \cong T_xS_x$. Moreover, for each sufficiently small slice [1, h. 3.5.25 and Lemma 3.5.26] state that the map $\varphi\colon S_x \to \mathcal{M}_{\mathrm{red}}, p \mapsto \pi(p)$ is a diffeomorphism onto its image, therefore $T_xS_x \cong T_{\pi(x)}\mathcal{M}_{\mathrm{red}}$. Putting everything together, we get:

$$T_{\pi(x)}\mathcal{M}_{\mathrm{red}} \cong T_x\ \mu^{-1}(0)/T_x(G\cdot x) \tag{4.1}$$

Where the isomorphism is given by the first isomorphism theorem. Indeed, $\ker \mathrm{d}_x\pi = T_x(G\cdot x)$, hence we get an isomorphism $\overline{\mathrm{d}_x\pi}\colon T_x\ \mu^{-1}(0)/T_x(G\cdot x) \to T_{\pi(x)}\mathcal{M}_{\mathrm{red}}$.

Observe as well that the fact that $\phi^{(x)}$ is an embedding means that $T_x(G\cdot x) = \mathrm{d}_e\phi^{(x)}(\mathrm{Lie}(G))$, thus, by equation (1.4)

$$T_x(G\cdot x) = \{X_\xi : \xi \in \mathrm{Lie}(G)\}. \tag{4.2}$$

Remark 4.3. Recall that the Hausdorff condition is something imposed to manifolds to avoid pathological situations, but it is not implied by the condition of being locally euclidean. When the action is proper, we have a topological group acting continuously and properly on a topological space. It is a standard result (see for example [36, Prop. 21.4]) that under those assumptions the quotient space is Hausdorff.

We can now state the reduction theorem for Banach manifolds.

Theorem 4.4 (Banach symplectic reduction). Let $\mathcal{M}$ be a symplectic manifold endowed with a symplectic form Ω, and a smooth, symplectic, left action of a Banach-Lie group G. Suppose that $\mu\colon \mathcal{M} \to \mathrm{Lie}(G)^*$ is a moment map for which 0 is a weak regular value, and satisfying the slice condition. Then, $\mathcal{M}_{\mathrm{red}} := \mu^{-1}(0)/G$ has the structure of a symplectic (possibly non-Hausdorff) Banach manifold, which is endowed with a symplectic form Ω_{red} and satisfying

$$\pi^*\Omega_{\mathrm{red}} = i^*\Omega$$

where $i\colon \mu^{-1}(0) \hookrightarrow \mathcal{M}$ is the inclusion and $\pi\colon \mu^{-1}(0) \to \mathcal{M}_{\mathrm{red}}$ the natural projection. Moreover, if the restricted action of G on $\mu^{-1}(0)$ is proper, then $\mathcal{M}_{\mathrm{red}}$ is Hausdorff.

Before giving the proof, let us state and show some lemmas we will need. Recall that, given a skew-symmetric bilinear map $\Omega\colon E \times E \to \mathbb{R}$ such that $\Omega(u,v) = 0$ for all $v \in E$ implies $u = 0$, (that is, $\ker \Omega^\flat = \{0\}$) we say that it is non-degenerate.

Lemma 4.5. Let E be a Banach space and let $\Omega\colon E\times E\to\mathbb{R}$ be a skew-symmetric, non-degenerate, continuous bilinear map. Consider, for each $v\in E$ the maps $p_v\colon E\to\mathbb{R}, u\mapsto|\Omega(u,v)|$. Then, $\{p_v\}_{v\in E}$ is a family of seminorms with respect to which E becomes a Hausdorff locally convex TVS. Moreover, given a linear map $\alpha\colon E\to\mathbb{R}$ continuous with respect to the locally convex topology, there exists $v\in E$ such that $\alpha(u)=\Omega(u,v)$ for all $u\in E$.

Proof. The fact that $\{p_v\}_{v\in E}$ is a family of seminorms on E follows immediately from the fact that Ω is bilinear. Moreover, non-degeneracy of Ω implies that if $p_v(u)=0$ for all $v\in E$, then $u=0$. Endow E with the initial topology with respect to the natural projections $q_i\colon E\to E/p_i^{-1}(0)$. This turns E into a Hausdorff TVS.

For the second statement, let $\alpha\colon E\to\mathbb{R}$ be continuous with respect to the TVS topology we just defined. It is a general fact of locally convex vector spaces that a linear map $\alpha\colon E\to\mathbb{R}$ is continuous if and only if there are seminorms $\{p_{i_1},\dots p_{i_n}\}$ and a constant $C>0$ such that $|\alpha(v)|\leq C\cdot\max_{1\leq j\leq n}\left\{p_{i_j}(v)\right\}$ (see [31, Lemma 1.4.2 and Lemma 3.4.2]). Back to our setting, this means that there exist $v_1,\dots v_n\in E$ and a constant $C>0$ such that

$$|\alpha(v)|\leq C\cdot\max_{1\leq j\leq n}\left\{p_{v_j}(v)\right\}=C\cdot\max_{1\leq j\leq n}\left\{|\Omega(v_j,v)|\right\}.$$

Observe that α vanishes on $F=\bigcap_{i=1}^n\ker\Omega^\flat(v_j)=(\operatorname{span}(v_1,\dots v_n))^\Omega$. Moreover, since each $\Omega^\flat(v_j)$ is continuous, F is closed with respect to the norm-topology and its codimension (the dimension of the quotient E/F) is at most n. Choose any algebraic complement W of F in E. That is, as vector spaces, $W\oplus F=E$. Since $W\cong E/F$, it must be finite dimensional, hence closed in E with respect to the Banach topology. This means that W^* is also finite-dimensional, and spanned by $\Omega^\flat(v_1)|_W,\dots\Omega^\flat(v_n)|_W$. Then $\alpha|_W\in W^*$ can be written as $\alpha|_W=\sum_{i=1}^n\lambda_n\cdot\Omega^\flat(v_n)|_W$ for some real coefficients λ_n. Since in F both sides are identically 0, this linear combination can be extended to all of E, thus

$$\alpha=\sum_{i=1}^n\lambda_i\cdot\Omega^\flat(v_i)=\Omega^\flat\left(\sum_{i=1}^n\lambda_i\cdot v_i\right)=\Omega^\flat(v),$$

where we defined $v=\sum_{i=1}^n\lambda_i\cdot v_i$. This proves the result. □

Lemma 4.6. Let E be a Banach space, and let $\Omega\colon E\times E\to\mathbb{R}$ be a skew-symmetric, non-degenerate, continuous bilinear map. For any closed subspace $Y\subset E$, define the Ω*orthogonal complement* of Y by

$$Y^\Omega:=\{v\in E:\Omega(u,v)=0\ \text{ for all }\ v\in Y\}.$$

Then, $\left(Y^\Omega\right)^\Omega=Y$.

Proof. The inclusion $Y\subset(Y^\Omega)^\Omega$ follows directly from the definition. For the reverse inclusion, let $v\notin Y$. By a standard consequence of the Hanh-Banach theorem for locally convex TVS, there exists $\alpha\colon E\to\mathbb{R}$ (continuous with respect to the locally convex topology) such that $\alpha|_Y\equiv 0$ and $\alpha(v)\neq 0$. By the previous lemma, there is an element $z\in E$ such that $\alpha(u)=\Omega(u,z)$. This means that $\Omega(u,v)=\alpha(v)\neq 0$. However, $z\in Y^\Omega$ because for every $y\in Y$, $0=\alpha(y)=\Omega(y,z)=0$. Therefore, $\alpha(v)=\Omega(v,z)\neq 0$, so $v\notin(Y^\Omega)^\Omega$. □

Lemma 4.7. Let $\mathcal{M}$ and $\mathcal{N}$ be Banach manifolds, and let $f\colon\mathcal{M}\to\mathcal{N}$ be a surjective smooth submersion. Let $\alpha\in\Omega^k_{\mathbb{R}}(N)$. If $f^*\alpha=0$, then $\alpha=0$.

Proof. $\alpha=0$ if and only if it vanishes when applied to any set of tangent vectors at any point. Let $p\in N$ and $q\in M$ such that $f(q)=p$. Choose any vectors $v_1,\dots v_k\in T_qN$, and $w_1,\dots w_k\in T_pM$ such that $\mathrm{d}_qf(w_j)=v_j$. Then:

$$\alpha_p(v_1,\dots v_j)=\alpha_{f(q)}(\mathrm{d}_pf(w_1),\dots\mathrm{d}_pf(w_k))=(f^*\alpha)_q(w_1,\dots w_k)=0.$$

□

Lemma 4.8. Let μ be a moment map on a Banach manifold $\mathcal{M}$ for the left smooth action of a Banach-Lie group G. Then, for each $p \in \mu^{-1}(0)$, we have $(T_p\ \mu^{-1}(0))^{\Omega_p} = T_p(G\cdot p)$ and $T_p\ \mu^{-1}(0) = (T_p(G\cdot p))^{\Omega_p}$.

Proof. By lemma 4.6 it suffices to check the second identity. Let $u \in T_p\mathcal{M}$ by such that $\Omega_p(\hat{X}_p, u) = 0$ $\hat{X}_p \in T_p(G\cdot p)$ (see (4.2)), and let us see that $u \in T_p\ \mu^{-1}(0)$. Observe that

$$\Omega(\hat{X}_p, u) = \left(i_{\hat{X}}\Omega\right)_p(u) = \mathrm{d}_p\mu^X(u) = \langle \mathrm{d}_p\mu(v), X\rangle$$

the latter being zero for all $X \in \mathrm{Lie}(G)$ if and only if $u \in \ker \mathrm{d}_p\mu = T_p\ \mu^{-1}(0)$. □

Proof of theorem 4.4. We are now ready to prove the theorem of symplectic reduction for Banach manifolds. First, observe that the fact that $\mathcal{M}_{\mathrm{red}}$ is a Banach manifold (Hausdorff if G acts properly on $\mu^{-1}(0)$) is taken care of by the slice condition, so we just need to concern ourselves with obtaining the symplectic form.

The requirement that π be a smooth surjective submersion defines Ω_{red} uniquely. Firstly, there is an isomorphism $T_{\pi(p)}\mathcal{M}_{\mathrm{red}} \cong T_p(\mu^{-1}(0))/T_p(G\cdot p)$ for all $p \in \mu^{-1}(0)$ by (4.1) let us denote equivalence classes using brackets $[\,\cdot\,]$. Define, pointwise

$$(\Omega_{\mathrm{red}})_{\pi(p)}([u],[v]) := \Omega_p(u,v) \qquad u, v \in T_p\ \mu^{-1}(0).$$

Let us check that the previous definition does not depend on the representatives chosen. To that extent, let $[v] = [v']$ and $[u] = [u']$, where all vectors are elements of $T_p(\mu^{-1}(0))$. Explicitly, by (4.2), this means that there exist $X, Y \in \mathrm{Lie}(G)$ such that $u' = u + \hat{X}_p$ and $v' = v + \hat{Y}_p$, where $\hat{X}$ and $\hat{Y}$ are the fundamental vector fields associated to X and $Y \in \mathrm{Lie}(G)$. Then

$$\begin{aligned}\Omega_p(u',v') &= \Omega_p(u+\hat{X}_p, v+\hat{Y}_p) = \Omega_p(u,v) + \Omega_p(u,\hat{Y}_p) + \Omega(\hat{X}_p, v) + \Omega(\hat{X}_p,\hat{Y}_p)\\ &= \Omega_p(u,v),\end{aligned}$$

where we used that $\Omega_p(\hat{X}_p,\hat{Y}_p) = \Omega_p(u,\hat{Y}_p) = \Omega_p(\hat{X}_p, v) = 0$ because, by lemma 4.8 $T_p(\mu^{-1}(0))$ is the Ω_p-orthogonal complement of $T_p(G\cdot p)$. Observe that $T_p(G\cdot p) \subset T_p\ \mu^{-1}(0) = (T_p(G\cdot p))^{\Omega_p}$ so elements of $T_p(G\cdot p)$ are Ω_p-orthogonal to themselves. Let us now see that the definition is also independent of the point p chosen to define Ω_p. Suppose that $\pi(p) = \pi(p')$. Since π is the natural projection induced by the action $\phi\colon G\times M \to M$, this means that there is some $g \in G$ such that $\phi_g(p) = p'$. Now, since $\pi(p) = \pi(p')$, we obviously have $T_{\pi(p)}\mathcal{M}_{\mathrm{red}} = T_{\pi(p')}\mathcal{M}_{\mathrm{red}}$, but then, the isomorphism (4.1) applies for both p and p'. We get that given $u \in T_p\ \mu^{-1}(0)$ and $u' \in T_{p'}\ \mu^{-1}(0)$, $[u]$ and $[u']$ correspond to the same element in $T_{\pi(p)}\mathcal{M}_{\mathrm{red}}$ if and only if $\mathrm{d}_p\pi(u) = \mathrm{d}_p\pi(u')$ where $p' = \phi_g(p)$ (see the comments bellow equation (4.1)). Then, since the action is symplectic

$$\begin{aligned}(\Omega_{\mathrm{red}})_{\pi(p')}\left([\mathrm{d}_p\phi_g(u)],[\mathrm{d}_p\phi_g(v)]\right) &= \Omega_{p'}(\mathrm{d}_p\phi_g(u),\mathrm{d}_p\phi_g(v)) = \Omega_{\phi_g(p)}(\mathrm{d}_p\phi(u),\mathrm{d}_p\phi(v)) = \left(\phi_g^*\Omega\right)_p(u,v)\\ &= \Omega_p(u,v) = (\Omega_{\mathrm{red}})_{\pi(p)}([u],[v]).\end{aligned}$$

Where we used the fact that $\mathrm{d}_{p'}\pi\left(\mathrm{d}_p\phi_g(u)\right) = \mathrm{d}_p(\pi\circ\phi_g)(u) = \mathrm{d}_p\pi(u)$ to deduce that $[u]$ and $[\mathrm{d}_p\phi_g(u)]$ represent the same element in $T_{\pi(p)}\mathcal{M}_{\mathrm{red}}$.

This means that Ω_{red} is well-defined. It is smooth because the definition implies that $\pi^*\Omega_{\mathrm{red}} = i^*\Omega$, and the right-hand side is smooth. Moreover, it is closed because d commutes with pullbacks

$$\pi^*(\mathrm{d}\Omega_{\mathrm{red}}) = \mathrm{d}\left(\pi^*\Omega_{\mathrm{red}}\right) = \mathrm{d}\left(i^*\Omega\right) = i^*\left(\mathrm{d}\Omega\right) = 0,$$

and π is a surjective smooth submersion, so we may apply lemma 4.7. It remains to show that Ω is non-degenerate. This is a linear issue, so choose $p \in \mu^{-1}(0)$, and suppose that there is some $[u] \in T_{\pi(p)}\mathcal{M}_{\mathrm{red}}$ such that for every $[v] \in T_{\pi(p)}\mathcal{M}$ we have $(\Omega_{\mathrm{red}})_{\pi(p)}([u],[v]) = 0$, let us see that $[u] = [0]$. The definition of Ω_{red} allows us to translate this problem to one on $T_p(\mu^{-1}(0))$. That is, given $u \in T_p(\mu^{-1}(0))$ such that for every $v \in T_p(\mu^{-1}(0))$ we have $\Omega_p(u,v) = 0$, we have to show that $u \in T_p(G\cdot p)$. Let us apply lemma 4.5. Take T_pM as the Banach space (it is a Banach space by lemma C.3), and take $T_p(G\cdot p)$ as the closed subspace. Now, lemma 4.5 implies that

$$T_p(G\cdot p) = \left((T_p(G\cdot p))^{\Omega}\right)^{\Omega} = (T_p\ \mu^{-1}(0))^{\Omega} = \left\{u \in T_pM : \Omega_p(u,v) = 0 \quad \text{for all} \quad v \in T_p\ \mu^{-1}(0)\right\},$$

which is what we wanted to show. □

4.2 The gauge group of a principal bundle as a Hilbert Lie group

The aim of this section is to show how the gauge group of a principal bundle can be seen as a Banach manifold. To so, we shall make use of the isomorphism $\mathscr{G} \cong \Gamma(M; \mathrm{Ad}(P))$ given by theorem 3.13. It is the completion of $\Gamma(M; \mathrm{Ad}(P))$ with respect to a Sobolev norm what actually *is* a Hilbert Lie group.

4.2.1 Sobolev spaces

Let us consider a vector bundle E, endowed with a positive-definite inner product $\langle \cdot, \cdot \rangle$, which induces a norm. Fix a connection ∇^E on E, that is compatible with the fiber metric. This connection induces a covariant derivative in all tensor bundles over M as we show bellow (we follow the exposition in [68]).

As we know, the connection ∇^E, defines a map $\nabla^E \colon \Gamma(E) \to \Gamma(T^*M \otimes E)$. On the other hand, since (M, g) is a riemannian metric, TM is naturally endowed with the Levi-Civita connection ∇^g. This induces a connection on the dual bundle ∇^{g*}. The bundle $T^*M \otimes E$ is therefore endowed with the connection $\nabla^{g*} \otimes \nabla^E \colon \Gamma(T^*M \otimes E) \to \Gamma(T^*M \otimes T^*M \otimes E)$. Now, $\otimes^2 T^*M \otimes E$ is endowed with a connection $\nabla^{g*} \otimes \nabla^E$, and we may use it to define a connection on $T^*M \otimes \left(\otimes^2 T^*M\right) \otimes E \cong \otimes^3 T^*M \otimes E$, namely, we define the connection $\nabla^{g*} \otimes \left(\nabla^{g*} \otimes \nabla^E\right)$ (see section 2.2). Iterating this process, we may define a connection on each $\otimes^k T^*M \otimes E$ by

$$\nabla^{g*} \otimes \overset{k}{\cdots} \otimes \nabla^{g*} \otimes \nabla^E.$$

Making an abuse of notation, we will denote by ∇ all the so-defined connections (including ∇^E). Explicitly, if $\alpha \in \Gamma(\otimes^k T^*M \otimes E)$, then $\nabla\alpha \in \Gamma(\otimes^{k+1} T^*M \otimes E)$ is defined by [68, Eq. B2]

$$\nabla\alpha(X_0, \dots X_k) = \nabla^E(\alpha(X_1, \dots X_k)) - \alpha(\nabla^g_{X_0} X_1, \dots X_k) - \dots - \alpha(X_1, \dots \nabla^g_{X_0} X_k),$$

For $k \in \mathbb{N}_{\geq 1}$ and any two sections $\phi, \psi \in \Gamma(E)$, we define the L^2_k inner product

$$\langle \phi, \psi \rangle_k := \int_M \left(\langle \phi, \psi \rangle + \langle \nabla\phi, \nabla\psi \rangle + \dots + \langle \nabla^k \phi, \nabla^k \psi \rangle \right) \mathrm{d}V_g. \tag{4.3}$$

Remark 4.9. Each term in the sum inside the integral is a smooth function, so it makes sense to integrate it with respect to the volume form $\mathrm{d}V_g$, and the compactness of M ensures that the integral will be finite. As for the first claim, the symbol $\langle \nabla^\ell \phi, \nabla^\ell \psi \rangle$ *hides* the fact that we need to define an inner product on tensor products. First, the fact that g is non-degenerate (it is a riemannian metric) allows us to define an inner product of one-forms by $\langle \alpha, \beta \rangle_g := g\left(\alpha^\#, \beta^\#\right)$, where $\#$ is the *sharp* isomorphism (see [34, Ch. 2]). On the other hand, whenever $V_1, \dots V_n$ are vector spaces endowed with inner products $\langle \cdot, \cdot \rangle_i$, we may define an inner product by extending by linearity the following expression

$$\langle v_1 \otimes v_2 \otimes \dots \otimes v_n, w_1 \otimes w_2 \otimes \dots \otimes w_n \rangle = \langle v_1, w_1 \rangle_1 \cdot \langle v_2, w_2 \rangle_2 \cdots \langle v_n, w_n \rangle_n.$$

This defines an inner product on each $\otimes^k T^*M \otimes E$ (see [52, Lemma 1.7] for a very detailed proof that the previous expression defines an inner product).

The map

$$\begin{array}{ccc} \Gamma(E) \times \Gamma(E) & \longrightarrow & \mathbb{R} \\ (\phi, \psi) & \mapsto & \langle \phi, \psi \rangle_k \end{array}$$

defines an inner product on $\Gamma(E)$. Of course, everything works out fine if we work with complex vector bundles and a hermitian fiber metric, in which case we can define a hermitian inner product on $\Gamma(E)$. Since $\Gamma(E)$ is a vector space with an inner product, we can always consider its Hilbert completion, which we shall denote $L^2_k(M; E)$ or $L^k_2(E)$. These spaces are called *Sobolev spaces*.

Remark 4.10. Recall that the topology of a Hilbert space is the metric topology with respect to the distance induced by the norm. The inner product we defined depends on g, ∇ and $\langle \cdot, \cdot \rangle$. However, different choices of these parameters lead to equivalent norms. This means that, as a topological space, $L^2_k(M; E)$ is well-defined and independent of any choice of these parameters.

Elements of a Hilbert completions are, formally, equivalence classes of Cauchy sequences. The following lemma provides a more tangible way of looking at these spaces.

Lemma 4.11 (Sobolev). Let $E \to M$ be a vector bundle over a compact smooth manifold, and let $k, p \in \mathbb{N}_{\geq 0}$ be such that $k > \dim M/2 + p$. Then, there is a continuous embedding $W^k(M;E) \hookrightarrow \mathscr{C}^p(M;E)$.

Proof. See [68, Th. B.2] Here, $\mathscr{C}^p(M;E)$ is endowed with the metric topology induced by the norm

$$||\phi||_\infty := \sup_{j\leq p} \left\{ \max_{p\in M} \langle \nabla^j \phi(p), \nabla^j \phi(p) \rangle \right\}.$$

□

That is, Sobolev spaces of sections can be seen as spaces of usual sections but with the caveat that some regularity may be lost.

We present now a sketch of the construction that allows us to see $\mathscr{G}$ as a Hilbert Lie group. For the construction, we consider G a compact, connected, semisimple *matrix Lie group*. That is, $G \subset \mathfrak{gl}(n,\mathbb{C}) = \mathrm{M}_n(\mathbb{C})$ for some $n \in \mathbb{N}$. We suppose that M is a compact smooth oriented riemannian manifold. Consider the vector bundle $E^n_{\mathbb{C}} := P \times_G \mathfrak{gl}(n,\mathbb{C})$ where G acts on $\mathfrak{gl}(n,\mathbb{C})$ by conjugation. We may see $\mathrm{Ad}(P)$ as a subbundle of $E^n_{\mathbb{C}}$, i.e. it is an embedded submanifold, and its projection is the restriction of the projection of $E^n_{\mathbb{C}}$. Observe that $\mathrm{Ad}(P)$ is also constructed using the action of G by conjugation. Using the riemannian metric and any hermitian fiber metric on $E^n_{\mathbb{C}}$ (those always exist, and they can be constructed through a partition of unity argument, see [35, Page 196]). Hence, we can consider $\Gamma(M;\mathrm{Ad}(P)) \subset \Gamma(M;E^n_{\mathbb{C}}) \subset L^2_k(M;E^n_{\mathbb{C}})$. We define $\mathscr{G}_k := \overline{\Gamma(M;\mathrm{Ad}(P))}$, i.e. the closure of $\Gamma(M;\mathrm{Ad}(P))$ in $L^2_k(M;E^n_{\mathbb{C}})$ with respect to the Sobolev topology of the latter. If we choose $k \in \mathbb{N}_{\geq 0}$ satisfying $k > \dim M/2 + 1$, then elements of $\mathscr{G}_k$ can be seen as $\mathscr{C}^2$ sections of the bundle $\mathrm{Ad}(P)$. Hence, $\mathscr{G}_k$ has a natural group structure where the group operation is defined pointwise. Actually, $L^2_k(M;E^n_{\mathbb{C}})$ has a natural product structure. This product, call it $m\colon L^2_k(M;E^n_{\mathbb{C}}) \times L^2_k(M;E^n_{\mathbb{C}}) \to L^2_k(M;E^n_{\mathbb{C}})$ is bilinear and continuous (see [40, Eq. 2.2]). Hence, $m(\overline{\Gamma(M;\mathrm{Ad}(P))} \times \Gamma(M;\mathrm{Ad}(P))) \subset \overline{\Gamma(M;\mathrm{Ad}(P))}$, and $\mathscr{G}_k$ is closed under products.

Theorem 4.12. Let $P(M,G)$ be a principal G-bundle over a compact manifold, where G is a compact, connected, matrix Lie group. Then, for $k > \dim M/2 + 1$, $\mathscr{G}_k$ is a Hilbert Lie group whose lie algebra is $\mathrm{Lie}(\mathscr{G}_k) \cong L^2_k(M;\mathrm{ad}(P))$, and whose exponential map is given by

$$\exp_{\mathscr{G}_k}(\xi)(p) = \big[p, \exp_G(\xi(p))\big], \qquad \xi \in \mathrm{Lie}(\mathscr{G}_k),\ p \in M$$

Proof. See [40].

□

Remark 4.13. $\Gamma(M;\mathrm{ad}(P))$ is a Lie algebra endowed with the pointwise Lie bracket it gets from $\mathfrak{g}$. Given $\boldsymbol{\xi}, \boldsymbol{\zeta} \in \Gamma(M;\mathrm{ad}(P_G))$ we define $[\boldsymbol{\xi}, \boldsymbol{\zeta}]$ as the section

$$[\boldsymbol{\xi}, \boldsymbol{\zeta}](x) := \big[\ [p,\xi],[p,\zeta]\ \big] = [p,[\xi,\zeta]\ ] \qquad \text{where } \boldsymbol{\xi}(x) = [p,\xi], \boldsymbol{\zeta}(x) = [p,\zeta].$$

This is well-defined independently of the chosen representatives because Ad_g is a Lie algebra isomorphism for all $g \in G$.

Observe that the smooth gauge group $\Gamma(M;\mathrm{Ad}(P))$ can be seen as a subgroup of $\mathscr{G}_k$ for each k. Moreover, since the exponential map is smooth, it maps $\Gamma(M;\mathrm{ad}(P))$ to $\mathscr{G}$, hence the former may be seen as the Lie algebra of $\mathscr{G}$. It is quite common in the literature to omit the Sobolev completions and directly state that $\Gamma(M;\mathrm{ad}(P))$ is the Lie algebra of G. It is useful to see $\Gamma(M;\mathrm{ad}(P))$ from different points of view. To that end, observe that this Lie algebra is isomorphic to the following

$$\mathfrak{G} := \{X \in \mathfrak{X}(P) : (\phi_g)_* X = X \ \ \forall\, g \in G \ \text{ and } \ X \ \text{ is vertical}\} \qquad \mathsf{G} := \left\{\zeta \in \mathscr{C}^\infty(P;\mathfrak{g}) : \zeta(p\cdot g) = \mathrm{Ad}_{g^{-1}}(\zeta(p))\right\}.$$

The first in endowed with the restriction of the Lie bracket it $\mathfrak{X}(P)$, and the latter with the pointwise Lie bracket induced by $\mathfrak{g}$. See [55, Sec. 3.5]

Chapter 5

The Hitchin-Kobayashi correspondence

Up until now, the different topics we have treated seem unrealted to each other. In this chapter, we shall see how everything comes together in the Hitchin-Kobayashi correspondence. The objetive of this chapter is to present an exposition to the Hitchin-Kobayashi correspondence. Our main interest lies in the principal bundle version of the correspondence, but we will give an expository account to the vector bundle both for the sake of completeness, and so that we can compare them. We will also try to emphasize the role played by moment maps in the theory, specially in the construction of moduli spaces. We will refer to the Hitchin-Kobayashi correspondence by HK.

5.1 The HK correspondence for vector bundles

The Hitchin-Kobayasi correspondence states that a holomorphic vector bundle over a compact Kähler manifold is polystable if and only if it admits a Hermite-Einstein metric. To make sense of this definition, we need to introduce several concepts. We will mainly follow [28], however, we should remark that when this book was originally published, the result had only been proven for Riemann surfaces, so it does not include a proof of the more general statement. In fact, the result is true under the weaker assumption that M is just a compact complex manifold. The book [37] gives a very detailed proof of the theorem under this relaxed assumption.

We introduce now the following notation (which is the same we introduced for principal bundles in section 3.8). Whenever we have a holomorphic vector bundle, we will use `mathcal` letters to name it, and we will use the corresponding plain letter for the underlying complex smooth vector bundle. For example, we will use $\mathcal{E}$ when we work with the holomorphic structure, and E to refer to the underlying complex smooth vector bundle.

5.1.1 Stability of vector bundles

Let us start with the definition of stability.

Definition 5.1. Let $\mathcal{E} \to M$ be a holomorphic vector bundle over a Kähler manifold M. We define the *degree* of $\mathcal{E}$ by

$$\deg(\mathcal{E}) := \int_M c_1(\mathcal{E}) \wedge \omega^{n-1}.$$

Remark 5.2.

1. The degree is well-defined *real* number. It should be remarked that $c_1(\mathcal{E})$ is a cohomology class, so the integral above is understood as the integral of some representative of the class. Accordingly, we must show that it is independent of the representative chosen. First, since M is compact, and we integrate a top-degree form, the integral is finite. Moreover, we may choose any two representatives of $c_1(\mathcal{E})$, say $c_1(\nabla)$ and $c_1(\nabla')$ for two different connections. Since they determine the same cohomology class, $c_1(\nabla) = c_1(\nabla') + \mathrm{d}\alpha$ for some one-form. Then,

$$\int_M c_1(\nabla) \wedge \omega^{n-1} = \int_M c_1(\nabla') \wedge \omega^{n-1} + \int_M \mathrm{d}\alpha \wedge \omega^{n-1}.$$

   The last summand is zero because $\mathrm{d}\alpha \wedge \omega^{n-1} = \mathrm{d}(\alpha \wedge \omega^{n-1})$ (the Kähler form is closed) and we may apply Stokes' theorem.

2. If $M = X$ is a Riemann surface, the degree is $\deg \mathcal{E} = \int_X c_1(\mathcal{E})$.

For Riemann surfaces, the notion of stability, or slope-stability, is quite straightforward. The definition of stability is originally due to Mumford [43].

Definition 5.3 (Stability). Let $\mathcal{E} \to X$ be a holomorphic vector bundle over a Riemann surface. We say that $\mathcal{E}$ is *stable* (resp. *semistable*) if for every proper holomorphic subbundle $\mathcal{E}'$ of $\mathcal{E}$ with $0 < \operatorname{rank}(\mathcal{E}') < \operatorname{rank}(\mathcal{E})$, one has

$$\mu(\mathcal{E}') := \frac{\deg(\mathcal{E}')}{\operatorname{rank}(\mathcal{E}')} < \frac{\deg(\mathcal{E})}{\operatorname{rank}(\mathcal{E})} \qquad \left(\text{resp. } \frac{\deg(\mathcal{E}')}{\operatorname{rank}(\mathcal{E}')} \leq \frac{\deg(\mathcal{E})}{\operatorname{rank}(\mathcal{E})}\right).$$

μ is called the *slope* of the vector bundle $\mathcal{E}$. We say that $\mathcal{E}$ is *polystable* if $\mathcal{E} = \oplus_{i=1}^n \mathcal{E}_i$ where each $\mathcal{E}_i$ is a stable holomorphic vector bundle, and all of them have the same slope.

For Riemann surfaces, i.e. Kähler manifolds of complex dimension one, the definition of stability in terms of subbundles works well. However, once we try to generalize to higher dimensional Kähler manifolds, it turns out that looking as subbundles is not enough. The correct definition (i.e. the one that works) is given in terms of coherent sheaves. Introducing coherent sheaves would take us another whole appendix, so we refer the reader to [28] for the relevant definitions.

5.1.2 Hermite-Einstein vector bundles

Besides stability, the other concept in the statement if the Hitchin-Kobayashi correspondence that is new to us is that of a Hermite-Einstein metric. A Hermite-Einstein metric is a usual hermitian fiber metric (in the sense of definition 2.5) which is asked to fulfill a particular equation.

Definition 5.4. Let $\mathcal{E} \to M$ be a holomorphic vector bundle over a compact Kähler manifold M. Let h be a hermitian fiber metric. We say that a h satisfies the *weak Hermite-Einstein condition* if its Chern connection ∇ satisfies $\mathrm{i}\,\Lambda_\omega F_\nabla = \varphi \cdot \mathrm{id}_E$ for some $\varphi \in \mathscr{C}^\infty(M;\mathbb{R})$, and we will say that it satisfies the *Hermite-Eisntein condition* if

$$\mathrm{i}\,\Lambda_\omega F_\nabla = c \cdot \mathrm{id}_{\mathcal{E}} \tag{5.1}$$

for some constant value c. We say that h is a *Hermite-Einstein metric*.

Even though the definition above is given in terms of an arbitrary real constant c, if equation (5.1) holds, then, the value of c is determined uniquely by the topology of the bundle and the manifold, as the following lemma shows. If a connection ∇ (not necessarily the Chern connection) satisfies equation (5.1), we say that has *constant central curvature*.

Lemma 5.5. Let $\mathcal{E} \to M$ be a holomorphic vector bundle over a compact Kähler manifold, and suppose that ∇ is connection that has constant central curvature. Then, the constant c is determined uniquely by

$$c := \frac{2\pi \cdot \deg(E)}{(n-1)! \cdot \operatorname{rank}(E) \cdot \operatorname{Vol}(M)} = \frac{2\pi \cdot \mu(E)}{(n-1)! \cdot \operatorname{Vol}(M)}.$$

Proof. Wedge ω^n and equation (5.1) on both sides. This yields $\mathrm{i}\,\Lambda_\omega F_\nabla \otimes \omega^n = c \cdot \omega \otimes \mathrm{id}_E$. Applying lemma A.34 we get $\mathrm{i}\,\Lambda_\omega F_\nabla \otimes \omega^n = \mathrm{i}\, n \cdot L_\omega^{n-1} F_\nabla = \mathrm{i}\, n \cdot \omega^{n-1} \wedge F_\nabla = c \cdot \omega^n \otimes \mathrm{id}_E$. Taking the trace, $n \cdot 2\pi \cdot \mathrm{i}\,/2\pi \cdot \operatorname{Tr}(F_\nabla) = c \cdot \omega^n \cdot \operatorname{Tr}(\mathrm{id}_E)$. Observe that $\operatorname{Tr}(\mathrm{id}_E) = r = \operatorname{rank}(E)$. Finally, using the expression for $c_1(\nabla)$ and integrating the top degree forms on both sides

$$2\pi n \cdot \deg(E) = n \cdot 2\pi \cdot \int_M c_1(\nabla) \wedge \omega^{n-1} = \int_M c \cdot \operatorname{rank}(E) \cdot n! \cdot \frac{\omega^n}{n!} = c \cdot \operatorname{rank}(E) \cdot n! \cdot \operatorname{Vol}(M).$$

□

Observe that this constant only depends on the topology of the bundle. Indeed, it only depends on the cohomology classes of $c_1(\nabla)$ (that is, $c_1(E)$) and ω.

Even though the weak Hermite-Einstein condition may seem more general at first glance, it is not. Indeed, one can show that, after a suitable conformal change, i.e. after rescaling the hermitian fiber product $h' := a \cdot h$ for some real constant, if (E, h) has a connection satisfying the weak Hermite-Einstein condition, then, the same connection satisfies the Hermite-Einstein condition on (E, h') [28, Prop. 4.2.4].

Remark 5.6. Observe that the condition of having constant central curvature can be seen as an attempt to relax the condition of being flat (recall that flat connections satisfy $F_\nabla = 0$). Indeed, by definition of the degree of a vector bundle, only vector bundles of degree zero admit flat connections. Since asking for the degree to be zero is very restrictive, one may think about a more relaxed condition.

Definition 5.7 (Hermitian Yang-Mills connection). Let $E \to M$ be a smooth complex vector bundle over a compact Kähler manifold M. A unitary connection ∇ is said to be *Hermitian-Yang-Mills* if it satisfies

$$\begin{cases} F_\nabla^{(0,2)} &= 0 \\ \mathrm{i}\,\Lambda_\omega F_\nabla &= c \cdot \mathrm{id}_E \end{cases}$$

for a constant c.

Observe that we do not ask for E to be holomorphic here. This definition is useful because it allows for a shift in our point of view. More precisely, as mentioned at the beginning of the chapter, and as stated in theorem 5.9 bellow, we fix a holomorphic vector bundle $\mathcal{E}$ (or more precisely, we fix the *holomorphic structure*), and we *look for a fiber metric* whose Chern connection satisfies (5.1). Nonetheless, we could think about this differently: we could ignore the holomorphic structure, but *fix a hermitian fiber metric* on E. Then, we can look for a Hermitian-Yang-Mills connection. If we happen to find one, ∇, the condition $F_\nabla^{(0,2)} = 0$ combined with lemma 2.17 and theorem 2.16 imply that $\nabla^{(0,1)}$ induces a holomorphic structure on E. If $\mathcal{E}$ is the corresponding holomorphic structure, its Chern connection is ∇, and the metric we had originally fixed becomes a Hermite-Einstein metric. The following theorem provides a solid basis for the ideas discussed above.

Theorem 5.8. Let $\mathcal{E} \to M$ be a holomorphic vector bundle over a compact Kähler manifold, and fix a hermitian fiber metric h. Let $\overline{\partial}_E \in \mathscr{H}$ be its Cauchy-Riemann operator. Then, there is a Hermite-Einstein fiber metric h' if and only if there is a gauge transformation $f \in \mathscr{G}^{\mathbb{C}}$ such that h is a Hermite-Einstein fiber metric with respect to the complex structure induced by $\overline{\partial}_E \cdot f$.

Proof. We present a very schematic proof. For more details see [57, Sec. 1.1]. Let h' be a Hermite-Einstein fiber metric, and let ∇' be its Chern connection. Every pair of hermitian fiber metrics is related as follows $h'(\xi, \zeta) = h(\xi, f(\zeta))$ for some $f \in \mathscr{G}^{\mathbb{C}}$ which coincides with its hermitian conjugate: $f = f^*$ with respect to both h and h'. This implies that $(\nabla')^{(1,0)} = f^{-1} \circ \nabla^{(1,0)} \circ f$. Finally, there is some $g \in \mathscr{G}^{\mathbb{C}}$ satisfying $g^2 = f$. We find $\nabla \cdot g^{-1} = g \circ \nabla' \circ g^{-1}$ and $F_{\nabla \cdot g^{-1}} = g \circ F_\nabla \circ g^{-1}$. Finally, let ∇_h be the Chern connection with respect to the pair $(h, \overline{\partial}_E \cdot g^{-1})$. One may show that $F_{\nabla_h} = g \circ F_{\nabla'} \circ g^{-1}$. This connection is Hermite-Yang-Mills if and only if $\mathrm{i}\,\Lambda_\omega F_{\nabla_h} = c \cdot \mathrm{id}_E$, which is equivalent to

$$c \cdot \mathrm{id}_E = \mathrm{i}\,\Lambda_\omega \left(g \circ F_{\nabla'} \circ g^{-1} \right) = g \circ (\mathrm{i}\,\Lambda_\omega F_{\nabla'}) \circ g^{-1} \iff \mathrm{i}\,\Lambda_\omega F_{\nabla'} = c \cdot \mathrm{id}_E,$$

which is equivalent to h' being Hermite-Einstein with respect to $\overline{\partial}_E$. □

To recapitulate, there are two ways of tackling the existence of a Hermite-Einstein metric: we either *fix the holomorphic structure* and *look for the hermitian fiber metric*, or we *fix the fiber metric* and *look for a Hermite-Yang-Mills connection*. By theorem 5.8 above these are equivalent: it states that if we fix both a holomorphic structure $\overline{\partial}_E$ and a hermitian fiber metric h, then we can find a (new) Hermite-Einstein metric h' for $\overline{\partial}_E$ if and only if we can find a (new) holomorphic structure *which is isomorphic to the original one* making the metric we initially had Hermite-Einstein. Observe that we obtain isomorphic holomorphic vector bundle structures because two integrable Dolbeault operators in the same $\mathscr{G}^{\mathbb{C}}$-orbit produce isomorphic structures (see section 2.6). Hence, an equivalent way to phrase the problem of finding a Hermite-Einstein fiber metric is as follows: if we are given a holomorphic vector bundle $\mathcal{E} \to M$ whose holomorphic structure is given by $\overline{\partial}_E$, and we wonder if

it has a Hermite-Einstein metric, a way to decide is to fix a hermitian fiber metric (recall that these always exist) and see if there is a gauge transformation $f \in \mathscr{G}^{\mathbb{C}}$ such that the Chern connection of the holomorphic structure induced by $f \cdot \overline{\partial}_E$ satisfies (5.1). Finally, by the Chern correspondence, this is equivalent to finding a connection of type $(1,1)$ (of the type $f \cdot \nabla_h$, where ∇_h is the original Chern-connection) satisfying (5.1).

5.1.3 The theorem

Theorem 5.9 (HK, vector bundle version). Let $\mathcal{E} \to M$ be a holomorphic vector bundle over a Kähler manifold M. Then, $\mathcal{E}$ admits a Hermite-Einstein metric if and only if it is polystable.

The proof of the fact the polystability implies the existence of a Hermite-Einstein metric uses some heavy analysis, and it would require several techniques we have not mentioned. We refer the reader to the original proof by Uhlenbeck and Yau [67], or to the notes by Lars Martin [57]. The proof of the converse is more accessible, but we have not introduced coherent sheaves. For completeness, we will give an outline of the proof for Riemann surfaces. Hence, we will actually give a proof of the theorem by Narashiman and Seshadri, but following Donaldson and Kobayashi. We will need some work before giving the proof.

Before getting into the proof, consider the following situation: let $\mathcal{E} \to M$ a holomorphic vector bundle, and $\mathcal{S}$ a subbundle. We can consider the quotient bundle (see example A.7) $\mathcal{Q} := \mathcal{E}/\mathcal{E}'$ which is a holomorphic vector bundle and fits into the following short exact sequence

$$0 \to \mathcal{S} \to \mathcal{E} \to \mathcal{Q} \to 0. \tag{5.2}$$

Let h be a hermitian fiber metric on $\mathcal{E}$, and let $h_{\mathcal{S}}$ be its restriction to $\mathcal{S}$. This is a hermitian fiber metric as well. On every fiber $\mathcal{E}_m$ the metric h defines an orthogonal complement to the subspace $\mathcal{S}_m$, which we shall denote $S^{\perp}_m$. We define $S^{\perp} := \sqcup_{m\in M} S^{\perp}_m$. This is a complex smooth vector bundle, but not holomorphic in general. Then (as smooth vector bundles) we have the following decomposition $E = S \oplus S^{\perp}$. In addition, $Q = E/S \cong S^{\perp}$ (again only as *smooth* vector bundles). Furthermore, h induces a hermitian fiber metric $h_{S^{\perp}}$ on $S^{\perp}$, which under the isomorphism, induces a hermitian fiber metric h_Q on Q (hence on $\mathcal{Q}$).

Let ∇ be the Chern connection on $\mathcal{E}$. For any local section of S $\xi \in \Gamma(S)$, $\nabla\xi$ can be projected onto S and $S^{\perp}$. We define ∇_S and $\alpha \in \Omega^1_{\mathbb{C}}(M; \mathrm{Hom}_{\mathbb{C}}(S, S^{\perp}))$ by

$$\nabla\xi = \nabla_S\xi + \alpha(\xi).$$

Then, [28, Prop. 1.6.4], ∇_S is the Chern connection in $\mathcal{S}$, and α is of type $(1,0)$. We call it the *second fundamental form* of S. Since $Q \cong S^{\perp}$, we can see $\alpha \in \Omega^{(1,0)}_{\mathbb{C}}(M; \mathrm{Hom}_{\mathbb{C}}(S, Q))$.

Similarly, given a local section $\eta \in \Gamma(S^{\perp})$, we define $\beta \in \Omega^1_{\mathbb{C}}(M; \mathrm{Hom}(S^{\perp}, S))$ and $\nabla_{S^{\perp}}$ by the projections onto S and $S^{\perp}$:

$$\nabla\eta = \beta(\eta) + \nabla_{S^{\perp}}\eta.$$

Under the isomorphism $S^{\perp} \cong Q$, $\nabla_Q \equiv \nabla_{S^{\perp}}$ is the Chern connection of $\mathcal{Q}$, and $\beta \in \Omega^{(0,1)}_{\mathbb{C}}(M; \mathrm{Hom}_{\mathbb{C}}(Q, S))$ [28, Prop. 1.6.6]. After a computation in coordinates, we may write (see section 2.2)

$$\nabla = \begin{pmatrix} \nabla_S & \beta \\ \beta^* & \nabla_Q \end{pmatrix} \qquad F_{\nabla} = \begin{pmatrix} F^{(\mathcal{S})} - \beta \wedge \beta^* & \nabla^{(1,0)}_{\mathrm{Hom}(Q,S)}\beta \\ -\nabla^{(0,1)}_{\mathrm{Hom}(S,Q)}\beta^* & F^{(\mathcal{Q})} - \beta^* \wedge \beta \end{pmatrix}. \tag{5.3}$$

There β^* is the adjoint of β, i.e. the only one-form satisfying

$$h_S(\beta(\xi), \eta) = h_Q(\xi, \beta(\eta)) \qquad \text{for} \qquad \xi \in \Gamma(M; Q)\ \eta \in \Gamma(M; S).$$

One can show that $\beta^* = -\alpha$. $F^{(\mathcal{S})}$ and $F^{(\mathcal{Q})}$ are the curvatures of the Chern connection. Equation (5.3) is to be interpreted as follows: for any section $\xi \in \Gamma(M; E)$, we consider its orthogonal decomposition $\xi = \xi_S + \xi_Q$ (the latter under the isomorphism $Q \cong S^{\perp}$). Then,

$$\nabla(\xi) = \begin{pmatrix} \nabla_S & \beta \\ \beta^* & \nabla_Q \end{pmatrix} \cdot \begin{pmatrix} \xi_S \\ \xi_Q \end{pmatrix} = \begin{pmatrix} \nabla_S\xi_S + \beta(\xi_Q) \\ \beta^*(\xi_S) + \nabla_Q\xi_Q \end{pmatrix},$$

and analogously for the curvature.

Proof of theorem 5.9 for a Riemann surface X. $\Rightarrow$). Suppose that $\mathcal{E}$ admits a Hermite-Einstein fiber metric h, and let ∇ be the corresponding Chern-connection. Let $\mathcal{S} \subset \mathcal{E}$ be a holomorphic subbundle. Since X is a Riemann surface, from remark A.33, equation (5.1) may be rewritten as $\star F_\nabla = -\mathrm{i}\, c \cdot \mathrm{id}_E$, where $\star$ acts only on the differential form. Since $\star \circ \star = 1$ acting on two-forms, and $\star 1 = \omega$, applying the Hodge star operator to the previous equation yields an equivalent equation: $F_\nabla = -\mathrm{i}\, c \cdot \omega \otimes \mathrm{id}_E$. From (5.3), we may write (5.1) as

$$F_\nabla = \begin{pmatrix} F^{(\mathcal{S})} - \beta \wedge \beta^* & \nabla^{(1,0)}_{\mathrm{Hom}(S;Q)}\beta \\ -\nabla^{(0,1)}_{\mathrm{Hom}(Q,S)}\beta^* & F^{(Q)} - \beta^* \wedge \beta \end{pmatrix} = -\mathrm{i}\, c \cdot \begin{pmatrix} \omega \otimes \mathrm{id}_{\mathcal{S}} & 0 \\ 0 & \omega \otimes \mathrm{id}_Q \end{pmatrix}. \tag{5.4}$$

This implies

$$F^{(S)} - \beta \wedge \beta^* = -\mathrm{i}\, c \cdot \omega \otimes \mathrm{id}_{\mathcal{S}}. \tag{5.5}$$

Observe that taking $n = 1$ in lemma 5.5, $c = 2\pi \cdot \mu(\mathcal{E})/\mathrm{Vol(X)}$. Taking the trace of equation (5.5) and integrating over X, we get

$$\int_X \mathrm{Tr}\left(F^{(\mathcal{S})} - \beta \wedge \beta^*\right) = -\mathrm{i}\, c \cdot \int_X \omega \otimes \mathrm{Tr}(\mathrm{id}_{\mathcal{E}'}) = -\mathrm{i}\, \frac{2\pi \cdot \mu(\mathcal{E})}{\mathrm{Vol(X)}} \cdot \mathrm{rank}(\mathcal{S}) \cdot \mathrm{Vol(X)}$$

Simplifying

$$\frac{1}{\mathrm{rank}(\mathcal{E}')} \int_X \frac{\mathrm{i}}{2\pi} \cdot \mathrm{Tr}\left(F^{(\mathcal{S})}\right) - \frac{\mathrm{i}}{2\pi \cdot \mathrm{rank}(\mathcal{E}')} \int_X \mathrm{Tr}\,(\beta \wedge \beta^*) = \mu(\mathcal{S}') - \frac{\mathrm{i}}{2\pi \cdot \mathrm{rank}(\mathcal{S})} \int_X \mathrm{Tr}\,(\beta \wedge \beta^*) = \mu(\mathcal{E}).$$

So it suffices to show that the second summand in the left-hand side is non-negative. To see this, choose orthonormal frames $(s_1, \dots s_r)$ and $(\sigma_1, \dots \sigma_\ell)$ of S and Q respectively, and let $(t^1, \dots t^r)$ and $(\vartheta^1, \dots \vartheta^\ell)$ be the corresponding dual frames. Then, β and β^* can be written as

$$\beta = \beta^i_j \otimes t^j \otimes \sigma_i \qquad \beta^* = (\beta^*)^\ell_m \otimes \vartheta^m \otimes s_\ell \qquad \text{for some } \ \beta^i_j, (\beta^*)^\ell_m \in \Omega^1_{\mathbb{C}}(X).$$

Since the frame is orthonormal, one may show that $(\beta^*)^i_k = \overline{\beta^k_i}$. Hence,

$$\beta \wedge \beta^* = \left(\beta^i_j \otimes t^j \otimes \sigma_i\right) \wedge \left((\beta^*)^\ell_m \otimes \vartheta^m \otimes s_\ell\right) = \beta^i_j \wedge (\beta^*)^\ell_m \cdot \otimes \left(\vartheta^m(\sigma_i) \cdot t^j \otimes s_\ell\right) = \beta^i_j \wedge (\beta^*)^\ell_i \otimes t^j \otimes s_\ell$$

Hence,

$$\begin{aligned} \mathrm{Tr}(\beta \wedge \beta) &= \sum_{i,j} r \cdot \beta^i_j \wedge \overline{\beta^j_i} = -r \cdot \sum_{i,j} \beta^i_j \wedge \star\star\, \overline{\beta^j_i} = -\mathrm{i}\, r \cdot \sum_{i,j} \beta^i_j \wedge \star \overline{\beta^j_i} \\ &= -\mathrm{i}\, r \cdot \sum_{i,j} \left\langle \beta^i_j, \overline{\beta^j_i} \right\rangle \cdot \omega = -\mathrm{i} \sum_{i,j} ||\beta^i_j||^2 \cdot \omega \end{aligned}$$

Hence, $\int_X \mathrm{Tr}(\beta \wedge \beta^*)$ is locally the integral of a non-negative multiple of ω multiplied by $-\mathrm{i}$. This implies that $\mathrm{i} \int_X \mathrm{Tr}(\beta \wedge \beta^*) \geq 0$, with equality if and only if $\beta = 0$. We deduce that $\mu(\mathcal{S}) \leq \mu(\mathcal{E})$.

It follows that $\mathcal{E}$ is semistable. Now we separate cases. If $\mu(\mathcal{S}) < \mu(\mathcal{E})$ for all holomorphic proper subbundle, then $\mathcal{E}$ is in fact stable and we are done. Else, there is some proper holomorphic subbundle $\mathcal{S}$ for which the equality holds, hence $\beta = 0$. Thus, $\mathcal{E} = \mathcal{S} \oplus \mathcal{Q}$ where $\mathcal{Q} \equiv S^\perp$ is now holomorphic (see [28, Prop. 1.6.14]). Observe that (5.4) implies that both $\mathcal{S}$ and $\mathcal{Q}$ satisfy the Hermite-Einstein condition with the same factor, and by the previous calculation, they have the same slope. By [28, Prop. 4.1.4], $\mathcal{E}$ satisfies the Hermite-Einstein condition for the same constant value, hence lemma 5.5 implies that $\mu(\mathcal{E}) = \mu(\mathcal{S}) = \mu(\mathcal{Q})$. Iterating this process, we decompose each subbundle until we can no longer decompose them. We are guaranteed to reach that point because each decomposition decreases the rank of the resulting bundles and $\mathrm{rank}(\mathcal{E})$ is finite. Thus, we obtain $\mathcal{E} = \oplus_{i=1}^\ell \mathcal{E}_i$ for some ℓ, all of them with the same slope, and the bundle is polystable. □

For the converse we refer the reader to the paper by Donaldson [14].

5.1.4 Moduli spaces and moment map interpretation

The aim of this section is to give an expository account of the moduli space interpretation of the previous result, and the role played by moment maps and the Marsden-Weinstein quotient. We want to foccus on the core ideas, so we will not give many details and we will omit most proofs. More importantly, we will ignore, for the moment, the fact that the space of connections and the gauge group must be completed with suitable Sobolev norms to make sense of the quotients that will appear. This is an important technicality to make all the statements in this section rigorous, but since we will deal with that when we present the principal bundle version of the correspondence, we will give heuristic constructions here. We follow the exposition in [17]. For the sake of simplicity, let us consider the case where $M = X$ is a Riemann surface. We will make some comments at the end about the way one generalizes this result for higher dimension Kähler manifolds, but for know, it frees us from having to consider only connections of type $(1,1)$.

Let $\mathscr{A}$ the set of all unitary connections on E. Let $\mathscr{G}^{\mathbb{C}} = \mathrm{GL}(E)$ be the gauge group of E, and $\mathscr{G} = \mathrm{U}(E,h)$ the group of unitary automorphisms (the notation comes from the fact that one can be seen as a complexification of the other). There is an action of $\mathscr{G}$ on $\mathscr{A}$ given by $g \cdot \nabla = g \circ \nabla \circ g^{-1}$. Since satisfying the Hermite-Einstein condition is invariant under the $\mathscr{G}$-action, we may restrict that action to $\mathscr{A}_0$, this being the set of connections satisfying the Hermite-Einstein equation. Now, the quotient $\mathscr{A}_0/\mathscr{G}$ is called the *moduli space of Hermite-Einstein connections*. Observe that fixing a holomorphic structure on E and then looking for a Hermite-Einstein fiber metric is equivalent to fixing a hermitian fiber metric and then looking for a Hermite-Einstein connection (see [17, Page 251]). Finally, a connection ∇ is said to be reducible if $(E,h) = (E_1,h_1) \oplus (E_2,h_2)$ and $\nabla = \nabla_1 \oplus \nabla_2$ and *irreducible* otherwise. Let $\mathscr{A}^*$ and $\mathscr{A}_0^*$ be the sets irreducible and irreducible Hermite-Einstein connections.

Theorem 5.10. The moduli space $\mathscr{A}_0^*/\mathscr{G}$ of irreducible Hermite-Einstein connections on (E,h) has the structure of a *smooth real manifold* of finite dimension equal to $2 + 2n^2 \cdot (g-1)$.

Observe that every Dolebault operator on a Riemann surface is integrable because $\Omega^{(0,2)}_{\mathbb{C}}(X;E) = \{0\}$. Thus, the Chern correspondence simplifies greatly, and yields a bijection $\mathscr{A} \leftrightarrow \mathscr{H} = \mathscr{D}$ (where $\mathscr{H}$ is the set of Dolbeault operators). Observe that, fixing the hermitian metric h, the proposition above implies that the Chern correspondence restricts to a map $\mathscr{A} \to \mathscr{H}^{ps}$, $\mathscr{H}^{ps}$ being the set of integrable Dolbeault operators that induce polystable holomorphic vector bundle structures. Let $\mathscr{H}^s$ be the set of stable holomorphic structures. Thus, that proposition may be reformulated as follows:

Theorem 5.11. There is a homeomorphism $\mathscr{A}_0/\mathscr{G} \cong \mathscr{H}^{ps}/\mathscr{G}^{\mathbb{C}} = \mathscr{C}^{ps}$, which restricts to another homeomorphism $\mathscr{A}_0^*/\mathscr{G} \cong \mathscr{H}^s/\mathscr{G}^{\mathbb{C}} = \mathscr{C}^s$.

By taking the quotient with respect to $\mathscr{G}^{\mathbb{C}}$ we get the moduli spaces of polystable holomorphic structures and stable structures, accounting for biholomorphisms. Now, to be completely rigorous, everything we just said, as it is stated is false, because our sets do not have manifold structures. They have to be completed with respect to appropriate Sobolev norms, and then everything makes sense and the constructions are independent of the chosen completions.

For Kähler manifolds, everything remains true, but with some adjustments: we have to restrict to connections of type $(1,1)$, and the notions of stability and polystability must be formulated in terms of coherent sheaves.

Let us see now the moment map picture for the Hitchin-Kobayashi correspondence. As we shall see, this is related to the Kempf-Ness theorem, but on an infinite-dimensional setting. Again, ignoring Sobolev completions, $\mathscr{A}$ is an affine space modelled on $\Omega^1_{\mathbb{R}}(\mathrm{End}(E,h))$, and it is a symplectic manifold whose symplectic form is given by

$$\Omega(a,b) = \int_M \mathrm{Tr}(a \wedge b), \qquad \nabla \in \mathscr{A}, a, b \in T_\nabla \mathscr{A} \cong \Omega^1_{\mathbb{R}}(\mathrm{End}(E,h)).$$

Similarly, the set $\mathscr{H}$ is also endowed with an affine space structure modelled on $\Omega^{(0,1)}_{\mathbb{C}}(\mathrm{End}(E))$, and

it has a complex structure $J_{\mathscr{H}}(\alpha) = i\alpha$ for $\overline{\partial}_e \in \mathscr{H}$ and $\alpha \in T_{\overline{\partial}_e}\mathscr{H} \cong \Omega^{(0,1)}_{\mathbb{C}}(X; \mathrm{End}(E))$. Through the Chern correspondence, this gives an almost complex structure on $\mathscr{A}$.

The $\mathscr{G}$ action on $\mathscr{A}$ is symplectic, and idenitifying $\mathrm{Lie}(\mathscr{G}) = \Gamma(X; \mathrm{End}(E, h))$ it admits a moment map given by

$$\langle \mu_{\mathscr{G}}(\nabla), a \rangle := \int_M \mathrm{Tr}(a \circ (i\Lambda_\omega F_\nabla - c \cdot \mathrm{id}_E))$$

With this moment map, $\mu^{-1}(0)$ is the set of Hermite-Einstein connections, and $\mu^{-1}(0)/\mathscr{G} = \mathscr{A}_0/\mathscr{G} \cong \mathscr{H}^{ps}/\mathscr{G}^{\mathbb{C}}$. This is *formally* an infinite-dimensional version of Kempf-Ness' theorem: we get a differential quotient using a moment map. How is stability of vector bundles related to GIT stability?

5.2 The HK correspondence for principal bundles

The Hitchin-Kobayashi correspondence can be formulated in the context of principal bundles. One then needs to formulate both the polystability condition and the Hermite-Yang-Mills equation. For the remainder of the section, we fix a complex connected reductive Lie group G and K a maximal compact subgroup. Write $\mathfrak{g}$ and $\mathfrak{k}$ for their respective Lie algebras, and recall that $\mathfrak{g} = \mathfrak{k}^{\mathbb{C}}$. Let $\mathcal{P}_G$ a holomorphic principal G-bundle over some compact Riemann surface X. For any smooth reduction of the structure group to K, say P_K, there is a "preferred" connection: the Chern connection. Recall from section 3.9 that this connection is obtained as follows: since $\mathcal{P}$ is a holomorphic principal bundle, it is endowed with an integrable almost complex G-structure. Hence, by the Chern correspondence, we assign the connection $A := \mathsf{CS}(J)$ on P_K. The statement of the Hitchin-Kobayashi correspondence remains almost the same: a holomorphic principal G-bundle over a compact Kähler manifold is polystable if and only if its Chern connection satisfies $\Lambda_\omega F_A = -\mathrm{i}\, c$, which we call the *Hermitian-Yang-Mills equation*, for some $c \in \mathrm{i}\,\mathfrak{z}(\mathfrak{g}) \cap \mathfrak{k}$ **(6)**. However, we need to make sense of both, the equation, and the stability condition. We will use extensively the notions of parabolic subgroups introduced in appendix B. It is important to remark that one may find several definitions of stability in the literature. They are all equivalent, even though it is not obvious at first glance. Our definitions will follow [49] and [18]. From now on, we fix $\langle \cdot, \cdot \rangle \colon \mathfrak{g} \times \mathfrak{g} \to \mathbb{C}$ a $\mathbb{C}$-bilinear Ad-invariant pairing on $\mathfrak{g}$ that restricts to a non-degenerate pairing on $\mathfrak{k}$. We fix the notation $\mathscr{G} := \Gamma(M; P_K \times_K K)$ and $\mathscr{G}^{\mathbb{C}} = \Gamma(M; P_G \times_G G)$. Since $G = K^{\mathbb{C}}$, $\mathscr{G}^{\mathbb{C}}$ is fiberwise the complexification of $\mathscr{G}$, hence the notation. We will write $\exp_{\mathscr{G}}$ and $\exp_{\mathscr{G}^{\mathbb{C}}}$ their respective exponential maps.

Remark 5.12. The Ad-invariant bilinear form induces a bilinear form on the space of sections of the associated bundle $P_G \times_G \mathfrak{g} = \mathrm{ad}(P_G)$. Indeed, given two sections $\boldsymbol{\xi}, \boldsymbol{\zeta} \in \Gamma(X; \mathrm{ad}(P_G))$, we define $\langle \boldsymbol{\xi}(x), \boldsymbol{\zeta}(x) \rangle := \langle \xi, \zeta \rangle$ where $\boldsymbol{\xi}(x) = [p, \xi], \boldsymbol{\zeta}(x) = [p, \zeta]$ and $\pi(p) = x$. This is well-defined because taking another pair of representatives $[p \cdot g, \mathrm{Ad}_{g^{-1}}\xi]$ and $[p \cdot g, \mathrm{Ad}_{g^{-1}}\zeta]$, Ad-invariance implies that $\langle \mathrm{Ad}_{g^{-1}}\xi, \mathrm{Ad}_{g^{-1}}\zeta \rangle = \langle \xi, \zeta \rangle$. Moreover, this extends to $\mathrm{ad}(P_G)$-valued differential forms. More precisely, we define

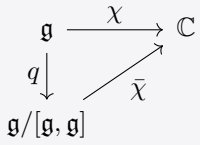
$$\langle \cdot \wedge \cdot \rangle \colon \quad \Omega^p_{\mathbb{R}}(X; \mathrm{ad}(P_G)) \times \Omega^q_{\mathbb{R}}(X; \mathrm{ad}(P_G)) \quad \longrightarrow \quad \Omega^{p+q}_{\mathbb{R}}(X)$$

by extension of $\langle \alpha \otimes \boldsymbol{\xi} \wedge \beta \otimes \boldsymbol{\zeta} \rangle = \langle \boldsymbol{\xi}, \boldsymbol{\zeta} \rangle \cdot \alpha \wedge \beta$. The same construction serves for $\Omega^1_{\mathrm{Ad},\ \mathrm{hor}}(P; \mathfrak{g})$. We may also endow $\Gamma(X, \mathrm{ad}(P_G))$ with a Lie algebra structure as in remark 4.13, and extend it to differential forms by linear extension of $[\alpha \otimes \boldsymbol{\xi} \wedge \beta \otimes \boldsymbol{\zeta}] := (\alpha \wedge \beta) \otimes [\boldsymbol{\xi}, \boldsymbol{\zeta}]$.

Since the notion of stability for vector bundles depends on a notion of degree, it seems reasonable to try to define a degree for holomorphic principal bundles, and use that as a basis to define a notion of stability. This is done by means of characters.

Definition 5.13. Let $\mathfrak{g}$ be a complex Lie algebra.

A *character* is a complex linear map $\chi \colon \mathfrak{g} \to \mathbb{C}$ that factorizes through the quotient $q \colon \mathfrak{g} \to \mathfrak{g}/[\mathfrak{g}, \mathfrak{g}]$. That is, there is a complex linear map $\bar{\chi} \colon \mathfrak{g} \to \mathfrak{g}/[\mathfrak{g}, \mathfrak{g}]$ such that diagram is commutative.



The following lemma gives examples of characters on parabolic algebras of the form $\mathfrak{p}_s$ for $s \in \mathrm{i}\,\mathfrak{k}$. Let $\langle \cdot, \cdot \rangle$ be a $\mathbb{C}$-bilinear non-degenerate pairing on $\mathfrak{g}$ invariant under the adjoint action of G.

Lemma 5.14. Let $\zeta \in \mathfrak{k} \subset \mathfrak{g}$. We define $\chi_\zeta \in \mathfrak{g}^*$ by

$$\chi_\zeta(\xi) = \langle \zeta, \xi \rangle \qquad \xi \in \mathfrak{g}.$$

Then, the restriction of χ_ζ to $\mathfrak{p}_\zeta$ (which we will denote by the same symbol) is a character on $\mathfrak{p}_s^*$.

Proof. χ_ζ is clearly $\mathbb{C}$-linear, and so is its restriction. Observe that $\zeta \in \mathfrak{p}_\zeta$. This follows from the definition of $\mathfrak{p}_\zeta$, and the fact that $\mathrm{Ad}_{\exp(t\zeta)}(\zeta) = \zeta$ for all $t \in \mathbb{R}$. To check that it factors through $[\mathfrak{p}_\zeta, \mathfrak{p}_\zeta]$, we must see that $\langle \zeta, [x,y] \rangle = 0$ for $x, y \in \mathfrak{p}_\zeta$. Observe that $\langle \zeta, [x,y] \rangle = \langle [x,\zeta], y \rangle$. This follows from Ad-invariance, and from the fact that $\mathrm{ad}(x)(y) = [x,y] = \mathrm{d}_e\mathrm{Ad}(x)(y)$ [36, Th. 20.27]:

$$\langle \zeta, [x,y] \rangle = \langle \zeta, \mathrm{ad}(x)(y) \rangle = \langle \zeta, \mathrm{d}_e\mathrm{Ad}(x)(y) \rangle = \left\langle \zeta, \frac{\mathrm{d}}{\mathrm{d}t}\Big|_{t=0} \mathrm{Ad}_{\exp(tx)}(y) \right\rangle$$
$$= \left\langle \frac{\mathrm{d}}{\mathrm{d}t}\Big|_{t=0} \mathrm{Ad}_{\exp(-tx)}(\zeta), y \right\rangle = \langle [-x, \zeta], y \rangle = \langle [\zeta, x], y \rangle.$$

Moreover, $\mathfrak{p}_\zeta = \mathfrak{l}_\zeta \oplus \mathfrak{u}_\zeta$ where $\mathfrak{u}_\zeta$ is the nilpotent radical obtained from the Levi factor $\mathfrak{l}_\zeta$ (c.f. section B.3.2 and proposition B.13). Hence, by definition of $\mathfrak{l}_\zeta$ if $x \in \mathfrak{l}_\zeta$, $\chi_\zeta([x,y]) = \langle [\zeta, x], y \rangle = 0$. By the direct sum decomposition above, it suffices to see that $\chi_\zeta([x,y])$ for $x \in \mathfrak{u}_\zeta$. Now, since $\mathfrak{u}_\zeta$ is an ideal of $\mathfrak{p}_\zeta$, we have $[\mathfrak{u}_\zeta, \mathfrak{p}_\zeta] \subset \mathfrak{u}_\zeta$, hence $[x,y] \in \mathfrak{u}_\zeta$ for $x \in \mathfrak{u}_\zeta, y \in \mathfrak{p}_\zeta$. Hence, we need only show that $\chi_\zeta(x) = 0$ for all $x \in \mathfrak{u}_\zeta$, but this is trivially true because $\mathfrak{u}_\zeta$ is the radical of the restriction of $\langle \cdot, \cdot \rangle$ to $\mathfrak{p}_\zeta$ (that is $\langle \xi, x \rangle = 0$ for all $\xi \in \mathfrak{p}_\zeta, x \in \mathfrak{u}_\zeta$, see [27, Page 273]). □

Consider now $\mathfrak{p}$ any parabolic subalgebra, and $\chi \in \mathfrak{p}^*$ a character. The non-degenerate pairing $\langle \cdot, \cdot \rangle$ induces a linear isomorphism $\mathfrak{g} \cong \mathfrak{g}^*, \zeta \mapsto \langle \zeta, \cdot \rangle$. Recall that we have a direct sum decomposition $\mathfrak{g} = \mathfrak{k} \oplus \mathrm{i}\mathfrak{k}$, which induces a projection $\mathfrak{g} \twoheadrightarrow \mathrm{i}\mathfrak{k}$. Finally, the pairing $\langle \cdot, \cdot \rangle$ induces a direct sum decomposition $\mathfrak{g} = \mathfrak{p} \oplus \mathfrak{p}^\perp$ **(7)**. This comes with a natural orthogonal projection $p_\perp \colon \mathfrak{g} \to \mathfrak{p}$, whose dual map gives an injection $\mathfrak{p}^* \hookrightarrow \mathfrak{g}^*, \chi \mapsto \chi \circ \mathrm{pr}_\perp$. We thus have the following chain of maps

$$\mathfrak{p}^* \hookrightarrow \mathfrak{g}^* \xrightarrow{\cong} \mathfrak{g} \to \mathrm{i}\mathfrak{k}.$$

Given $\chi \in \mathfrak{p}^*$ we denote $\zeta_\chi \in \mathrm{i}\mathfrak{k}$ its image through the previous chain of maps.

Definition 5.15. Let $\mathfrak{p} \subset \mathfrak{g}$ be a parabolic subalgebra. A character $\chi \in \mathfrak{p}^*$ is said to be *antidominant* if $\mathfrak{p}_{s_\chi} \subset \mathfrak{p}$ and *strictly antidominant* if $\mathfrak{p}_{\zeta_\chi} = \mathfrak{p}$.

Remark 5.16. For each $\zeta \in \mathrm{i}\mathfrak{k}$, we have $\zeta_{\chi_\zeta} = \zeta$, thus the character χ_ζ of $\mathfrak{p}_\zeta$ is always strictly antidominant. This follows from the fact that $\chi_\zeta \circ \mathrm{pr}_\perp = \chi_\zeta$ (recall that $\zeta \in \mathfrak{p}_\zeta$) the way $\langle \cdot, \cdot \rangle$ induces the pairing.

We now prove a lemma we will need to define the notion of degree.

Lemma 5.17. Let $P(M,G)$ be a principal bundle, and let A be a connection. Let $\alpha \in \Omega^1_{\mathbb{R}}(M; \mathrm{ad}(P_G))$. Then

$$F_{A+\alpha} = F_A + \mathrm{d}_A\alpha + \frac{1}{2} \cdot [\alpha, \alpha]$$

Proof. Let $\hat{\alpha} := \Upsilon^{-1}(\alpha) \in \Omega^1_{\mathrm{Ad},\ \mathrm{hor}}(P; \mathfrak{g})$. By $A + \alpha$ we mean the connection one-form $A + \hat{\alpha}$. From remark 3.26 we apply Υ to equation (3.12). Then,

$$F_{A+\alpha} = \Upsilon\left(\mathrm{D}_A A + \mathrm{D}_A\hat{\alpha} + \frac{1}{2} \cdot [\hat{\alpha}, \hat{\alpha}]\right) = F_A + \Upsilon(\mathrm{D}_A\hat{\alpha}) + \frac{1}{2} \cdot \Upsilon([\hat{\alpha} \wedge \hat{\alpha}])$$
$$= F_A + \mathrm{d}_A\Upsilon(\hat{\alpha}) + \frac{1}{2} \cdot [\Upsilon(\hat{\alpha}) \wedge \Upsilon(\hat{\alpha})].$$

It remains to check that $\Upsilon([\hat{\alpha}, \hat{\alpha}]) = [\Upsilon(\hat{\alpha}), \Upsilon(\hat{\alpha})]$. To do so, let us unveil the different definitions. Let $\alpha = \alpha^j \otimes \boldsymbol{\xi}_j$ for $\alpha^j \in \Omega^1_{\mathbb{R}}(M)$ and $\boldsymbol{\xi}_j \in \Gamma(M; \mathrm{ad}(P_G))$. A simple calculation shows that $\Upsilon^{-1}(\alpha) =$

$\pi^*\alpha^j \otimes \hat{\xi}_j$, where $\hat{\xi}_j := \Upsilon^{-1}(\boldsymbol{\xi}_j)\colon P \to \mathfrak{g}$ is G-equivariant (see [**51**, Remark 1.2.13]). Then, letting $p \in P$, $v_1, v_2 \in T_pP$

$$\begin{aligned}\big(\Upsilon([\hat{\alpha}\wedge\hat{\alpha}])\big)_{\pi(p)}(\mathrm{d}_p\pi(v_1),\mathrm{d}_p\pi(v_2)) &= q_p\big((\pi^*(\alpha^j\wedge\alpha^\ell))_p(v_1,v_2)\cdot[\hat{\xi}_j(p),\hat{\xi}_\ell(p)]\big)\\ &= \big[p,(\pi^*(\alpha^j\wedge\alpha^\ell))_p(v_1,v_2)\cdot[\hat{\xi}_j(p),\hat{\xi}_\ell(p)]\big]\end{aligned}$$

Whereas

$$\begin{aligned}[\Upsilon(\hat{\alpha})\wedge\Upsilon(\hat{\alpha})]_{\pi(p)}(\mathrm{d}_p\pi(v_1),\mathrm{d}_p\pi(v_2)) &= [\alpha^j\otimes\xi_j\wedge\alpha^\ell\otimes\xi_\ell]_{\pi(p)}(\mathrm{d}_p\pi(v_1),\mathrm{d}_p\pi(v_2))\\ &= (\alpha^j\wedge\alpha^\ell)_{\pi(p)}(\mathrm{d}_p\pi(v_1),\mathrm{d}_p\pi(v_2))\cdot[p,\xi_j(\pi(p)),\xi_\ell(\pi(p))].\end{aligned}$$

And these coincide due to the way the Lie bracket on $\Gamma(X;\mathrm{ad}(P))$ is defined (c.f. remark 4.13) and the way that Υ acts on sections of $\mathrm{ad}(P_G)$ (c.f. section 3.5). □

Consider $\mathcal{P}$ a holomorphic principal G-bundle over a Riemann surface X. Let $\zeta \in \mathrm{i}\,\mathfrak{k}$ and let $P_\zeta \subset G$ be the parabolic subgroup defined by ζ as in proposition B.13. Consider a reduction of the structure group of $\mathcal{P}$ to P_ζ given by a holomorphic section $\sigma \in \mathcal{O}(X;\mathcal{P}/P_\zeta)$, and let $\mathcal{P}_\sigma := \sigma^*\mathcal{P}$ be the corresponding holomorphic P_ζ-principal bundle over X. As noted in remark 5.16, the character $\chi_s \in \mathfrak{p}^*$ is antidominant. Consider L_ζ the Levi subgroup of P_ζ defined by B.13. Define $K_\zeta := K \cap L_\zeta$ which is a Lie group whose Lie algebra is given by $\mathfrak{k}_\zeta := \mathfrak{k} \cap \mathfrak{l}_\zeta$. Then, K_ζ is a maximal compact subgroup of L_ζ and the inclusion $K_\zeta \subset L_\zeta$ is a homotopy equivalence. The inclusion $L_\zeta \subset P_\zeta$ is a homotopy equivalence as well, so K_ζ and P_ζ are homotopy equivalent. This implies that any reduction σ induces another reduction (unique up to homotopy equivalence) of the structure group to K_ζ. Call P'_σ the resulting smooth principal K_ζ-bundle. See [**30**, Example 5.6] and the results surrounding it for a justification of this fact.

Definition 5.18. Let $\mathcal{P}(X,G)$ be a holomorphic principal bundle and $\zeta \in \mathrm{i}\,\mathfrak{k}$. Keeping the notations of the previous paragraph, consider a connection A on $\mathcal{P}'_\sigma$, and let $F_A \in \Omega^2(X;P'_\sigma \times_{K_\zeta} \mathfrak{k}_\zeta)$ its curvature, We define the *degree of $\mathcal{P}$ with respect to ζ and the reduction σ* by

$$\deg \mathcal{P}(\sigma,\zeta) = \frac{\mathrm{i}}{2\pi}\cdot\int_X \chi_\zeta(F_A).$$

Remark 5.19. Let us check that $\deg\mathcal{P}(\sigma,\chi)$ is a well-defined real number. First, observe that $\chi_\zeta(\mathfrak{k}_\zeta) \subset \mathrm{i}\,\mathbb{R}$. Indeed, $\zeta = \mathrm{i}\,\zeta'$ for some $\zeta' \in \mathfrak{k}$, hence $\chi_\zeta(\xi) = \langle \mathrm{i}\,\zeta', \xi\rangle = \mathrm{i}\,\langle\zeta',\xi\rangle$ for $\zeta \in \mathfrak{k}_\zeta \subset \mathfrak{k}$ using $\mathbb{C}$-bilinearity. Moreover, since $\mathfrak{g} = \mathfrak{k}\oplus\mathrm{i}\,\mathfrak{k}$, $\mathfrak{g}$ is endowed with a natural conjugation map. With respect to this conjugation map, and due to $\mathbb{C}$-bilinearity, a direct computation shows that $\overline{\langle\zeta,\xi\rangle} = \langle\overline{\zeta},\overline{\xi}\rangle$ for any pair $\zeta,\xi \in \mathfrak{g}$. Hence, if $\zeta,\xi\in\mathfrak{k}$, they coincide with their complex conjugates and their product is real.

On the other hand, F_A takes values in the associated bundle $P'_\sigma \times_{K_\zeta} \mathfrak{k}_\zeta$. We define the composition of χ_ζ and any p-form in $\Omega^p(X;P'_\sigma\times_{K_\zeta}\mathfrak{k}_\zeta)$ by linear extension of $\chi_s(\alpha\otimes\boldsymbol{\xi}) = \chi_s(\boldsymbol{\xi})\cdot\alpha$, where $\boldsymbol{\xi} \in \Gamma(X;P'_\sigma\times_{K_s}\mathfrak{k}_s)$. Finally, if $\boldsymbol{\xi}(x) = [p,\xi] \in P'_\sigma\times_{K_\zeta}\mathfrak{k}_\zeta$, with $\pi(p)=x$, we define $\chi_\zeta(\boldsymbol{\xi}(x)) = \chi_\zeta(\xi)$. This does not depend on the representative chosen. Indeed, $[p,\xi] = [p\cdot k, \mathrm{Ad}_{k^{-1}}(\xi)]$ for $k \in K_\zeta$, but

$$\chi_\zeta(\mathrm{Ad}_{k^{-1}}(\xi)) = \langle\zeta,\mathrm{Ad}_{k^{-1}}(\xi)\rangle = \langle\mathrm{Ad}_k(\zeta),\xi\rangle = \langle\zeta,\xi\rangle = \chi_\zeta(\xi)$$

using the Ad-invariance of the product and the fact that $K_\zeta \subset L_\zeta = \{g\in G : \mathrm{Ad}_g(\zeta)=\zeta\}$. Finally, let us see that the definition is independent of the chosen connection A. First, recall that Since the space of connections $\mathscr{A}(P'_\sigma)$ is an affine space modelled on $\Omega^1_{\mathbb{R}}(X;\mathrm{ad}(P'_\sigma))$, hence if we fix a connection $A\in\mathscr{A}(P'_\sigma)$, any other connection will be given by $A+\alpha$ for $\alpha\in\Omega^1_{\mathbb{R}}(X;P'_\sigma)$. Observe that by our definitions, and due to the fact that everything is smooth, it follows that $\chi_\zeta(\alpha)\in\Omega^1_{\mathbb{R}}(X)$. Let us assume that $\chi_\zeta(\mathrm{d}_A\alpha) = \mathrm{d}\chi_\zeta(\alpha)$ (we give a proof of this fact in lemma 5.20 bellow). The curvature two-forms of A + $A+\alpha$ relate as $F_{A+\alpha} = F_A + \mathrm{d}_A\alpha + 1/2\cdot[\alpha\wedge\alpha]$, (see lemma 5.17). Since χ_ζ is a character of $\mathfrak{p}_\zeta$, it vanishes on $[\mathfrak{p}_\zeta,\mathfrak{p}_\zeta]$, hence, it vanishes on $[\mathfrak{k}_\zeta,\mathfrak{k}_\zeta]$ as well, and that is where $[\alpha\wedge\alpha]$ takes values. Hence, applying Stokes' theorem, we find

$$\int_X \chi_\zeta(F_{A+\alpha}) = \int_X \chi_\zeta(F_A) + \int_X \mathrm{d}\chi_\zeta(\alpha)$$

the latter being zero because $\partial X = \varnothing$.

Lemma 5.20. In the conditions of definition 5.18, we have $\chi_\zeta(\mathrm{d}_A\alpha) = \mathrm{d}\chi_\zeta(\alpha)$ for all $\zeta \in \mathrm{i}\,\mathfrak{k}$ and $\alpha \in \Omega^1_{\mathbb{R}}(X; \mathrm{ad}(P'_\sigma))$.

Proof. The calulation is similar to that of lemma 5.17. Let $\pi\colon P'_\sigma \to X$ be the projection, and $\alpha \in \Omega^1_{\mathbb{R}}(X; P'_\sigma \times_{K_\zeta} \mathfrak{k}_\zeta)$. Recall from theorem 3.28 that $\Upsilon^{-1}(\alpha) \in \Omega^1_{\text{Ad, hor}}(P'_\sigma; \mathfrak{k}_\zeta)$. Expand $\alpha = \alpha^j \otimes \boldsymbol{\xi}_j$ for some $\alpha^j \in \Omega^1_{\mathbb{R}}(X)$ and $\boldsymbol{\xi}_j \in \Gamma(X; P'_\sigma(\mathfrak{k}_\zeta))$. As in lemma 5.17 $\Upsilon^{-1}(\alpha) = \pi^*\alpha^j \otimes \hat{\xi}_j$, where $\hat{\xi}_j = \Upsilon^{-1}(\boldsymbol{\xi}_j)\colon P'_\sigma \to \mathfrak{k}_\zeta$ is K_ζ-equivariant. After setting this notation up, the result follows from a simple computation. Observe that the definition of Υ forces us to set a point. Hence, let $p \in P'_\sigma$ and $v_1, v_2 \in T_pP'_\sigma$. On the one hand, it follows from our definitions that $\mathrm{d}\chi_\zeta(\alpha) = \chi_\zeta(\boldsymbol{\xi}_j) \cdot \mathrm{d}\alpha^j$. Hence,

$$\mathrm{d}_{\pi(p)}\ \chi_\zeta(\alpha)(\mathrm{d}_p\pi(v_1), \mathrm{d}_p\pi(v_2)) = \mathrm{d}_{\pi(p)}\alpha(\mathrm{d}_p\pi(v_1), \mathrm{d}_p\pi(v_2)) \cdot \chi_\zeta(\boldsymbol{\xi}_j(\pi(p))) = (\pi^*\alpha^j)_p(v_1, v_2) \cdot \chi_\zeta(\boldsymbol{\xi}_j(\pi(p))).$$

On the other hand,

$$\begin{aligned}\chi_\zeta((\mathrm{d}_A\alpha)_p(\mathrm{d}_p\pi(v_1), \mathrm{d}_p\pi(v_2))) &= \chi_\zeta(q_p(\mathrm{D}_A\Upsilon^{-1}(\alpha))_p(v_1^H, v_2^H)) = \chi_\zeta\big(q_p(\mathrm{d}_p(\pi^*\alpha^j)(v_1^H, v_2^H) \cdot \hat{\xi}_j(p))\big)\\ &= (\pi^*\mathrm{d}\alpha^j)_p(v_1, v_2) \cdot \chi_\zeta(\boldsymbol{\xi}_j(\pi(p))).\end{aligned}$$

In the last step we used the fact that $\xi_j(\pi(p)) = [p, \hat{\xi}_j(p)]$ and the fact that $\mathrm{d}_p\pi(v) = \mathrm{d}_p\pi(v^H)$ because vertical vectors are annihilated by the projection. We thus get the identity $\mathrm{d}\chi_\zeta(\alpha) = \chi_\zeta(\mathrm{d}_A\alpha)$ but written pointwise. □

Observe the clear parallelism between this definition of degree and the definition for vector bundles (definition 5.1). For vector bundles, we integrate the first Chern class, which is (up to some constants) the cohomology class of $\mathrm{Tr}(F_\nabla)$. Here, the role of the trace is played by the antidominant character. We are now ready to state the definition of stability. First, observe that parabolic subgroups P_ζ are complex closed subgroups, and the action of P_ζ on G by left multiplication $P_\zeta \times G \to G, (p, g) \mapsto p \cdot g$ is smooth free and proper (the reasoning in the proof of [**36**, Th. 21.17] hods), hence, by theorem B.5, G/P_ζ has the structure of a complex manifold. Then, given holomorphic principal bundle $\mathcal{P}$, the associated bundle $\mathcal{P}(G/P_\zeta)$ is holomorphic as well, and we can consider its holomorphic sections.

Definition 5.21 (Stability condition). Let $\mathcal{P}_G$ be a holomorphic principal bundle over a Riemann surface, and let $c \in \mathrm{i}\,\mathfrak{z}(\mathfrak{g}) \cap \mathfrak{k}$. We say that $\mathcal{P}_G$ is **(8)**

1. *c-stable* (resp. *c-semistable*) if for any $\zeta \in \mathrm{i}\,\mathfrak{k}$, and any $\sigma \in \mathcal{O}(X; \mathcal{P}(G/P_\zeta))$, we have
$$\deg \mathcal{P}(\sigma, \zeta) - \langle c, \zeta\rangle > 0 \qquad (\text{resp. } \deg \mathcal{P}(\sigma, \zeta) - \langle c, \zeta\rangle \geq 0).$$
2. *c-polystable* if it is c-semistable and or any $\zeta \in \mathrm{i}\,\mathfrak{k}$ and any $\sigma \in \mathcal{O}(X; \mathcal{P}(G/P_\zeta))$ such that $\deg \mathcal{P}(\sigma, \zeta) - \langle c, \zeta\rangle = 0$, there is a holomorphic reduction of the structure group $\sigma' \in \mathcal{O}(X; \mathcal{P}_\sigma(P/L_\zeta))$.

The arguments of remark 5.19 apply to show that $\langle c, \zeta\rangle \in \mathbb{R}$, hence the inequalities above make sense. With the correct notion of stability, we can now state the correspondence.

Theorem 5.22 (Hitchin-Kobayashi for principal bundles). Let G be a connected complex reductive Lie group, $K \subset G$ a maximal compact subgroup, and $\mathcal{P}$ a holomorphic principal G-bundle over a compact Riemann surface X. Then, $\mathcal{P}$ is c-polystable if and only if there is a reduction of the structure group to K, P_K whose Chern connection A satisfies the equation

$$\Lambda_\omega F_A = -\mathrm{i}\,c \tag{5.6}$$

for some $c \in \mathrm{i}\,\mathfrak{z}(\mathfrak{g}) \cap \mathfrak{k}$. In that case, the reduction is unique.

Equation (5.6) is called the *Hermite-Yang-Mills equation for principal bundles*. We insist on the fact that $F_A \in \Omega^2(X; P_K \times_K \mathfrak{k})$ is the curvature of the Chern connection on the *reduced* bundle.

Remark 5.23. Observe that $\Lambda_\omega F_A = -\mathrm{i}\,c$ should be understood as an equality between smooth sections of the bundle $P_K \times_K \mathfrak{k}$. The right-hand side then corresponds to a constant section $m \mapsto$

$[p, c]$ for $\pi(p) = m$. One should check that this section is well-defined. Indeed, choose another representative $[p \cdot g, \mathrm{Ad}_{k^{-1}}(c)]$. Since $c \in \mathfrak{z}(\mathfrak{g}) \cap \mathfrak{k}$ *and* K *is connected*, $\mathrm{Ad}_k(c) = c$ for each $c \in \mathfrak{z}(\mathfrak{g}) \cap \mathfrak{k}$. It is very important to remark that this only holds for connected Lie groups. A quick proof is as follows:

$$\mathrm{Ad}_k(c) = \frac{\mathrm{d}}{\mathrm{d}t}\Big|_{t=0} L_k \circ R_{k^{-1}} \circ \exp(tc) = \frac{\mathrm{d}}{\mathrm{d}t}\Big|_{t=0} k \exp(tc) k^{-1} = \frac{\mathrm{d}}{\mathrm{d}t}\Big|_{t=0} kk^{-1} \exp(tc) = c.$$

This uses the fact that $\mathfrak{z}(\mathfrak{g}) \cap \mathfrak{k} \subset \mathfrak{z}(\mathfrak{k})$, that the exponential on a compact connected Lie group is surjective [**23**, Cor. 12.2.3] and the Baker-Campbell-Hausdorff formula.

Moreover, as happens for vector bundles, if equation (5.6) is satisfied, then the constant c is determined uniquely by the topology of the bundle. Uniqueness is not difficult to see. Recall that $\langle \cdot, \cdot \rangle$ is chosen so that its restriction to $\mathfrak{k}$ is negative definite ?. Hence, let $\{\zeta_1, \dots \zeta_r\}$ be an orthonormal basis. $-\mathrm{i}\, c \in \mathfrak{k}$, hence $-\mathrm{i}\, c = \sum_{j=1}^r \langle -\mathrm{i}\, c, \zeta_j \rangle \cdot \zeta_j$. Moreover, $\langle \zeta_j, -\mathrm{i}\, c \rangle = \langle \zeta_j, \Lambda_\omega F_A \rangle$ where this is an equality between smooth real-valued maps. Integrating on X with respect to the volume form ω, we get

$$\int_X \langle \zeta_j, -\mathrm{i}\, c \rangle \cdot \omega = \langle \zeta_j, -\mathrm{i}\, c \rangle \cdot \mathrm{Vol}(X) = \int_X \langle \zeta_j, \Lambda_\omega F_A \rangle \cdot \omega = \int_X \langle \zeta_j, F_A \rangle$$

Using lemma A.34 in the last identity and extending the inner product to differential forms the usual way. Then,

$$-\mathrm{i}\, c = \sum_{j=1}^r \frac{1}{\mathrm{Vol}(X)} \cdot \int_X \langle \zeta_j, \Lambda_\omega F_A \rangle. \tag{5.7}$$

Hence, we have shown that the Chern connection may only satisfy (5.6) for a single value of c. For a proof of the fact that 5.7 depends only on the topology of the manifold and the bundle, see [**19**, Prop. 3.4] or [**2**, Remark 1.2] and the references therein.

In theorem 5.22 we have phrased things as follows: we have fixed a holomorphic principal bundle structure on P_G, and we "are looking" for a reduction of the structure group whose Chern connection satisfies the Hermite-Einstein-Yang-Mills equation. We find that this reduction exists if and only if $\mathcal{P}_G$ is polystable. If we compare this with the vector bundle version, it is analogous to the point of view where we fix the holomorphic structure and look for a Hermite-Einstein fiber metric. We could think is this differently: instead of fixing the holomorphic structure, we could start with a principal G-bundle, and fix a reduction of the structure group (this corresponds to fixing the hermitian fiber metric). Then, we look for a connection on the reduction satisfying $F_A^{(0,2)} = 0$ and equation 5.6. Through the Chern-Singer correspondence, this induces an integrable almost complex G-structure on P_G with respect to which A is the Chern connection. In that case, by the theorem above, the bundle $\mathcal{P}_G$ with the correspondence induced by the Chern-Singer correspondence will be c-polystable.

5.2.1 The moment map picture

In this section we will lay the foundations of the moment map picture of the Hitchin-Kobayashi correspondence. We will specify how the various spaces involved have to be completed with suitable Sobolev norms to actually get infinite-dimensional manifolds. Our main references for this section are [**40**], [**65**], [**11**] [**57**] and [**51**].

Consider first a compact Lie group K and a principal K-bundle P_K over a compact Kähler manifold M (for this part of the exposition we may work over a compact Kähler manifold instead of a Riemann surface without significant added difficulties). From example 3.26 and remark 3.29, we know that $\mathscr{A}(P_K)$ is an affine space modelled on $\Omega^1(X; P_K \times_K \mathfrak{K})$. We consider $L_k^2(X; T^*M \otimes \mathrm{ad}(P_K))$ its Sobolev completion with respect to the connection ∇_A that any connection $A \in \mathscr{A}(P_K)$ induces on the associated bundle $\mathrm{ad}(P_K)$ (see section 4.2.1). **(4)** Since $\mathscr{A}(P_K)$ is an affine space, we can fix an element $A_0 \in \mathscr{A}$. We define the space of L_2^k-connections by

$$\mathscr{A}_k(P_K) := \left\{ A_0 + \alpha : \alpha \in L_k^2(T^*M \otimes \mathrm{ad}(P_K)) \right\}.$$

This is again an affine space, and the set is independent of the smooth connection A_0 we use to define it. In fact, since $\mathbf{\Omega}_k^1(M; \mathrm{ad}(P_K)) := L_k^2(M; T^*M \otimes \mathrm{ad}(P_K))$ is a Hilbert space by construction, this

endows $\mathscr{A}_k(P_K)$ with the structure of a Banach manifold. Indeed, charts are given by $\alpha \mapsto \alpha + A$ for some smooth connection A, and the transition maps are given by $\alpha \mapsto \alpha + (A_0 - A_1)$ for two smooth connections $A_0, A_1 \in \mathscr{A}(P_K)$. This is clearly a smooth map in the sense of Banach manifolds. Finally, choosing k sufficiently large, in particular, $k > \dim_{\mathbb{R}} M/2 + 1$ implies that $\mathscr{A}_k(P_G)$ is embedded in the set of $\mathscr{C}^1$ connections (those whose connection one form is of class $\mathscr{C}^1$ instead of smooth, see lemma 4.11). One should remark that for affine spaces, the tangent space in the sense of section C.1.1 coincides (is isomorphic to) with the model vector space. To define the curvature associated to an L^2_1-connection, we may use the structure equation; for any $A \in \mathscr{A}(P_K)$, $\mathrm{D}_A A := \mathrm{d}A + 1/2 \cdot [A \wedge A]$. This is a $\mathfrak{k}$-valued two form but, in general, it will only be continuous. It still makes sense to define $F_A := \Upsilon(\mathrm{D}_A A)$, which is a continuous two form on the associated bundle $P_K \times_K \mathfrak{k}$.

When K is a compact *connected, matrix* Lie group **(9)**, then $\mathscr{G}_k := \overline{\Gamma(X; \mathrm{ad}(P_K))}$ is a Hilbert Lie group, see section 4.2. Choose $k+1$ to endow the gauge group with a smooth structure. Then, the action by pullback

$$\Phi : \mathscr{G}_{k+1} \times \mathscr{A}_k \longrightarrow \mathscr{A}_k, \quad (A, \phi) \mapsto (\phi^{-1})^* A \tag{5.8}$$

is smooth in the sense of Banach manifolds [**40**, Prop. 3.12]. See equation (3.26) for an explicit expression of the action. To set some notation, given $\phi \in \mathscr{G}_{k+1}$, we will write $\tau \colon P \to G$ the map corresponding to ϕ by the isomorphism of theorem 3.13. Observe that the construction of that theorem is still well-defined is we restrict the regularity of our maps. Observe as well that we consider the pullback with respect to the inverse to get a left-action. Had we not used the inverse, we would ahve gotten a right action. This is not particularly relevant, but we define a left action to be coherent with the discussion of chapter 4. To be very explicit, elements of $\mathscr{G}_k$ can be viewed as $\mathscr{C}^2$ sections of the bundle $\mathrm{ad}(P_K)$. The group isomorphisms of theorem 3.13 hold if we consider maps of lower regularity. Under these isomorphisms, every element of $\mathscr{G}_k$ corresponds to a $\mathscr{C}^2$ automorphism of P_K covering the identity, which can be used to define the pullback.

Back to $\mathscr{A}_k$, this Hilbert manifold is in fact symplectic. To endow it with a symplectic structure, we define

$$\Omega_A(\alpha, \beta) := \int_X \Lambda_\omega \langle \alpha \wedge \beta \rangle \, \mathrm{d}V_g = \int_X \langle \alpha \wedge \beta \rangle \wedge \frac{\omega^{n-1}}{(n-1)!} \qquad \alpha, \beta \in T_A \mathscr{A}_k = \Omega^1_k(M; \mathrm{ad}(P_K)) \tag{5.9}$$

where the second expression follows from A.34. $\langle \cdot \wedge \cdot \rangle$ is defined as in remark 4.13. Here $\langle \cdot, \cdot \rangle$ is an Ad-invariant inner product on $\mathfrak{k}$, which always exists because K is compact (this is done by averaging any positive define inner product on $\mathfrak{k}$ by means of the Haar measure, see [**23**, Lemma 10.4.14]).

Remark 5.24. It is true, and we shall assume it henceforth, that almost every property of differential forms is still true when we restrict ourselves to differential forms with reduced regularity. For example, in [**32**], the theory of differential forms is constructed without requiring smoothness.

Lemma 5.25. Let K be a compact Lie group, and $P(M, K)$ a principal K-bundle over a compact Kähler manifold, and consider the Sobolev completion of its space of connections $\mathscr{A}_k(P_G)$ for $k > \dim_{\mathbb{R}} M/2 + 1$. Then, the form Ω defined by (5.9) is symplectic. Moreover, the $\mathscr{G}_k$-action is symplectic.

Proof. For each $A \in \mathscr{A}$, Ω_A is clearly bilinear and skew-symmetric. $\mathrm{d}\Omega = 0$ follows from the fact that Ω does not depend on the point (see proposition C.3). Since this is the first time we encounter the exterior differential applied to an infinite-dimensional form, let us be more explicit. First, for a two-form, the definition given in (C.3) reads

$$\begin{aligned} \mathrm{d}\Omega(X_1, X_2, X_3) =& X_1(\Omega(X_2, X_3)) - X_2(\Omega(X_1, X_3)) + X_3(\Omega(X_1, X_2)) - \Omega([X_1, X_2], X_3) \\ &+ \Omega([X_1, X_3], X_2) - \Omega([X_2, X_3], X_1). \end{aligned}$$

Since Ω does not depend on the point we evaluate it at, this expression depends only on the values of the vector fields at a point. Since all the tangent spaces can be canonically idenitified with $\mathbf{\Omega}^1_k(M; \mathrm{ad}(P_G))$ we can calculate the right-hand side using constant vector fields. Their commutator is then zero (the proof given [**57**, Prop. 2.6] can be adapted to this case). Hence, it suffices to show that each of the first three summands vanishes. To do so, consider $\alpha, \beta, \vartheta \in \mathbf{\Omega}^1_k(M; \mathrm{ad}(P_K))$, which

we will see as constant vector fields. Observe that we may interpret $\Omega(\alpha,\beta)$ as a map $\mathscr{A}\to\mathbb{R}$. We then consider the vector field defined by assigning ϑ to every point, which we still denote ϑ. To apply the comments on remark C.5, we identify it with the map that to any $A\in\mathscr{A}$ assigns the equivalence class of the curve $t\mapsto A+t\cdot\vartheta$. Hence, at any point $A\in\mathscr{A}$, we have

$$\big(\vartheta(\Omega(\alpha,\beta))\big)_A = \mathrm{d}_A\big(\Omega(\alpha,\beta)\big)([t\mapsto A+t\vartheta]) = \left.\frac{\mathrm{d}}{\mathrm{d}t}\right|_{t=0}\Omega_{A+t\vartheta}(\alpha,\beta)=0$$

the latter being zero because the map $\Omega(\alpha,\beta)\colon\mathscr{A}\to\mathbb{R}$ is constant. Finally, to check non-degeneracy, let $\alpha\in\Omega^1_k(M;\mathrm{ad}(P_K))$. Observe that the assignment

$$(\alpha,\beta):=\int_M\langle\alpha\wedge\star\beta\rangle\qquad\qquad\alpha,\beta\in\boldsymbol{\Omega}^1_k(M;\mathrm{ad}(P_K))$$

defines an inner product on $\Omega^1_k(M;\mathrm{ad}(P_G))$ (see the comment bellow (A.31)). Now, for a fixed α, let $\beta:=\Lambda_\omega^{n-1}\star\alpha$. Hence,

$$\begin{aligned}\int_M\langle\alpha\wedge\Lambda_\omega^{n-1}\star\alpha\rangle\wedge\omega^{n-1}&=\int_M\langle\alpha\wedge L_\omega^{n-1}\Lambda_\omega^{n-1}\star\alpha\rangle=\int_M\langle L_\omega^{n-1}\Lambda_\omega^{n-1}\star\alpha\wedge\star\star\alpha\rangle\\&=(L_\omega^{n-1}\Lambda_\omega^{n-1}\star\alpha,\star\alpha)=(\Lambda_\omega^{n-1}\star\alpha,\Lambda_\omega^{n-1}\star\alpha)^2>0\end{aligned}$$

and we conclude that the latter is positive using that $\Lambda_\omega^{n-1}\star\alpha=\star L_\omega^{n-1}\alpha$ and L_ω^{n-1} is injective [**35**, Lemma 9.39]. Finally, the $\mathscr{G}_k$ action is symplectic. Let $\phi(p)=p\cdot\tau(p)$, and consider $\Phi_\phi\colon\mathscr{A}_k\to\mathscr{A}_k, A\mapsto(\phi^{-1})^*A$. Let $\alpha\in\boldsymbol{\Omega}^1_k(M;\mathrm{ad}(P_G))$. We have (see (3.27))

$$\mathrm{d}_A\Phi_\phi(\alpha)=\left.\frac{\mathrm{d}}{\mathrm{d}t}\right|_{t=0}(\phi^{-1})^*(A+t\alpha)=\left.\frac{\mathrm{d}}{\mathrm{d}t}\right|_{t=0}\Big(\mathrm{Ad}_\tau(A+t\alpha)+(\tau^{-1})^*\theta\Big)=\mathrm{Ad}_\tau(\alpha).$$

Here, d_A is the exterior differential evaluated at a connection A, not the covariant exterior differential induced on an associated bundle. Finally,

$$(\Phi_\phi^*\Omega)_A(\alpha,\beta)=\Omega_{\phi^*A}(\mathrm{d}_A\Phi_\phi(\alpha),\mathrm{d}_A\Phi_\phi(\beta))=\Omega_{\phi^*A}(\mathrm{Ad}_\tau(\alpha),\mathrm{Ad}_\tau(\beta))=\Omega_A(\alpha,\beta)$$

using the fact that $\langle\cdot,\cdot\rangle$ is Ad-invariant, and the adjoint action applies to the "section of $\mathrm{ad}(P_K)$" part of the differential form. □

Once shown that the action is symplectic, let us see that it is Hamiltonian. Hence, we define $\mu\colon\mathscr{A}_k\to\mathrm{Lie}(\mathscr{G}_{k+1})^*$ by

$$(\mu(A),\boldsymbol{\xi}):=\int_M\langle\boldsymbol{\xi},(\Lambda_\omega F_A-\mathrm{i}\,c)\rangle\,\frac{\omega^n}{n!}\qquad\qquad A\in\mathscr{A}_k,\ \boldsymbol{\xi}\in\mathrm{Lie}(\mathscr{G}_{k+1})=L^2_{k+1}(X;\mathrm{ad}(P_G)).\tag{5.10}$$

Here, the symbol $(\cdot,\cdot)$ is the pairing between $\mathrm{Lie}(\mathscr{G}_{k+1})$ and its dual as in (1.3).

Lemma 5.26. Let $P(M,G)$ be a principal bundle over a compact Kähler manifold M. The map μ defined by (5.10) is a moment map for the action defined by (5.8).

Proof. For each $A\in\mathscr{A}_k$ it is obvious that $\mu(A)$ is linear. Moreover, $F_{\phi^*A}=\mathrm{Ad}_\tau\circ F_A$ (see [**51**, Page 462]). Hence

$$\begin{aligned}(\mu(\phi^*A),\boldsymbol{\xi})&=\int_M\langle\boldsymbol{\xi},(\Lambda_\omega F_{(\phi^{-1})^*A}-\mathrm{i}\,c)\rangle\,\mathrm{d}V_g=\int_M\langle\boldsymbol{\xi},\mathrm{Ad}_\tau(\Lambda_\omega F_A-\mathrm{i}\,c)\rangle\,\mathrm{d}V_g\\&=\int_M\langle\mathrm{Ad}_{\tau^{-1}}(\boldsymbol{\xi}),\Lambda_\omega F_A-\mathrm{i}\,c\rangle\,\mathrm{d}V_g=\langle\mu(A),\mathrm{Ad}_{\tau^{-1}}(\boldsymbol{\xi})\rangle\end{aligned}$$

Finally, given $\boldsymbol{\xi}\in\mathrm{Lie}(\mathscr{G}_{k+1})$, the infinitesimal action is $X_{\boldsymbol{\xi},A}:=\mathrm{d}/\mathrm{d}t|_{t=0}\ \exp_{\mathscr{G}_{k+1}}(t\boldsymbol{\xi})\cdot A=-\mathrm{d}_A\boldsymbol{\xi}$, where d_A is connection that A induces in the associated bundle $\mathrm{ad}(P_G)$. Here, we identify $\exp_{\mathscr{G}_{k+1}}(t\boldsymbol{\xi})$ with a gauge transformation by means of theorem 3.13. For a proof of the claim, see [**65**, Lemma 2.2]

revisar. Let us check that $\mathrm{d}\mu^{\boldsymbol{\xi}} = i_{X_{\boldsymbol{\xi}}}\Omega$. To do so, we use the expression for $F_{A+\alpha}$ proven in lemma 5.17. Now, (again, d_A is the exterior differential evaluated at point A)

$$\begin{aligned}\mathrm{d}_A\,\mu^{\boldsymbol{\xi}}(\alpha) &= \frac{\mathrm{d}}{\mathrm{d}t}\Big|_{t=0}\int_M \langle \boldsymbol{\xi}, (\Lambda_\omega F_{A+t\alpha} - c)\rangle\,\mathrm{d}V_g = \frac{\mathrm{d}}{\mathrm{d}t}\Big|_{t=0}\int_M \left\langle \boldsymbol{\xi}, \Lambda_\omega F_A + t\Lambda_\omega \mathrm{d}_A\alpha + \frac{t^2}{2}\Lambda_\omega[\alpha\wedge\alpha]\right\rangle \mathrm{d}V_g \\ &= \int_M \langle \boldsymbol{\xi}, \Lambda_\omega \mathrm{d}_A\alpha\rangle\cdot\frac{\omega^n}{n!} \overset{(1)}{=} \int_M \langle \boldsymbol{\xi}\wedge\mathrm{d}_A\alpha\rangle\wedge\frac{\omega^{n-1}}{(n-1)!} \overset{(2)}{=} \int_M \langle -\mathrm{d}_A\xi\wedge\alpha\rangle\wedge\frac{\omega^{n-1}}{(n-1)!} \\ &= i_{X_{\boldsymbol{\xi},A}}\Omega_A(\alpha).\end{aligned}$$

Hence, $\mathrm{d}\mu^{\xi} = i_{X_{\boldsymbol{\xi}}}\Omega$. In (1) we used lemma A.34 and in (2) we Stokes' theorem with the fact that

$$\mathrm{d}\left(\langle\boldsymbol{\xi}\wedge\alpha\rangle\wedge\omega^{n-1}\right) = \langle \mathrm{d}_A\boldsymbol{\xi}\wedge\alpha\rangle\wedge\omega^{n-1} + \langle\boldsymbol{\xi}\wedge\mathrm{d}_A\alpha\rangle\wedge\omega^{n-1}.$$

Indeed, $\mathrm{d}\left(\langle\boldsymbol{\xi}\wedge\alpha\rangle\wedge\omega^{n-1}\right) = \mathrm{d}\big(\langle\boldsymbol{\xi}\wedge\alpha\rangle\big)\wedge\omega^{n-1} + \langle\boldsymbol{\xi}\wedge\alpha\rangle\wedge\mathrm{d}\omega^{n-1}$, the latter being zero. For the former, one can show that $\mathrm{d}\langle\boldsymbol{\xi},\alpha\rangle = \langle\mathrm{d}_A\boldsymbol{\xi}\wedge\alpha\rangle + \langle\boldsymbol{\xi},\mathrm{d}_A\alpha\rangle$. To do this, first observe that a straightforward computation shows that $\langle\Upsilon^{-1}(\alpha)\wedge\Upsilon^{-1}(\beta)\rangle = \pi^*\langle\alpha\wedge\beta\rangle$ for all $\mathrm{ad}(P_K)$-valued differential forms α and β. Let $\tilde{\alpha} := \Upsilon^{-1}(\alpha)$ and $\hat{\xi} := \Upsilon^{-1}(\xi)$. Then,

$$\begin{aligned}\pi^*\langle\xi\wedge\mathrm{d}_A\alpha\rangle &= \langle\hat{\xi}\wedge\mathrm{D}_A\tilde{\alpha}\rangle = \langle\hat{\xi}\wedge(\mathrm{d}\tilde{\alpha}+[A,\tilde{\alpha}])\rangle = \mathrm{d}\langle\hat{\xi}\wedge\tilde{\alpha}\rangle - \langle\mathrm{d}\hat{\xi}\wedge\tilde{\alpha}\rangle + \langle\hat{\xi}\wedge[A\wedge\tilde{\alpha}]\rangle \\ &\overset{(3)}{=} \pi^*\mathrm{d}\langle\xi\wedge\alpha\rangle - \langle(\mathrm{d}\hat{\xi}+[A,\hat{\xi}])\wedge\tilde{\alpha}\rangle = \pi^*\mathrm{d}\langle\xi\wedge\alpha\rangle - \langle\mathrm{D}_A\hat{\xi}\wedge\tilde{\alpha}\rangle \\ &= \pi^*\big(\mathrm{d}\langle\xi\wedge\alpha\rangle - \langle\mathrm{d}_A\xi\wedge\alpha\rangle\big).\end{aligned}$$

In (3) we used the invariance of the product (see the proof of lemma 5.14). The equality holds when removing the pullbacks because π is a surjective smooth submersion (c.f. lemma 4.7). □

Observe that, by construction, $\mu^{-1}(0)$ is precisely the set of connections in $\mathscr{A}_k(P_G)$ that satisfy the Hermite-Yang-Mills equation.

Remark 5.27. The map

$$\begin{array}{ccc}\mathrm{Lie}(\mathscr{G}_{k+1}) & \longrightarrow & \mathrm{Lie}(\mathscr{G}_{k+1})^* \\ \boldsymbol{\xi} & \mapsto & \int_M\langle\boldsymbol{\xi},\cdot\rangle\,\mathrm{d}V_g\end{array}$$

is a vector space isomorphism. This justifies the fact that one commonly finds in the literature the statement that the moment map is given by $\mu\colon \mathscr{A}_k \to \mathrm{Lie}(\mathscr{G}_{k+1}), A\mapsto \Lambda_\omega F_A - \mathrm{i}\,c$. When writing this, we are seeing the moment map under that isomorphism.

5.2.2 Infinite-dimensional Kempf-Ness theorem

We have claimed several times that the Hitchin-Kobayashi correspondence can be seen as an infinite-dimensional version of the Kempf-Ness theorem. The previous section has been devoted to stablish, with some level of detail, the moment map picture. In this section, we shall explore how this situation can be seen as a formal analogue of the Kempf-Ness theorem. Proofs of most of our claims are beyond the scope of this thesis, so this section will be rather expository. Moreover, even though the discussion on the moment map picture above holds for arbitrary compact Kähler manifolds, we shall restrict ourselves to the case of a Riemann surface in this section to simplify the discussion. Thus, we will be working in the context of the theorem by Ramanathan. The general case requires dealing with some aditional technicalities, but the core ideas are the same. To be clear, the proof does become way more complicated if one works with a compact Kähler manifold, but the interpretation in terms of the Kempf-Ness theorem is what does not change that much. The main reference for this section is [65].

Let us fix K a compact Lie group, $G = K^{\mathbb{C}}$ its complexification, P_K a principal K-bundle and $P_G = P_K\times_K G$. Observe that P_K is a reduction of the structure group of P_G to K. Let $\mathscr{G}_k$ and $\mathscr{G}_k^{\mathbb{C}}$ be the corresponding gauge groups. Since we are dealing with Riemann surfaces, every connection $A\in\mathscr{A}(P_K)$ is of type $(1,1)$ and induces a holomorphic structure on P_G. We know that $\mathscr{A}_k$ is a Banach manifold, and $\mathscr{G}_{k+1}$ is a Banach Lie group whose action on $\mathscr{A}_k$ is smooth and Hamiltonian for the moment map given in (5.10). We will assume that the slice condition is satisfied ?. Moreover, the zeroes of the moment map are precisely the connections satisfying the Hermite-Yang-Mills equation. By the Banach manifold version of the Marsden-Meyer-Weisntein theorem, we know that the

quotient $\mu^{-1}(0)/\mathscr{G}_{k+1}$ has the structure of a Banach symplectic manifold. We have thus constructed the *moduli space* of Hermite-Yang-Mills connections on the bundle P_K. Smoothness??

The discussion of the previous paragraph was relatively clear from the theory we developed in section 5.2.1. It is now time to see the algebraic side of the story. Looking at the definition of stability we gave for the finite-dimensional version of the Kempf-Ness theorem (definition **??**), it is not clear how this relates to stability of principal bundles. In other words, had we not called the conditions of definition **??** stability, semistability and polystability, it would not be obvious to think about the Kempf-Ness theorem. To remedy this, we shall give an equivalent definition of stability which resembles definition **??** more clearly. Recall that the action of the gauge group $\mathscr{G}$ on $\mathscr{A}$ can be extended to an action of $\mathscr{G}^{\mathbb{C}}$ by means of the Chern-Singer correspondence and $\mathscr{J}(P_G)/\mathscr{G}^{\mathbb{C}}$ parametrizes the set of holomorphic structures on P_G up to isomorphism. This can be extended so that we get an action of $\mathscr{G}^{\mathbb{C}}_{k+1}$ on $\mathscr{A}_k(P_K)$. ?????? Moreover, for every connection $A \in \mathscr{A}_k(P_K)$ there is a complex gauge transformation $\phi \in \mathscr{G}^{\mathbb{C}}_{k+1}$ such that $\phi \cdot A \in \mathscr{A}(P_K)$ (that is, $\phi \cdot A$ is a smooth connection). This means a connection still determines a holomorphic structure on P_G up to (complex) gauge transformation.

Consider the following definition.

Definition 5.28. Under the following assumptions

5.2.3 A sketch of the proof

We will sketch a proof of theorem 5.22 following Mundet i Riera, García Prada and Gothen [**49**] and [**18**]. The results stated in these articles are more general: they are given in the context of Kähler fibrations and Higgs bundles respectively, but we may reduce their arguments to ours by ignoring the extra structures. For an alternate proof this result which relies heavily on the vector bundle theorem, we refer the reader to [**3**]. Our strategy for the proof is to reduce the structure group further to another subgroup in such a way that another stronger condition is satisfied: simplicity. Then, the results will follow from the arguments on [**49**]. The results in these articles are quite involved, giving detailed proofs would take a lot of work, and is beyond the scope of this thesis. We include a sketch for the sake of completeness.

First of all, we say that an element $\boldsymbol{\xi} \in \mathrm{Lie}(\mathscr{G}^{\mathbb{C}}) = \Gamma(X; P_G \times_G \mathfrak{g})$ is *semisimple* if, for any $x \in X$, $\boldsymbol{\xi}(x) \in \mathfrak{g}$ is semisimple (i.e. its adjoint diagonalizes). We see $\boldsymbol{\xi}(x) \in \mathfrak{g}$ through any of the isomorphisms $(P_G \times_G \mathfrak{g})_x \cong \mathfrak{g}$ (see proposition 3.8). The condition of being semisimple does not depend on the chosen representative (because any other representative will be its image through the adjoint action, and Ad_g is an isomorphism for every $g \in G$). Finally, recall that the action of $\mathscr{G}$ on $\mathscr{A}$ can be extended to an action of $\mathscr{G}^{\mathbb{C}}$ using the Chern-Singer correspondence (see section 3.9). Then, a connection $A \in \mathscr{A}(P_G)$ is said to be *simple* if no element $\boldsymbol{\xi} \in \mathrm{Lie}(\mathscr{G})$ fixes A under the infinitesimal action of $\mathscr{G}^{\mathbb{C}}$. More explicitly, given $\boldsymbol{\xi} \in \mathrm{Lie}(\mathscr{G}^{\mathbb{C}})$ we can construct the associated vector field $X_{\boldsymbol{\xi}}$. Then, the *infinitesimal action* is the vector $X_{\boldsymbol{\xi},A} \in \Omega^1(M; \mathrm{ad}(P_G))$. As shown in [**65**, Prop. 2.3], this is given, explicitly, by

$$X_{\zeta_1 + \mathrm{i}\,\zeta_2, A} = -\mathrm{d}_A \xi - \star\, \mathrm{d}_A \zeta$$

for $\boldsymbol{\xi} = \zeta_1 + \mathrm{i}\,\zeta_2$ and $\zeta_1, \zeta_2 \in \mathrm{Lie}(\mathscr{G})$. Then, [**49**] shows the following version of the Hitchin-Kobayashi correspondence.

Theorem 5.29 (Hitchin-Kobayashi for simple connections). Let $\mathcal{P}_G$ be a holomorphic principal G-bundle over a Riemann surface, and $A \in \mathscr{A}(P_K)$ a simple connection. Let $c \in \mathrm{i}\,\mathfrak{z}(\mathfrak{g}) \cap \mathfrak{k}$. Then, there is a gauge transformation $\psi \in \mathscr{G}^{\mathbb{C}}$ such that

$$\Lambda_\omega F_{\psi\cdot A} = -\mathrm{i}\, c \tag{5.11}$$

if and only if A is c-stable. If $\psi \cdot A$ and $\psi' \cdot A$ solve equation (5.11), then there is some $\phi \in \mathscr{G}$ such that $\psi' = \phi \circ \psi$.

Sketch of proof [49]. The main idea is to construct a functional $\Psi\colon \mathscr{A}_k(P_K) \times \mathscr{G}^{\mathbb{C}}_{k+1} \to \mathbb{R}$ whose critical points are solutions of (5.11) after completing these spaces with suitable Sobolev norms as in section 5.2.1. Since everything works fine when working in the completed spaces, we will ignore that technicality. We fix a *hermitian metric* $\langle\cdot,\cdot\rangle$ metric in $\mathfrak{g}$.

First, one defines *weight*

$$\lambda_t(A;\boldsymbol{\zeta}) := \mu^{\boldsymbol{\zeta}}(\exp_{\mathscr{G}}(t\ \boldsymbol{\zeta})\cdot A) = \big(\mu(\exp_{\mathscr{G}}(t\ \boldsymbol{\zeta})\cdot A, \boldsymbol{\zeta})\big) \in \mathbb{R},$$

and the *maximal weight*

$$\lambda(A;\boldsymbol{\zeta}) = \lim_{t\to 0} \lambda_t(A;\boldsymbol{\zeta}) \in \mathbb{R}\cup\{\infty\}\,.$$

Then, one shows that there is a unique functional $\Psi\colon \mathscr{A}(P_K)\times\mathscr{G}^{\mathbb{C}} \to \mathbb{R}$, called *the integral of the moment map* satisfying:

1. For any $A \in \mathscr{A}(P_G)$, the critical points of the restriction Ψ_A of Ψ to $\{A\}\times\mathscr{G}$ coincide with the points of the orbit $A\cdot\mathscr{G}$ on which μ vanishes.
2. The restriction of Ψ_A to lines of the form $\{\exp_{\mathscr{G}}(t\boldsymbol{\zeta}) : t\in\mathbb{R}\}$ for $\boldsymbol{\zeta} \in \mathrm{Lie}(\mathscr{G})$ is convex.

One then shows that an element $\phi \in \mathscr{G}$ is a critical point of Ψ_A for some $A \in \mathscr{A}(P_K)$ if and only if $\mu(A\cdot\phi) = 0$, i.e. if $A\cdot\phi$ is a solution of (5.11). Then, one shows that the integral of the moment map may be written as

$$\Psi_A(\exp(i\boldsymbol{\zeta})) = \int_X \langle \Lambda_\omega F_A + \mathrm{i}\, c, \boldsymbol{\zeta}\rangle + \int_0^1 (1-\ell)\cdot|\exp(i\ell\boldsymbol{\zeta})\ \overline{\partial}_A(\boldsymbol{\zeta})\ \exp(-i\ell\boldsymbol{\zeta})|^2\ \mathrm{d}\ell.$$

In the previous expression, $\overline{\partial}_A = \Pi^{(0,1)}\circ \mathrm{d}_A$ is the $(0,1)$ component of the induced connection d_A on the associated bundle $P_K\times_K\mathfrak{k}$, and $|\cdot|^2$ is the norm associated to the $\mathbb{C}$-bilinear product $(\cdot,\cdot)$. One may show that $\Psi(A,\phi\circ\psi) = \Psi(A,\psi)$ for $\phi\in\mathscr{G}$ and $\mathscr{G}^{\mathbb{C}}$ [49, Prop. 3.4]. Then, Ψ_A factors through $\mathscr{G}$:

$$\Psi_A\colon \mathscr{G}^{\mathbb{C}}/\mathscr{G} \to \mathbb{R}.$$

We want to minimize that functional. To do so, since $\mathfrak{g}/\mathfrak{k} \cong \mathrm{i}\,\mathfrak{k}$, we may use the exponential map to identify $\mathscr{G}^{\mathbb{C}}/\mathscr{G}$ with $\Gamma(M; P_K\times_K \mathrm{i}\,\mathfrak{k})$. For an appropriate choice of p, we define $\mathscr{M}^p_2 = L^p_2(P_K\times_K \mathrm{i}\,\mathfrak{k})$ (the L^p_2 norm is defined as usual), which we shall identify with the quotient of the Sobolev completions. When A is a simple connection, one can restrict to looking for minima on the subset

$$\mathscr{M}^p_{2,B} := \{\boldsymbol{\zeta}\in\mathscr{M}^p_2 : ||\mu(\exp(\boldsymbol{\zeta}\cdot A))||_{L_p} \le B\}$$

for some positive constant B. We see the moment map as taking values in $\mathrm{Lie}(\mathscr{G})$ to make sense of the norm (c.f. remark 5.27). Finally, one shows that there is an element of $\mathscr{M}^p_{2,B}$ that minimizes Ψ_A and this minimizing element is smooth. Then, it corresponds to an equivalence class of the quotient $\mathscr{G}^{\mathbb{C}}/\mathscr{G}$, so there is a solution, and any two different solutions are related by a transformation of $\mathscr{G}$.

Reciprocally, choose a simple connection A and a gauge transformation $\psi\in\mathscr{G}^{\mathbb{C}}$ such that $A\cdot\psi$ satisfies (5.11). To show that A is stable, let $s\in\mathrm{i}\,\mathfrak{k}$, and choose a holomorphic reduction of the structure group to P_s, $\sigma\in\mathcal{O}(X;\mathcal{P}(G/P_s))$. By definition, one should check that $\deg\mathcal{P}(\sigma,s) - \langle c,s\rangle > 0$, however, [49, Lemma 4.3] shows that one may equivalently check that $\lambda(A;-\mathrm{i}\,g_{\sigma,s})$, where $g_{\sigma,s}\in \mathrm{i}\,\mathrm{Lie}(\mathscr{G}) = \Gamma(X;P_K\times_K\mathrm{i}\,\mathfrak{k})$ is fiberwise dual to the character χ_s (see [49, Sec. 2.7]). One can show the following bound [49, Sec. 6.3]

$$t\sup|g_{\sigma,s}| \le C_1\cdot\Psi_A(\exp_{\mathscr{G}}(tg_{\sigma,s})) + C_2, \qquad C_1, C_2 > 0, \tag{5.12}$$

and for sufficiently large $t > 0$. Moreover, one can show that $\Psi_A(\exp_{\mathscr{G}}(\mathrm{i}\, t g_{\sigma,s})) = \int_0^1 \lambda_{t\cdot\ell}(A; g_{\sigma,s})\, \mathrm{d}\ell$. If $\lambda(A; -\mathrm{i}\, g_{\sigma,s}) < 0$, then the previous integral would be negative for every t. Finally, (5.12) would imply $t \cdot \sup |g_{\sigma,s}| < C_2$, but the left-hand side tends to $+\infty$, so we get a contradiction. □

Since our goal is to use theorem 5.29 to prove theorem 5.22, we will need a way to translate the stability condition into a condition dealing with simplicity. We do this by means of the following theorem. The following theorem associates to each c-polystable principal bundle a c-stable principal bundle over a different group.

Theorem 5.30 (Jordan-Hölder reduction). Let $\mathcal{P}_G$ be a holomorphic principal G-bundle over a compact Riemann surface, and let $c \in \mathrm{i}\,\mathfrak{z}(\mathfrak{g}) \cap \mathfrak{k}$. Then, there is a compact subgroup $K' \subset K$ and a holomorphic reduction of the structure group of $\mathcal{P}_G$ to $G' := (K')^{\mathbb{C}}$, that is c-stable. Any two such reductions $\mathcal{P}^{(1)}_{G'}$ and $\mathcal{P}^{(2)}_{G'} \subset \mathcal{P}_G$ are related by $\mathcal{P}^{(1)}_{G'} = \mathcal{P}^{(2)}_{G''} \cdot g$ for some $g \in G$.

Proof. The proof is not trivial at all. We refer the reader to section 2.10 of [18]. **(3)** □

Sketch of proof of theorem 5.22. Suppose first that $\mathcal{P}_G$ is c-stable, and fix a reduction to the structure group to K, P_K then one can show that admits a simple connection $A \in \mathscr{A}(P_K)$ **(1)**. The holomorphic structure induced by A on P_G via the Chern-Singer correspondence coincides with that of $\mathcal{P}_G$ **(2)**. By theorem 5.29, there is $\varphi \in \mathscr{G}^{\mathbb{C}}$ such that $A' := A \cdot \varphi$ satisfies equation 5.6. Now, since A and A' are on the same $\mathscr{G}^{\mathbb{C}}$-orbit, they induce the same structure of holomorphic principal bundle on P_G (see (3.28)). Finally, A' is the Chern connection with respect to $J \cdot g$, $J = \mathsf{CS}^{-1}(A)$, and $J \cdot g$ induces the same structure of holomorphic principal bundle on P_G as J.

Suppose now that $\mathcal{P}_G$ is c-polystable, but not stable. We use theorem 5.30 and obtain a c-stable reduction of the structure group to some subgroup G'. Now, $\mathcal{P}_{G'}$ is stable, and the previous reasoning applies. Hence, there is a reduction of the structure group to $P_{K'}$ whose Chern connection satisfies (5.11). Now we define $P_K := P_{K'} \times_{K'} K \subset P_{K'} \times_{K'} G = P_G$ (c.f. 3.35) for the action of K' on K by conjugation. This is a reduction of the structure group to K, and one can show that its Chern connection satisfies (5.6).

For the converse, one shows first that if $\mathcal{P}_G$ admits a reduction of the structure group to K satisfying (5.6), then it is c-semistable. ?.

Now, to prove polystability, let $\sigma \in \Gamma(X; P(G/K))$ be a reduction of the structure group satisfying (5.6). We know that the bundle is semistable, thus, for some parabolic subgroup $P_s \subset G$, we consider a holomorphic reduction of the structure group $\sigma \in \mathcal{O}(X, \mathcal{P}(G/P_s))$ such that $\deg \mathcal{P}(\sigma, s) = \langle s, c\rangle$. We seek a further restriction of the structure group to the Levi subgroup L_s. Let P_K be the reduction of the structure group whose Chern connection satisfies (5.6). This reduction produces a section ζ of the bundle $P_K \times_K \mathrm{i}\,\mathfrak{k}$. Since $\mathrm{Lie}(\mathscr{G}) = \Gamma(X; P_K \times_K \mathfrak{k})$, we can identify $-\mathrm{i}\,\zeta$ with an element of $\mathrm{Lie}(\mathscr{G})$. Now, let J be the almost complex structure of $\mathcal{P}_G$, and let $A = \mathsf{CS}(J) \in \mathscr{A}(P_K)$. One can see that the condition $\deg \mathcal{P}(\sigma, s) = \langle \alpha, s\rangle$ is equivalent to

$$\lambda(A, -\mathrm{i}\,\zeta) := \lim_{t\to\infty} \lambda_t(A, -\mathrm{i}\,\zeta) = \lim_{t\to\infty} (\mu(\exp_{\mathscr{G}}(t\zeta)\cdot A), -\mathrm{i}\,\zeta) = 0.$$

From [49, Lemma 2.2], we know that the function $\lambda_t(A, -\mathrm{i}\,\zeta)$ is non-decreasing. Moreover, by the way the moment map μ is defined, equation (5.6) is equivalent to the vanishing of the moment map. Hence, at $t = 0$, we have $\mu(\exp_{\mathscr{G}}(t\zeta)\cdot A)|_{t=0} = \mu(A)$, and the latter vanishes because the Chern connection A satisfies (5.6). Therefore, we have $\lambda_0(A, -\mathrm{i}\,\zeta) = 0$. Now, the fact that the weight is non-decreasing implies that it is constant and equal to 0 for any $t \geq 0$. One may deduce from this fact that $\exp_{\mathscr{G}_k}(t\zeta)\cdot J = J$ for all $t \geq 0$. **(5)** This implies that ζ induces a holomorphic section $\sigma_s \in \mathcal{O}(M; E_\sigma(P_s/L_s))$, hence a reduction of the structure group to L_s, and this is precisely the definition of polystability. □

5.3 The HK correspondence for non-connected structure groups

This section contains the most original part of the thesis. It formulates a HK correspondence for principal bundles with non-connected complex reductive structure groups. Most of the section

is based on the unpublished article [8] that my supervisor kindly shared with me. His work is more general, but with his ideas, we try to give a generalization of the theorems by Narasimahn and Seshadri [46] and Ramanathan [47]. This section relies heavily on the concepts introduced in appendix B.5. For the whole section we fix a complex reductive Lie group $\hat{G}$, and $\hat{K} \subset \hat{G}$ a maximal compact subgroup. K and G will be their respective identity components. We also set $\Gamma := \hat{G}/G \cong \hat{K}/K$ the group of connected components. Then, K is a maximal compact subgroup of G. We fix $\langle \cdot, \cdot \rangle$ a non-degenerate $\mathbb{C}$-bilinear pairing on $\mathfrak{g}$ invariant under the adjoint $\hat{G}$ action. If $\hat{G}$ is semisimple, then the Killing form $\kappa_{\mathfrak{g}}$ does the job. If $\hat{G}$ is only reductive, an invariant pairing still exists [6]. Indeed, reductive Lie algebras can be decomposed as $\mathfrak{g} = \mathfrak{z}(\mathfrak{g}) \oplus [\mathfrak{g}, \mathfrak{g}]$. On the semisimple part, the Killing form gives a $\mathbb{C}$-bilinear, $\hat{G}$-invariant, non-degenerate pairing. On the zenter, we may choose *any* non-degenerate bilinear from, and it will be invariant uncer the adjoint action because the center is abelian. The direct sum of these forms yields the desired pairing. **(11)** If a $\hat{G}$-invariant pairing exists, we can make it invariant under the Γ-action on $\mathfrak{g}$ (c.f. equation (B.12)) through a standard averaging procedure, that is, we define a new $\mathbb{C}$-bilinear pairing:

$$\langle \cdot, \cdot \rangle' := \frac{1}{|\Gamma|} \sum_{\gamma \in \Gamma} \langle \cdot \gamma, \cdot \gamma \rangle.$$

Observe that Γ is finite, hence so is the sum. Recall from the appendix that the action of Γ on G and K is given by a lift of the characteristic homomorphism $\bar{\theta} \colon \Gamma \to \mathrm{Out}(K)$, and its extension to G. In particular, K is invariant under the Γ action. Observe that the differential of each θ_γ induces an action on the Lie algebra $\mathfrak{k}$ (c.f. equation (B.12)). Since the differential of any Lie group homomorphism is a Lie algebra homomorphism, the action leaves the Lie algebra of $Z(K)$, $\mathfrak{z}_K$ invariant. Indeed, given $\zeta \in \mathfrak{z}_K, \xi \in \mathfrak{k}$, we have

$$\left[\mathrm{d}_e\theta_{\gamma^{-1}}(\zeta), \xi\right] = \mathrm{d}_e\theta_{\gamma^{-1}}\left(\left[\zeta, \mathrm{d}_e\theta_\gamma(\xi)\right]\right) = \mathrm{d}_e\theta_{\gamma^{-1}}(0) = 0. \tag{5.13}$$

5.3.1 The Hermite-Einstein-Yang-Mills equation

In our quest to generalize the Hitchin-Kobayashi correspondence to the case of a non-connected structure group, a naïve guess we could try is to take the statement of theorem 5.22 and try to remove directly the hypothesis of connectedness. However, looking at remark 5.23, we soon realize that this is not going to work: the equation $\Lambda_\omega F_A = c$ no longer makes sense because now we cannot define a constant section by setting $m \mapsto [p, c]$. This already gives us a hint that some things need to be changed. Let $\mathfrak{z}_K$ be the Lie algebra of $Z(K)$. Since K is connected (c.f. remark B.18), this coincides with $\mathfrak{z}(\mathfrak{k})$ revisar que $K = \hat{K} \cap G$ sea conexo.

A calculation analogous to equation (5.13) shows that $\mathfrak{z}_K$ is invariant under the Ad-action of $\hat{K}$. This does not mean that the adjoint action is *trivial* on $\mathfrak{z}_K$; that claim does need connectedness.

Definition 5.31. Let $\mathcal{P}_{\hat{G}} \to X$ be a holomorphic principal $\hat{G}$-bundle, and consider a reduction $P_{\hat{K}}$ of the structure group to $\hat{K}$. Let A be the Chern connection A associated to the reduction and $a \in \Gamma(X; P_{\hat{K}} \times_{\hat{K}} \mathrm{i}\,\mathfrak{z}_K)$ satisfying $\mathrm{d}_A a = 0$. We say that A satisfies the *a-Hermitian-Yang-Mills equation* if

$$\Lambda_\omega F_A = -\mathrm{i}\, a, \tag{5.14}$$

where $F_A \in \Omega^2(X; P_{\hat{K}} \times_{\hat{K}} \mathfrak{k})$ is the curvature of the Chern connection.

Equation (5.14) should be understood as an equality between sections of the vector bundle $P_{\hat{K}} \times_{\hat{K}} \mathfrak{k}$ (observe that it contains $P_{\hat{K}} \times_{\hat{K}} \mathfrak{z}_K$ as a vector subbundle). **Nota:** En el artículo en preparación ponéis que a debe ser una sección de $P_{\hat{K}} \times_{\hat{K}} \mathfrak{z}_K$ sin el factor i, pero luego en la ecuación si que ponéis el factor i, entiendo que es una errata y que si que debe aparecer el factor i en $P_{\hat{K}} \times_{\hat{K}} \mathrm{i}\,\mathfrak{z}_K$ para que luego en la ecuación tengamos una igualdad entre secciones de $P_{\hat{K}} \times_{\hat{K}} \mathfrak{k}$.

Remark 5.32.

1. When G is connected (that is $\hat{G} = G$), equation (5.14) reduces to the connected version of the Hermite-Einstein-Yang-Mills equation (5.6). Indeed, when $\hat{G}$ is connected, so is $\hat{K} = K$, and the adjoint action of a connected Lie group on the center of its Lie algebra is trivial. Therefore, $P_{\hat{K}} \times_{\hat{K}} \mathfrak{z}_K = X \times \mathrm{i}\,\mathfrak{z}_K$ is a trivial bundle. Hence, sections of this bundle can be identified with

smooth maps $X \to \mathrm{i}\,\mathfrak{z}_K$. From section 3.5, to $a\colon X \to \mathrm{i}\,\mathfrak{z}_K$ we assign a K-equivariant (invariant actually, because the K-action on $\mathrm{i}\,\mathfrak{z}_K$ is trivial) map $\tilde{a}(p) := \Upsilon^{-1}(a)(p) = a(\pi(p))$. Hence, $\mathrm{d}_A a = \Upsilon(\mathrm{D}_A\tilde{a}) = 0$, and since Υ is a vector space isomorphism, we have $\mathrm{D}_A\tilde{a} = (\mathrm{d}\tilde{a})^H = 0$. However, $\mathrm{d}\tilde{a} = (\mathrm{d}\tilde{a})^H$. To see this, it suffices to check that $\mathrm{d}\tilde{a}$ vanishes when acting on vertical vectors. Indeed, given a vertical vector $X_{\xi,p}$ for $\xi \in \mathfrak{k}, p \in P$, we have

$$\mathrm{d}_p\tilde{a}(X_{\xi,p}) = \mathrm{d}_p(a\circ\pi)(\mathrm{d}_e\phi^{(p)}(\xi)) = \mathrm{d}_e(a\circ\pi\circ\phi^{(p)})(\xi) = 0$$

because $\pi\circ\phi^{(p)}$ is a constant map identically equal to p. Hence, $\mathrm{d}\tilde{a} = 0$, and this implies that $\tilde{a}$ is constant. By construction, a must be also constant. Hence, if G is connected, we recover $\Lambda_\omega F_A = -\mathrm{i}\,a$ for $a \in \mathrm{i}\,\mathfrak{z}_K = \mathrm{i}\,\mathfrak{z}(\mathfrak{k}) \subset \mathrm{i}\,\mathfrak{z}(\mathfrak{g}) \cap \mathfrak{k}$.

2. If $\hat{G}$ is not connected but $\mathcal{P}_{\hat{G}}$ has surjective monodromy, then the section $a \in \Gamma(X; P_{\hat{K}} \times_{\hat{K}} \mathfrak{z}_K)$ can be identified with a single element $\alpha \in \mathfrak{z}_K^\Gamma$. Then, solving (5.14) is equivalent to solving $\Lambda_\omega F_A = -\mathrm{i}\,\alpha$. If the monodromy of $\mathcal{P}_{\hat{G}}$ is not surjective, we may always find a reduction $\mathcal{P}_{\hat{G}'}$ of the structure group to another group $\hat{G}' \subset \hat{G}$ that has surjective monodromy. To see why, first observe that, sine $\hat{G}$ acts on $P_{\hat{G}}$, proposition B.25 implies that there is a natural Γ-action on $Y := P_{\hat{G}}/G$. In fact, $Y \to X$ has teh structure of a principal Γ-bundle. We thus have a natural vector bundle isomorphism given by

$$P_{\hat{K}} \times_{\hat{K}} \mathfrak{z}_K \longrightarrow Y \times_\Gamma \mathfrak{z}_K,\ [p,\zeta] \mapsto [\hat{\pi}(p),\zeta]\ ?$$

where $\hat{\pi}\colon P_{\hat{G}} \to Y$ is the natural projection (which we restrict to $P_{\hat{K}}$). No entiendo como seguimos. A reduction of the structure group to $\hat{K}$ satisfying $\Lambda_\omega F_A = -\mathrm{i}\,\alpha$ is called an *α-Hermite-Yang-Mills reduction of the structure group*.

3. If (5.14) holds, then α is determined uniquely by the topology of the bundle and the manifold. justificción ? **(17)**

5.3.2 Stability condition

The first step towards the generalization of the correspondence is to give a notion of stability. In fact, we will need two notions: one for principal bundles with non-connected structure group, and another for twisted equivariant bundles.

Observe that the statement of lemma 5.14 holds in the non-connected case. Moreover, if the pairing $\langle\cdot,\cdot\rangle$ is Γ-invariant, and $\zeta \in \mathrm{i}\,\mathfrak{k}^\Gamma$, then χ_ζ is Γ-invariant, i.e. $\chi_\zeta(\xi\cdot\gamma) = \chi_\zeta(\xi)$ for all $\xi \in \mathfrak{g}$. Observe that the definition of the degree of a holomorphic principal bundle with respect to a reduction and an element of $\mathrm{i}\,\mathfrak{k}$ applies verbatim.

Before giving the definition of stability, observe that if we choose $\zeta \in \mathrm{i}\,\mathfrak{k}^\Gamma$, then the parabolic subgroup P_ζ defined by proposition B.13 is a Γ-invariant subgroup of G. From proposition B.29 and the comments bellow, we deduce that G/P_ζ is a (θ,c)-twisted (g,Γ)-equivariant manifold. Finally, by proposition B.26, we deduce that, given a (θ,c)-twisted Γ-equivariant bundle, $P \to X$, the associated bundle $P(G/P_\zeta) \to X$ is a Γ-equivariant principal bundle. Hence, we may consider Γ-equivariant sections or, equivalently, Γ-equivariant reductions of the structure group to P_ζ. Similarly, consider $L_\zeta \subset P_\zeta$ is Γ-invariant, and given a Γ-invariant reduction fo the structure group of P to P_ζ, call it P_σ, we know from proposition B.30 that P_σ is endowed with the structure of a (θ,c)-twisted Γ-equivariant principal P_ζ-bundle. A partir de ahora no sé si esto es verdad Since $L_\zeta \subset P_\zeta$ is also Γ-invariant, and $\hat{L}_\zeta \subset \hat{P}_\zeta$ meets all the connected components of the latter, we have an isomorphism $P_\zeta/L_\zeta \cong \hat{P}_\zeta/\hat{L}_\zeta$. Thus, the same argumets apply, $P_\sigma(P_\zeta/L_\zeta)$ is a Γ-equivariant principal, and it makes sense to consider Γ-equivariant reductions of the structure group to L_ζ. With this in mind, we are now ready to define stability.

Definition 5.33. Let $\mathcal{P}$ be a (θ, c)-twisted Γ-equivariant principal bundle over a Riemann surface Y, and let $\alpha \in \mathfrak{i}\mathfrak{z}_K^\Gamma$. We say that $\mathcal{P}$ is

1. Γ-*invariantly* α-*stable (resp. semistable)* if for any $\zeta \in \mathfrak{i}\mathfrak{k}^\Gamma$ and any Γ-invariant holomorphic section $\sigma \in \mathcal{O}(Y; \mathcal{P}(G/P_\zeta))$, we have
$$\deg \mathcal{P}(\sigma, \zeta) - \langle \alpha, \zeta \rangle > 0 \qquad (\text{resp.} \quad \deg \mathcal{P}(\sigma, \zeta) - \langle \alpha, \zeta \rangle \geq 0)\,.$$
2. α-*invariantly* Γ-*polystable* if it is Γ-invariantly α-semistable and, for any $\zeta \in \mathfrak{k}^\Gamma$ and any Γ-invariant holomorphic section $\sigma \in \mathcal{O}(Y; \mathcal{P}(G/P_s))$ such that $\deg \mathcal{P}(\sigma, \zeta) - \langle \alpha, \zeta \rangle = 0$, there is a Γ-invariant holomorphic reduction of the structure group $\sigma' \in \mathcal{O}(X; \mathcal{P}_\sigma(P_\zeta/L_\zeta))$.

Now we need a notion of stability for principal bundles with non-connected structure group. The definition we are about to give is quite similar to the definition for twisted equivariant principal bundles, and as we shall wee, these definitions are closely related.

Definition 5.34. Let X be a compact Riemann surface, and let $\mathcal{P}(X, \hat{G})$ be a holomorphic principal $\hat{G}$-bundle with surjective monodromy, and let $\alpha \in \mathfrak{i}\mathfrak{z}_K^\Gamma$. We say that the principal bundle is

1. α-*stable (resp- semistable)* if for any $\zeta \in \mathfrak{i}\mathfrak{k}^\Gamma$ and any holomorphic section $\sigma \in \mathcal{O}(X; \mathcal{P}(\hat{G}/\hat{P}_\zeta))$, we have
$$\deg \mathcal{P}(\sigma, \zeta) - \langle \alpha, \zeta \rangle > 0 \qquad (\text{resp.} \quad \deg \mathcal{P}(\sigma, \zeta) - \langle \alpha, \zeta \rangle > 0)$$
2. α-*polystable* if it is α-semistable and for any $\zeta \in \mathfrak{i}\mathfrak{k}^\Gamma$ and $\sigma \in \mathcal{O}(X; \mathcal{P}(\hat{G}/\hat{P}_\zeta))$ such that $\deg \mathcal{P}(\sigma, \zeta) - \langle \alpha, \zeta \rangle = 0$, there is a holomorphic reduction of the structure group $\sigma' \in \mathcal{O}(X; \mathcal{P}_\sigma(\hat{P}_\zeta/\hat{L}_\zeta))$.

When $G = \hat{G}$ is connected, we recover the usual stability criteria (see definition 5.21).

5.3.3 The theorem

The aim of this section is to state and proof the non-connected version of the Hitchin-Kobayashi correspondence. For this section, we fix a complex reductive non connected Lie group $\hat{G}$, its identity component G, $\hat{K}$ a maximal compact subgroup, and $K := \hat{K} \cap G$ is identity component. We thus have the following diagram of extensions

$$\begin{array}{ccccccccc} 1 & \longrightarrow & G & \longrightarrow & \hat{G} & \longrightarrow & \Gamma & \longrightarrow & 1 \\ & & \uparrow & & \uparrow & & \| & & \\ 1 & \longrightarrow & K & \longrightarrow & \hat{K} & \longrightarrow & \Gamma & \longrightarrow & 1 \end{array}$$

We also fix a Riemann surface X. Under this hypothesis, the non-connected version of the Hitchin-Kobayashi correspondence reads as follows.

Theorem 5.35 (Hitchin-Kobayashi, non-connected version). Let $\alpha \in \mathfrak{i}\mathfrak{z}_K^\Gamma$. A holomorphic principal $\hat{G}$-bundle $\mathcal{P}_{\hat{G}}$ admits a reduction of the structure group to $\hat{K}$, say $P_{\hat{K}}$, satisfying equation (5.14) for a section $a \in \Gamma(X; P_{\hat{K}} \times_K \mathfrak{z}_K)$ satisfying $\mathrm{d}_A a = 0$, where A is the Chern connection in $\hat{K}$ if and only if it is α-polystable for $\alpha \in \mathfrak{i}\mathfrak{z}_K$ defined as in remark 5.32.

Our strategy for the proof is to translate the problem into one concerning principal bundles with connected structure groups. This is done by means of the correspondence of categories given by theorem B.31. We then show that the statement of the correspondence for non-connected structure group is equivalent to a HK correspondence for twisted equivariant bundles over a covering, for which we sketch (?) a proof.

Proposition 5.36. Let $\mathcal{P}_{\hat{G}}$ be a holomorphic principal $\hat{G}$-bundle, and let $\mathcal{P}_G$ the corresponding (θ, c)-twisted Γ-equivariant principal bundle over $Y := \mathcal{P}_{\hat{G}}/G$ given by theorem B.31. Let $\alpha \in \mathfrak{i}\mathfrak{z}_K^\Gamma$, then $\mathcal{P}_{\hat{G}}$ is α-polystable if and only if $\mathcal{P}_G$ is Γ-invariantly α-polystable.

Before given the proof, we will need some lemmas.

Lemma 5.37. Let $\hat{G}$ be an $\mathcal{S}$-Lie group, G its identity component, and $\hat{H} \subset \hat{G}$ a closed embedded $\mathcal{S}$-Lie subgroup that meets every connected component of $\hat{G}$. Let $H := \hat{H} \cap G$. Then, $\hat{G}/\mathcal{H}$ is $\mathcal{S}$-diffeomorphic to G/H.

Proof. Define $G/H \to \hat{G}/\hat{H}$, $gH \mapsto g\hat{H}$. This map is well-defined: if $gH = g'H$, then $g^{-1}g' \in H \subset \hat{H}$, hence $g\hat{H} = g'\hat{H}$. It is also injective: if $g\hat{H} = g'\hat{H}$, then $g^{-1}g' \in \hat{H} \cap G = H$, and surjective. To see surjectivity, observe that all the connected components of $\hat{G}$ are given by the cosets $\hat{g}G$ for $g \in \hat{G}$. Let $\hat{g}\hat{H} \in \hat{G}/\hat{H}$. We want to see that there is $g \in G$ such that $\hat{g}\hat{H} = g\hat{H}$. The connected component containing $\hat{g}$ is $\hat{g}G$. Since $\hat{H}$ meets every connected component of $\hat{G}$, $\hat{g}G \cap \hat{H} \neq \varnothing$, hence there is $h \in H$ such that $\hat{g}g = h$. Since $G \triangleleft \hat{G}$, $\hat{g}G = G\hat{g}$, so there is $g_2 \in G$ such that $\hat{g}g = g_2\hat{g} = h$. Choosing $g := g_2^{-1}$, we have $g\hat{H} = g_2^{-1}h\hat{H} = \hat{g}H$. Finally the map is an $\mathcal{S}$-isomorphism by the characteristic property of surjective submersions and the equivariant rank theorem [**35**, or. 2.6] and [**36**, Th. 7.25]. □

Lemma 5.38. Let F be a (θ, c)-twisted (G, Γ)-manifold and $\pi\colon P \to X$ be a (θ, c)-twisted Γ-equivariant principal G-bundle. Let $\mathscr{C}^\infty(P, F)^{G,\Gamma}$ the space of (θ, c)-twisted (G, Γ)-equivariant smooth maps (B.18), and let $\Gamma_{\text{inv}}(X, P(F))$ the space of Γ-invariant sections on the twisted equivariant bundle $P \to X$. Then, there is a bijection $\mathscr{C}^\infty(P, F)^{G,\Gamma} \leftrightarrow \Gamma_{\text{inv}}(X, P(F))$. The result holds form holomorphic sections and maps as well.

Proof. See [**5**, Prop. 4.8] □

Proof of proposition 5.36. First, observe that on both definitions of stability, we ask for $\alpha \in \mathfrak{i}\mathfrak{z}_K^\Gamma$, and we choose elements $\zeta \in \mathfrak{i}\mathfrak{k}^\Gamma$. Moreover, given $\zeta \in \mathfrak{i}\mathfrak{k}^\Gamma$, the group $\hat{P}_\zeta$ meets all the connected components of $\hat{G}$ [**8**, Prop. 2.12]. Since $P_\zeta = \hat{P}_\zeta \cap G$, lemma 5.37 implies that $G/P_\zeta = \hat{G}/\hat{P}_\zeta$. This implies that any holomorphic section of the associated bundle $\mathcal{P}_{\hat{G}}(\hat{G}/\hat{P}_\zeta) \to X$ induces a Γ-invariant section of the Γ-equivariant bundle $\mathcal{P}_G(G/P_\zeta) \to Y$ and vice-versa. To see this, first observe that $\mathcal{P}_{\hat{G}} = \mathcal{P}_G$ as manifolds (c.f. remark B.32). Observe that $G/P_\zeta \cong \hat{G}/\hat{P}_\zeta$ is a (θ, c)-twisted (G, Γ)-equivariant manifold ref. Moreover, a map between twisted equivariant manifolds is (θ, c)-twisted (G, Γ)-equivariant if and only if it is $\hat{G}$-equivariant. Sections of $\mathcal{P}_{\hat{G}}(\hat{G}/\hat{P}_\zeta) \to X$ are in one-to-one correspondence with $\hat{G}$-equivariant maps $\mathcal{P}_{\hat{G}} \to G/P_\zeta$, and these are in one-to-one correspondence with (θ, c)-twisted (G, Γ)-equivariant maps $\mathcal{P}_{\hat{G}} \to \hat{G}/\hat{P}_\zeta$, observe that these coincide with (θ, c)-twisted (G, Γ)-equivariant maps $\mathcal{P}_G \to G/P_\zeta$ because the sets are the same. Moreover, by proposition B.26 we know that $\mathcal{P}_{\hat{G}}(\hat{G}/\hat{P}_\zeta) \to X$ is a Γ-equivariant fiber bundle. By lemma Applying 5.38, to each (θ, c)-twisted (G, Γ)-equivariant map $\mathcal{P}_G \to G/P_\zeta$ corresponds a Γ-invariant section of the Γ-equivariant bundle $\mathcal{P}_G \to Y$. Under some abuse of notation, we shall use the same notation for both sections. Observe that by proposition B.30 this means that there is a one-to-one correspondence between holomorphic reductions of the structure group of $\mathcal{P}_{\hat{G}} \to X$ to a parabolic subgroup $\hat{P}_\zeta$, and Γ-invariant reductions of the twisted equivariant bundle $\mathcal{P}_G \to Y$ to the connected component P_ζ of $\hat{P}_\zeta$.

The same discussion applies to Levi subgroups: $\hat{L}_\zeta$ meets every connected component of $\hat{P}_\zeta$, P_ζ is the identity component of $\hat{P}_\zeta$, and $L_\zeta = \hat{L}_\zeta \cap P_\zeta$. Hence, lemma 5.37 holds, and $\hat{P}_\zeta/\hat{L}_\zeta \cong P_\zeta/L_\zeta$ esto no tengo ni idea de si es verdad, en el paper solo pone "the same applies to the Levi subgroup". By the same reasoning, a section of the bundle $(\mathcal{P}_{\hat{G}})_\sigma(\hat{P}_\zeta/\hat{L}_\zeta) \to X$ is equivalent to a Γ-invariant section of the Γ-equivariant bundle $(\mathcal{P}_G)_\sigma(P_\zeta/L_\zeta) \to Y$. Observe as well that the Γ-invariance of ζ ensures that the integral descends to an integral over Y. In particular, we have $\deg \mathcal{P}_{\hat{G}}(\sigma, \zeta) = \deg \mathcal{P}_G(\sigma, \zeta)$.

Aquí en el paper solo afirmáis que la integral desciende a una integral en X porque siempre se puede encontrar una conexión A que sea Γ-invariante. Esto no lo entiendo. Primero, aunque $Y/\Gamma = X$, no tengo muy claro como se relacionan las integrales entre variedades y sus cocientes.

He encontrado algo llamado "integración a lo largo de fibras" que es una especie de Fubini para variedades cociente, aunque no sé si es lo que estamos aplicando aquí. Según ese concepto de integración a lo largo de fibras, si tenemos un recubrimiento, es como un Γ-fibrado, entonces para las formas α en Y que son invariantes, existen formas β en X de manera que

$$\int_Y \alpha = |\Gamma| \int_X \beta$$

Consideramos el fibrado twisted $P_G \to Y$. Suponiendo que K_ζ sea Γ-invariante tiene sentido hablar de una reducción $(P_G)'_\sigma$ que sea Γ-invariante ?. Entonces si la 2-forma $\alpha = \chi_\zeta(F_A)$ se correspondiera con una 2-forma $\chi_\zeta(F_{\hat{A}})$ para alguna conexión $\hat{A}$ en el fibrado $P_{\hat{G}} \to X$, tendríamos

$$\int_Y \chi_\zeta(F_A) = |\Gamma| \cdot \int_X \chi_\zeta(F_{\hat{A}}).$$

Esto sigue sin ser lo que queremos porque aparece un factor de proporcionalidad $|\Gamma|$, pero si de alguna forma nos pudiéramos quitar el factor de proporcionalidad sí que tendríamos igualdad de los grados, que entiendo que es lo que queremos. En cualquier caso, no sé si con la frase de que la integral desciende a una en X os referís a esto o a otra cosa.

Having said that, suppose that $\mathcal{P}_{\hat{G}}$ is α-polystable for $\alpha \in \mathrm{i}\,\mathfrak{z}_K^\Gamma$, and let $\zeta \in \mathrm{i}\,\mathfrak{k}^\Gamma$. Let $\sigma \in \mathcal{O}(Y; \mathcal{P}_G(G/P_\zeta))$ be Γ-invariant. We have seen that this induces a section (which we cal the same) $\sigma \in \mathcal{O}(X; \mathcal{P}_{\hat{G}}(\hat{G}/\hat{P}_\zeta))$. Now, since $\mathcal{P}_{\hat{G}}$ is α-polystable, it is α-semistable, hence

$$\deg \mathcal{P}_{\hat{G}}(\sigma, \zeta) - \langle \alpha, \zeta \rangle = \deg \mathcal{P}_G(\sigma, \zeta) - \langle \alpha, \zeta \rangle \geq 0, \tag{5.15}$$

so $\mathcal{P}_G$ is Γ-invariantly α-semistable. Now, take $\zeta \in \mathrm{i}\,\mathfrak{k}^\Gamma$ and a Γ-invariant $\sigma \in \mathcal{O}(Y; \mathcal{P}_G(G/P_\zeta))$ such that the equality holds in (5.15). Again, this induces a section $\sigma \in \mathcal{O}(X; \mathcal{P}_{\hat{G}}(\hat{G}/\hat{P}_\zeta))$ such that the equality holds. Since $\mathcal{P}_{\hat{G}}$ is α-polystable, there is a holomorphic section $\sigma' \in \mathcal{O}(X; \mathcal{P}_{\hat{G}}(\hat{P}_\zeta/\hat{L}_\zeta))$ which induces a Γ-invariant section (again, we keep the name) $\sigma' \in \mathcal{O}(Y; \mathcal{P}_G(P_\zeta/L_\zeta))$. The converse follows from the same reasoning in the opposite direction. □

Theorem 5.39 (Hitchin-Kobayashi for twisted bundles). Let $P \to X$ a (θ, c)-twisted Γ-equivariant principal G-bundle, and let $\alpha \in \mathrm{i}\,\mathfrak{z}_K^\Gamma$. Then, P is Γ-invariantly α-polystable if and only if there exists a Γ-equivariant reduction of the structure group $\sigma \in \Gamma(X; P(G/K))$ satisfying $\Lambda_\omega F_A = -\mathrm{i}\,\alpha$, where A is the Chern connection on P_K.

A Γ-equivariant reduction of the structure group satifying $\Lambda_\omega F_A = -\mathrm{i}\,\alpha$ is called a *Γ-equivariant α-Hermite-Yang-Mills reduction of the structure group* to K. Moreover, observe that a Γ-invariant reduction of the structure group is, in particular, a reduction of the structure group, hence it makes sense to consider its Chern-connection.

Proof. De momento no sé por donde tirar con esto. □

Proposition 5.40. Let $P_{\hat{G}}$ be a principal $\hat{G}$-bundle over a compact Riemann surface X, and consider the associated (θ, c)-twisted Γ-equivariant principal bundle $P_G \to P/G =: Y$. Then, $P_{\hat{G}}$ admits an α-Hermite-Yang-Mills reduction of the structure group to $\hat{K}$ if and only if P_G admits a Γ-equivariant α-Hermite-Yang-Mills reduction of structure group to K.

Proof. Since $\hat{K}$ meets all the connected components of $\hat{G}$, lemma 5.37 implies that $G/K \cong \hat{G}/\hat{K}$, Moreover, $P_G(G/K)/\Gamma \cong P_{\hat{G}}(\hat{G}/\hat{K})$ implies that there is a bijection between the set of sections of the bundle $P_{\hat{G}}(G/K) \to X$ and the set of Γ-equivariant sections $P_G(G/K) \to Y$ (recall that this in Γ-equivariant ref) ??. Hence, suppose that $P_{\hat{G}} \to X$ admits an α-Hermite-Yang-Mills reduction of the structure group to $\hat{K}$. This reduction is given by a section $\sigma \in \Gamma(X; P_{\hat{G}}(\hat{G}/\hat{K}))$, its Chern connection satisfies $\Lambda_\omega F_A = -\mathrm{i}\,\alpha$. Through the bijection above, the reduction corresponds to a Γ-equivariant section of the Γ-equivariant bundle $P_G(G/K) \to Y$ no sé como seguir la demostración, más que nada porque no acabo de ver como se relaciona la conexión de Chern en $P_{\hat{K}}$ con la conexión de Chern en P_K. Desde luego, la $\Lambda_\omega F_A = -\mathrm{i}\,\alpha$ no es *exactamente la misma* en ambos casos y ya está, porque según si estamos en un fibrado o en otro, tenemos (a ambos lados de la igualdad) secciones de $P_K \to Y$ o de $P_{\hat{K}} \to X$, que obviamente son fibrados distintos. □

With all the work we have done, the proof of the non-connected version of the HK correspondence is just a matter of putting all the pieces together. Observe that the weight of proof is carried by the correspondence for twisted bundles.

Proof of theorem 5.35. First, by proposition 5.36, $\mathcal{P}$ is α-polystable if and only if P_G is Γ-invariantly α-polystable. By theorem 5.39, this is equivalent to the existence of a Γ-equivariant reduction of the structure group to K satisfying $\Lambda_\omega F_A = -\mathrm{i}\,\alpha$, i.e. a Γ-equivariant α-Hermite-Yang-Mills reduction of the structrue group. Finally, by proposition 5.40, this is equivalent to the existence of an α-Hermite-Yang-Mills reduction of the structure group to $\hat{K}$, i.e. a reduction satisfying $\Lambda_\omega F_A = -\mathrm{i}\,\alpha$. □

5.3.4 Moment map picture for the non-connected version

This version of the correspondence also has a moment map interpretation. Let $P_{\hat{G}}$ be a smooth principal $\hat{G}$-bundle, and consider a reduction of the structure group $P_{\hat{K}}$ to a principal $\hat{K}$-bundle. By the Chern-Singer correspondence, each $A \in \mathscr{A}\left(P_{\hat{K}}\right)$ defines a holomorphic structure on $P_{\hat{G}}$. Let $\mathscr{G}$ the gauge group of $P_{\hat{K}}$. Upon completing these spaces with Sobolev norms, $\mathscr{A}\left(P_{\hat{K}}\right)$ and $\mathscr{G}$ are a Banach manifold and a Banach Lie group respectively. Then, given a section $a \in \Gamma(X; P_{\hat{K}} \times_K \mathfrak{z}K)$ satisfying $\mathrm{d}_A a = 0$, the action of $\mathscr{G}$ on $\mathscr{A}\left(P_{\hat{K}}\right)$ given by (5.8) is Hamiltonian, and the associated moment map is

$$\begin{array}{rccc} \mu\colon & \mathscr{A}\left(P_{\hat{K}}\right) & \longrightarrow & \mathrm{Lie}(\mathscr{G}) \\ & A & \mapsto & \Lambda_\omega F_A + \mathrm{i}\,a \end{array}$$

is a moment map (we are idenitifying $\mathrm{Lie}(\mathscr{G})$ and its dual as in remark 5.27). Hence, the moduli space of solutions to the Hermite-Yang-Mills equations for $P_{\hat{G}}$ is given by

$$\mu^{-1}(0)/\mathscr{G}.$$

Problemas/Dudas:

(1) En [18] dice: "Then, by proposition 2.14 $\mathrm{aut}^{ss}(E,\varphi) = H^0(E(\mathfrak{z}))$, so (E,φ) is simple in the sense of Definition 3.8 in [9]". Pero la definición que se da de simple es de un par (A,Φ) simple, no de que el propio par principal (E,φ) sea simple. No he visto la definición de que el propio par principal sea simple en ningún sitio, así que he asumido que quiere decir que tiene una conexión simple.

(2) Esto directamente no se si es verdad.

(3) Este enunciado no aparece explícitamente en el paper, intenta ser un resumen de lo que llamáis descomposición de Jordan-Holder dando los mínimos detalles posibles. No sé si estará bien.

(4) En la sección de espacios de Sobolev (sección 4.2.1) defino esas normas de una manera concreta, pero no estoy seguro de que esté bien. Primero, en la mayoría de sitios donde he visto esto hecho simplemente escriben

$$\langle \psi.\phi \rangle_k = \int_X \sum_{i=1}^{r} \langle \nabla^i \psi, \nabla^i \phi \rangle$$

y no dicen que es ∇^i. En el TFM de Diego he visto que define la norma para formas a valores en un fibrado vectorial cualquiera con una conexión a partir del operador d_∇ como

$$||\alpha||_k := \sqrt{\sum_{i=0}^{r} ||\mathrm{d}_\nabla^i \alpha||_{L_2}^2}.$$

No sé si la definición que he dado yo es equivalente a la anterior para espacios de formas. Lo que hago yo lo he visto en el libro [68]. He mirado el libro de Donalson [15] y yo diría que hace lo mismo que yo, pero es de esos que no parece decir explícitamente a que se refiere con ∇^i.

(5) Esto está sacado de la demostración de [18] pero no entiendo bien ni como se define la sección que yo he llamado ζ y en el paper llamáis $s_{h,\sigma,\chi}$ ni cuál es la sección holomorfa σ_s. A partir de ahora sigo vuestra notación en el paper. Tampoco entiendo bien a qué llamáis en el paper [18] $\overline{\partial}_A$ porque decís que es una estructure holomorfa en el fibrado principal diferenciable subyacente a la reducción E_h, pero citáis el paper de Ignasi [49], y en él $\overline{\partial}_A$ siempre denota un operador de Dolbeault en un fibrado vectorial asociado.

(6) La condición de $c \in \mathrm{i}\,\mathfrak{z}(\mathfrak{g}) \cap \mathfrak{k}$ la he sacado del paper de Ignasi [49]. En vuestro paper sobre grupo de estructura no conexo, se toma $c \in \mathrm{i}\,\mathfrak{z}_K^\Gamma$ donde $\mathfrak{z}_K = \mathfrak{z}(\mathfrak{k})$ es el álgebra de Lie del centro de K. En general, no es cierto que $\mathfrak{z}(\mathfrak{g}) \cap \mathfrak{k} = \mathfrak{z}_K$, aunque sí que tenemos una inclusión $\mathfrak{z}(\mathfrak{g}) \cap \mathfrak{k} \subset \mathfrak{z}_K$. Por otro lado, en su artículo [3], Biswas toma c directamente en $\mathfrak{z}(\mathfrak{g})$, y vosotros en [18], tomáis $c \in \mathfrak{z}_{\mathbb{R}}$ que es el álgebra de Lie del compacto maximal de Z, donde $Z \subset G$ es el centro de G. Mi pregunta es dónde tengo que tomar el elemento c en la ecuación. Lo más "coherente" con lo que hago para el caso no conexo sería tomar $c \in \mathfrak{z}_K = \mathfrak{z}(\mathfrak{k})$, pero como en cada paper me parece que se hace una cosa, prefiero preguntar.

(7) He visto que en varios de los artículos habláis de tomar projecciones ortogonales con respecto a la forma bilineal $\langle \cdot, \cdot \rangle$ en $\mathfrak{g}$. El problema es que en general, una forma bilineal simétrica compleja en un espacio V no induce descomposiciones ortogonales ni aún siendo no degenerada. El criterio para que sí que induzca una descomposición ortogonal en un subespacio W es que su restricción a $W \times W$ sea no degenerada, pero esto no se sigue de que sea no degenerada en V (por ejemplo, en $\mathbb{C}^2$ la forma bilineal dada en la base canónica por la identidad es no degenerada en todo $\mathbb{C}^2$, pero no lo es en $W := \mathrm{span}_{\mathbb{C}}(i, 1)$, de hecho, la restricción de la forma bilineal en cuestión se anula en W).

(8) He puesto esta definición de estabilidad buscando ser un poco coherente con las que ponéis en los papers. Los critetios de estabilidad no aparecen escritos exactamente igual en todos los papers, así que no sé si esta estará bien (también he quitado todas las condiciones relativas a la sección φ en el fibrado adjunto, pero esa parte debería estar bien). Más concretamente, he cogido la definición que dais de estabilidad en el paper sin publicar y en el paper [18]. En otros papers como el de Biswas o el Ignasi las condiciones de estabilidad parecen algo diferentes, y no me parece claro de primeras que sean equivalentes. En alguna reunión que hemos tenido me has dicho que sí que lo son, pero aún así preferiría que me confirmaras que esta definición está bien.

(9) El resultado de que $\mathscr{G}_{k+1}$ tenga una estructura de grupo de Lie de dimensión infinita lo he encontrado solo para G compacto, conexo, semisimple y matricial. En el paper de Mitter [40] dicen que el grupo tiene que ser conexo y compacto aunque yo no he sido capaz de ver donde utilizan la conexión en su prueba (quizás sea alguna propiedad que no sé de la exponencial?). El caso es que lo de que sea un grupo matricial si que me parece muy necesario porque se utiliza muy fuertemente en la construcción. No sé si se suele considerar que todos los grupos con los que se trabaja son matriciales o si hay alguna forma de hacerlo para grupo no matricial, pero yo no la encontré. La verdad es que traté de entender la demostración cuando hice el trabajo y no fui capaz. No tengo nada claro donde necesita todas esas condiciones sobre el grupo de Lie, y no he encontrado más pruebas de que $\mathscr{G}_k$ sea un grupo de Lie de Hilbert en ningún sitio, así que no sé ni siquiera si es cierto cuando eliminamos todas esas hipótesis y solo pedimos que el grupo sea reductivo.

(10) Lo de que cada órbita contenga una conexión $\mathscr{C}^\infty$ lo he visto en el articulo de Atiyah-Bott para superficies de Riemann, y para la acción de $\mathscr{G}^{\mathbb{C}}$ sobre $\mathscr{A}^{(1,1)}$. Para la acción de $\mathscr{G}$ sobre $\hat{\mu}^{-1}(0)$ no sé si es cierto, aunque si queremos que el cociente sea el moduli de conexiones de Einstein-Hermite salvo transformación de gauge, tiene sentido. Por otro lado, el otro día hablamos de que todo lo del moment map y el teorema de Kempf-Ness es *lo que motiva* la correspondencia de Hitchin-Kobayashi, o al menos lo que nos lleva a preguntarnos si dicha correspondencia existencia. La parte de la aplicación momento desde luego que veo como se relaciona con el caso finito-dimensional, al fin y al cabo tenemos una aplicación momento y una acción simpléctica. Lo que no veo claro es la parte algebraica. En el kempf-Ness clásico tenemos también nociones de semiestabilidad y estabilidad, pero más allá de que nosotros damos definiciones de estabilidad poliestabilidad etc para fibrados principales, y decidimos darles esos nombres, no veo como esto se tendría que relacionar con la noción finito-dimensional

de estabilidad. Es decir, no entiendo por qué uno vería las definiciones que hemos dado de estabilidad, semiestabilidad y poliestabilidad y, si se llamasen de otra forma, pensaría que son análogas a las de dimensión finita (definición 1.13).

(11) Que exista un pairing invariante bajo la acción adjunta es algo que afirmáis en el paper pero no dais muchos detalles. He buscado mucho en internet y lo más explícito que he encontrado es una mención a que dichas formas bilineales no degeneradas existes existen en la introducción de este artículo [6]. De todas formas, no sé si está bien porque la acción adjunta de $\hat{G}$ en el centro del álgebra de Lie no tendría que ser trivial en princio si $\hat{G}$ no es conexo. En resumen, no entiendo de donde sale que exista ese pairing.

(12) No tengo nada claro que eso esté bien, pues de momento no sé si es cierta la relación $\chi_\zeta(\mathrm{d}_A\alpha) = \mathrm{d}\chi_\zeta(\alpha)$. Otro problema que noto es que no sé donde hace falta en nada de lo que hago que al caracter sea antidominante. Sé que tiene que ser necesario en algún momento, no sé si para definir el grado o en algún otro sitio.

(13) Pongamos que tenemos la situación usual $1 \to G \to \hat{G} \to \Gamma \to 1$. Dado un elemento $s \in \mathrm{i}\,\mathfrak{k}$, se define el grupo parabólico asociado como

$$\hat{P}_s := \left\{ g \in \hat{G} : \exp(ts)\ g\ \exp(-ts) \text{ está acotado cuando } t \to \infty \right\} \subset \hat{G}$$

y no entiendo con respecto a qué está acotado. Si trabajamos siempre con grupos de matrices, entonces lo vemos en algún $\mathrm{GL}(n,\mathbb{C})$, que es un espacio normado y ahí ya tiene perfecto sentido hablar de acotación (y de límites), pero para grupos de Lie abstractos no acabo de ver como lo hacemos. En el texto, lo he interpretado en términos de representaciones unitarias. Esto no lo he visto dicho explícitamente en ningún sitio, pero en [65] definen las cosas de un modo un poco diferente, pues piden que el límite exista, no que haya acotación. Para interpretar el límite sí que toman representaciones. No se dice que lo que se hace sea independiente de la representación elegida, cosa que, en caso de ser esta la interpretación correcta, entiendo que debe ser cierta.

(14) Imagino que los subgrupos parabólicos tienen que ser grupos de Lie complejos y subgrupos cerrados de G, pues en los papers tomáis secciones holomorfas de fibrados del tipo G/P_s, de modo que G/P_s tiene que ser una variedad compleja, y por tanto P_s debería ser un grupo complejo, pero no he encontrado ningún sitio en que eso esté dicho explícitamente. De hecho, me está constanto mucho encontrar referencias que hablen de subgrupos parabólicos en el contexto de los grupos de Lie. Creo que es cierto que todo grupo reductivo complejo tiene estructure de grupo algebraico, pero en los grupos que tratan con grupos algebraicos como Humprheys [25] hacen todo con la topología de Zariski y no me estoy enterando de como pasar estas propiedades a la topología usual. Por ejemplo, para Humprheys $P \subset G$ es parabólico si y solo si G/P es una variedad proyectiva *completa*. Completa y compacta en principio no es lo mismo: completa quiere decir, según su propia definición, que para toda variedad proyectiva Y la segunda proyección $G/P \times Y \to Y$ es una aplicación cerrada. Según el propio Humprheys, si los espacios son suficientemente buenos y de Hausdorff, y dotamos a todo de la topología producto, entonces esa condición equivale a compacidad, pero él en principio considera todo con la topología de Zariski. Lo mismo en el paper de Ramanathan [47] que me recomendaste en su momento, dice que el hecho de que P sea parabólico es equivalente a que G/P sea completo, no compacto.

(15) De estas airmaciones hay bastantes cosas de las que no estpy seguro. Casi todo está sacado del remark 3.21 del paper en preparación y de la sección 3.3 del paper de Geometría Dedicata [5]. En primer lugar, que $G/H \cong \hat{G}/\hat{H}$ no lo tengo muy claro. Si $\hat{H}$ interseca a todas las componentes conexas de $\hat{G}$ entonces sí (lo tengo probado en el lema 5.37), pero no sé si es cierto. Por otro lado, entiendo que hacemos esto porque queremos que G/H tenga una acción de $\hat{G}$, o equivalentemente, una (θ, c)-twisted (G, Γ) action, para poder aplicar así la proposición B.26 y decir que el fibrado $P(G/H)$ es Γ-equivariante, y que así tenga sentido hablar de secciones Γ-equivariantes (o Γ-invariantes, que creo que son lo mismo por como definís la Γ-acción, pero creo que no se dice explícitamente).

(16) Lo que pongo en ese remark no lo tengo muy claro. Cuando consideramos $\hat{G}$ un grupo reductivo, complexificación de $\hat{K}$, G es la componente conexa de la identidad de $\hat{G}$ y definimos

$K := \hat{K} \cap G$ entiendo que sea compacto, pero no estoy seguro ni de que sea conexo ni de que sea cierto $K^{\mathbb{C}} = G$. Tampoco sé si es muy relevante.

(17) He leído la parte del artículo en que habláis de monodromía y demás, pero no entiendo como deducís que se puede pasar de la ecuación $\Lambda_\omega F_A = -\mathrm{i}\, a$ donde $a \in \Gamma(P_{\hat{K}} \times_{\hat{K}} \mathfrak{k})$ es una sección con $\mathrm{d}_A a = 0$ a la misma ecuación con $\Lambda_\omega F_A = -\mathrm{i}\, \alpha$ y $\alpha \in \mathfrak{z}_K^\Gamma$.

Appendix A

Vector bundles and complex manifolds

This chapter presents a brief introduction to the theory of complex and Kähler manifolds, both of them will constitute the basic frame in which we will work. We also devote part of the chapter to introducing abstract vector bundles. In basic differential geometry, one usually learns about the tangent and cotangent bundles, as well as tensor bundles. These ideas can be generalized to abstract vector bundles, which constitute the basic framework we will need for chapter 2. Since this is understood as preliminary material, we will omit most of the proofs. We refer the reader to the books [35] and [26], for everything related to complex and Kähler manifolds. For the basics of vector bundles, [36, Ch. 10] is a good reference. It only deals with real smooth vector bundles, but most things apply to complex and holomorphic vector bundles with minor modifications. Since we will use i as an index for coordinate expressions very often, we set the notation $\mathrm{i} := \sqrt{-1}$. We introduce in this section the Dolbeault complex, i.e. complex differential forms. To differentiate them from real differential forms, let us set the notation $\Omega^p_{\mathbb{R}}(M)$ for the $\mathscr{C}^\infty(M;\mathbb{R})$-module of p-differential forms on a smooth manifold M.

A.1 Complex manifolds

Let us start by recalling some facts from complex analysis, and introducing some concepts from multivariable complex calculus. Recall that if $f\colon W \to \mathbb{C}$ is a function, where $W \subset \mathbb{C}$ is open, it is *holomorphic* if it has a complex derivative at any point of W, or, equivalently, if, when writing $f = u + \mathrm{i}\, v$, as seeing it as a map from $\mathbb{R}^2$ to $\mathbb{R}^2$, its real and imaginary parts satisfy the *Cauchy-Riemann* equations, i.e $\partial_x u = \partial_y v$ and $\partial_y u = -\partial_x v$. Equivalently, if we introduce the operators

$$\frac{\partial}{\partial z} := \frac{1}{2}\left(\frac{\partial}{\partial x} - \mathrm{i}\,\frac{\partial}{\partial y}\right) \qquad \frac{\partial}{\partial \bar{z}} := \frac{1}{2}\left(\frac{\partial}{\partial x} + \mathrm{i}\,\frac{\partial}{\partial y}\right), \tag{A.1}$$

then f is holomorphic if and only if $\partial_{\bar{z}} f = 0$ and *antiholomorphic* (that is, $\bar{f}$ is holomorphic) if and only if $\partial_z f = 0$. These definitions can be adapted to functions of several complex variables: if $W \subset \mathbb{C}^n$ is open and $f\colon W \to \mathbb{C}^m$ is continuous in W and, for each $p \in W$, the following limit exists

$$\frac{\partial f}{\partial z^j} := \lim_{h\to 0} \frac{f(p^1,\dots,p^j+h,\dots p^n) - f(p^1,\dots p^n)}{h} \tag{A.2}$$

then we say that f is holomorphic in W. The Cauchy-Riemann equations still give a characterization of holomorphicity (but now they have to be satisfied for all coordinates). Extending the definition of (A.1) to several complex variables, we have

$$\frac{\partial f}{\partial z^j} = \frac{\partial f}{\partial x^j} = \frac{1}{\mathrm{i}} \cdot \frac{\partial f}{\partial y^j},$$

which coincides with the limit in equation (A.2) when calculated at a point where f is holomorphic. We will use, when needed, the identification $\mathbb{R}^{2n} \equiv \mathbb{C}^n$ via $(x^1, y^1, \dots, x^n, y^n) \leftrightarrow (x^1 + \mathrm{i}\, y^1, \dots x^n + \mathrm{i}\, y^n)$, and we set $z^j := x^j + \mathrm{i}\, y^j$.

Now, the definition of a complex manifold is the obvious modification of the definition of a smooth manifold:

Definition A.1. A *complex manifold* M is a second countable Hausdorff topological space, endowed with a complex atlas, that is, a family $\mathscr{A}$ of pairs (complex charts) (U, φ) where each $U \subset M$ is open, the open sets cover M, $\varphi\colon U \to \mathbb{C}^n$ is a homeomorphism, and whenever $(U, \varphi), (V, \psi) \in \mathscr{A}$ satisfy, $U \cap V \neq \varnothing$ the maps,

$$\psi \circ \varphi^{-1}\colon \varphi(U \cap V) \to \psi(U \cap V) \qquad \varphi \circ \psi^{-1}\colon \psi(U \cap V) \to \varphi(U \cap V)$$

are holomorphic.

Then, a map between complex manifolds $f\colon M \to N$ is holomorphic, if its *localization* is holomorphic, that is, for all charts (U, φ) and (V, ψ) of M and N respectively, the map $\psi \circ f \circ \varphi^{-1}$ is holomorphic. A holomorphic bijective map whose inverse is also holomorphic is called a *biholomorphism*. Whenever M is a complex manifold, $\mathcal{O}_U$ denotes the ring of all holomorphic functions on the open set U. Observe that all complex manifolds are even-dimensional when considered as real manifolds. Moreover, one may show that complex manifolds are always orientable (see [**35**, Prop. 1.49] for a proof).

The concepts of *submersion, immersion* and *embedding* admit direct generalizations to complex manifolds: a holomorphic map is said to have either of these properties if it has them when considered as a smooth map between the smooth underlying manifolds.

Example A.2. Some examples of complex manifolds are $\mathbb{C}^n$ and its open subset and $\mathbb{CP}^n$. More generally, every open subset of a complex manifold inherits the structure of a complex manifold.

In an analogy with smooth manifolds, we will often say "let $(z_1, \dots z_n)$ be local holomorphic coordinates" of a complex manifold M, meaning we have chosen a complex chart (U, φ), where the components of φ are called z^j.

Remark A.3. The notion of Lie group admits a direct generalization to complex manifolds, which is as follows: A *complex Lie group* is a complex manifold G such that the group structure is compatible with the complex structure, i.e the maps $G \times G \to G, (g, h) \mapsto g \cdot h^{-1}$ and $G \to G, g \mapsto g^{-1}$ are holomorphic. For example, $\mathrm{GL}(n, \mathbb{C})$ is a complex manifold (it can be seen as the open set $\det^{-1}(\mathbb{C}^*) \subset \mathbb{C}^{n^2}$), and the entries of the product and inverse are either polynomial or rational maps, both of which are holomorphic. Of course, in this case, Lie group homomorphisms are required to be *holomorphic* group homomorphism (not merely smooth).

A.2 Vector bundles

We will use both smooth and holomorphic vector bundles, but given how similar these concepts are, we will introduce them simultanously. To do so, we shall mimic the notation used in [**17**]. We will use the symbol $\mathcal{S}$ to denote either smooth or holomorphic functions, sections, maps etc. For example, $\mathcal{S}(M)$ denotes either the set of smooth functions on a manifold, $\mathscr{C}^\infty(M; \mathbb{K})$ ($\mathbb{K} = \mathbb{R}$ or $\mathbb{C}$, depending on the context), or the set of holomorphic functions on a complex manifold, $\mathcal{O}_M$. Similarly, an $\mathcal{S}$-manifold is either a smooth manifold, or a holomorphic one, an $\mathcal{S}$-morphism between $\mathcal{S}$-manifolds, is either a smooth map, or a holomorphic one, and an $\mathcal{S}$-isomorphism is either a diffeomorphism or a biholomorphism. Finally, an $\mathcal{S}$-manifold is either a smooth or complex manifold.

With these ideas in mind, we are ready to define the notion of vector bundle. Vector bundles appear as a generalization of the tangent bundle of a smooth manifold. Intuitively, the tangent bundle of a manifold is obtained by gluing the tangent space to the manifold at a point, T_pM to the corresponding point, p, and then considering the union of all those spaces. Vector bundles try to generalize this concept by replacing the tangent spaces T_pM by arbitrary vector spaces E_p (all of them of the same dimension). Therefore, now we have to glue a vector space E_p to each point p, and consider the union of all those spaces. Of course, the added generality means that now there is not a canonical topology or smooth structure (in general), so those should be given in advance.

Definition A.4 (Vector bundle). Let M be an $\mathcal{S}$-manifold. A real (or complex) $\mathcal{S}$-vector bundle of rank k over M is another $\mathcal{S}$-manifold, E, together with a surjective $\mathcal{S}$-morphism $\pi\colon E \to M$ satisfying the following conditions

1. For each $p \in M$, $E_p := \pi^{-1}(p)$ is a k-dimensional real (or complex) vector space.
2. For each $p \in M$ there is an open neighborhood U of p, and an $\mathcal{S}$-isomorphism $\Phi\colon \pi^{-1}(U) \to U \times \mathbb{R}^k$ (or $\Phi\colon \pi^{-1}(U) \to U \times \mathbb{C}^k$), called a *local trivialization* satisfying the following conditions:
   (a) $\pi_U \circ \Phi = \pi$, where $\pi_U\colon U \times \mathbb{R}^k \to U$ is the usual projection;
   (b) $\Phi|_{E_q}\colon E_q \to \{q\} \times \mathbb{R}^k$ (or $\Phi|_{E_q}\colon E_q \to \{q\} \times \mathbb{C}^k$) is a linear isomorphism.

The space M is often called the *base space*, and E is called the *total space*. The map $\pi\colon E \to M$ is called the *projection*. From now on, the symbol $\mathbb{K}$ will be used to denote either $\mathbb{R}$ or $\mathbb{C}$. Let us make clear that the field may still be $\mathbb{C}$ even when we consider smooth not holomorphic bundles. Indeed, there are a lot of interesting complex smooth vector bundles that do not admit a holomorphic structure such as $T^*_{\mathbb{C}}M$ (see section A.3 below). Finally, observe that we do not restrict ourselves to consider only holomorphic vector bundles over complex manifolds, sometimes we need to study complex smooth bundles over those as well. The dimension of each fiber on a vector bundle is called its *rank*, and it is denoted $\operatorname{rank}(E)$.

Example A.5. Let M be an $\mathcal{S}$-manifold. Then, $M \times \mathbb{K}^r$ is a $\mathbb{K}$-vector bundle over M. The projection is the trivial one, and it has one single trivialization, $\Phi = \mathrm{id}_{M\times\mathbb{K}^r}$.

The previous example is a particular case of what we call a *trivial bundle*. A trivial bundle is a bundle for which there is a global trivialization, that is, a trivialization defined over all of M. This of course implies that if a bundle E over M is trivial, then it is $\mathcal{S}$-isomorphic to $M \times \mathbb{R}^k$ (or $\mathbb{C}^k$ if the bundle was complex). Once we define the right notion of morphism between vector bundles, we will see that this in fact implies that E and $M \times \mathbb{R}^k$ are isomorphic.

The maps $\tau\colon U \cap V \to \mathrm{GL}(k, \mathbb{K})$ are called *transition functions*, even though, strictly speaking, they are not functions, but maps whose codomain is $\mathrm{GL}(n, \mathbb{K})$.

It is often the case that we want to create a vector bundle by gluing a family of vector spaces we choose to each point of a manifold. However, this alone does not provide the union with the structure of an $\mathcal{S}$-vector bundle, as we still have to define a topology and an $\mathcal{S}$-structure on the union. If we were to this directly every time, we would have to define the topology, the $\mathcal{S}$-structure and the trivializations, and verify all the requirements of the definition of vector bundle. The following lemma provides a shortcut; if we fix the trivializations, and they have transition functions that are $\mathcal{S}$-morphisms, then both the topology and the $\mathcal{S}$-structure are determined unequivocally.

Before stating the theorem, let us briefly recall that if we are given an index set A, and a family of sets indexed by A, $\{W_a\}_{a\in A}$, the disjoint union is defined as

$$\bigsqcup_{a\in A} W_a := \bigcup_{a\in A} \{a\} \times W_a. \tag{A.3}$$

In particular, elements of the disjoint union are pairs (a, w) such that $w \in W_a$. Under some abuse of notation, it is quite common to omit the indexing element, and not write elements of the disjoint union.

Lemma A.6 (Vector bundle chart lemma). Let M be an $\mathcal{S}$-manifold, and suppose we are given a vector space E_p for each $p \in M$, all of them of the same dimension r. Define $E := \bigsqcup_{p\in M} E_p$, and let $\pi\colon E \to M, (p,v) \mapsto v$. Suppose as well that we are given the following data:

1. an open cover $\{U_\alpha\}_{\alpha\in A}$ of M,
2. for each $\alpha \in A$, a bijection $\Phi_\alpha\colon \pi^{-1}(U_\alpha) \to U_\alpha \times \mathbb{K}^r$ that restricts to a vector space isomorphism $\Phi_\alpha|_{E_p}\colon E_p \to \{p\} \times \mathbb{K}^r$ for each $p \in U_\alpha$,
3. for each pair $\alpha, \beta \in A$ such that $U_\alpha \cap U_\beta \neq \varnothing$, an $\mathcal{S}$-morphism, $\tau_{\alpha\beta}\colon U_\alpha \cap U_\beta \to \mathrm{GL}(k,\mathbb{K})$, such that
$$\begin{array}{rccc} \Phi_\alpha \circ \Phi_\beta^{-1}\colon & (U_\alpha \cap U_\beta) \times \mathbb{K}^r & \longrightarrow & (U_\alpha \cap U_\beta) \times \mathbb{K}^r \\ & (p,v) & \mapsto & (p, \tau(p)v). \end{array} \tag{A.4}$$

Then, E has a unique topology and smooth structure making it into an $\mathcal{S}$-manifold, and a rank r $\mathcal{S}$-vector bundle over M such that (U_α, Φ_α) is a trivialization for every $\alpha \in A$.

Proof. See [36, Lemma 10.6] □

Example A.7. Lemma A.6 is very useful because it allows to create a wide range of vector bundles out of existing ones. More precisely, it can be used to "elevate" a lot of constructions that can be done with vector spaces, such as sums, tensor products etc, to bundles. We will present the explicit construction of the tensor product bundle. All other bundles in this example can be constructed by similar procedures.

1. Let us show how the *tensor product bundle* is defined. Let E, F be two $\mathcal{S}$-vector bundles over the same base space M, of respective ranks r and s. We want to define another vector bundle whose at each point is $E_p \otimes F_p$. To do so, we can use lemma A.6 as follows. Define $E \otimes F := \bigsqcup_{p\in M} E_p \otimes F_p$ and the obvious projection. For the open cover, consider all intersections $U \cap V$ such that (U,Φ), (V,Ψ) are local trivializations of $E \to M$ and $F \to M$ respectively. Define trivializations $\Gamma_{\Phi,\Psi}$ for $E \otimes F$ by linear extensions in the fibers of
$$\begin{array}{ccc} \pi^{-1}(U \cap V) & \longrightarrow & (U \cap V) \times \mathbb{K}^{r\cdot s} \\ (p,v,w) & \mapsto & (p, \pi_r(\Phi(p,v)), \pi_s(\Psi(p,v))), \end{array}$$
where $\pi_r\colon U \times \mathbb{K}^r \to \mathbb{K}^r$ is the usual projection. These are bijections because $\mathbb{K}^{r\cdot s} \cong \mathbb{K}^r \otimes \mathbb{K}^s \cong E_p \otimes F_p$, and $\Gamma_{\Phi,\Psi}$ restricts to an isomorphism on each fiber by definition. Only the third condition of lemma A.6 needs to be verified. Since E and F are $\mathcal{S}$-vector bundles over M, if we are given (U_1,Φ_1) and (U_2,Φ_2) two local trivializations such that $U_1 \cap U_2 \neq \varnothing$, then there is an $\mathcal{S}$-morphism $\tilde{\tau}\colon U_1 \cap U_2 \to \mathrm{GL}(k,\mathbb{K}^r)$ satisfying the condition of equation (A.4) (see lemma [**intro_leeSM**]), that is,
$$\left(\Phi_1 \circ \Phi_2^{-1}\right)(p,v) = (p, \tau(p)v)$$
and the same applies for two local trivializations (V_1,Ψ_1) and (V_2,Ψ_2) such that $V_1 \cap V_2 \neq \varnothing$ and a map $\hat{\tau}$. Then, if $(U_1 \cap U_2) \cap (V_1 \cap V_2) \neq \varnothing$, we have
$$\Gamma_{\Phi_1\Psi_1} \circ \Gamma^{-1}_{\Phi_2\Psi_2}(p,u) = (p, [\tau(p) \otimes \hat{\tau}(p)]\,(u)),$$
where $\tau(p) \otimes \hat{\tau}(p)$ is the *tensor product of linear maps*, i.e. the linear extension of the map acting on decomposable elements as $[\tau(p) \otimes \hat{\tau}(p)]\,(v \otimes w) = [\tau(p)v] \otimes [\hat{\tau}(p)w]$ (this map is well-defined by the universal property of tensor products). Indeed,
$$\begin{aligned} \Gamma_{\Phi_1\Psi_1} \circ \Gamma^{-1}_{\Phi_2\Psi_2}(p, v\otimes w) &= \Gamma_{\Phi_1\Psi_2}(p, \Phi_2^{-1}(p,v) \otimes \Psi_2^{-1}(p,v)) \\ &= \Gamma_{\Phi_1\Psi_1}(p, \pi_r(\Phi_1 \circ \Phi_2^{-1}(p,v)) \otimes \pi_2(\Psi_1 \circ \Psi_2(p,w))) \\ &= (p, [\tau(p)v] \otimes [\hat{\tau}(p)w]) \end{aligned}$$
and by linear extension, we see that this works for every vector in $\mathbb{K}^s \otimes \mathbb{K}^r$. Finally, $\tau \otimes \hat{\tau}\colon U \cap V \to \mathrm{GL}(r,\mathbb{K}) \otimes \mathrm{GL}(s,\mathbb{K}) \cong \mathrm{GL}(r\cdot s,\mathbb{K})$, and it is an $\mathcal{S}$-morphism (this is essentially due to the way the tensor product of matrices is defined see [16, Section 11.2]).
2. Given a vector bundle E, its *dual vector bundle*, E^* is the vector bundle whose fiber at each point $p \in M$ is E_p^*. Again, one uses lemma A.6 and checks that the transition functions for this bundle are the transpose inverse of those of E.

3. Given two vector bundles, E and F, $\mathrm{Hom}(E,F)$ is the bundle whose fiber at each point is the vector space of linear homomorphisms $\mathrm{Hom}(E_p,F_p)$. When $E=F$, we denote $\mathrm{Hom}(E,E)=\mathrm{End}(E)$.

4. Given two vector bundles E and F one may construct its *Whitney sum*, i.e the vector bundle $E\oplus F$ whose fiber at each point $p\in M$ is $E_p\oplus F_p$. The construction follows the same ideas as the tensor product. Nonetheless, it is done explicitly in [36, Example 10.7]

5. Let E be a complex vector bundle, we may define its *complex conjugate vector bundle*, $\bar{E}$ as follows. By definition, the fibers of $\bar{E}$ are the complex conjugate vector spaces to the fibers of E. More precisely, for each $x\in E$, $\bar{E}_x$ coincides with E_x as a real vector space, but we see it as a complex vector space by defining its product by a complex number as follows

$$\lambda *_{\bar{E}} v : \overline{\lambda}\cdot v,$$

   where the second product is the usual product by a complex number in E_x. This turns $\bar{E}_x$ into a complex vector space, and the set $\sqcup_{x\in M}\{x\}\times\bar{E}_x$ with the obvious projection into a smooth complex vector bundle over M.

6. The quotient of two $\mathcal{S}$-bundles in the bundle whose fiber at each point is the quotient of fibers.

There is a natural notion of subbundle.

Definition A.8 (Subbundle). Let $\pi\colon E\to M$ be an $\mathcal{S}$-bundle over an $\mathcal{S}$-manifold M. An $\mathcal{S}$-subbundle is an embedded submanifold $D\subset E$ such that $D_p:=D\cap E_p$ is a vector subspace of E_p for each $p\in M$, all of them of the same dimension, together with the restriction $\pi|_D\colon D\to M$, such that $\pi|_D\colon D\to M$ is a vector bundle.

Finally, there is natural notion of pullback as well. Consider two $\mathcal{S}$-manifolds M and N, with an $\mathcal{S}$-morphism $\varphi\colon M\to N$, and let $E\to N$ be an $\mathcal{S}$-vector bundle. Then, we may define the *pullback vector bundle*

$$\varphi^*E:=\{(m,v)\in M\times E : \varphi(m)=\pi(v)\}. \tag{A.5}$$

This has a natural vector bundle structure with the projection given by $(m,v)\mapsto m$. The fiber over each point is $\{m\}\times E_{\varphi(m)}$, and applying the vector bundle chart lemma (lemma A.6) we may see that it is an $\mathcal{S}$-vector bundle over M. See [35, Page 75] for details.

A.2.1 Local and global sections

In the usual tangent bundle, vector fields are thought of as maps sending each point to a vector and varying smoothly. More formally, they are sections of TM. The analogous concept for vector bundles is that of a section.

Definition A.9 (Section). Let $\pi\colon E\to M$be an $\mathcal{S}$-bundle. A *global section* of E is a continuous map $\sigma\colon M\to E$ such that $\pi\circ\sigma=\mathrm{id}_M$. A local section is a continuous map defined on an open set $U\subset M$, $\sigma\colon U\to M$ such that $\pi\circ\sigma=\mathrm{id}_U$. $\mathcal{S}$-sections are sections that are $\mathcal{S}$-morphisms.

Since we will usually work with $\mathcal{S}$-sections, the word section will mean $\mathcal{S}$-section unless otherwise stated. Following the notation in [36], a *rough section* is a section that needs not be continuous. We will denote $\Gamma(M;E)$ and $\Gamma(U;E)$ the set of global and local smooth sections on a smooth manifold, and $\mathcal{O}(U;E)$ and $\mathcal{O}(M;E)$ the corresponding sets for holomorphic sections. We make the distinction now because we will need it later. However, for the basic definitions and properties, there is no difference between smooth and holomorphic sections, so we will use $\mathscr{E}(M;E)$ and $\mathscr{E}(U;E)$ to denote either of them. We will use the notations $\sigma(p)$ or σ_p interchangeably to denote the value of the section at $p\in M$.

Observe that $\mathscr{E}(M;E)$ is a $\mathbb{K}$-vector space under pointwise addition and scalar multiplication: if $\lambda,\mu\in\mathbb{K}$ and $\sigma_1,\sigma_2\in\mathscr{E}(M;E)$, then we can define

$$(\lambda\sigma_1+\mu\sigma_2)(p):=\lambda\sigma_1(p)+\mu\sigma_2(p).$$

The sum takes place on each E_p, so it is well-defined, and the result is still a section. Observe that we get an $\mathcal{S}$-section. Indeed, the map is an $\mathcal{S}$-morphism because we may always compose it with a trivialization (U, Φ). In that case, since Φ is linear isomorphism on each E_p, we get

$$\Phi(\lambda\sigma_1 + \mu\sigma_2) = (\mathrm{id}_U, \lambda\Phi(\sigma_1) + \mu\Phi(\sigma_2)).$$

This is an $\mathcal{S}$-morphism because each component is, and since Φ is an $\mathcal{S}$-isomorphism, we conclude that $\lambda\sigma_1 + \mu\sigma_2$ is an $\mathcal{S}$-morphism as well. Similarly, we can show that $\mathscr{E}(M; E)$ is an $\mathcal{S}(M)$-module. Again, the product of a function $f \in \mathcal{S}(M)$ and an $\mathcal{S}$-section is defined pointwise; $(f\sigma)(p) := f(p)\cdot\sigma(p)$, and an analogous procedure to the one we just did shows the claim.

As happens with vector fields, it is very convenient to have a set of sections that span the vector space E_p at each point p on some open set. This motivates the definition of local and global frames.

Definition A.10 (Frame). Let $\pi\colon E \to M$ be a rank r $\mathcal{S}$-bundle. A set of r global (resp. local on U) $\mathcal{S}$-sections $s_1, \dots s_r \in \mathscr{E}(M; E)$ form a *global frame* (resp. *local frame* on U) if for each $p \in M$ (resp. $p \in U$), $\{s_1(p), \dots s_r(p)\}$ is a basis of E_p.

Example A.11.

1. Let M be an $\mathcal{S}$-bundle, and consider a trivial bundle $E = M \times \mathbb{K}^r$. The standard basis $\{e_1, \dots e_n\}$ on $\mathbb{K}^r$ yields a global frame as follows: define $\tilde{e}_i\colon M \to M \times \mathbb{K}^r, p \mapsto (p, e_i)$. This clearly defines $\mathcal{S}$-morphism, and since $E_p = \{p\} \times \mathbb{K}^r$, $\{\tilde{e}_1(p), \dots \tilde{e}_n(p)\}$ is a basis for each $p \in M$.
2. Let $\pi\colon E \to M$ be a rank r $\mathcal{S}$-vector bundle, and let (U, Φ) be a local trivialization. Define $s_i := \Phi^{-1} \circ \tilde{e}_i$ for $1 \le i \le r$. Then, $\{s_1, \dots s_r\}$ is a local frame for E over U.

The local frames defined in the previous example are all possible local frames, as shown by the following proposition.

Proposition A.12 (Local frames associated with local trivializations). Let $\pi\colon E \to M$ be an $\mathcal{S}$-vector bundle. Then, there is a one-to-one correspondence between local frames and local trivializations.

Proof. Given a local trivialization (U, Φ), we define sections as in example A.11. Reciprocally, given a local frame $(s_1, \dots s_r)$, a trivialization sends $\lambda_1 \cdot s_1(p) + \dots + \lambda_r \cdot s_r(p)$ to $(p, (\lambda_1, \dots \lambda_r))$. Details may be found in [**36**, Prop. 10.19]

□

Changes of coordiantes with respect to local trivializations induce changes of coordinates with respect to local frames as follows.

Proposition A.13. Let $(s_1, \dots s_r)$ and $(\tilde{s}_1, \dots \tilde{s}_r)$ be two $\mathcal{S}$-local frames defined on overlapping open domains. Then, there are $\mathcal{S}$-morphisms τ_k^j such that $\tilde{s}_k = \tau_k^j \cdot s_j$ and the matrix $\tau = (\tau_k^j)_{1 \le j,k \le r}$ is non-singular.

Proof. By proposition A.12, the frames $(s_1, \dots s_r)$ and $(\tilde{s}_1, \dots \tilde{s}_r)$ are associated to local trivializations (U, Ψ) and (V, Φ) respectively. U and V are the definition domains of the frames, so $U \cap V \neq \varnothing$. By lemma [**36**, Lemma 10.5] there is an $\mathcal{S}$-morphism $\tau\colon U \cap V \to \mathrm{GL}(k, \mathbb{K})$ such that $\Phi \circ \Psi^{-1}(p, v) = (p, \tau(p)v)$. Since each frame defines a basis for each fiber, there are functions λ_k^j satisfying $\hat{s}_k = \lambda_k^j \cdot s_j$. Now, applying the trivializations, we get

$$\tilde{e}_k = (\mathrm{id}_{U\cap V}, e_k) = \Psi(\tilde{s}_k) = \Psi(\lambda_k^j \cdot s_j) = \Psi \circ \Phi^{-1}(\mathrm{id}_{U\cap V}, \lambda_k^j \cdot e_j) = (\mathrm{id}_{U\cap V}, \lambda_k^j \cdot \tau^{-1} \cdot e_j)$$

hence, $\lambda_k^j \cdot \left(\tau^{-1}\right)_j^i = \delta_k^i$, so $\lambda_k^j = \tau_k^j$, which is an $\mathcal{S}$-morphism. □

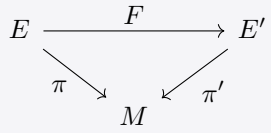
Finally, let us study frames on the dual bundle E^* of an $\mathcal{S}$-bundle E. Recall that every basis $\{e_1, \dots e_n\}$ of a finite-dimensional vector space admits a dual basis $\{e_1^*, \dots e_n^*\}$ defined by $e_i^*(e_j) = \delta_{ij}$. We seek a similar construction for vector bundles: given a local frame $\{s_1, \dots s_r\}$ on E, we look for a dual frame $\{t_1, \dots t_n\}$ on E^* such that $t^j(s_i) = \delta_i^j$. Of course, the fact that the sections t^j exist is clear, we can just define rough sections that define the dual basis to $\{s_1(p), \dots s_r(p)\}$ at each fiber.

A.2.2 Bundle homomorphisms

Since the key idea of vector bundles is to attach a vector space to each point of a manifold, bundle morphisms to respect that linear structure, i.e. restrict to linear maps between fibers.

Definition A.14 (Vector bundle morphism). Let $\pi\colon E \to M$ and $\pi'\colon E' \to M$ be two $\mathcal{S}$-vector bundles over an $\mathcal{S}$-manifold M. A morphism of $\mathcal{S}$-vector bundles, F, is an $\mathcal{S}$-morphism $F\colon E \to E'$ satisfying:

1. $\pi' \circ F = \pi$
2. $F|_{E_p}\colon E_p \to E'_{F(p)}$ is a linear map.



Of course, we will say that $F\colon E \to E'$ is an $\mathcal{S}$-vector bundle isomorphism if it is bijective and its inverse also is a bundle morphism. In that case, we say that E and E' are isomorphic.

Remark A.15.

1. Given a vector bundle E, the identity map $\mathrm{id}_E\colon E \to E$ is trivially a vector bundle morphism (an isomorphism in fact).

2. Observe that an $\mathcal{S}$-bundle over M is trivial if and only if it is isomorphic to $M \times \mathbb{K}^r$. Indeed, an isomorphism $F\colon M \times \mathbb{K}^r \to E$ is a global trivialization. Just observe that F is bijective because it is an isomorphism, and it is an $\mathcal{S}$-morphism and commutes with the projections by definition A.14, so it satisfies all the properties of the definition of a global trivialization. Reciprocally, any global trivialization will define a vector bundle isomorphism using the same reasoning.

3. The set of bundle homomorphisms $E \to E'$ can be identified with the space of sections of the vector bundle $\mathrm{Hom}(E, E')$. Indeed, each bundle morphism $F\colon E \to E'$ defines a section $\hat{F} \in \mathscr{E}(M; \mathrm{Hom}(E, E'))$ by setting $\hat{F}_p := F|_{E_p}$. Reciprocally, a section $\hat{F}$ defines a vector bundle morphism F by setting $F(v) := \hat{F}_p(v)$ if $v \in E_p$. As a particular case of this construction, there is a vector bundle isomorphism $\mathrm{End}(E) \cong E^* \otimes E$.

Example A.16.

1. A well-known example comes from riemannian geometry: if M is a riemannian geometry, the *musical isomorphims* define a vector bundle isomorphism between TM and T^*M. Analogously, a symplectic form defines a bundle isomorphim between the same vector bundles.

2. Given two $\mathcal{S}$-vector bundles E, E' over the same $\mathcal{S}$-manifold, M, we have an $\mathcal{S}$-bundle isomorphism $\mathrm{Hom}(E, E') \cong E^* \otimes E'$. This eventually comes down to the fact that, for every pair of finite-dimensional $\mathbb{K}$-vector spaces V and W, there is an isomorphism $\mathrm{Hom}(V, W) \cong V^* \otimes W$. Indeed, define $\Phi\colon V^* \times W \to \mathrm{Hom}_{\mathbb{K}}(V, W), (\alpha, w) \mapsto \phi^w_\alpha$, where $\phi^w_\alpha(v) := \alpha(v) \cdot w \in W$. Φ is clearly bilinear, so by the universal property of the tensor product, it descends to a linear map $\hat{\Phi}\colon V^* \otimes W \to \mathrm{Hom}_{\mathbb{K}}(V, W)$. Since both spaces are finite-dimensional, $\dim_{\mathbb{K}}(V^* \otimes W) = \dim_{\mathbb{K}}(\mathrm{Hom}_{\mathbb{K}}(V, W))$, so it suffices to show that $\hat{\Phi}$ is surjective. Indeed, choose a basis $\{e_1, \dots e_n\}$ of V, and take its dual basis $\{e_1^*, \dots e_n^*\}$. Choose $\phi \in \mathrm{Hom}_{\mathbb{K}}(V, W)$, and let $w_i := \phi(e_i)$. Then, $\phi = \hat{\Phi}(\sum_{i=1}^n e_i^* \otimes w_i)$.

A.2.3 Subbundles

Subbundles are a way to formalize the natural notion of "taking a subspace of a vector bundle" that one would come up with. More precisely, if we are to take a reasonable subset of a vector bundle, and we want to still call it a bundle, the first reasonable think to ask is that the intersection of the subset we have taken and each of the fibers of the original vector bundle is a vector subspace of the fiber. Since we want to make a bundle out of it, we must ask for all the so defined vector subspaces so have the same dimension. Furthermore, since we work with vector bundles that are $\mathcal{S}$-manifolds, we should ask subbundles to be embedded $\mathcal{S}$-submanifolds.

Definition A.17 (Subbundle). Let $\pi\colon E \to M$ be an $\mathcal{S}$-bundle over an $\mathcal{S}$-manifold M. An $\mathcal{S}$-subbundle is an embedded submanifold $D \subset E$ such that $D_p := D \cap E_p$ is a vector subspace of E_p for each $p \in M$, all of them of the same dimension, together with the restriction $\pi|_D\colon D \to M$, such that $\pi|_D\colon D \to M$ is a vector bundle.

The following provides a very useful way to recognize subbundles.

Lemma A.18 (Local frame criterion for subbundles). Let $\pi\colon E \to M$ be a rank r, $\mathcal{S}$-bundle, and suppose that, for each $m \in M$, we are given an ℓ-dimension subspace $D_m \subset E_m$. Then, $D = \cup_{m\in M} D_m \subset E$ is an $\mathcal{S}$-subbundle if and only if each point $m \in M$ has a neighborhood U on which there are $\mathcal{S}$-local sections $\{\sigma_1, \dots \sigma_\ell\}$ such that $\{\sigma_1(m), \dots \sigma_\ell(p)\}$ is a basis of D_m.

Proof. See [36, Lemma 10.32] for a proof in the smooth category, which can be adpated verbatim to the holomorphic category. □

A.3 The complexified Tangent and Cotangent Bundles

Observe that under the identification $\mathbb{R}^{2n} \equiv \mathbb{C}^n$, every complex manifold is a smooth manifold. Hence, it makes sense to speak about vector fields and covector fields (one-forms) in any complex manifold. However, we would like to extend those definitions to complex vector fields and one-forms (we will introduce higher-order complex differential forms later on).

A.3.1 Complexification of real vector spaces

Firstly, given a real vector space V, we may always consider its *complexification*, which is the complex vector space $V_{\mathbb{C}} := V \otimes_{\mathbb{R}} \mathbb{C}$. Its elements can always be written as $u \otimes 1 + v \otimes \mathrm{i}$, and we shall omit the tensor product and simply write $u + \mathrm{i}\, v \in V_{\mathbb{C}}$. Every $\mathbb{R}$-linear map $f\colon V \to W$ defines a $\mathbb{C}$-lineal map $f_{\mathbb{C}}\colon V_{\mathbb{C}} \mapsto W_{\mathbb{C}}$ (which we will usualy denote f as well) by linear extension of $f_{\mathbb{C}}(u \otimes \alpha) := (f(u)) \otimes \alpha$ for $\alpha \in \mathbb{C}$. Complexifications are endowed with natural conjugation maps $\overline{u + \mathrm{i}\, v} := u - \mathrm{i}\, v$. This allows us to speak about real vectors, which are those $w \in V_{\mathbb{C}}$ for which $\overline{w} = w$. Obsserve that if we set $(V_{\mathbb{C}})^{\mathbb{R}}$ the underlying real subspace, then $(V_{\mathbb{C}})^{\mathbb{R}} \neq V$ (for example, $\mathbb{R}_{\mathbb{C}} \cong \mathbb{C}$, but $\mathbb{C}^{\mathbb{R}} \cong \mathbb{R}^2$). Moreover, one has $(V_{\mathbb{C}})^{\mathbb{R}} = V \oplus \mathrm{i}\, V$. We usually write $V_{\mathbb{C}} = V \oplus \mathrm{i}\, V$ making some abuse of notation, but the right-hand side is not a direct sum decomposition of the left-hand side as complex vector spaces.

A.3.2 Complexification of real vector bundles

Let E be a real vector bundle over a smooth manifold M. The *complexification* of E is the vector bundle $E_{\mathbb{C}} := E \otimes \mathbb{C}$. More precisely, we consider the tensor product of the real vector bundle E and the trivial bundle $M \times \mathbb{C}$ (seen as a real vector bundle). Then, $E \otimes (M \times \mathbb{C})$ defined as above is a real vector bundle whose fiber at each $p \in M$ is just $E_p \otimes_{\mathbb{R}} \mathbb{C}$, that is, $(E_p)_{\mathbb{C}}$. Thus, $E_{\mathbb{C}}$ is endowed with the structure of a complex vector bundle. Observe that it is smooth, but not necessarily holomorphic. To sum up, $E_{\mathbb{C}}$ is the complex vector bundle whose fiber at each point is the complexification of the fiber of the original bundle (see [35, Ch. 1] for a more explicit construction).

The *complexified tangent and cotangent bundles* are the complexifications of TM and T^*M respectively. They are denoted $T_{\mathbb{C}}M$ and $T^*_{\mathbb{C}}M$ respectively. Complex vector and covector fields are just sections of these bundles. Observe that $T_{\mathbb{C}}M$ can be interpreted as a very natural object. Recall from basic differential geometry, that for a smooth manifold, T_pM is the vector space of derivations acting on (germs of) smooth real-valued maps. Then, there is a complex vector space isomorphism between $T_pM \otimes_{\mathbb{R}} \mathbb{C}$ and the space of $\mathbb{C}$-linear derivations acting on germs of complex-valued smooth maps revisar. What that means for practical purposes is that, given $Z = X + \mathrm{i}\, Y \in \Gamma(M; T_{\mathbb{C}}M)$ and $f = u + \mathrm{i}\, v \in \mathscr{C}^\infty(M; \mathbb{C})$, we have

$$Z(f) = (X(u) - Y(v)) + \mathrm{i}\, (Y(u) + X(v)).$$

We can define a Lie bracket on $\Gamma(M; T_{\mathbb{C}}M)$ by complex-linear extension of the original Lie bracket on TM:

$$[X_1 + \mathrm{i}\, Y_1, X_2 + \mathrm{i}\, Y_2] := \big([X_1, X_2] - [Y_1, Y_2]\big) + \mathrm{i}\, \big([Y_1, X_2] + [X_1, Y_2]\big),$$

which turns $(\Gamma(M;T_{\mathbb{C}}M),[\cdot,\cdot])$ into a Lie algebra.

Similarly, elements of $\Gamma(M;T^*_{\mathbb{C}}M)$ are called complex one-forms. Since $V^*_{\mathbb{C}}$ is naturally isomorphic to $(V^*)_{\mathbb{C}}$ (see [26, Lemma 1.2.6]), for each $p\in M$, we have that the fiber of $T^*_{\mathbb{C}}M$ at p, $T^*_pM\otimes_{\mathbb{R}}\mathbb{C}$ is naturally isomorphic to $(T_pM\otimes_{\mathbb{R}}\mathbb{C})^*$. In the language of vector bundles, this means that $T^*_{\mathbb{C}}M\cong \mathrm{Hom}_{\mathbb{C}}(T_{\mathbb{C}}M,\mathbb{C})$. Then the action of a complex one form $\alpha+\mathrm{i}\,\beta$ on a complex vector field $X+\mathrm{i}\,Y$ is just given by $(\alpha+\mathrm{i}\,\beta)(X+\mathrm{i}\,Y)=(\alpha(X)-\beta(Y))+\mathrm{i}\,(\alpha(Y)+\beta(X))$. Any $f\in\mathscr{C}^\infty(M;\mathbb{C})$ defines now a complex one-form $\mathrm{d}f:=\mathrm{d}u+\mathrm{i}\,\mathrm{d}v$ (we will generalize the exterior differential when we introduce higher order complex differential forms).

A.4 Complex coordinate frames

Let M be a complex manifold and choose holomorphic coordinates $(z^1,\dots z^n)$. Each coordinate map z^j can be decomposed into its real and imaginary part x^j and y^j, thus we get real coordinates $(x^1,y^1,\dots,x^n,y^n)$. This yields a local smooth frame of TM given by $\{\partial/\partial x^1,\partial/\partial y^1,\dots\partial/\partial x^n,\partial/\partial y^n\}$. Defining $\partial/\partial z^j$ and $\partial/\partial\bar z^j$ as in equation (A.1), we obtain a new local frame for $T_{\mathbb{C}}M$ which is given by

$$\left\{\frac{\partial}{\partial z^1},\frac{\partial}{\partial\bar z^1},\dots\frac{\partial}{\partial z^n},\frac{\partial}{\partial\bar z^n}\right\}.$$

A function $f\in\mathscr{C}^\infty(M;\mathbb{C})$ is holomorphic if and only if $\partial f/\partial\bar z^j=0$ and antiholomorphic if and only if $\partial f/\partial z^j=0$ for $1\le j\le n$ (see [35, Prop. 1.4.2]). Analogously, a local coframe for the complexified cotangent bundle is given by $\{\mathrm{d}z^1,\dots\mathrm{d}z^n,\mathrm{d}\bar z^1,\dots\mathrm{d}\bar z^n\}$, where we have defined ($\mathrm{d}x^j$ and $\mathrm{d}y^j$ are extended by complex linearity)

$$\mathrm{d}z^j:=\mathrm{d}x^j+\mathrm{i}\,\mathrm{d}y^j\qquad \mathrm{d}\bar z^j:=\mathrm{d}x^j-\mathrm{i}\,\mathrm{d}y^j\qquad 1\le j\le n.$$

Then, we may express $\mathrm{d}f\in\Gamma(M;T^*_{\mathbb{C}}M)$ in either frame

$$\mathrm{d}f=\frac{\partial f}{\partial x^j}\cdot\mathrm{d}x^j+\frac{\partial f}{\partial y^j}\cdot\mathrm{d}z^j=\frac{\partial f}{\partial z^j}\cdot\mathrm{d}z^j+\frac{\partial f}{\partial\bar z^j}\cdot\mathrm{d}\bar z^j. \tag{A.6}$$

Observe that the complexified bundles may be defined for smooth manifolds as well, but the local frames defined above only make sense for complex manifolds (because they require holomorphic coordinates).

Finally, consider two complex manifolds, M and N, and let $F\colon M\to N$ be a smooth map. From real differential geometry, we know that we mat consider its differential at a point $m\in M$, which is an $\mathbb{R}$-linear map $\mathrm{d}_mF\colon T_mM\to T_{F(m)}N$. This map can be extended by complexification $\mathrm{d}_mF_{\mathbb{C}}\colon (T_mM)_{\mathbb{C}}\to(T_{F(m)}N)_{\mathbb{C}}$. We will usually denote it by the same symbol d_mF. If we now write $D'F(p)=(\partial F^k/\partial z^j)$, holomorphicity is characterized by the fact that, holomorphic map are those whose differential has the following matrix form

$$\begin{pmatrix} D'F(p) & 0\\ 0 & \overline{D'F(p)}\end{pmatrix}$$

with respect to any pair of holomorphic frames (associated to holomorphic coordinates of the manifold). See [35, Ch. 1] for more details.

A.5 Almost complex structures

Let V be a real vector space. An endomorphism $J\colon V\to V$ is called an *almost complex structure* if $J^2:=J\circ J=-\mathrm{id}_V$. Almost complex structures receive their name because, when they exist, they endow V with the structure of a complex vector space. Indeed, for $a+\mathrm{i}\,b\in\mathbb{C}$ and $v\in V$ their product can be defined as $(a+\mathrm{i}\,b)*v:=a\cdot v+b\cdot J(v)$. As seen in proposition A.19 bellow, only even-dimensional real vector spaces can have almost complex structures.

Given a real vector space endowed with an almost complex structure, the fact that $J^2=-\mathrm{id}_V$ means that J has no real eigenvalues. If we extend J to the complexified space, then the only

possible eigenvalues of J are i and $-\mathrm{i}$. If we use the symbol J acting on vectors of a complexified vector space, we will always understand that we are working with the complexified map.

Proposition A.19. Let J be a complex structure on a real vector space V. Then $V_{\mathbb{C}}$ has a complete eigenspace decomposition $V_{\mathbb{C}} = V^{(1,0)} \oplus V^{(0,1)}$, where $V^{(1,0)}$ is the i-eigenspace and $V^{(0,1)}$ is the $(-\mathrm{i})$-eigenspace. Moreover, the decomposition is given, for $w \in V_{\mathbb{C}}$ by

$$w^{(1,0)} = \frac{1}{2} \cdot (w - \mathrm{i}\, J(w)) \qquad w^{(0,1)} = \frac{1}{2} \cdot (w + \mathrm{i}\, J(w)).$$

If $\dim_{\mathbb{R}} V < \infty$, then $\dim_{\mathbb{C}} V^{(1,0)} = \dim_{\mathbb{C}} V^{(0,1)}$, and therefore, $\dim_{\mathbb{C}} V_{\mathbb{C}} = \dim_{\mathbb{R}} V$ is even.

Proof. See [35, Prop. 1.52]. □

Remark A.20. Let V be an even-dimensional real vector space endowed with an almost complex structure J_V. We have seen that this induces on V the structure of a complex vector space. On the other hand, the complexification $V_{\mathbb{C}}$ induces another linear complex structure on V. *These structures are not the same!* Indeed, $\dim_{\mathbb{C}} V_{\mathbb{C}} = \dim_{\mathbb{R}} V = 2 \cdot \dim_{\mathbb{C}}(V, J_V)$. Another way of seeing this is that for $v \in V^{(0,1)}$ $J_V(v)$ and $\mathrm{i}\, v$ are opposites, even though both of them can be thought of as "multiplying by the imaginary unit".

These constructions can be applied to vector bundles. Given a real vector bundle E over a smooth manifold, a vector bundle endomorphism $J\colon E \to E$ is an *almost complex structure* if $J^2 = -\mathrm{id}_E$.

Remark A.21. Observe that $\mathbb{C}^n$ admits an almost complex structure. Indeed, $T\mathbb{C}^n$ (seen as a real vector bundle) has an almost complex structure, which is given by the endomorphism

$$J\left(\frac{\partial}{\partial x^j}\right) = \frac{\partial}{\partial y^j} \qquad J\left(\frac{\partial}{\partial y^j}\right) = -\frac{\partial}{\partial x^j}. \tag{A.7}$$

Every complex manifold admits a complex structure on TM, which is given, in terms of any holomorphic coordinate system $(z^1 = x^1 + \mathrm{i}\, y^1, \dots z^n = x^n + \mathrm{i}\, y^n)$ by equation (A.7) (see [35, Prop. 1.55]). In that case, J induces a decomposition in the fibers of $T_{\mathbb{C}}M$ by proposition A.19. This can be elevated to a vector bundle decomposition: indeed, for every complex manifold there is a Whitney sum decomposition (see example A.7)

$$T_{\mathbb{C}}M = T^{(1,0)}M \oplus T^{(0,1)}M, \tag{A.8}$$

where the fibers at each point of $T^{(1,0)}M$ and $T^{(0,1)}M$ are the i and $(-\mathrm{i})$-eigenspaces of J at each point. Moreover, local frames for those complex smooth subbundles are

$$\left\{\frac{\partial}{\partial z^1}, \dots \frac{\partial}{\partial z^n}\right\} \qquad \text{and} \qquad \left\{\frac{\partial}{\partial \bar{z}^1}, \dots \frac{\partial}{\partial \bar{z}^n}\right\}.$$

Observe that, indeed we have

$$J\left(\frac{\partial}{\partial z^j}\right) = J\left(\frac{1}{2} \cdot \left(\frac{\partial}{\partial x^j} - \mathrm{i}\, \frac{\partial}{\partial y^j}\right)\right) = \frac{1}{2} \cdot \left(\frac{\partial}{\partial y^j} + \mathrm{i}\, \frac{\partial}{\partial x^j}\right) = \mathrm{i}\, \frac{\partial}{\partial z^j},$$

and analogously, $J(\partial/\partial \bar{z}^j) = -\mathrm{i}\, \partial/\partial \bar{z}^j$. In that case, $T^{(1,0)}M$ has the structure of a holomorphic vector bundle, but $T^{(0,1)}M$ is only a smooth (complex) vector bundle. This justifies their names: $T^{(1,0)}M$ is referred to as the *holomorphic tangent bundle* and $T^{(0,1)}M$ is the *antiholomorphic tangent bundle*. Proofs of these claims may be found in [35, Prop. 156 and Example 3.37]. Finally, we may also consider $T_J M$, which is the usual tangent bundle, but seen as a complex vector bundle through the complex structure J. It is isomorphic as a holomorphic vector bundle to $T^{(0,1)}M$.

Remark A.22. Let $(z^1, \dots z^n)$ be holomorphic coordinates. Every $Z \in T^{(0,1)}M$ can be written as $Z = Z^j \cdot \partial/\partial \bar{z}^j$. If $f \in \mathcal{O}_M$, then $Z(f) = Z^j(f) \cdot \partial f/\partial \bar{z}^j = 0$.

The canonical almost complex structures on complex manifolds characterize holomorphicity of smooth maps as follows.

Theorem A.23. Let M, N be two complex manifolds, and let J_M and J_N be their canonical almost complex structures. Let $F\colon M \to N$ be a smooth map. Then, F is holomorphic if and only if $\mathrm{d}F \circ J_M = J_N \circ \mathrm{d}F$.

Proof. See [35, Prop. 1.55]. □

Given an even-dimensional smooth manifold M it makes sense to ask whether it admits a complex structure. It does if it is complex, but that is not a necessary condition: there are smooth manifolds that admit almost complex structures, but do not admit a complex manifold structure. Let M be a smooth manifold. We say that M is an *almost complex manifold* if TM is endowed with an almost complex structure. The decomposition given in equation (A.8) remains valid when M is a $2n$-smooth manifold (see [35, Lemma 1.60]). We say that an almost complex structure J on a smooth manifold is integrable if, for every $X, Y \in \Gamma(M; T^{(1,0)}M)$, $[X,Y] \in \Gamma(M; T^{(1,0)}M)$. Then, there is a theorem by Newlander and Niremberg that establishes that an almost complex structure comes from a complex manifold structure on M if and only if it is integrable. Equivalently, an almost complex structure J is integrable if and only if its *Nijenhuis tensor* N_J vanishes, i.e. if the following tensor is identically zero:

$$\begin{array}{rccc} N_J\colon & \Gamma(T_{\mathbb{C}}M)\otimes\Gamma(T_{\mathbb{C}}M) & \longrightarrow & \Gamma(T_{\mathbb{C}}M) \\ & (X,Y) & \mapsto & [X,Y]-[JX,JY]+J[JX,Y]+J[X,JY]. \end{array}$$

A.6 The Dolbeault complex

An almost complex structure J on a vector space V induces an almost complex structure on its dual by $\big[J(f)\big](v) := f(J(v))$, which induces a natural decomposition on $V^*_{\mathbb{C}} = (V^*)^{(1,0)} \oplus (V^*)^{(0,1)}$ and there are isomorphisms $(V^*)^{(1,0)} \cong \big(V^{(1,0)}\big)^*$ and $(V^*)^{(0,1)} \cong \big(V^{(0,1)}\big)^*$. This decomposition carries over to the complexified cotangent bundle, and yields $T^*_{\mathbb{C}}M = (T^*M)^{(1,0)} \oplus (T^*M)^{(0,1)} \cong (T^{(1,0)}M)^* \oplus (T^{(0,1)}M)^*$. Moreover, local frames for $(T^*M)^{(1,0)}$ and $(T^*M)^{(0,1)}$ are given by

$$\big\{\mathrm{d}z^1, \dots \mathrm{d}z^n\big\} \qquad \text{and} \qquad \big\{\mathrm{d}\bar{z}^1, \dots \mathrm{d}\bar{z}^n\big\} \tag{A.9}$$

respectively.

Let M be a complex manifold we define

$$\bigwedge^{(p,q)}_{\mathbb{C}} M := \left(\bigwedge^p (T^{(1,0)}M)^*\right) \otimes_{\mathbb{C}} \left(\bigwedge^q (T^{(0,1)}M)^*\right).$$

A complex (p,q)-form is a section of this complex vector bundle. We will denote $\Omega^{(p,q)}_{\mathbb{C}}(M) := \Gamma(M; \bigwedge^{(p,q)}_{\mathbb{C}} M)$. Its elements are called *complex differential forms*. Revisar Moreover, each $\bigwedge^{(p,q)}_{\mathbb{C}} M$ can be seen in a canonical way as a vector subbundle of $\bigwedge^{p+q} T^*_{\mathbb{C}}M$. This follows from the fact that, at the vector space level, (all definitions for real vector spaces endowed with an almost complex structure are the same) we have a canonical linear injection

$$\bigwedge^p V^{(1,0)} \otimes_{\mathbb{C}} \bigwedge^q V^{(0,1)} \longrightarrow \bigwedge^{p+q} V_{\mathbb{C}}$$

given by the linear extension of $\alpha\otimes\beta \mapsto \alpha\wedge\beta$ (to do this more rigorously we would have to invoke the universal property of the tensor product). What is more, we actually have a vector bundle Whitney sum decomposition

$$\bigwedge^k_{\mathbb{C}} M := \bigwedge^k T^*_{\mathbb{C}}M = \bigoplus_{p+q=k} \left(\bigwedge^{(p,q)}_{\mathbb{C}} M\right). \tag{A.10}$$

By setting $\Omega^{(p,q)}_{\mathbb{C}}(M) := \Gamma(M; \bigwedge^{(p,q)}_{\mathbb{C}} M)$, whose elements are called (p,q)*-forms* or forms of *bidegree* (p,q), the Whitney sum decomposition carries over to a Whitney sum decomposition of the spaces of complex differential forms

$$\Omega^k_{\mathbb{C}}(M) = \bigoplus_{p+q=k} \Omega^{(p,q)}_{\mathbb{C}}(M).$$

There is another canonical isomorphism $\bigwedge^k_{\mathbb{C}} M \cong (\bigwedge^k T^*M) \otimes_{\mathbb{R}} \mathbb{C}$, which means that every complex k-form can be seen as a sum $\alpha + \mathrm{i}\,\beta$, where $\alpha, \beta \in \Omega^k_{\mathbb{R}}(M)$. In particular, this means that there is a conjugation map in the space of complex differential forms, which is given by $\overline{\alpha + \mathrm{i}\,\beta} := \alpha - \mathrm{i}\,\beta$. By convention, we set $\Omega^{(p,q)}_{\mathbb{C}}(M) = \{0\}$ except if $0 \le p, q \le n$. We define the graded algebra of differential

forms by $\Omega^*_{\mathbb{C}}(M) := \oplus_{k\geq 0}\Omega^k_{\mathbb{C}}(M)$. Observe that we also have $\Omega^k_{\mathbb{C}}(M) = \Omega^k_{\mathbb{R}}(M)\otimes_{\mathbb{R}}\mathbb{C}$.

Observe that every vector space can be embedded into its complexification as a real vector subspace $V \hookrightarrow V_{\mathbb{C}}, v \mapsto v\otimes 1$. In particular, we may see $\bigwedge^k T^*M$ as a real subbundle of $\bigwedge^k T^*_{\mathbb{C}}M = \bigwedge^k_{\mathbb{C}} M$. This means that we can see $\Omega^k_{\mathbb{R}}(M)$ as a real vector subspace of $\Omega^k_{\mathbb{C}}(M)$ by simply extending real forms to complex vectors by complex bilinearity.

Remark A.24. We could relax our hypothesis and ask M to be almost complex, but since we will work almost exclusively with complex manifolds, the added generality is pointless. Moreover, when M is a complex manifold, we have holomorphic coordinates $(z^1,\dots z^n)$ the local coframes of equation (A.9) are available. More importantly, (p,q)-forms are characterized by the fact that they can be written as

$$\omega = \sum_{I\bar{J}}{}' \omega_{I,\bar{J}}\cdot \mathrm{d}z^{i_1}\wedge\ldots\wedge \mathrm{d}z^{i_p}\wedge \mathrm{d}\bar{z}^{j_1}\wedge\ldots\wedge \mathrm{d}\bar{z}^{j_q}, \tag{A.11}$$

where $I=(i_1,\dots i_p)$ and $\bar{J}=(j_1,\dots j_q)$ are increasing sets of indices, and $\omega_{I\bar{J}}$ are smooth complex-valued maps (see [**35**, Lemma 4.1]). The symbol $\sum'$ means that we sum over ordered increasing sets of indices. Another useful characterization is that a complex $p+q$-form ω is of type (p,q) if and only if it vanishes whenever it is applied to more that p sections of $T^{(1,0)}M$ or more than q sections of $T^{(0,1)}M$.

Decomposition (A.10) induces canonical projections

$$\Pi^{(p,q)}\colon \bigwedge^{p+q}_{\mathbb{C}} M \longrightarrow \bigwedge^{(p,q)}_{\mathbb{C}} M.$$

Moreover, the exterior differential d can be extended by complex linearity to $\Omega^k_{\mathbb{C}}(M)$. Then, a simple coordinate calculation shows that $\mathrm{d}\left(\Omega^{(p,q)}_{\mathbb{C}}(M)\right)\subset \Omega^{(p+1,q)}_{\mathbb{C}}(M)\oplus\Omega^{(p,q+1)}_{\mathbb{C}}(M)$ (see [**35**, Prop. 4.5])

It now makes sense to define the *Dolbeault operators*

$$\partial := \Pi^{(p+1,q)}\circ\mathrm{d}\colon \Omega^{(p,q)}_{\mathbb{C}}(M)\to\Omega^{(p+1,q)}_{\mathbb{C}}(M) \qquad \overline{\partial} := \Pi^{(p,q+1)}\circ\mathrm{d}\colon \Omega^{(p,q)}_{\mathbb{C}}(M)\to\Omega^{(p,q+1)}_{\mathbb{C}}(M)$$

Their coordinate expressions are given by

$$\partial\alpha = \sum_{I\bar{J}}{}' \frac{\partial\alpha_{I\bar{J}}}{\partial z^k}\cdot \mathrm{d}z^k\wedge\mathrm{d}z^{i_1}\wedge\ldots\wedge \mathrm{d}z^{i_p}\wedge \mathrm{d}\bar{z}^{j_1}\wedge\ldots\wedge \mathrm{d}\bar{z}^{j_q}$$

$$\overline{\partial}\alpha = \sum_{I\bar{J}}{}' \frac{\partial\alpha_{I\bar{J}}}{\partial \bar{z}^k}\cdot \mathrm{d}\bar{z}^k\wedge\mathrm{d}z^{i_1}\wedge\ldots\wedge \mathrm{d}z^{i_p}\wedge \mathrm{d}\bar{z}^{j_1}\wedge\ldots\wedge \mathrm{d}\bar{z}^{j_q}$$

The following properties also hold for $\alpha\in\Omega^k_{\mathbb{C}}(M)$ and $\beta\in\Omega^\ell_{\mathbb{C}}(M)$:

$$\partial(\alpha\wedge\beta) = \partial\alpha\wedge\beta + (-1)^k\cdot\alpha\wedge\partial\beta. \tag{A.12}$$

$$\overline{\partial}(\alpha\wedge\beta) = \overline{\partial}\alpha\wedge\beta + (-1)^k\cdot\alpha\wedge\overline{\partial}\beta. \tag{A.13}$$

$$\mathrm{d}\alpha = \partial\alpha + \overline{\partial}\alpha. \tag{A.14}$$

$$\overline{\partial}f = 0 \text{ if } f\in\mathcal{O}_U \tag{A.15}$$

$$\overline{\partial\alpha} = \overline{\partial}\bar{\alpha}. \tag{A.16}$$

$$\partial\partial\alpha = \overline{\partial}\overline{\partial}\alpha = 0 \tag{A.17}$$

$$\partial\overline{\partial}\alpha = -\overline{\partial}\partial\alpha \tag{A.18}$$

$$\partial f = 0 \text{ if } \bar{f}\in\mathcal{O}_U \tag{A.19}$$

The exterior differential produces de Rham cohomology groups

$$\{0\}\to\Omega^0_{\mathbb{C}}(M)\xrightarrow{\mathrm{d}}\Omega^1_{\mathbb{C}}(M)\xrightarrow{\mathrm{d}}\Omega^2_{\mathbb{C}}(M)\xrightarrow{\mathrm{d}}\cdots\xrightarrow{\mathrm{d}}\Omega^n_{\mathbb{C}}(M)\to\{0\},$$

which are called the *de Rham cohomology groups with complex coefficients*, and denoted $H^k_{\mathrm{dR}}(M;\mathbb{C})$. The real de Rham cohomology groups are denoted $H^k_{\mathrm{dR}}(M;\mathbb{R})$. They can be seen as real subspaces of the complex de Rham cohomology groups (classes of forms invariant under conjugation).

A.7 Bundle-Valued forms

Let M be a smooth manifold, and let $E\to M$ be a smooth real vector bundle. A *real E-valued p-form* is a section of the real vector bundle $\bigwedge^p T^*M\otimes E$. We define

$$\Omega^k_{\mathbb{R}}(U;E) := \Gamma\left(U;\bigwedge^p T^*M\otimes E\right)$$

the space of all E-valued p-forms defined on an open set $U \subset M$. Similarly, if E is a smooth complex vector bundle (observe that we do not require E to be holomorphic), we define a *complex E-valued k-form* as a section of the bundle $\bigwedge^k_{\mathbb{C}} M \otimes E$, and a *complex E-valued (p,q)-form* as a section of the bundle $\bigwedge^{(p,q)}_{\mathbb{C}} M \otimes E$. We define

$$\Omega^k_{\mathbb{C}}(U;E) := \Gamma\left(U; \bigwedge^k_{\mathbb{C}} M \otimes E\right) \qquad \text{and} \qquad \Omega^{(p,q)}_{\mathbb{C}}(U;E) := \Gamma\left(U; \bigwedge^{(p,q)}_{\mathbb{C}} M \otimes E\right)$$

for an open set $U \subset M$. If the open set is all of M, and it is clear from the context, we shall omit it. Observe an element of $\sigma \in \Omega^k_{\mathbb{C}}(U;E)$ is a sum of elements $\omega \otimes \sigma$, where $\omega \in \Omega^k_{\mathbb{C}}(M)$ and $\sigma \in \Gamma(E)$. The Dolbeault direct sum decomposition of the space of complex k-forms implies that ω can be uniquely decomposed as a sum of (p,q)-forms satisfying $k = p+q$. This then implies that any E-valued k-form can be uniquely decomposed as a sum of E-valued (p,q)-forms. In conclusion, we have the following Whitney sum decomposition

$$\Omega^k_{\mathbb{C}}(U;E) = \bigoplus_{p+q=k} \Omega^{(p,q)}_{\mathbb{C}}(U;E). \tag{A.20}$$

By convention, we set $\Omega^0_{\mathbb{C}}(M;E) := \Gamma(M;E)$.

There are several wedge product constructions on bundle-valued differential forms, and they are often used without any particular comment (specially in classical references). We devote this part of the section to the definition of those wedge products. Firstly, consider $\alpha \in \Omega^p_{\mathbb{C}}(M)$ and $\beta \in \Omega^q_{\mathbb{C}}(M;E)$ for a smooth complex vector bundle E. Suppose that $\beta = \tilde{\beta} \otimes s$ for a q-form $\tilde{\beta} \in \Omega^q_{\mathbb{C}}(M)$ and a section $s \in \Gamma(M;E)$ (note that every E-valued q-form is a linear combination of terms this kind of terms). We define

$$\alpha \wedge (\tilde{\beta} \otimes s) := (\alpha \wedge \tilde{\beta}) \otimes s \in \Omega^{p+q}_{\mathbb{C}}(M;E),$$

where the $\wedge$ symbol on the right-hand side is the usual exterior product of differential forms, and we extend it linearly to define a wedge product $\wedge \colon \Omega^p_{\mathbb{C}}(M) \times \Omega^q_{\mathbb{C}}(M) \to \Omega^{p+q}_{\mathbb{C}}(M;E)$.

Remark A.25. To be somewhat more rigorous, both the wedge and tensor products are actually defined pointwise. Let $m \in M$ and fix $\alpha \in \bigwedge^p (T^*_m M)_{\mathbb{C}}$. The map

$$\begin{array}{rccc} \wedge_\alpha \colon & (\bigwedge^q (T^*_m M)_{\mathbb{C}}) \otimes_{\mathbb{C}} E_m & \longrightarrow & \left(\bigwedge^{p+q} (T^*_m M)_{\mathbb{C}}\right) \otimes_{\mathbb{C}} E_m \\ & (\beta, s) & \mapsto & (\alpha \wedge \beta) \otimes s \end{array}$$

is bilinear. Thus, by the universal property of the tensor product, it extends to a uniquely defined linear map making the following diagram commutative

$$\begin{array}{ccc} \bigwedge^q (T^*_m M)_{\mathbb{C}} \times E_m & \xrightarrow{\wedge_\alpha} & \bigwedge^{q+p} (T^*_m M)_{\mathbb{C}} \otimes_{\mathbb{C}} E_m \\ \otimes \big\downarrow & \nearrow_{\tilde{\wedge}_\alpha} & \\ \bigwedge^q (T^*_m M)_{\mathbb{C}} \otimes_{\mathbb{C}} E_m & & \end{array}$$

Then, for a general element $\beta \in \left(\bigwedge^q (T^*_m M)\mathbb{C}\right) \otimes_{\mathbb{C}} E_m$, $\alpha \wedge \beta$ should be understood as $\tilde{\wedge}_\alpha(\beta)$. We will simply write $\tilde{\wedge}_\alpha(\beta)$ as $\alpha \wedge \beta$.

When $\beta \in \Omega^0_{\mathbb{C}}(M)$ we define $\alpha \wedge \beta := \alpha \otimes \beta$, and if $\alpha \in \Omega^0_{\mathbb{C}}(M) = \mathscr{C}^\infty(M;\mathbb{C})$, we define $\alpha \wedge \beta := \alpha \cdot \beta$.

We may also define a wedge product between E and E^*-valued forms by linear extension (a rigorous treatment is done as in remark A.25) of

$$(\alpha \otimes \varphi) \wedge (\beta \otimes s) := (\alpha \wedge \beta) \otimes (\varphi(s)) = \varphi(s) \cdot \alpha \wedge \beta \in \Omega^{p+q}_{\mathbb{C}}(M)$$

if $\alpha \otimes \varphi \in \Omega^p_{\mathbb{C}}(M;E^*)$ and $\beta \otimes s \in \Omega^q_{\mathbb{C}}(M;E)$. Observe that $\varphi(s) \in \mathscr{C}^\infty(M;\mathbb{C})$, so the above product makes sense. This defines a wedge product $\wedge \colon \Omega^p_{\mathbb{C}}(M;E^*) \times \Omega^q_{\mathbb{C}}(M;E) \to \Omega^{p+q}_{\mathbb{C}}(M)$. Similarly, if $\alpha \otimes A \in \Omega^p_{\mathbb{C}}(M;\mathrm{End}(E))$ and $\beta \otimes B \in \Omega^q_{\mathbb{C}}(M;\mathrm{End}(E))$, we can define a wedge product $\wedge \colon \Omega^p_{\mathbb{C}}(M;\mathrm{End}(E)) \times \Omega^q_{\mathbb{C}}(M;\mathrm{End}(E)) \to \Omega^{p+q}_{\mathbb{C}}(M;\mathrm{End}(E))$ by linear extension of

$$(\alpha \otimes A) \wedge (\beta \otimes B) := (\alpha \wedge \beta) \otimes (A \circ B).$$

Finally, $\wedge\colon \Omega^p_{\mathbb{C}}(M;\mathrm{End}(E)) \times \Omega^q_{\mathbb{C}}(M;E) \to \Omega^{p+q}_{\mathbb{C}}(M;E)$ can be defined by linear extension of

$$(\alpha \otimes A) \wedge (\beta \otimes s) := (\alpha \wedge \beta) \otimes (A(s)).$$

For more details, see [35, Ch. 4].

When the bundle E is holomorphic, then there is a very special operator, called the *Cauchy-Riemann operator*.

Proposition A.26 (Cauchy-Riemann operator). Let M be a complex manifold and $\pi\colon E \to M$ be a holomorphic vector bundle. Then, there are unique operators $\overline{\partial}_E\colon \Omega^{(p,q)}_{\mathbb{C}}(M;E) \to \Omega^{(p,q+1)}_{\mathbb{C}}(M;E)$ satisfying the following properties:

1. If $s \in \Gamma(M;E)$, then s is holomorphic if and only if $\overline{\partial}_E s = 0$.
2. If $\alpha \in \Omega^{(p,q)}_{\mathbb{C}}(M)$ and $\beta \in \Omega^{(p',q')}_{\mathbb{C}}(M;E)$, then
$$\overline{\partial}_E(\alpha \wedge \beta) := \overline{\partial}\alpha \wedge \beta + (-1)^{p+q}\alpha \wedge \overline{\partial}_E\beta. \tag{A.21}$$
3. $\overline{\partial}_E \circ \overline{\partial}_E = 0$.

Proof. As usual for this kind of constructions, the key idea is to define the operator in terms of a local frame, and then show that the definition is independent of the chosen frame. Thus, let $(s_1, \dots s_r)$ be a local holomorphic frame. Let $\sigma \in \Omega^{(p,q)}_{\mathbb{C}}(M;E)$. Locally, we may write it as $\sigma = \alpha^j \otimes s_j$ for some $\alpha^j \in \Omega^{(p,q)}_{\mathbb{C}}(M)$. If $\overline{\partial}_E$ existed and satisfied conditions 2 and 3, we would have

$$\overline{\partial}_E\sigma = \overline{\partial}\alpha^j \otimes s_j. \tag{A.22}$$

Therefore, we define $\overline{\partial}_E$ in the domain of th holomorphic frame we chose by equation (A.22). To check that this defines $\overline{\partial}_E$ globally on M, choose another local frame $(\tilde{s}_1, \dots \tilde{s}_r)$ whose domain intersect that of the original frame we had. With respet to that frame, we may write $\sigma = \tilde{\alpha}^k \otimes \tilde{s}_k$. By proposition A.13, there are holomorphic functions τ^j_k such that $\tilde{s}_k = \tau^j_k \cdot s_j$. Then, $\tau^j_k \cdot \tilde{\alpha}^k = \alpha^j$.

$$\begin{aligned}\overline{\partial}_E\sigma = \overline{\partial}\alpha^j \otimes s_j = \overline{\partial}(\tau^j_k \cdot \tilde{\alpha}^k) \otimes s_j = \left(\overline{\partial}(\tau^j_k) \wedge \tilde{\alpha}^k + \tau^j_k \cdot \overline{\partial}\tilde{\alpha}^k\right) \otimes s_j \overset{(1)}{=} \overline{\partial}\tilde{\alpha}^k \otimes \left(\tau^j_k \cdot s_j\right) \\ = \overline{\partial}\tilde{\alpha}^k \otimes \tilde{s}_k,\end{aligned}$$

hence $\overline{\partial}_E$ is well-defined. Observe that in (1) we have used that τ^j_k is holomorphic, so $\overline{\partial}\tau^j_k = 0$ (see (A.15)). Now, 3 follows immediately from the fact that $\overline{\partial}^2 = 0$. To show 1, let $\sigma = f^j \cdot s_j$ for some functions $f^j \in \mathscr{C}^\infty(M;\mathbb{C})$. We have $\overline{\partial}_E\sigma = \overline{\partial}f^j \otimes s_j$, which is 0 if and only if $\overline{\partial}f^j = 0$ for each $1 \le j \le r$ again by (A.15). Finally, to check 2, consider $\beta \in \Omega^{(p',q')}_{\mathbb{C}}(M)$ and $\alpha \in \Omega^{(p,q)}_{\mathbb{C}}(M)$. With respect to some local holomorphic frame, we may write $\beta = \beta^j \otimes s_j$, $\beta^j \in \Omega^{(p',q')}_{\mathbb{C}}(M)$. Then,

$$\begin{aligned}\overline{\partial}_E(\alpha \wedge \beta) = \overline{\partial}_E((\alpha \wedge \beta^j) \otimes s_j) = \overline{\partial}(\alpha \wedge \beta^j) \otimes s_j = \left[\ \overline{\partial}\alpha \wedge \beta^j + (-1)^{p+q} \cdot \alpha \wedge \overline{\partial}\beta^j\right] \otimes s_j \\ = \overline{\partial}\alpha \wedge (\beta^j \otimes s_j) + (-1)^{p+q} \cdot \alpha \wedge \left[\ \overline{\partial}\beta^j \otimes s_j\right] \\ = \partial\alpha \wedge \beta + (-1)^{p+q} \cdot \alpha \wedge \overline{\partial}_E\beta.\end{aligned}$$

□

We will get back to this operator.

A.8 Kähler manifolds

Let M be a complex manifold, and recall that its canonical almost complex structure J induces a complex vector bundle structure on T_JM. A hermitian fiber metric in T_JM that case is a map $h\colon T_JM \times T_JM \to \mathscr{C}^\infty(M;\mathbb{C})$ satisfying

1. h is $\mathscr{C}^\infty(M;\mathbb{R})$-bilinear.
2. $h(JX,Y) = \mathrm{i}\, h(X,Y)$.
3. $h(X,JY) = -\mathrm{i}\, h(X,Y)$.
4. $h(Y,X) = \overline{h(X,Y)}$.
5. $h(X,X) > 0$ for $X \neq 0$.

where $X, Y \in \Gamma(M; T_J M)$. A more general definition is given in chapter 3 (definition 2.5) Observe that every hermitian fiber metric on $T_J M$ defines a riemannian metric $g := \mathrm{Re}(h)$. Something slightly less obvious is the fact that $\omega = -\mathrm{Im}(h)$ is a real form of type $(1,1)$, that is $\omega \in \Omega^2_{\mathbb{R}}(M) \cap \Omega^{(1,1)}_{\mathbb{C}}(M)$ [35, Lemma 8.2].

A *hermitian metric* on a complex manifold M is a riemannian metric g for which J is orthogonal, i.e. $g(JX,JY) = g(X,Y)$. If g is a hermitian metric, we say that M is a *Hermitian manifold*. If that is the case, then, we can *define* $\omega(\cdot,\cdot) := g(J\cdot,\cdot)$, and $h := g - \mathrm{i}\,\omega$ and it turns out that h is a hermitian fiber metric as defined above. Under these assumptions, we call ω the *fundamental 2-form* of h. The forms ω and g are uniquely determined by each other. Indeed $\omega(\cdot,\cdot) = g(J\cdot,\cdot)$ and $g(\cdot,\cdot) = \omega(\cdot, J\cdot)$ (see [35, lemma 8.3 and lemma 8.7]). It also follows that every complex manifold admits a hermitian metric (see [35, lemma 8.5]).

A $(1,1)$-form ω on a complex manifold M is said to be *positive* if $\omega_p(v, J_p v) > 0$ for every $v \in T_p M \backslash \{0\}$ and $p \in M$. Given a complex manifold M endowed with a two-form ω, if we define $g(\cdot,\cdot) := \omega(\cdot, J\cdot)$, then g is a hermitian form whose fundamental form is ω if and only if ω is a positive $(1,1)$-form (see [35, Prop. 8.9])

Definition A.27 (Kähler manifold). Let M be a complex manifold, and g be a hermitian metric on M. We say that g is a *Kähler metric* if its fundamental two-form ω is closed. In that case, we say that M is a Kähler manifold.

Observe that every Kähler manifold is a symplectic manifold, the fundamental form being its symplectic form. Indeed, ω_p defines an injective map $\omega_p^\flat \colon T_p M \to T_p^* M, v \mapsto \omega(v,\cdot)$ because if $\omega_p^\flat(v) = 0$, then $\omega_p(v, Jv) = 0$, so $v = 0$

Example A.28. $\mathbb{C}^n$ endowed with its usual symplectic structure (when seen as $\mathbb{R}^{2n}$) is an example of a Kähler manifold. Other examples are $\mathbb{CP}^n$ and all Riemann surfaces (complex manifolds of complex dimension equal to 1). Indeed, one simply fixes a hermitian metric g (this can be done by [35, Lemma 8.5]), and then defines $\omega(\cdot,\cdot) = g(\cdot, J\cdot)$. This is a positive form of type $(1,1)$ [35, Prop. 8.9] and it is closed because every two-form is closed on a Riemann surface (its dimension as a smooth manifold is 2).

Remark A.29. In order to work with coordinates more efficiently, it is quite common to introduce the following convention. Let M be a complex manifold, and consider holomorphic coordiantes $(z^1, \dots z^n)$. As we have seen, this yields two frames for the complexified tangent and cotangent bundles, namely,

$$\left\{ \frac{\partial}{\partial z^1}, \dots \frac{\partial}{\partial z^n}, \frac{\partial}{\partial \bar{z}^1}, \dots \frac{\partial}{\partial \bar{z}^n} \right\} \qquad \text{and} \qquad \left\{ \mathrm{d}z^1, \dots \mathrm{d}z^n, \mathrm{d}\bar{z}^1, \dots \mathrm{d}\bar{z}^n \right\}.$$

If we number these vector and covector fields from 1 to $2n$ as we have ordered them, then, for each $1 \leq j \leq n$ we will interpret $\bar{j}$ to mean $j + n$. We define $\partial_j := \partial/\partial z^j$ and $\partial_{\bar{j}} := \partial/\partial \bar{z}^j$. Then, we may write $X \in \Gamma(T_{\mathbb{C}} M)$ with respect to that frame $X = X^j \partial_j + X^{\bar{j}}\, \partial_{\bar{j}}$. Here, $X^{\bar{j}}$ is interpreted as a $\mathscr{C}^\infty(M;\mathbb{C})$ function totally independent from X^j.

The same applies to the coframe for $T^*_{\mathbb{C}} M$. For example, if M is a Kähler manifold, let g be its riemannian metric. By abusing notation, we extend it by complex bilinearity to act on complex vector fields, but keep the same notation. Then, g is a section of the bundle $T^*_{\mathbb{C}} M \otimes T^*_{\mathbb{C}} M$. In local holomorphic coordiantes, it can be written as

$$g = g_{jk} \cdot \mathrm{d}z^j \otimes \mathrm{d}z^k + g_{j\bar{k}} \cdot \mathrm{d}z^j \otimes \mathrm{d}\bar{z}^k + g_{\bar{j}\, k} \cdot \mathrm{d}\bar{z}^j \otimes \mathrm{d}z^k + g_{\bar{j}\, \bar{k}} \cdot \mathrm{d}\bar{z}^j \otimes \mathrm{d}\bar{z}^k, \tag{A.23}$$

where $g_{j\bar{k}} = g(\partial_j, \partial_{\bar{k}})$ and analogously for all other possible combinations. The coordinate expression actually simplifies a lot in this local frame, because $g(X,Y) = 0$ if $X, Y \in T^{(1,0)} M$ (see [35, Prop.

8.4]) and the indices satisfy the following symmetry conditions $\overline{g_{jk}} = g_{\bar{j}\,\bar{k}} = g_{\bar{k}\bar{j}}$ and $\overline{g_{j\bar{k}}} = g_{\bar{j}\,k} = g_{k\bar{j}}$. Then, equation (A.23) simplifies to

$$g = g_{j\bar{k}} \cdot \mathrm{d}z^j \otimes \mathrm{d}\bar{z}^k + g_{\bar{j}k} \cdot \mathrm{d}\bar{z}^j \otimes \mathrm{d}z^k.$$

Finally, observe that the relation $\omega(X,Y) = g(JX,Y)$ provides a coordinate expression for ω in terms of the coframe

$$\omega = \mathrm{i}\, g_{j\bar{k}} \cdot \mathrm{d}z^j \wedge \mathrm{d}\bar{z}^k.$$

A.9 Hodge Theory

A.9.1 The Hodge inner product

The aim o this section is to define a hermitian inner product on the set of differential forms on a complex manifold. This can be used to define the Lefschetz operator and its dual, which is essential to define Einstein-Hermitian connections. This can be done directly on vector spaces, but we may also give all definitions directly for manifolds. For the first definitions, there is no need to consider complex manifolds, riemannian manifolds suffice.

Let (M,g) be an n-dimensional compact, oriented riemannian manifold. The riemannian metric can be complexified by extending it by complex bilinearity so that it acts on the complexified tangent bundle $T_{\mathbb{C}}M$. We can use the musical isomorphism $\sharp\colon T^*_{\mathbb{C}}M \to T_{\mathbb{C}}M$ to induce a hermitian fiber metric on $\Omega^1_{\mathbb{C}}(M)$, which is given as follows:

$$\langle \alpha, \beta \rangle := g\left(\alpha^\sharp, \overline{\beta^\sharp}\right). \tag{A.24}$$

Firtsly, we will extend this inner product to higher order forms. For smooth complex-valued functions, we simply define $\langle f, h\rangle := f \cdot \overline{h}$. For higher degree forms, consider any local orthonormal frame $\{\varepsilon^1, \dots \varepsilon^n\}$ for the hermitian fiber metric (A.24). The hermitian inner product on $\bigwedge^k_{\mathbb{C}} M$ is defined such that $\{\varepsilon^{i_1} \wedge \ldots \wedge \varepsilon^{i_k} : i_1 < \ldots < i_k\}$ is orthonormal. More precisely, there is a unique hermitian fiber product on $\bigwedge^k_{\mathbb{C}} M$ which satisfies

$$\langle \alpha_1 \wedge \ldots \wedge \alpha_k,\ \beta_1 \wedge \ldots \wedge \beta_k \rangle = \det\left(\left(\langle \alpha_i, \beta_j \rangle\right)_{1 \le i,j \le k}\right),$$

for any locally defined one-forms [**35**, Lemma 9.1]. We will write $||\beta||^2 = (\beta, \beta)$. When M is compact, this yields an inner product on $\Omega^k_{\mathbb{C}}(M)$ given as follows:

$$(\alpha, \beta) := \int_M \langle \alpha, \beta \rangle \ \mathrm{d}V_g, \qquad \alpha, \beta \in \Omega^k_{\mathbb{C}}(M). \tag{A.25}$$

Compactness of M ensures the convergence of the integral above. Equation (A.25) defines a hermitian product on the complex vector space of complex k-forms. This is called the *(global) Hodge inner product* or L_2 product. The symbol $\mathrm{d}V_g$ in equation (A.25) denotes the riemannian volume on M. For a Kähler manifold, $\mathrm{d}V_g = \omega^n/n!$, and the orientation with respect to which we integrate is determined by ω ref.

Remark A.30. The previous construction can be generalized to any complex vector bundle over M endowed with a hermitian fiber metric h. In that case, the space of smooth sections of the bundle can be endowed with an inner product as follows:

$$(\xi, \eta) := \int_M h(\xi, \eta) \ \mathrm{d}V_g.$$

If we did not ask for compactness, we would restrict ourselves to compactly supported sections. We will write $|\xi| := (\xi, \xi)^{1/2}$.

A.9.2 The Hodge star operator

Again, consider a compact, oriented, riemannian manifold and set $n = \dim M$. The Hodge star operator is a vector bundle isomorphism taking k-forms to $n-k$-forms. Just as the Hodge inner

product, the construction does not use anything particular to vector bundles: it is purely linear, and everything can be done in euclidean vector spaces with a fixed orientation. The operator is defined in the following proposition (again, compactness is not needed for the definition).

Proposition A.31. Let (M, g) be an oriented riemannian n-dimensional manifold. Then, there is a unique smooth vector bundle isomorphism

$$\star\colon \bigwedge_{\mathbb{C}}^{k} M \longrightarrow \bigwedge_{\mathbb{C}}^{n-k} M,$$

called the *Hodge star operator* satisfying, for every $\alpha, \beta \in \bigwedge_{\mathbb{C}}^{k} M$, the following relation

$$\alpha \wedge \star\overline{\beta} = \langle \alpha, \beta \rangle \, \mathrm{d}V_g. \tag{A.26}$$

Proof. First, we define the operator locally in terms of a coframe, and we then check that the definition does not depend on the choice of the coframe. Therefore, let $\{\varepsilon^1, \dots \varepsilon^n\}$ be a local (real) frame of T^*M, so, after extending by complex linearity, we get an orthonormal frame $\{\varepsilon^{i_1} \wedge \ldots \wedge \varepsilon^{i_k} : i_1 < \ldots < i_k\}$ of $\bigwedge_{\mathbb{C}}^{k} M$. Let $I = (i_1, \dots i_k)$, and define $\varepsilon^I := \varepsilon^{i_1} \wedge \ldots \wedge \varepsilon^{i_k}$. Then, define

$$\star\, \varepsilon^I = \operatorname{sgn}(\sigma(I, I')) \cdot \varepsilon^{I'}. \tag{A.27}$$

Here, $\sigma(I, I')$ is the permutation taking $\{1, \dots n\}$ to $(i_1, \dots i_k, i'_1, \dots i'_{n-k})$. Let us see that this definition does not depend on the choice of local coframe. To do so, we shall see that definition (A.27) satisfies (A.26), and that the latter determines a bundle isomorphism uniquely. Since both sides of (A.26) are linear and conjugate linear in α and β, it suffices to check it for basis forms. Hence, let J and K be two sets of ascending indices of length k, and $\alpha := \varepsilon^J, \beta := \varepsilon^K$. Then, $\overline{\beta} = \beta$ because we choose a real coframe, and $\star\beta = \operatorname{sgn}(\sigma(K, K')) \cdot \varepsilon^{K'}$ by definition. Thus, $\beta \wedge \star\overline{\beta} = \varepsilon^K \wedge \varepsilon^{K'} = \mathrm{d}V_g$. Finally, we separate cases:

1. If $J = K$ then $\alpha = \beta$, $\langle \alpha, \beta \rangle = 1$ and we just showed that (A.26) is satisfied.
2. If $J \neq K$ then $\langle \alpha, \beta \rangle = 0$ because the frame ε^I, for I increasing and on length k is orthonormal. The right-hand side vanishes as well $\alpha \wedge \beta$ has some repeated element.

Finally, to show uniqueness, suppose that $\tilde{\star}$ satisfies (A.26) as well, and let $\delta = \star - \tilde{\star}$. Observe that for every $\alpha, \beta \in \Omega^q_{\mathbb{C}}(M)$ we have $\alpha \wedge \delta\overline{\beta} = 0$. Therefore,

$$|\delta\beta|^2 \cdot \mathrm{d}V_g = \delta\beta \wedge \star\overline{\delta\beta} = \pm \star \overline{\delta\beta} \wedge \delta\beta = 0,$$

so $|\delta\beta|^2 = 0$ for any β, and since we have a hermitian fiber metric, $\delta \equiv 0$. □

Moreover, the star operator satisfies [35, Prop. 9.8]:

$$\overline{\star\, \alpha} = \star\, \overline{\alpha}. \tag{A.28}$$

$$\star\star\ \alpha = (-1)^{k\cdot(N-k)}\alpha. \tag{A.29}$$

$$\langle \star\, \alpha, \star\, \beta \rangle = \langle \alpha, \beta \rangle. \tag{A.30}$$

By (A.26), the global inner product can be equivalently written as

$$(\alpha, \beta) = \int_M \alpha \wedge \star\overline{\beta}. \tag{A.31}$$

This can be done in a Riemannian manifold using real forms. In that case, an inner product on $\Omega^p_{\mathbb{R}}(M)$ is given by $\int_M \alpha \wedge \star\beta$.

Remark A.32. Everything we have done can be done in riemannian manifolds without requiring complexification of the cotangent bundle. In that case, the pointwise inner product is defined directly on T^*M by $\langle \alpha, \beta \rangle := g(\alpha^\sharp, \beta^\sharp)$ (no conjugation), the Hodge star operator satisfies $\alpha \wedge \star\beta = \langle \alpha, \beta \rangle \cdot \mathrm{d}V_g$, and the global inner product can be defined as $(\alpha, \beta) = \int_M \alpha \wedge \star\beta$.

In fact, one can show that, on complex manifolds, for every $\alpha \in \Omega_{\mathbb{C}}^{(p,q)}(M)$, we have $\star\alpha \in \Omega_{\mathbb{C}}^{(n-p,n-q)}(M)$ [35, Lemma 9.23-(b)].

For bundle-valued the L_2 inner product admits a direct generalization. First, for any complex vector bundle $E \to M$, endowed with a hermitian fiber metric h, the bundles $\bigwedge_{\mathbb{C}}^k M \otimes E$ admit a unique hermitian fiber metric which is given by sesguilinear extension of the fiber metric whose action on decomposable pairs is as follows

$$\langle \alpha \otimes \xi, \beta \otimes \eta \rangle_{q,E} = \langle \alpha, \beta \rangle \cdot h(\xi, \eta).$$

Abusing notation, we shall write $\langle \cdot, \cdot \rangle$ also for the fiber metric on $\bigwedge_{\mathbb{C}}^k M \otimes E$. By remark A.30, we get a global inner product as well.

With these definitions in mind, given a Kähler manifold (M, ω) we define the *Lefschetz operator* as the complex-linear bundle homomorphism given by

$$\begin{array}{rccc} L_\omega \colon & \bigwedge^{(p,q)} M & \longrightarrow & \bigwedge^{(n-p,n-q)} M \\ & \alpha & \mapsto & \omega_p \wedge \alpha \end{array} \qquad \alpha \in \left(\bigwedge^{(p,q)} M \right)_p$$

When it acts on differential forms (which is the way one usually deals with it), it acts pointwise. There is a unique operator, called the *dual Lefschetz operator* or the *contraction with the Kähler form*, denoted Λ_ω or $L_\omega^\star$, which is defined uniquely by the property

$$\langle L_\omega \alpha, \beta \rangle = \langle \alpha, \Lambda_\omega \beta \rangle \qquad \alpha \in \Omega_{\mathbb{C}}^q(M), \beta \in \Omega_{\mathbb{C}}^{q+2}(M).$$

Explicitly, we have $\Lambda_\omega = \star^{-1} \circ L_\omega \circ \star$. Finally, these operators can be extended to bundle valued forms [26]. Consider a complex vector bundle $E \to M$, and define $L_\omega \otimes \mathrm{id}_E$ and $\Lambda_\omega \otimes \mathrm{id}_E$ on the spaces $\bigwedge^{(p,q)} M$. That is, these maps are the complex-bilinear extension of

$$(L_\omega \otimes \mathrm{id}_E)(\alpha \otimes \xi) = (L_\omega \alpha) \otimes \xi \qquad (\Lambda_\omega \otimes \mathrm{id}_E)(\beta \otimes \eta) = (\Lambda_\omega \beta) \otimes \eta.$$

Remark A.33. For two-forms on a Riemann surface X, one has $\star\alpha = \Lambda_\omega \alpha$. This follows from the fact that $\star\omega = 1$ (see [44, p. 6.2.1] and recall that $\mathrm{d}V_g = \omega$ for Riemann surfaces), so for $f \in \mathscr{C}^\infty(X; \mathbb{C}), \alpha \in \Omega_{\mathbb{C}}^2(X)$

$$\langle L_\omega f, \alpha \rangle = \langle f \cdot \omega, \alpha \rangle = f \cdot \langle \star 1, \star\star\alpha \rangle = f \cdot \langle 1, \star\alpha \rangle = f \cdot \overline{\star\alpha} = \langle f, \star\alpha \rangle.$$

Finally, a lemma we will need.

Lemma A.34. Let M be an Kähler manifold of compelx dimension n and $\alpha \in \Omega_{\mathbb{C}}^2(M)$. Then,

$$\Lambda_\omega \alpha \cdot \omega^n = n \cdot L_\omega^{n-1} \alpha.$$

Proof. From [35, Lemma 9.39] we know that $[L_\omega, \Lambda_\omega] = (k-n) \cdot \mathrm{id}$ acting on k-forms, where $[\cdot, \cdot]$ is the commutator of operators. Inductively, one shows, $[L_\omega^p, \Lambda_\omega]\alpha = p \cdot (k - n + p - 1) \cdot L_\omega^{p-1}\alpha$ for each k-form α. Indeed, $p = 1$ is the lemma. For the inductive step (we do not write the k-form)

$$\begin{aligned} [L_\omega^p, \Lambda_\omega] &= L_\omega^p \Lambda_\omega - \Lambda_\omega L_\omega^p = L_\omega^p \Lambda_\omega - \Lambda_\omega L_\omega L_\omega^{p-1} \overset{(1)}{=} L_\omega^p \Lambda_\omega - (L_\omega \Lambda_\omega - (k + 2(p-1) - n) \cdot \mathrm{id}) L_\omega^{p-1} \\ &= L_\omega (L_\omega^{p-1} \Lambda_\omega - \Lambda_\omega L_\omega^{p-1}) + (k + 2(p-1) - n) \cdot L_\omega^{p-1} \\ &= (p-1) \cdot (k - n + p - 2) L_\omega^{p-1} + (k + 2(p-1) - n) L_\omega^{p-1} \\ &= p \cdot (k - n + p - 1) L_\omega^{p-1} \end{aligned}$$

In (1) we used that $\Lambda_\omega L_\omega$ now acts on a $k + 2(p-1)$-form because it acts on $L_\omega \alpha$. Once this formula has been proven, we apply it to a two-form and $p = n$. Thus, $[L_\omega^n, \Lambda_\omega]\alpha = n \cdot L_\omega^{n-1}\alpha$. Hence (observe that $\omega^n \wedge \alpha = 0$ because it is a form whose degree is higher than the dimension of the manifold)

$$L_\omega^n \Lambda_\omega \alpha - \Lambda_\omega L_\omega^n \alpha = \omega^n \wedge \Lambda_\omega \alpha = \Lambda_\omega \alpha \cdot \omega^n = n \cdot \omega^{n-1} \wedge \alpha.$$

□

Appendix B

Some Lie theory

We devote this appendix to give a brief exposition on some aspects of Lie theory that we will need for the development of the main text. First, we give the definition of a complex Lie group, and we recall the notion of a complexification of a Lie group. We then give a brief introduction to parabolic subgroups, as these will be essential to generalize to principal bundles de notion if stability, recalling briefly root systems and introducing parabolic subalgebras. The chapter ends introducing some tools from the theory of non-connected Lie groups and the way these can be applied to bundle theory. We assume basic familiarity with the theory of Lie groups. The main references we used are [23] for basic Lie theory, and [5] for everything related to twisted equivariant bundles. Even though in other chapter we have used greeks letters ξ, ζ to name elements of the Lie algebra of a Lie group, we will use in this appendix the more standard notation of roman capital letters X, Y. We will write e_G for the unit element of the group G. We will, however, use the notation 1 for the trivial group.

B.1 Complex Lie groups

Unsurprisingly, complex Lie groups are defined as follows:

> **Definition B.1 (Complex Lie group).** A *complex Lie group* is a complex manifold G which is endowed with a group structure such that the product $G \times G \to G, (g, h) \mapsto g \cdot h$ and inversion $G \to G, g \mapsto g^{-1}$ maps are holomorphic.

Clearly, every complex Lie group is a (real) Lie group in the usual sense. Observe that, G being a complex manifold, it is endowed with an almost complex structure J_G. This induces a linear almost complex structure on its tangent spaces, in particular on T_eG (considering T_eG as the tangent space of the underlying smooth manifold). Since $T_eG \cong \mathfrak{g}$, the Lie algebra is endowed with a complex structure $J_\mathfrak{g}$, i.e. $\mathfrak{g}$ has the structure of a complex vector space with the product by complex scalars defined by $(a + \mathrm{i}\, b) \cdot X := a \cdot X + b \cdot J_\mathfrak{g}(X)$. In fact, $\mathfrak{g}$ is a complex Lie algebra (equipped with the Lie bracket it gets from the smooth Lie group structure of G) [27, Page 55]. This amounts to check that

$$J_\mathfrak{g}[X, Y] = [J_\mathfrak{g} X, Y] = [X, J_\mathfrak{g} Y] \qquad X, Y \in \mathfrak{g}. \tag{B.1}$$

Of course, there is an underlying real Lie algebra, which we shall denote $\mathfrak{g}^\mathbb{R}$.

To sum up, we have chosen to endow $\mathfrak{g}$ with a complex structure using the almost complex structure on T_eG. We could have proceeded differently. If we use the complex structure on G to obtain $T_\mathbb{C}G = T^{(1,0)}G \oplus T^{(0,1)}G$, then, it is a fact that left-invariant vector fields are holomorphic (i.e. sections of $T^{(1,0)}G$). The Lie bracket of a left-invariant vector field yields another, and since they are sections of $T^{(1,0)}G$ multiplication by i obtained by the complexification and the action of J coincide (see proposition A.19). Hence, we can equivalently define the Lie algebra of G as the set of left-invariant complex vector fields. Another important remark is the fact that the adjoint representation $\mathrm{Ad}\colon G \to \mathrm{Aut}(\mathfrak{g})$ is holomorphic. A thorough treatment of complex Lie groups is given in [33].

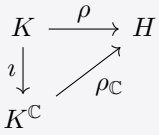
Example B.2. For any $n \in \mathbb{N}$, $\mathrm{GL}(n, \mathbb{C})$ and $\mathrm{SL}(n, \mathbb{C})$ are examples of complex Lie groups. Indeed, $\mathrm{GL}(n, \mathbb{C})$ is an open set of $\mathbb{C}^{n^2}$, and $\mathrm{SL}(n, \mathbb{C})$ is a complex submanifold of $\mathrm{GL}(n, \mathbb{C})$ because 1 is a regular value of the determinant. In fact, $\mathrm{SL}(n, \mathbb{C})$ is a closed subgroup (or a *complex Lie subgroup*) of $\mathrm{GL}(n, \mathbb{C})$.

Complex Lie groups come equipped with an exponential map, which is simply the exponential map the have as real Lie groups. It turns out that the exponential of a complex Lie group is a holomorphic map.

Every Lie algebra $\mathfrak{g}$ can be complexified. This is done by setting $\mathfrak{g}^{\mathbb{C}} := \mathfrak{g} \otimes_{\mathbb{R}} \mathbb{C} \cong \mathfrak{g} \oplus \mathrm{i}\,\mathfrak{g}$ and extending the Lie bracket by $[X \otimes \alpha, Y \otimes \beta] := \alpha\beta \cdot [X, Y]$ for $\alpha, \beta \in \mathbb{C}$ and $X, Y \in \mathfrak{g}$. Actually, complexifications can be extended to Lie groups, although the definition is not as straightforward.

Definition B.3 (Complexification of a Lie group). Let K be a real Lie group. A *complexification* of K is complex Lie group, denoted $K^{\mathbb{C}}$, together with an injective Lie group homomorphism (a smooth group homomorphism) $\imath\colon K \to K^{\mathbb{C}}$ satisfying the following universal property:

For every complex Lie group H, and every (real) Lie group homomorphism $\rho\colon K \to H$, there exists a unique holomorphic Lie group homomorphism $\rho_{\mathbb{C}}\colon K^{\mathbb{C}} \to H$ such that $\rho = \rho_{\mathbb{C}} \circ \imath$. That is, the following diagram is commutative



The previous definition is given in terms of a universal property, which automatically means that, if such a complexification exists, it is unique up to isomorphism [23, Remark 15.1.3]. With some work, one can show that every real Lie group admits a complexification [23, Th. 15.1.4].

When K is compact, the Lie algebra of the complexification is the complexification of the Lie algebra of K. That is, let $G := K^{\mathbb{C}}$ with Lie algebra $\mathfrak{g}$, and let $\mathfrak{k}$ be the Lie algebra of K. Then, $\mathfrak{g} = \mathfrak{k}^{\mathbb{C}} \cong \mathfrak{k} \oplus \mathrm{i}\,\mathfrak{k}$ (we should actually write $\mathfrak{g}^{\mathbb{R}} = \mathfrak{k} \oplus \mathrm{i}\,\mathfrak{k}$, see section A.3.1). In fact, the case in which K is compact is the most interesting: under those assumptions, the inclusion is injective, and $\imath(K) \subset K^{\mathbb{C}}$ is a maximal compact subgroup of $K^{\mathbb{C}}$ [23, Prop. 15.2.1]. In particular, $\imath(K) \subset G$ is closed, so it is a real Lie subgroup of G. Hence, K can be seen as a real Lie subgroup of G.

Example B.4. Consider the unitary group $\mathrm{U}(n) := \{A \in \mathrm{M}_n(\mathbb{C}) : A \cdot A^* = \mathrm{I}_n\}$. It is a real Lie group, but not a complex group. In fact, one may show that $\mathrm{U}(n)^{\mathbb{C}} \cong \mathrm{GL}(n, \mathbb{C})$. More generally, consider any compact subgroup $K \subset \mathrm{U}(n)$. One may show that

$$K^{\mathbb{C}} = K \cdot \exp(\mathrm{i}\,\mathfrak{k}) := \{k \cdot \exp(\mathrm{i}\,X) : k \in K, X \in \mathfrak{g}\} \subset \mathrm{GL}(n, \mathbb{C}).$$

In fact, the map

$$K \times \mathfrak{k} \longrightarrow K^{\mathbb{C}}, (g, Y) \mapsto g \cdot \exp(\mathrm{i}\,Y)$$

is a diffeomorphism. As a consequence, $\dim G = 2 \cdot \dim K$ when G is seen as a real manifold. See [23, Prop. 15.2.1] for proofs of these statements.

Finally, there is a quotient manifold theorem for holomorphic actions on complex manifolds.

Theorem B.5 (Complex quotient manifold theorem). Let G be a complex Lie group and let M be a complex manifold. Suppose that there is a free and proper holomorphic action on M. Then, the quotient space M/G has a unique complex manifold structure such that the natural projection $\pi\colon M \to M/G$ is a holomorphic submersion.

Proof. See [26, Prop. 2.1.13] or [22, Lemma 3.2.6]. □

Remark B.6. For real Lie groups, the closed subgroup theorem [23, Th. 9.3.7] states than whenever H is a closed subgroup of a Lie group G, then, it is an embedded Lie subgroup of G. By an embedded Lie subgroup, we mean a subgroup that is also an embedded submanifold of G and whose Lie group structure is given by restricting the group operations in G. For complex Lie groups, this is no longer true, we may have closed subgroups of a complex Lie group that embedded *real* Lie groups, but not complex Lie groups. However, if $H \leq G$ is a closed subgroup, and $\mathfrak{h}$ is a complex Lie subalgebra (i.e. it is $J_{\mathfrak{g}}$-invariant), then it is a complex Lie subgroup. That is, a complex Lie group that is an embedded complex submanifold and whose group operations are obtained by restricting those of G. See [29, Example 2.1].

B.2 Linearly reductive groups

A very important class of complex Lie groups is the class of linearly reductive groups. There are several characterizations of those groups, and we shall explore some of them.

Definition B.7 (Linearly reductive complex Lie group). A complex Lie group G is *(linearly) reductive* if it is the complexification of a real compact Lie group.

We will often class linearly reductive just reductive Lie groups. This is not standard, as some authors reserve reductive for a more general type of group, but since we will only refer to linearly reductive groups, no confusion will arise.

Remark B.8. Observe that this definition does not involve connectedness. For example, $\mathrm{O}(n)$ is a compact (non-connected) Lie group whose complexification is $\mathrm{O}(n)^{\mathbb{C}} = \mathrm{O}(n,\mathbb{C})$, which is linearly reductive and non-connected. On the other hand, $\mathrm{SO}(n)$ is compact and connected, and $\mathrm{SO}(n)^{\mathbb{C}} = \mathrm{SO}(n,\mathbb{C})$ is linearly reductive and connected.

Complex reductive Lie groups (with finitely many connected components) are characterized by the fact that they admit faithful holomorphic finite-dimensional representations, and they are all completely reducible [23, Th. 15.3.11]. This fact can be used to endow every reductive complex Lie group with the structure of an algebraic group [10, p. III. 8].

B.3 Parabolic subgroups

This section discusses parabolic subgroups of Lie groups. These type of groups will be essential to define polystability on principal bundles. We first present a brief account on root spaces. Beware, most of the exposition bellow can be further generalized.

B.3.1 A brief summary about root systems

Let $\mathfrak{g}$ be a semisimple Lie algebra over $\mathbb{C}$, i.e. its Killing form $\kappa_{\mathfrak{g}}$ is non-degenerate, and choose a nilpotent subalgebra $\mathfrak{h}$. Recall that, by definition, $\mathfrak{h}$ is nilpotent if $C^k(\mathfrak{h}) = \{0\}$ for some $k \in \mathbb{N}$, where $C^k(\mathfrak{h})$ is defined inductively by $C^0(\mathfrak{h}) := \mathfrak{h}$ and $C^k(\mathfrak{h}) = [\mathfrak{h}, C^{k-1}(\mathfrak{h})]$. Consider the adjoint representation $\mathrm{ad}|_{\mathfrak{h}} \colon \mathfrak{h} \to \mathfrak{gl}(\mathfrak{g}) =: V$, and consider a linear functional $\lambda \colon \mathfrak{h} \to \mathbb{K}$. Define the *root space*

$$\mathfrak{g}_\lambda := \bigcap_{H\in\mathfrak{h}} V_{\lambda(H)}(\mathrm{ad}(H)) = \{X \in \mathfrak{g} : [H,X] = \mathrm{ad}(H)(X) = \lambda(H)\cdot X\}$$

where $V_{\lambda(H)}(\mathrm{ad(H)})$ is the eigenspace of $\mathrm{ad}(H)$ corresponding to the eigenvalue $\lambda(H)$. If $\mathfrak{g}_\lambda \neq \{0\}$, we say that λ is a *root*. The set of all roots is denoted $\Delta(\mathfrak{g},\mathfrak{h})$ or simply Δ if no confusion arises.

Recall that we say that a subalgebra $\mathfrak{h} \subset \mathfrak{g}$ is a *Cartan subalgebra* if it is nilpotent and self-normalizing (that is $N_{\mathfrak{g}}(\mathfrak{h}) := \{X \in \mathfrak{g} : [X,\mathfrak{h}] \subset \mathfrak{h}\}$). Cartan subalgebras of semisimple Lie algebras satisfy $\mathfrak{g}_0 = \mathfrak{h}$.

When $\mathfrak{g}$ is semisimple, and $\mathfrak{h} \subset \mathfrak{g}$ is a Cartan subalgebra, every $\mathrm{ad}(H)$ is diagonalizable [23, Prop. 6.3.2], and we get *root space decomposition*

$$\mathfrak{g} = \mathfrak{h} \oplus \left(\bigoplus_{\lambda\neq 0} \mathfrak{g}_\lambda\right).$$

Recall that a Lie algebra is *reductive* if for every ideal $\mathfrak{a} \subset \mathfrak{g}$, there is another ideal $\mathfrak{b}$ such that $\mathfrak{g} = \mathfrak{a} \oplus \mathfrak{b}$. In particular, if $\mathfrak{g}$ is reductive, and $\mathfrak{z}(\mathfrak{g})$ is the *center* of $\mathfrak{g}$, i.e.

$$\mathfrak{z}(\mathfrak{g}) := \{X \in \mathfrak{g} : [X,Y] = 0 \ \ \forall\, Y \in \mathfrak{g}\}.$$

Then we can always decompose $\mathfrak{g}$ as a direct sum $\mathfrak{g} = \mathfrak{z}(\mathfrak{g}) \oplus [\mathfrak{g},\mathfrak{g}]$, the latter being semisimple [23, Lemma 5.7.2]. The Lie algebra of a reductive Lie group is always reductive [23, Remark 15.2.8].

Putting everything together, the Lie algebra of a reductive Lie group can always be decomposed as

$$\mathfrak{g} = \mathfrak{z}(\mathfrak{g}) \oplus [\mathfrak{g},\mathfrak{g}] = \mathfrak{z}(\mathfrak{g}) \oplus \mathfrak{h} \oplus \left(\bigoplus_{\lambda \neq 0} \mathfrak{g}_\lambda \right).$$

Keeping the same notations, the Killing form $\kappa_\mathfrak{g}$ restricts to a non-degenerate bilinear form on the Cartan subalgebra $\mathfrak{h}$ [23, Prop. 6.3.2]. This implies that every root $\alpha \in \Delta$ determines uniquely an element H'_α satisfying $\kappa(H, H'_\alpha) = \alpha(H)$ for all $H \in \mathfrak{h}$ [23, Th. 6.3.4]. We then define $\mathfrak{h}_\mathbb{R} := \mathrm{span}_\mathbb{R}\tilde{\Delta}$, where $\tilde{\Delta} := \{H'_\alpha : \alpha \in \Delta\}$. One may show that the killing form restricted to $\mathfrak{h}_\mathbb{R}$ is real-valued, and positive definite, hence makes $\mathfrak{h}_\mathbb{R}$ into a euclidean space, and that $\tilde{\Delta}$ is a reduced abstract root system for that euclidean space. See [23, Ch. 6] for details. Since the Killing form is a euclidean inner product, it provides an isomorphism $\mathfrak{h}_\mathbb{R} \cong \mathfrak{h}^*_\mathbb{R}$. Moreover, one may show that for each $\alpha \in \Delta$, $\alpha(\mathfrak{h}_\mathbb{R}) \subset \mathbb{R}$, hence $\mathrm{span}_\mathbb{R}\Delta \subset \mathfrak{h}^*_\mathbb{R}$, and in fact they are equal. The isomorphism endows $\mathfrak{h}^*_\mathbb{R}$ with a euclean inner product which satisfies $(\alpha, \beta) = \kappa(H'_\alpha, H'_\beta)$ for each $\alpha, \beta \in \Delta$. Then, Δ is a reduced root system for the euclidean space $\mathfrak{h}^*_\mathbb{R}$. Explicitly, this means that the following 3 properties are satisfied.

1. $\Delta \cap \mathrm{span}_\mathbb{R}\alpha = \{\pm\alpha\}$ for each $\alpha \in \Delta$.
2. $\beta - (\beta, \alpha) \cdot \check{\alpha} \in \Delta$ for all $\alpha, \beta \in \Delta$, where $\check{\alpha} := 2 \cdot \alpha/(\alpha, \alpha)$.
3. $(\beta, \check{\alpha}) \in \mathbb{Z}$ for all $\alpha, \beta \in \Delta$.

Finally, a subset $\Pi \subset \Delta$ is a *basis* or *set of simple roots* of the root system if any other root $\beta \in \Delta$ can be written as $\beta = \sum_{\alpha \in \Pi} k_\alpha \cdot \alpha$, where all the coefficients k_α are either all positve (or zero) or all negative (or zero). Roots for which the coefficients are positive are called *positive roots*, and roots for which they are negative are called *negative roots*. The sets of positive and negative roots are denoted, respectively, Δ^+ and Δ^-.

B.3.2 Parabolic subalgebras

Let $\mathfrak{g}$ be a complex semisimple Lie algebra and $\mathfrak{h}$ a Cartan subalgebra. Fixing a basis or set of simple roots, it makes sense to speak about the set of positive roots, which we shall denote Δ^+. In that situation, we say that a *Borel subalgebra* is a subalgebra:

$$\mathfrak{b} := \mathfrak{h} \oplus \left(\bigoplus_{\alpha \in \Delta^+} \mathfrak{g}_\alpha \right). \tag{B.2}$$

Observe that $\mathfrak{b}$ is indeed a subalgebra. This follows from two facts. First, for a complex semisimple Lie algebra, if α, β are two positive roots (or zero), then $[\mathfrak{g}_\alpha, \mathfrak{g}_\beta] \subset \mathfrak{g}_{\alpha+\beta}$, and $\alpha + \beta \in \Delta^+$ [23, Lemma 6.3.5] and $\mathfrak{h} = \mathfrak{g}_0$. Observe that Borel subalgebras are not unique; they depend on the chosen basis.

Definition B.9 (Parabolic subalgebra). Let $\mathfrak{g}$ be a complex semisimple Lie algebra. A *parabolic subalgebra* is a Lie subalgebra $\mathfrak{p}$ of $\mathfrak{g}$ containing a Borel subalgebra.

All the parabolic subalgebras containing $\mathfrak{b}$ (defined as in (B.2)) are given by

$$\mathfrak{p} = \mathfrak{h} \oplus \left(\bigoplus_{\lambda \in \Gamma} \mathfrak{g}_\lambda \right), \tag{B.3}$$

for some $\Delta^+ \subset \Gamma \subset \Delta$. More precisely, let Π be a basis of the root system. For every subset $\Pi' \subset \Pi$, define

$$\Gamma := \Delta^+ \cup \{\alpha \in \Delta \cap \mathrm{span}_\mathbb{C}(\Pi')\}. \tag{B.4}$$

Then, there is a one-to-one correspondence between all parabolic subalgebras containing $\mathfrak{b}$ and sets Γ defined as above, which associates to each set Γ the algebra (B.3) [27, Prop. 5.90].

Let $-\Gamma$ be the set of opposites of elements in Γ. Observe that the second set in (B.4) is closed under taking opposites. Each parabolic subalgebra admits a decomposition $\mathfrak{p} = \mathfrak{l} \oplus \mathfrak{u}$ where

$$\mathfrak{l} := \mathfrak{h} \oplus \left(\bigoplus_{\lambda \in \Gamma \cap -\Gamma} \mathfrak{g}_\lambda \right) \qquad \text{and} \qquad \mathfrak{u} := \bigoplus_{\lambda \in \Gamma \cap (\Gamma \setminus -\Gamma)} \mathfrak{g}_\lambda. \tag{B.5}$$

$\mathfrak{u}$ is called the *nilpotent radical*, and $\mathfrak{l}$ is called the *Levi factor*. $\mathfrak{u}$ is an ideal, and a nilpotent Lie algebra, and $\mathfrak{l}$ is a reductive Lie algebra [27, Cor. 5.94].

Remark B.10. When the Lie algebra $\mathfrak{g}$ is not semisimple, the definition is generalized as follows: consider the decomposition $\mathfrak{g} = \mathfrak{z}(\mathfrak{g}) \oplus [\mathfrak{g}, \mathfrak{g}]$. $[\mathfrak{g}, \mathfrak{g}]$ is semisimple, so the previous discussion applies. Then, parabolic subalgebras are

$$\mathfrak{p} = \mathfrak{z}(\mathfrak{g}) \oplus \mathfrak{h} \oplus \left(\bigoplus_{\lambda \in \Gamma} \mathfrak{g}_\lambda \right),$$

for subsets Γ defined as above. The center is included in the Levi factor $\mathfrak{l} := \mathfrak{z}(\mathfrak{g}) \oplus \mathfrak{h} \oplus (\oplus_{\lambda \in \Gamma \cap -\Gamma} \mathfrak{g}_\lambda)$.

Observe that both parabolic and borel subalgebras are complex subalgebras. We are now ready to define parabolic subgroups.

B.4 Definition and first properties of parabolic subgroups

Definition B.11 (Parabolic subgroups). Let G be a connected complex reductive Lie group. A *borel subgroup* of G is any connected complex embedded Lie subgroup B whose lie algebra is a Borel subalgebra. A *parabolic subgroup* P is any connected subgroup of G whose Lie algebra is parabolic.

If we fix a parabolic subalgebra $\mathfrak{p}$ and consider the Lie subgroup P such that $\mathfrak{p}$ is its Lie algebra, then, any other parabolic subgroup is obtained from this one by conjugation. Observe that parabolic subgroups are complex closed subgroups of G, hence embedded Lie subgroups because their Lie algebras are complex subalgebras of $\mathfrak{g}$ **(14)**. Equivalently, parabolic subgroups are precisely those containing a Borel subgroup. A very useful characterization of parabolic subgroups is given by the following result:

Theorem B.12. Let G be a connected complex reductive Lie group. A subgroup $P \subset G$ is parabolic if and only if the quotient space G/P is compact.

Proof. See [25, Sec. 29.3] □

Every parabolic subgroup P of a connected complex reductive Lie group G can be written as $P = LU$ where L and U are the connected Lie groups whose Lie algebras are given by $\mathfrak{l}$ and $\mathfrak{u}$ in (B.5) respectively. L is called the *Levi subgroup* and U the *unipotent radical*. U is a normal subgroup of P, see [12, Page 316] and [18, Sec. 2.1].

Moreover, parabolic subgroups of a complex reductive Lie group G are parametrized precisely by elements of the Lie algebra $\mathrm{i}\,\mathfrak{k}$. The precise statement is as follows.

Proposition B.13. Let $G = K^{\mathbb{C}}$ be a complex reductive Lie group, and $\mathfrak{g} = \mathfrak{k} \oplus \mathrm{i}\,\mathfrak{k}$ its Lie algebra. For any $s \in \mathrm{i}\,\mathfrak{k}$, define

$$P_\zeta := \{g \in G : \exp(t\zeta)\ g\ \exp(t\zeta) \ \text{ is bounded as } \ t \to \infty\} \qquad L_\zeta := \{g \in G : \mathrm{Ad}_g(\zeta) = \zeta\}$$

and

$$\mathfrak{p}_\zeta := \left\{\xi \in \mathfrak{g} : \mathrm{Ad}_{\exp(t\zeta)}(\xi) \text{ is bounded as } t \to \infty\right\} \qquad \mathfrak{l}_\zeta := \{\xi \in \mathfrak{g} : [\xi, \zeta] = 0\}.$$

Then P_ζ and L_ζ are connected, closed subgroups of G whose Lie algebras are P_ζ and l_ζ respectively. Moreover, p_ζ is parabolic (in the sense of (B.3)) so P_ζ is a parabolic subgroup, and L_ζ is reductive.

Proof. See [18, Lemma 2.5] and [65, Lemma 2.12]. □

L_ζ is what we call a *Levi subgroup of* P_ζ. Its Lie algebra l_ζ, is a Levi factor in the sense of (B.5) (see [65, Lemma 2.14]).

Remark B.14. Since $\mathfrak{g}$ is a finite-dimensional vector space, and all norms on finite-dimensional vector spaces are equivalent, the notion of being bounded is well-defined. Now, for the definition of P_ζ, observe that there is no natural notion of boundedness on G. To interpret it correctly, we use Weyl's unitary trick [23, Th. 15.2.10] combined with the existence of faithful holomorphic representations of reductive complex Lie groups [23, Th. 15.3.11] to obtain a faithful holomorphic representation $\rho\colon G \to \mathrm{GL}(n,\mathbb{C})$ such that $\rho(K) \subset \mathrm{U}(n)$. Then, boundedness means boundedness in $\mathrm{GL}(n,\mathbb{C})$ of $\rho(\exp(t\zeta)\ g\ \exp(t\zeta))$, which now makes sense as the latter is endowed with many equivalent norms. This does not depend on the chosen representation. **(13)**

We finish the section by drawing attention to the fact that, in general, parabolic subgroups *are not* normal subgroups.

B.4.1 Examples

Let us finish by giving some examples of parabolic subgroups. Let $K = \mathrm{U}(n)$ (which is compact [23, Lemma 2.1.4]), and $G = \mathrm{GL}(n,\mathbb{C})$ its complexification. Even though proposition B.13 characterizes all parabolic subgroups of G, it is always instructive to look at a concrete example. To describe these parabolic subgroups, we briefly introduce the concept of flag. Given a complex vector space, say $\mathbb{C}^n$ for concretness, a *flag* is a chain of vector subspaces $0 \subsetneq V_1 \subsetneq V_2 \subsetneq \ldots \subsetneq V_r = \mathbb{C}^n$. Then, one may show that every parabolic subgroup $P \subset G$ is the stabilizer of a partial flag [25, Sec. 21.3 Cor. B]. That is, $p \cdot V_i = V_i$ for each $1 \leq i \leq r$ and $p \in P$. Hence, up to an adequate choice of a basis (that is, up to conjugation) all parabolic subgroups of G are given by

$$P = \left\{ \begin{pmatrix} A_{11} & A_{12} & \ldots & A_{1k} \\ 0 & A_{22} & \ldots & A_{2k} \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \ldots & A_{kk} \end{pmatrix} : A_{ii} \in \mathrm{GL}(n_i,\mathbb{C}), A_{jk} \in \mathrm{M}_{n_i \times n_j}(\mathbb{C}) \right\}$$

where $n_1 = \dim V_1$, and $n_i = \dim V_i - \dim V_{i-1}$ for $i \geq 2$. To look at an even simplier example, if we consider a partial flag $0 \subsetneq \mathrm{span}_{\mathbb{C}}\{e_1, \ldots e_\ell\} \subsetneq \mathbb{C}^n$ for some $\ell < n$, the parabolic subgroup that stabilizes it is

$$P = \left\{ \begin{pmatrix} A & U \\ 0 & B \end{pmatrix} : \begin{array}{c} A \in \mathrm{GL}(\ell,\mathbb{C}), B \in \mathrm{GL}(n-\ell,\mathbb{C}), \\ U \in \mathrm{M}_{\ell \times (n-\ell)}(\mathbb{C}) \end{array} \right\} = \underbrace{\left\{ \begin{pmatrix} \mathrm{I}_\ell & U \\ 0 & \mathrm{I}_{n-\ell} \end{pmatrix} \right\}}_{\mathcal{R}_U(P)} \rtimes \underbrace{\left\{ \begin{pmatrix} A & 0 \\ 0 & B \end{pmatrix} \right\}}_{L}.$$

Where the right-hand side gives the decomposition in terms of the unipotent radical and the Levi subgroup.

B.5 Non-connected Lie groups

Many important results in the theory of Lie groups rely heavily on connectedness to work. There are, however, some ways to deal with non-connected groups, which constitute the content of this section. These techniques will be used to generalize the Hitchin-Kobayashi correspondence to principal bundles over non-connected groups. References for this section are [23, Ch. 18] and specially [5]. In fact, later sections are a summary of the key results of the latter. Our main setting will be the following: consider a non-connected Lie group $\hat{G}$ with finitely-many connected components, and let G be the connected component containing the identity (identity component, from now on). Let $\Gamma = \hat{G}/G$, since the connected component of the identity is always a normal closed subgroup [36, Prop. 7.15], the quotient has a natural Lie group structure [36, Th. 21.26]. This is the group of connected components of $\hat{G}$, and it fits in an exact sequence of Lie groups

$$1 \to G \hookrightarrow \hat{G} \overset{q}{\twoheadrightarrow} \Gamma \to 1, \tag{B.6}$$

where the first map is the inclusion, and the second, the quotient map. The fact that Γ is the group of connected components of $\hat{G}$ follows from the fact that each L_g is a diffeomorphism. Hence, $L_g(G) = gG$ is a connected component of $\hat{G}$ for each $g \in G$. Since $gG = \tilde{g}G$ if and only if $g\tilde{g}^{-1} \in G$, the latter being the relation defining Γ, it follows that $gG = \tilde{g}G$ if and only if g and $\tilde{g}$ are in the same connected component of $\hat{G}$. Since $G \triangleleft \hat{G}$, for each $\hat{g} \in \hat{G}$, $C_{\hat{g}}$ restricts to a Lie group isomorphism of G. This defines the *characteristic homomorphism* $\bar{\theta}\colon \Gamma \to \mathrm{Out}(G)$. More precisely, first, for any Lie group, define $\mathrm{Aut}(G)$ the group of Lie group isomorphisms, and $\mathrm{Int}(G) := \{C_g : g \in G\}$, where $C_g(h) = ghg^{-1}$. Then $\mathrm{Int}(G) \triangleleft \mathrm{Aut}(G)$, and we may consider $\mathrm{Out}(G) := \mathrm{Aut}(G)/\mathrm{Int}(G)$. The characteristic homomorphism is defined as follows:

$$\begin{array}{rccc} \bar{\theta}\colon & \Gamma & \longrightarrow & \mathrm{Out}(G) \\ & \gamma = \hat{g}G & \mapsto & [C_{\hat{g}}], \end{array} \tag{B.7}$$

where $[\cdot]$ denotes the equivalence class in $\mathrm{Out}(G)$. Observe that for $\hat{g} \in \hat{G} \setminus G$, $C_{\hat{g}} \notin \mathrm{Int}(G)$.

Lemma B.15. Let $\hat{G}$ be a non-connected Lie group and consider an exact sequence as in (B.6). Then, the map $\bar{\theta}$ defined by (B.7) is well-defined.

Proof. Let $\hat{g}_1 G = \hat{g}_2 G$. Then, $\hat{g}_1^{-1}\hat{g}_2 \in G$. The result follows from $(C_{\hat{g}_1})^{-1} \circ C_{\hat{g}_2} = C_{(\hat{g}_1)^{-1}\hat{g}_2} \in \mathrm{Int}(G)$. □

We will assume for now that $\bar{\theta}$ admits a lift $\theta\colon \Gamma \to \mathrm{Aut}(G)$ such that the diagram is commutative ($\bar{q}$ denotes the quotient map). We will write $\theta_\gamma = \theta(\gamma)$ and $\theta_\gamma(g) = g^\gamma$.

$$\begin{array}{ccc} \Gamma & \xrightarrow{\theta} & \mathrm{Aut}(G) \\ & \searrow^{\bar{\theta}} & \downarrow{\scriptstyle \bar{q}} \\ & & \mathrm{Out}(G). \end{array}$$

The lift θ, together with an element $c \in Z^2_\theta(\Gamma, Z(G))$ will give us another way to look at $\hat{G}$. To make sense of this, we need a brief account of group cohomology (see [**23**, Appendix 18.3] for details).

If H is a group that acts through a group homomorphism on an abelian group A, i.e. there is a group homomorphism $\theta\colon H \to \mathrm{Aut}(A), h \mapsto \theta_h$, then we have an action of H on A given by $H \times A \to A, (h, a) \mapsto h \cdot a := \theta_h(a)$, and we say that A is a H-module. This allows to define cochains as

$$C^p_\theta(H, A) := \left\{ c\colon H \times \overset{p}{\cdots} \times H \to A : c(h_1, \dots h_p) = e_A \text{ if } \exists\, i : h_i = e_H \right\}.$$

We have a natural coboundary operator $\delta^p_H\colon C^p_\theta(H, A) \to C^{p+1}_\theta(H, A)$ given by (we use multiplicative notation even though A is abelian)

$$(\delta^p_H c)(h_0, \dots h_p) = (c(h_1, \dots h_p))^{h_0} \cdot \left(\prod_{j=1}^{p} c(h_0, \dots \hat{h}_j, \dots h_p)^{(-1)^j} \right) \cdot c(h_0, \dots h_{p-1})^{(-1)^{p+1}}. \tag{B.8}$$

After showing that $\delta^2_H := \delta^{p+1}_H \circ \delta^p_H \equiv e_A$ (we use the usual abuse of notation of writing δ_H for all operators), we define, as usual in coholomogy, the following abelian groups $Z^p_\theta(H, A) := \ker \delta^p_H$ and $B^p_\theta(H, A) = \mathrm{Im}(\delta^{p-1}_H)$. Then, the p-th *cohomology group* is $H^p_\theta(H, A) = Z^p_\theta(H, A)/B^p_\theta(H, A)$.

In our case, we take $H = \Gamma$ and $A = Z(G)$. The homomorphism θ restricts to a homomorphism in $Z(G)$. Indeed, given $g \in Z(G), h \in G$, since $\theta \in \mathrm{Aut}(G)$, $h\, \theta_\gamma(g)\, h^{-1} = \theta_\gamma(\tilde{h}\, g\, \tilde{h}^{-1}) = \theta_\gamma(g)$ for $\tilde{h} = \theta_\gamma^{-1}(h)$. Now, we choose $c \in Z^2_\theta(\Gamma, Z(G))$. The condition $\delta^2_\Gamma c = e_G$ reads

$$c(\gamma_1, \gamma_2)^{\gamma_0} c(\gamma_0, \gamma_1\gamma_2) = c(\gamma_0, \gamma_1) c(\gamma_0\gamma_1, \gamma_2), \qquad \gamma_0, \gamma_1, \gamma_2 \in \Gamma. \tag{B.9}$$

Definition B.16. Let G, θ and $c \in Z^2_\theta(\Gamma, Z(G))$ be as above, we define the group $G \times_{(\theta,c)} \Gamma$, whose underlying set is $G \times \Gamma$, and whose group operation is given by

$$(g_1, \gamma_1)\ (g_2, \gamma_2) := (g_1 g_2^{\gamma_1}\ c(\gamma_1, \gamma_2), \gamma_1\gamma_2).$$

A direct computation shows that it is indeed a group (associativity follows from (B.9) and the fact that c is $Z(G)$-valued) whose unit element is (e_G, e_Γ), and such that the inverse of (g, γ) is given

by $((g^{-1}\ c(\gamma,\gamma^{-1})^{-1})^{\gamma^{-1}},\gamma^{-1})$ [23, Lemma 18.1.9]. Observe that $c\equiv 1$ implies $G\times_{(\theta,c)}\Gamma = G\rtimes_\theta\Gamma$. This group fits in an exact sequence

$$1\to G\xrightarrow{f} G\times_{(\theta,c)}\Gamma\xrightarrow{k}\Gamma\to 1,$$

where $f(g)=(g,e_\Gamma)$ and $k(g,\gamma)=\gamma$. Equipped with the product structure we just defined, one may show that $G\times_{(\theta,c)}\Gamma$ is a Lie group. Indeed, $G\times\Gamma$ is a smooth manifold endowed with the product manifold structure (Γ is discrete hence a zero dimensional manifold, and θ and c are smooth). For adequate choices of c and θ, we actually have $G\times_{(\theta,c)}\Gamma\cong\hat{G}$. To construct such an isomorphism, we start by choosing a section $s\colon\Gamma\to\hat{G}$ (i.e. $q\circ s=\mathrm{id}_\Gamma$) that is *normalized*, meaning that the composition of the conjugation map $\hat{G}\to\mathrm{Aut}(G)$ with s is equal to θ. In particular, $s(e_\Gamma)=e_{\hat{G}}$. In other words $\theta_\gamma(g)=s(\gamma)\ g\ s(\gamma)^{-1}$ for all $\gamma\in\Gamma, g\in G$. The section can always be chosen like that because for each $\gamma\in\Gamma$, conjugation, seen as a map $q^{-1}(\gamma)\to\mathrm{Aut}(G)$ has as its image the whole class of $\bar{\theta}(\gamma)$. In that case, we can *define* the cocycle c by the relation

$$c(\gamma_1,\gamma_2)=s(\gamma_1)\ s(\gamma_2)\ s(\gamma_1\gamma_2)^{-1}.$$

Hence, c measures the failure of s to be a group homomorphism. This obviously satisfies $c(e_\Gamma,\gamma)=c(\gamma,e_\Gamma)=e_G$. A direct calculation shows that c satisfies (B.9). Moreover, it is $Z(G)$-valued:

$$\begin{aligned} g\ c(\gamma_1,\gamma_2)\ g^{-1} &= s(\gamma_1)\ s(\gamma_1)^{-1}g\ s(\gamma_1)\ s(\gamma_2)\ s(\gamma_1\gamma_2)^{-1}g^{-1}s(\gamma_1\gamma_2)\ s(\gamma_1\gamma_2)^{-1}\\ &= s(\gamma_1)\ \theta_{\gamma_1^{-1}}(g)\ s(\gamma_2)\ \theta_{(\gamma_1\gamma_2)^{-1}}(g^{-1})\ s(\gamma_1\gamma_2)^{-1}\\ &= s(\gamma_1)\ s(\gamma_2)\ \theta_{\gamma_2^{-1}}(\theta_{\gamma_1^{-1}}(g))\theta_{(\gamma_1\gamma_2)^{-1}}(g^{-1})\ s(\gamma_1\gamma_2)^{-1}\\ &= c(\gamma_1,\gamma_2). \end{aligned}$$

Then, the map

$$\Phi\colon G\times_{(\theta,c)}\Gamma\longrightarrow\hat{G},(g,\gamma)\mapsto g\ s(\gamma) \tag{B.10}$$

is a Lie group isomorphism [23, Remark 18.1.8].

Remark B.17. Observe that we can define $\hat{\Gamma}_{\theta,c}:=Z(G)\times_{(\theta,c)}\Gamma\subset\hat{G}$ by restriction of the group operations, and it is a group. We then get an extension $1\to Z(G)\to\hat{\Gamma}_{\theta,c}\to\Gamma\to 1$.

For what is to come, we fix $\hat{G}$ a complex reductive Lie group, and $\hat{K}\subset\hat{G}$ a maximal compact Lie subgroup. By a theorem of Mostow [41, Th. 3.1] $\Gamma\cong\hat{K}/K$ where $K:=\hat{K}\cap G$ is the identity component of $\hat{K}$, and it is a maximal compact subgroup of G. Since every compact manifold has finitely many connected componets, Γ is finite. Moreover, observe that the previous reasoning holds for any extension of Lie groups, for instance $1\to K\to\hat{K}\to\Gamma\to 1$. Given $\bar{\theta}\colon\Gamma\to\mathrm{Out}(K)$, By the results in [58, See Ch I, Sec 2], there is a lift $\theta\colon\Gamma\to\mathrm{Aut}(K)$. More explicitly, that theorem states that letting $K\subset G$ be a maximal compact subgroup, then the short exact subsequence

$$1\to\mathrm{Int}(K)\to\mathrm{Aut}(K)\to\mathrm{Out}(K)\to 1$$

splits. Hence, the composition of $\bar{\theta}\colon\Gamma\to\mathrm{Out}(K)$ with a left splitting $\mathrm{Out}(K)\to\mathrm{Aut}(K)$ yields a lift of θ. Using the universal property of complexifications, we may obtain a lift of $\bar{\theta}\colon\Gamma\to\mathrm{Out}(G)$. By the universal property of complexification, we may see it as $\theta\colon\Gamma\to\mathrm{Aut}(G)$. Moreover, again by the universal property of complexifications, $Z(K)\subset Z(G)$ so any cocycle in K can be seen as a cocycle in G. Then, upon fixing a normalized section, the Lie group isomorphism $\hat{G}\cong G\times_{(\theta,c)}\Gamma$ restricts to an isomorphism $\hat{K}\cong K\times_{(\theta,c)}\Gamma$ if we see $K\times_{(\theta,c)}\Gamma$ as a subgroup of $G\times_{(\theta,c)}\Gamma$ [5, Prop. 2.2]. In particular, $K\subset G$ is Γ-invariant because the lift θ comes from a lift of the homomorphism $\Gamma\to\mathrm{Out}(K)$.

Remark B.18. Let $\hat{G}$ be a complex reductive Lie group as before, which a complexification of a maximal compact subgroup $\hat{K}$, and let G the identity component of $\hat{G}$ and $K:=\hat{K}\cap G$. First, since the connected component of the identity is always closed [35, Lemma 7.11 and Prop. 7.15], and $\hat{K}$ is compact, it follows that K is compact. Moreover, from the universal property of complexifications, $K^{\mathbb{C}}\cong G$. Hence, G is diffeomorphic to $K\times\mathfrak{k}$. Since G is connected, K must be connected as well. **(16)**

B.5.1 Twisted equivariant actions

This section is devoted to introducing what we will call *twisted equivariant actions*. As we shall see, introducing this concept will allow us to obtain principal bundles with connected structure group out of principal bundles with non-connected structure group. Our main reference is [5]. The lift θ induces a right action of Γ on G given by

$$G \times \Gamma \longrightarrow G, (g, \gamma) \mapsto \theta_{\gamma^{-1}}(g). \tag{B.11}$$

Observe that this induces a right action on the corresponding Lie algebra.

$$\mathfrak{g} \times \Gamma \to \mathfrak{g}, (Y, \gamma) \mapsto \mathrm{d}_e\theta_{\gamma^{-1}}(Y). \tag{B.12}$$

If M is an $\mathcal{S}$-manifold and G an $\mathcal{S}$-Lie group, we will have $\mathcal{S}$-actions.

Definition B.19. Let M be an $\mathcal{S}$-manifold. A (θ, c)-*twisted left (resp. right)* (G, Γ)-*action* is a pair given by:

1. a left (resp. right) G-action by $\mathcal{S}$-isomorphisms,
2. an $\mathcal{S}$-isomorphism $\Gamma \times M \to M, (\gamma, m) \mapsto \gamma \cdot m$ satisfying, (for a left action)

$$e_\Gamma \cdot m = m \qquad \forall\ m \in M, \tag{B.13}$$
$$\gamma \cdot (g \cdot m) = g^\gamma \cdot (\gamma \cdot m) \qquad \forall\ m \in M, \gamma \in \Gamma, g \in G, \tag{B.14}$$
$$\gamma_1 \cdot (\gamma_2 \cdot m) = c(\gamma_1, \gamma_2) \cdot ((\gamma_1\gamma_2) \cdot m) \qquad \forall\ m \in M, \gamma_1, \gamma_2 \in \Gamma. \tag{B.15}$$

We also say that M is a (θ, c)*twisted* (G, Γ)-*manifold*.

For a right action, (B.13) is swapped for $m \cdot e_\Gamma = m$, (B.14) for

$$(m \cdot g^\gamma) \cdot \gamma = (m \cdot \gamma) \cdot g \qquad \forall\ m \in M, \gamma \in \Gamma, g \in G, \tag{B.16}$$

and (B.15) for

$$(m \cdot \gamma_1) \cdot \gamma_2 = (m \cdot c(\gamma_1, \gamma_2)) \cdot (\gamma_1\gamma_2) \qquad m \in M, \gamma_1, \gamma_2 \in \Gamma. \tag{B.17}$$

If $G = Z(G)$ is abelian, we simply speak of a (θ, c)-*twisted* Γ-*action* (sometimes we will restrict actions to $\hat{\Gamma}_{\theta,c}$). Given two $\mathcal{S}$-manifolds endowed with (θ, c)-twisted (G, Γ)-equivariant left actions, M and N, an $\mathcal{S}$-map $f\colon M \to N$ is called (θ, c)-*twisted* (G, Γ)-*equivariant* if it satisfies

$$f(g \cdot m) = g \cdot f(m) \quad \text{and} \quad f(\gamma \cdot m) = \gamma \cdot f(m) \qquad \forall\ m \in M, \gamma \in \Gamma. \tag{B.18}$$

When the data (θ, c) and (G, Γ) are clear from the context, we will speak directly of twisted equivariant actions and manifolds.

Since $\hat{G} \cong G \times_{(\theta,c)} \Gamma$, one may think that $\hat{G}$ actions and twisted actions are related. Indeed:

Proposition B.20. Let $\hat{G} = G \times_{(\theta,c)} \Gamma$. There is a bijective correspondence between $\hat{G}$-actions on an $\mathcal{S}$-manifold and (θ, c)-twisted (G, Γ)-actions given as follows:

1. Given a $\hat{G}$-action on M, we define the G-action by restriction and $\gamma \cdot m := (e_G, \gamma) \cdot m$ (c.f. (B.10)).
2. Given a (θ, c)-twisted (G, Γ)-action, we define $(g, \gamma) \cdot m = g \cdot (\gamma \cdot m)$ for a right action, and $m \cdot (g, \gamma) = (m \cdot g) \cdot \gamma$.

Where we used the isomorphism $\hat{G} \cong G \times_{(\theta,c)} \Gamma$. Moreover, a map $f\colon M \to N$ between sets with (θ, c)-twisted (G, Γ)-actions is (θ, c)-twisted (G, Γ)-equivariant if and only if it is $\hat{G}$ equivariant under the previous correspondence.

Proof. See [5, Prop. 2.7 and Prop. 2.12], the proof is just a sequence of lengthy computations. □

Remark B.21. In our discussion, the lift θ has been fixed, but it may not be unique. The choice of lift, however, does not matter in the end. Without further detail, different choices lead to equivalent categories. We refer the reader to [5, Sec. 4] for details.

B.5.2 Twisted equivariant bundles

Twisted equivariant actions allow us to define analogues to principal bundles, where instead of a regular action we now have a (θ, c)-twisted (G, Γ)-equivariant action. Chapter 3 contains the basic features of principal bundles needed to understand this section. If M is an $\mathcal{S}$-manifold endowed with a left Γ-action by $\mathcal{S}$-isomorphisms, we will say that it is a $(\mathcal{S}, \Gamma)$-manifold.

Both general fiber bundles and principal bundles admit generalizations to twisted equivariant actions.

Definition B.22 (Twisted equivariant fiber bundles). Let M be an $(\mathcal{S}, \Gamma)$-manifold. A (θ, c)-*twisted* Γ-*equivariant bundle* is an $\mathcal{S}$-fiber bundle $F \to M$ endowed with a (θ, c)-twisted action of Γ on F such that $Z(G)$ acts fiberwise (i.e. for $z \in Z(G), f \in \pi^{-1}(m)$, we have $z \cdot f \in \pi^{-1}(m)$) and $\pi \colon F \to M$ is Γ equivariant, that is $\pi(\gamma \cdot f) = \gamma \cdot \pi(f)$ for a left action, and $\pi(f \cdot \gamma) = \pi(f) \cdot \gamma$ for a right action.

We can then define *twisted* Γ-*equivariant sections* $s \colon M \to F$ as usual sections satisfying the Γ-equivariance condition: $s(m \cdot \gamma) = s(m) \cdot \gamma$ for a right action and $s(\gamma \cdot m) = \gamma \cdot s(m)$ for a left action. Observe that this is well-defined thanks to equivariance of the projection π.

Definition B.23 (Twisted equivariant principal G-bundles). Let M be an $(\mathcal{S}, \Gamma)$-manifold. A (θ, c)-*twisted* Γ-*equivariant principal* G-*bundle* is a right (θ, c)-twisted equivariant bundle $P \to M$ with a right G-action such that

1. The G and Γ actions make P into a (θ, c)-twisted (G, Γ)-manifold.
2. The right G-action makes $P \to M$ into an $\mathcal{S}$-principal G-bundle.

As usual, if we work with holomorphic bundles, both the Lie group and the base manifold must be complex. Let us remark that, in the previous definition, the G-action restricts to an action of $Z(G)$ that is the same as the one induced by the twisted equivariant Γ-action. Observe that by proposition B.20, P may be seen as a manifold endowed with a $\hat{G}$-action.

Remark B.24. Just to be very clear, a twisted principal G-bundle is a G-bundle right Γ-action, such that the projection map is Γ-equivariant, and the following relations hold

$$(p \cdot g^{\gamma}) \cdot \gamma = (p \cdot \gamma) \cdot g \qquad (p \cdot \gamma_1) \cdot \gamma_2 = (p \cdot c(\gamma_1, \gamma_2)) \cdot (\gamma_1 \gamma_2) \tag{B.19}$$

for all $p \in P, g \in G, \gamma \in \Gamma$. These are just (B.16) and (B.17) applied to our particular situation.

The notion of morphism has to be modified as well (c.f. definition 3.4). Given two twisted Γ-equivariant principal bundles with the same structure group G, say P and P', over two Γ manifolds M and N, a *morphism* $\vartheta \colon P \to P'$ is a twisted equivariant $\mathcal{S}$-morphism such that the induced $\mathcal{S}$-morphism $\vartheta' \colon M \to N$ is Γ equivariant. That is $\vartheta(p \cdot \gamma) = \vartheta(p) \cdot \gamma$, $\vartheta(p \cdot g) = \vartheta(p) \cdot g$ and $\vartheta'(m \cdot \gamma) = \vartheta'(m) \cdot \gamma$. By proposition B.20, we may ask ϑ to be $\hat{G}$-equivariant.

$$\begin{array}{ccc} P & \xrightarrow{\vartheta} & P' \\ {\scriptstyle\pi}\downarrow & & \downarrow{\scriptstyle\pi'} \\ M & \xrightarrow{\vartheta'} & N \end{array}$$

Pullbacks under Γ-equivariant maps behave well: if $P \to Y$ is a twisted equivariant principal G-bundle, and $\varphi \colon X \to Y$ is a Γ-equivariant $\mathcal{S}$-morphism between Γ-manifolds, then $\varphi^* P \to X$ defined as in (3.1) has a natural structure of twisted equivariant principal G-bundle. The Γ action on $\varphi^* P$ is defined component-wise, i.e. $(n, p) \cdot \gamma = (n \cdot \gamma, p \cdot \gamma)$.

Finally, something unexpected happens when we try to apply this construction to associated bundles: the twisting disappears and we get a "genuine" Γ-bundle, i.e. a fiber bundle with a Γ-action over a Γ-manifold such that the projection is Γ-invariant. Observe that (θ, c)-twisted Γ-equivariant fiber bundles are *not* Γ-equivariant bundles precisely because the twisted Γ-action is *not* an action in the usual sense. This result is based on the following proposition (we state it in general terms).

Proposition B.25. Let $1 \to G \xrightarrow{i} \hat{G} \xrightarrow{\pi} \Gamma \to 1$ be a $\mathcal{S}$-Lie group extension, and suppose that there is a free and proper $\mathcal{S}$-action of $\hat{G}$ on an $\mathcal{S}$-manifold M. Then, there is an $\mathcal{S}$-action of Γ on M/G and the quotient map $\hat{\pi}\colon M \to M/G$ is equivariant for the homomorphism $q\colon \hat{G} \to \Gamma$.

Proof. Consider right-actions and define $[m]\cdot\gamma := [m\cdot\hat{g}]$ for any $\hat{g} \in \hat{\pi}^{-1}(\gamma)$. Choosing $\hat{g}_1, \hat{g}_2 \in \pi^{-1}(\gamma)$, we have $\hat{g}_2^{-1}\hat{g}_1 \in \ker\pi = i(G)$ so $[m\cdot\hat{g}_1] = [m\cdot\hat{g}_2\hat{g}_2^{-1}\hat{g}_1] = [m\cdot\hat{g}_2]$ so the action does not depend on the representative of γwe choose. Similarly, $[m]\cdot\gamma = [m\cdot g]\cdot\gamma$ for $g \in G$ because $i(G) \triangleleft \hat{G}$. Indeed, $[m\cdot\hat{g}] = [m\cdot g\cdot\hat{g}]$ because $g\hat{g} = \hat{g}g'$ for some $g' \in i(G)$. Finally, $\pi\colon M \to M/G$ is equivariant by definition of the action. □

If $\hat{G} = G \times_{(\theta,c)} \Gamma$, and we have a (θ, c)-twisted (G,Γ)-action on M, then we also have a $\hat{G}$ action on M (see proposition B.20). The previous proposition applies and yields a Γ-action on M/G.

Proposition B.26. Let F be a (θ, c)-twisted right (G,Γ)-manifold and $\pi\colon P(M,G) \to M$ be a (θ, c)-twisted Γ-equivariant principal G-bundle over a Γ-manifold M. Then, the associated bundle $P \times_G F$ is a Γ-equivariant fiber bundle with the action $[p,f]\cdot\gamma = [p\cdot\gamma, f\cdot\gamma]$. That is, a fiber bundle with a Γ-action such that the projection is Γ-equivariant.

Proof. Endow $P\times F$ with the component-wise (θ, c)-twisted (G,Γ)-action obtained from that of P and G and apply proposition B.25 to the induced $\hat{G} = G \times_{(\theta,c)} \Gamma$-action. It follows that $[p,f]\cdot\gamma = [p\cdot\gamma, f\cdot\gamma]$ is a right Γ-action, and the projection is equivariant by definition (c.f. definition B.22 and section 3.1.2). □

Remark B.27. If F had been a (θ, c)-twisted *left* (G,Γ)-manifold, we would have needed to modify the left twisted equivariant action to obatin a right twisted equivariant action. This is done by setting $f\cdot g := g^{-1}\cdot f$ and $f\cdot\gamma := c(\gamma^{-1},\gamma)^{-1}\cdot(\gamma^{-1}\cdot f)$.

B.5.3 Reductions of the structure group on twisted equivariant bundles

The aim of this section is to generalize the notion of a reduction of the structure group of a principal bundle to twisted equivariant principal bundles. To do so, let H be a Lie group and suppose that there is a homomorphism $\tau\colon \Gamma \to \mathrm{Aut}(H)$. Suppose that $\psi\colon G \to H$ is a Γ-equivariant Lie group homomorphism, which in this context means $\psi(\theta_\gamma(g)) = \tau_\gamma(\psi(g))$ for each $g \in G, \gamma \in \Gamma$. Suppose as well that $\psi(c(\Gamma\times\Gamma)) \subset Z(H)$. Under those assumptions, $\psi\circ c\colon \Gamma\times\Gamma \to Z(H)$ is an element of $Z^2_\tau(\Gamma, Z(H))$ (this is obvious from the fact that ψ is a Γ-equivariant morphism and c a two-cocycle). With this setting in mind, we are ready to generalize extensions and reductions of structure groups.

Remark B.28. Let $P(M,G)$ be an $\mathcal{S}$-principal G-bundle, and let $\psi\colon G \to H$ be an $\mathcal{S}$-Lie group homomorphism. Then, H acts on G by left-translations $G\times H \to H, (g,h) \mapsto \psi(g)\,h$. This produces an associated fiber bundle over M, $P(H) := P \times_G H$ (c.f. section 3.1.2). There is a right H-action on $P(H)$ given by $[p,h]\cdot h' := [p, hh']$ whic is clearly independent of the representative chosen. This action is free, and the preimage of the projection of each pair $[p,h]$ is the H-orbit. Finally, $P(H)$ is locally trivial (see [63, Sec. 10.8]). To sum up, $P(H)$ has the structure of a principal H-bundle over M.

In the conditions of the previous remark, if P happened to be a twisted Γ-equivariant bundle, we can endow $P(H)$ with that structure under a pointwise action $[p,h]\cdot\gamma = [p\cdot\gamma, h\cdot\gamma]$. This is well-defined. Indeed, choosing another representative $[p,h] = [p\cdot g, \psi(g^{-1})\cdot h]$, we have

$$\begin{aligned}[p\cdot g, \psi(g^{-1})\cdot h]\cdot\gamma &= \left[p\cdot g\cdot\gamma, \left(\psi(g^{-1})\cdot h\right)\cdot\gamma\right] = \left[p\cdot\gamma\cdot\theta_{\gamma^{-1}}(g), \tau_{\gamma^{-1}}\left(\psi(g^{-1})\cdot h\right)\right] \\ &= \left[p\cdot\gamma\cdot\theta_{\gamma^{-1}}(g), \psi\left(\left(\theta_{\gamma^{-1}}(g)\right)^{-1}\right)\cdot\gamma\right].\end{aligned}$$

A straightforward calculation shows that (B.19) are satisfied for $(\psi\circ c)$ so $P(H)$ is a $(\tau, \psi\circ c)$-twisted Γ-equivariant principal H-bundle.

Similarly, we want to adapt the notion of reduction of the stucture group to twisted equivariant principal bundles. More precisely, we want to see how reductions of the structure group of a

principal G-bundle may be endowed with a twisted equivariant structure.We will work in the smooth category or definiteness, but everything carries over to the holomorphic category under obvious modifications and we refer the reader to section 3.6 for the basics on reductions of the structure group. Under our usual situation (B.6), let $H \subset G$ be a closed Lie subgroup, and suppose that it is invariant under the Γ-action, i.e. $\gamma \cdot H = H$ for all $\gamma \in \Gamma$. Under these conditions, there is a natural right Γ-action on G/H given by

$$G/H \times \Gamma \to G/H,\ (gH, \gamma) \mapsto \theta_{\gamma^{-1}}(g) \cdot H,$$

which is well-defined because $\theta_{\gamma^{-1}}(gh) \cdot H = \theta_{\gamma^{-1}}(g) \cdot H$ (Γ-invariance means that $\theta_{\gamma^{-1}}(h) \in H$). Let

Proposition B.29. Let $H \subset G$ be a Γ-invariant closed Lie subgroup. The action $G \times G/H \to G/H, (g, g'H) \mapsto gg'H$, and the map $\Gamma \times G/H \to G/H, (\gamma, gH) \mapsto \theta_\gamma(g) \cdot H$ define a θ-twisted (G, Γ)-action on G/H.

Proof. Let $g, g' \in G$ and $\gamma, \gamma' \in \Gamma$. We have $\gamma \cdot (g'gH) = \theta_\gamma(g') \cdot (\theta_\gamma(g)H) = (g')^\gamma \cdot (\gamma \cdot gH)$, and trivially $\gamma\gamma' \cdot (gH) = (\theta_\gamma \circ \theta_{\gamma'}(g))H = \gamma \cdot (\gamma' \cdot gH)$. □

The previous proposition can be phrased by saying that there is an action of $G \rtimes_\theta \Gamma$ on G/H. If $\hat{G} = G \times_{(\theta,c)} \Gamma$ and the cocycle c takes values in $Z(H)$ (this happens if we choose $H = K$ when the lift θ is obtained from a lift on K), and we define $\hat{H} \cong H \times_{(\theta,c)} \Gamma$ then $G/H \cong \hat{G}/\hat{H}$. On $\hat{G}/\hat{H}$ we have a natural $\hat{G}$ action. Under the isomorphism, this induces a $\hat{G}$-action on G/H, and by proposition B.20, this is equivalent to a (θ, c)-twisted (G, Γ)-action on G/H. Now, proposition B.26 implies that $P(G/H)$ is a Γ-equivariant fiber bundle. Then, there is a true action of Γ on $\Gamma(M; P(G/H))$ given by $(\gamma \cdot \sigma)(m) := \sigma(m \cdot \gamma^{-1}) \cdot \gamma$. This is Γ-equivariant with respect of the Γ ations on M and $P(G/K)$ if and only if σ is Γ-invariant for this action. Moreover, this action coincides with the $G \rtimes_\theta \Gamma$. **(15)**

Proposition B.30. Let $H \subset G$ be a θ-invariant subgroup, and $P(M, G)$ a (θ, c)-twisted Γ-equivariant principal G-bundle over a Γ-manifold M. Let $\sigma \in \Gamma(M; P(G/H))$ be a reduction of the structure group. The bundle $P_\sigma \subset P$ is Γ-invariant if and only if σ is Γ-equivariant. If any of these two equivalent conditions is met, then P_σ has a unique structure of (θ, c)-twisted Γ-equivariant H-principal H-bundle.

Proof. We already know that P_σ is a principal H-bundle. Recall that reductions of the structure group to H are given by sections of the associated bundle $P(G/H)$, and given such a section σ, we may see P_σ as an H-invariant subbundle of P. Moreover, these sections can be identified with G-equivariant maps $s\colon P \to G/H$ (the comments around (3.15) apply). Carefully following the isomorphisms of remark 3.35 and (3.15), we see that to a section $\sigma \in \Gamma(M; P(G/H))$ we assign a G-equivariant map $s\colon P \to G/H, p \mapsto g^{-1}H$ if and only if $p \in P_\sigma \cdot g$. Now, from the definition of twisted equivariant action (B.16), we have

$$p \cdot \gamma \in (P_\sigma \cdot g) \cdot \gamma = (P_\sigma \cdot \gamma) \cdot \theta_\gamma(p)^{-1}.$$

Hence, if s is Γ-equivariant, we find $s(p \cdot \gamma) = s(p) \cdot \gamma = \theta_{\gamma^{-1}}(g) \cdot H$, so $P_\sigma \cdot \gamma = P_\sigma$. Conversely, if P_σ is Γ-invariant, then $P_\sigma \cdot \gamma = P_\sigma$, hence $p \cdot \gamma \in P_\sigma \cdot \theta_{\gamma^{-1}}(g)$, and $s(g \cdot \gamma) = \theta_\gamma(g)^{-1} \cdot H$. Moreover, Γ-equivariance of σ is equivalent to Γ-equivariance of the associated G-equivariant map $s\colon P \to G/H$, hence equivalent to Γ-invariance of $P_\sigma \subset P$. Finally, if P_σ is Γ-invariant, then the restricton of the (θ, c)-twisted Γ-equivariant action on P results a twisted equivariant action on P_σ. □

B.5.4 Twisted equivariant bundles and coverings

We present here the most important result proven in [5]. Plainly stated, it says that whenever we have a non-connected complex reductive Lie group $\hat{G}$, and a principal $\hat{G}$-bundle $P_{\hat{G}}$ over an $\mathcal{S}$-manifold X, we may find a covering $Y \to X$, whose deck group is $\Gamma = \hat{G}/G$, and twisted equivariant principal bundle $P_G \to Y$. That is, at the expense of "jumping" to a covering Y of our original manifold, and acquiring a "twisting", we may get a principal bundle whose structure group is connected out of one whose structure group is non-connected.

Fix a connected $\mathcal{S}$-manifold Y endowed with an $\mathcal{S},\Gamma$-action $\Gamma \to \mathrm{Aut}(Y)$, and let $P(Y,G) \to Y$ be a (θ,c)-twisted equivariant principal G-bundle. By remark 3.2, $P/G \cong Y$, and the G-action on P induces a Γ-action on Y. Assuming this Γ-action to be free and proper, $P \to X := Y/\Gamma$ is a principal $\hat{G}$-bundle [5, Sec. 4.1]. In that case, $X := Y/\Gamma$ has unique smooth structure such that $q\colon Y \to X$ is an $\mathcal{S}$ covering map [36, Th. 21.13].

Thus, fix an $(\mathcal{S},\Gamma)$-covering $Y \to X$ (that is, a covering whose deck group is $\mathrm{Deck}_q(Y) = \Gamma$), and suppose that Γ acts on Y on the right. We define two categories $\mathcal{C}_1$ and $\mathcal{C}_2$ as follows.

1. $\mathcal{C}_1$ is the category whose
   - **objets** are principal $\hat{G}$-bundles P over X such that there is an isomorphism $P/G \cong Y$ covering the identity on X (see diagram B.1a) and whose
   - **morphisms** are morphisms of principal $\hat{G}$-bundles.
2. $\mathcal{C}_2$ is the category whose
   - **objets** are (θ,c)-twisted Γ-equivariant principal G-bundles $P \to Y$ such that the twisted Γ-action descends to the action of Γ as covering transformations of the fixed Γ-covering $P/G \cong Y \to X$ (see diagram B.1b, γ means the map induced by the Γ (twisted) action) and whose
   - **morphisms** are (θ,c)-twisted (G,Γ)-equivariant maps $\phi\colon P \to P'$ such that the diagram B.1c commutes and the induced map $\bar{\phi}\colon Y \to Y$ is a covering map that belongs to $Z(\Gamma) \subset \Gamma = \mathrm{Deck}_q(Y)$.

$$\begin{array}{ccc} P/G & \xrightarrow{\phi} & Y \\ {\scriptstyle\hat{\pi}}\downarrow & & \downarrow{\scriptstyle q} \\ X & \xrightarrow{\mathrm{id}_X} & X = Y/\Gamma \end{array}$$

(a) Objets on $\mathcal{C}_1$.

$$\begin{array}{ccc} P & \xrightarrow{\gamma} & P \\ {\scriptstyle\pi}\downarrow & & \downarrow{\scriptstyle\pi} \\ Y & \xrightarrow{\gamma} & Y \end{array}$$

(b) Objets on $\mathcal{C}_2$.

$$\begin{array}{ccc} P & \xrightarrow{\phi} & P' \\ {\scriptstyle\pi}\downarrow & & \downarrow{\scriptstyle\pi'} \\ Y & \xrightarrow{\bar{\phi}} & Y \end{array}$$

(c) Morphisms in $\mathcal{C}_2$.

Diagram B.1: Diagrams for the equivalence of categories.

In diagram B.1a, $\hat{\pi}$ is the $\mathcal{S}$-morphism definedd by $\hat{\pi}([p]) = \pi(p)$ where $[\cdot]$ is the equivalence class for the relation obtained from the quotient by G, and q is the quotient map for $Y \to Y/\Gamma$.

Then, the key theorem is the following.

Theorem B.31. There is an isomorphism of categories between $\mathcal{C}_1$ and $\mathcal{C}_2$.

Proof. See [5, Sec. 4.1]. □

Remark B.32. In practice, one usually wants to use this result as follows: let (X,ω_X) a compact Kähler manifold, and let $P_{\hat{G}} \to X$ be a principal $\hat{G}$-bundle. We define $Y := P_{\hat{G}}/G$. Then, we have a Γ-action on Y induced by the $\hat{G}$-action, and $p\colon Y \to X$ is a holomorphic covering (recall that Γ is finite). We endow Y with a Kähler structure via pullback $\omega_Y := p^*\omega_X$ is Y a compact Kähler manifold? Observe that Y is induced with a natural Γ-action (see proposition B.25). Then, the (θ,c)-twisted Γ principal G-bundle over Y is $P_{\hat{G}} \to Y$. It is common to write this new bundle as $P_G \to Y$ even though as sets, and as manifolds, $P_G = P_{\hat{G}}$.

B.5.5 Parabolic subgroups of non-connected groups

For non-connected groups $\hat{G}$, we want to extend te definition of paabolic subgroup to allow for non-connected parabolic subgroups. By theorem B.12, a reasonable definition is as follows: we say that a closed subgroup $\hat{P}$ of a complex reductive group $\hat{G}$ is *parabolic* if $\hat{G}/\hat{P}$ is compact. $\hat{P}$ is a parabolic subgroup of $\hat{G}$ if and only if $P := \hat{P} \cap G$ is a parabolic subgroup of G. We will work in the context of an extension of Lie groups given as in (B.6). For a non-connected Lie group $\hat{G}$, and any element $\zeta \in \mathrm{i}\mathfrak{k}$, we define $\hat{P}_\zeta, \hat{L}_\zeta, \mathfrak{p}_\zeta$ and $\mathfrak{l}_\zeta$ by the same expressions given in proposition B.13, but

allowing $g \in \hat{G}$ for the groups. Then, each $\hat{P}_\zeta$ is a parabolic subgroup whose Levi subgroup is $\hat{L}_\zeta$, and their respective Lie algebras are $\mathfrak{p}_\zeta$ and $\mathfrak{l}_\zeta$. Moreover, if we define $P_\zeta \subset G$ as in B.13, then P_ζ is the identity component of $\hat{P}_\zeta$.

Appendix C

The basics of Banach manifolds

The aim of this appendix is to set the basics of infinite-dimensional manifolds. This will be used in chapter 4 to given an infinite-dimensional version of the Marsden-Weinstein-Meyer theorem. This is in turn useful because it provides a rigorous framework to deal with the moment map picture of the Hitchin-Kobayashi correspondence, which is introduced in chapter 5. Some results in this appendix have proofs that are analogous to those for finite-dimensional manifolds, and we shall omit them. We refer the reader to [32], which is a book that develops the theory of smooth manifolds in a very general setting, giving most proofs and definitions for Banach manifolds, and particularizing results to finite-dimensional ones when there are substantial differences.

C.1 Basic definitions

When working with finite-dimensional manifolds, we sometimes have to work with some spaces that look locally like infinite-dimensional vector spaces. Since the key idea of smooth manifolds is to consider sets that locally look like $\mathbb{R}^n$ or $\mathbb{C}^n$, it seems reasonable to try to generalize the definition of manifold to an infinite-dimensional setting. There are several ways of doing that, depending on the level of generality we want to achieve. We shall only consider the case of Banach manifolds, which essentially, are manifolds that locally look like a Banach space.

The definition of smooth manifold not only requires the manifold to be covered by open sets homeomorphic to open sets of $\mathbb{R}^n$; it also requires the transition maps to be smooth whenever two open sets overlap. Therefore, if we want to replicate this notion in a Banach space, we need a notion of smoothness, and even before that, we need a notion of derivative. Let $\mathbf{E}, \mathbf{F}$ be two Banach spaces, we write $\mathcal{L}(\mathbf{E}, \mathbf{F})$ for the Banach space of linear maps defined from $\mathbf{E}$ to $\mathbf{F}$, which is a Banach space when endowed with the operator norm. We define $\mathrm{Aut}(\mathbf{E})$ the space of *toplinear isomorphisms*, that is, vector space isomorphisms of $\mathbf{E}$ that are continuous and have continuous inverses.

Definition C.1 (Fréchet derivative). Let $(\mathbf{E}, ||\cdot||_{\mathbf{E}}), (\mathbf{F}, ||\cdot||_{\mathbf{F}})$ be two Banach spaces, $U \subset \mathbf{E}$ an open set, and $f\colon U \to \mathbf{F}$ a continuous map. We say that f is *Fréchet differentiable* at $x \in U$ if there is a linear map $Df(x) \in \mathcal{L}(\mathbf{E}, \mathbf{F})$ such that

$$\lim_{h\to 0} \frac{||f(x+h) - f(x) - Df(x)(h)||_{\mathbf{F}}}{||h||_{\mathbf{E}}} = 0.$$

Observe that this definition is just the direct generalization to a Banach space of the usual differential of a map; that is, a continuous map is Fréchet differentiable if it admits a linear approximation that "approximates the value of f faster that the norm". Observe that if f is Fréchet differentiable at every point $x \in U$, then we may define a map $Df\colon U \to \mathcal{L}(\mathbf{E}, \mathbf{F})$. If this map is continuous, we say that f is a Fréchet-$\mathscr{C}^1$ map. We now have a continuous map $Df\colon U \to F'$, where $\mathbf{F}' = \mathcal{L}(\mathbf{E}, \mathbf{F})$, so we may ask whether this new map is Fréchet differentiable. If it is, we define $D^2 f := D(Df)$, and if this new map is continuous, we say that f is a Fréchet-$\mathscr{C}^2$ map. Inductively, we define $D^k f := D^{k-1}(Df)$, $k \in \mathbb{N}$, and we sat that f is Fréchet-smooth or a Fréchet-$\mathscr{C}^\infty$ map if it is a Fréchet-$\mathscr{C}^k$ map for every $k \in \mathbb{N}$. Whenever it is clear from the context that we refer to Fréchet regularity, we will omit the word Fréchet. Observe that this reduces to the usual definition of differential if the Banach spaces are euclean space. If a Fréchet derivative exists, then it is unique.

Definition C.2 (Banach manifold). Let $\mathcal{M}$ be a Hausdorff topological space. A $\mathscr{C}^k$ *atlas* on $\mathcal{M}$ is a collection of triplets $\{(U_\alpha, \varphi_\alpha, \mathbf{E}_\alpha)\}_{\alpha\in A}$, where each U_α is an open set, $(\mathbf{E}_\alpha, ||\cdot||_\alpha)$ is a Banach space, and $\varphi_\alpha \colon U_\alpha \to \mathbf{E}_\alpha$ is a continuous map satisfying:

1. $\{U_\alpha\}_{\alpha\in A}$ is an open covering of $\mathcal{M}$
2. Each $\varphi_\alpha \colon U_\alpha \to \varphi_\alpha(U_\alpha) \subset \mathbf{E}_\alpha$ is a homeomorphism onto its image, which is open
3. If $U_\alpha \cap U_\beta \neq \varnothing$, then
$$\varphi_\alpha \circ \varphi_\beta^{-1} \colon \varphi_\beta(U_\alpha \cap U_\beta) \to \varphi_\alpha(U_\alpha \cap U_\beta)$$
is a Fréchet $\mathscr{C}^k$ map.

Observe that we do not require the Banach spaces that model the manifold locally to be the same. However, whenever $U_\alpha \cap U_\beta \neq \varnothing$, it follows from the definition that $\mathbf{E}_\alpha$ and $\mathbf{E}_\beta$ are isomorphic (indeed, $D(\varphi_\alpha \circ \varphi_\beta^{-1})$ is a linear isomorphism between them). When all the spaces are the same (call it $\mathbf{E}$), we say that $\mathcal{M}$ is an $\mathbf{E}$-*manifold*. Each connected component of a Banach manifold is an $\mathbf{E}$-manifold, and usual finite-dimensional smooth manifolds are $\mathbb{R}^n$-manifolds. As in the finite-dimensional case, every atlas is contained within a maximal one. Obvious modifications to the previous definition allow us to define *Hilbert manifolds* and *locally convex manifolds* for manifolds that are locally modelled on those spaces. Observe that Hilbert manifolds are just a special case of Banach manifolds, but this is not the case for locally convex manifolds beacuse we do not have norms to define Fréchet derivatives there. Smooth maps are defined analogously: if M and N are two Banach manifolds, and $f \colon \mathcal{M} \to \mathcal{N}$ is a map, we say that it is smooth if whenever (U, ψ) and (V, ψ) are smooth charts satisfying $f(U) \subset V$, then the *localization of* f defined by $\psi^{-1} \circ f \circ \varphi \colon \varphi(U) \to \psi(V)$ is a $\mathscr{C}^\infty$ map (in the Fréchet sense). $\mathscr{C}^k$ maps are maps whose localizations are of class $\mathscr{C}^k$.

C.1.1 The tangent space to a Banach manifold

In the category of finite dimensional smooth manifolds, the tangent space to a manifold at a point can be defined as the space of derivations acting on germs of functions at that point, and as the set of equivalence classes of curves passing through that point. In the infinite-dimensional setting, these two definitions are no longer equivalent, and the definition chosen is the second one. With this idea in mind, let $I \subset \mathbb{R}$ be an interval, and $\mathcal{M}$ be a Banach manifold, a smooth curve is a smooth map $\alpha \colon I \to \mathcal{M}$. Let $\alpha \colon I \to \mathcal{M}$ and $\beta \colon I' \to \mathcal{M}$ be two smooth curves satisfying $\alpha(0) = \beta(0) = m \in \mathcal{M}$. We say that they are *equivalent* if there is a chart $(U, \psi, \mathbf{E})$ such that $(\psi \circ \alpha)'(0) = (\psi \circ \beta)'(0)$, were we defined

$$(\psi \circ \alpha)'(t) = [D(\psi \circ \alpha)(t)](1). \tag{C.1}$$

In that case, (C.1) holds for every chart. Observe that the right-hand-side makes sense; $D(\psi\circ\alpha)(t) \in \mathcal{L}(\mathbb{R}, \mathbf{E})$, so when evaluating at 1, the result is a vector in $\mathbf{E}$. This defines an equivalence relation on the set of all smooth curves passing through m at $t = 0$. A *tangent vector* is thus an equivalence relation of curves. We denote it by $[\alpha, m]$, or simply $[\alpha]$ if the point is clear from the context. The set of all tangent vectors is called the *tangent space* of $\mathcal{M}$ at m, and denoted $T_m\mathcal{M}$ as usual. This is indeed a vector space. To endow it with a linear structure, we can use any chart $(U, \phi, \mathbf{E})$, which defines a bijection $\bar{\phi} \colon T_m\mathcal{M} \to \mathbf{E}, [\alpha] \mapsto (\phi \circ \alpha)'(0)$ as the following proposition shows.

Lemma C.3. Let $\mathcal{M}$ be an infinite-dimensional Banach manifold.

1. Let $\phi \colon \mathcal{M} \to (\mathbf{E}, ||\cdot||_{\mathbf{E}})$ be a chart around $m \in \mathcal{M}$, and set $p_m := \phi(m) \in \mathbf{E}$. Then, the following maps are inverses
$$\begin{array}{rccc} \bar{\phi}\colon & T_m\mathcal{M} & \longrightarrow & \mathbf{E} \\ & [\alpha] & \mapsto & (\phi\circ\alpha)'(0) \end{array} \qquad \begin{array}{rccc} \bar{\phi}^{-1}\colon & \mathbf{E} & \longrightarrow & T_m\mathcal{M} \\ & v & \mapsto & [t \mapsto \phi^{-1}(p_m + t\cdot v)]. \end{array}$$
2. For all charts ϕ, ψ around m, we have $\bar{\psi} \circ \bar{\phi}^{-1} = D(\psi \circ \phi^{-1})(p_m)$.
3. $T_m\mathcal{M}$ admits a unique vector space structure and topology making each $\bar{\phi}$ into a linear isomorphism.

Proof. The proof on [54, Lemma 1.41] can be adapted to this situation. □

In fact, using any chart we wanted, we could endow the tangent space with the structure of a Banach space. However, different charts will lead to different (but equivalent) norms. We can thus consider the tanget space as a Banach space if needed, and the topology will be independent of the chosen chart.

Remark C.4. As happens in the finite-dimensional case, if $\mathbf{E}$ is a Banach space, then $\mathrm{id}_{\mathbf{E}}\colon \mathbf{E} \to \mathbf{E}$ defines a chart, and $\overline{\mathrm{id}_{\mathbf{E}}}$ allows us to identify $\mathbf{E}$ and $T_e\mathbf{E}$ for every $e \in \mathbf{E}$. In particular, if $\mathbf{E} = \mathbb{R}$ we define $\frac{\mathrm{d}}{\mathrm{d}t}|_t := (\overline{\mathrm{id}_{\mathbb{R}}})^{-1}(1) \in T_t\mathbb{R}$.

We can then define the differential of a smooth map; if $f\colon \mathcal{M} \to \mathcal{N}$ is a smooth map between smooth Banach manifolds, then we define $\mathrm{d}_m f\colon T_m\mathcal{M} \to T_{f(m)}\mathcal{N}, [\alpha, m] \mapsto [f \circ \alpha, f(m)]$. This map is linear. Using the identification above, the differential of a map whose codomain is a Banach space is a linear map of the tangent space taking values on the same Banach space. The map $\mathrm{d}_m f$ is often called the tanget map to f at m, and denoted $\mathrm{T}_m f$ in the literature.

Many of the usual constructions of finite-dimensional manifold theory carry to the Banach manifold setting. Firstly, the tangent bundle associated to a Banach manifold is defined as usual:

$$T\mathcal{M} := \bigsqcup_{m\in\mathcal{M}} T_m\mathcal{M},$$

and it is a Banach manifold, endowed with a smooth projection $\pi([\alpha, m]) = m$, whose Banach manifold structure is obtained by requiring that for every chart $(U, \phi, \mathbf{E})$, the map $(\pi^{-1}(U), \Phi)$ is a diffeomorphism, where $\Phi\colon \pi^{-1}(U) \to \phi(U) \times \mathbf{E}, u \mapsto (\pi(u), \bar{\phi}(u))$. These maps define an atlas (see [54, Sec. 1.1.5]). Now, vector fields are defined as smooth sections of $T\mathcal{M}$. We use the same notation as in the finite-dimensional case or the vector space of all vector fields $\mathfrak{X}(\mathcal{M})$.

Remark C.5. We still want to have vector fields acting on smooth real valued functions to provide a Lie algebra structure on $\mathfrak{X}(\mathcal{M})$. This requires a few comments.

1. Let $f\colon \mathcal{M} \to \mathbb{R}$ be a smooth map, then, $\mathrm{d}_m f([\alpha, m]) = [f \circ \alpha, f(m)] \in T_{f(m)}\mathbb{R} \cong \mathbb{R}$ where the last isomorphism is given by $\overline{\mathrm{id}_{\mathbb{R}}}$. Thus, to see it as a real number, we understand $\mathrm{d}_m f([\alpha, m]) := \overline{\mathrm{id}_{\mathbb{R}}}([f \circ \alpha, f(m)]) = (f \circ \alpha)'(0)$, seeing the latter as the derivative at 0 of a smooth real function. That is, we essentially recover the standard finite-dimensional result: the differential of a smooth map acting on a vector is the derivative at 0 of $f \circ \alpha$ where α is a smooth curve satisfying $\alpha(0) = 0$ and $\alpha'(0)$ is the vector.

2. A vector field does induce a derivation on the algebra of smooth real-valued functions, which is given by $X(f)(m) := \mathrm{d}_m f(X_m)$.

Defining the Lie bracket of vector fields is somewhat more involved in the infinite-dimensional setting. The usual definition of $[X, Y]$ as the vector field whose action on smooth maps is $[X, Y](f) = X(Y(f)) - Y(X(f))$ no longer works, because even though the previous expression gives a well-defined map, provided that $X(f)\colon \mathcal{M} \to \mathbb{R}$ is a well-defined smooth map, it is not obvious that the map $X(Y(f)) - Y(X(f))$ is generated by a vector field. It is true, nonetheless, as shown by [32, Prop. V-1.1 and Prop V-1.3]. The vector field $[X, Y]$ is called the *Lie bracket* of X and Y. Is is bilinear, antisymmetric, and satisfies the Jacobi identity, and hence defines a Lie algebra structure on $\mathfrak{X}(\mathcal{M})$ (see [32, Cor. 1.4]).

C.1.2 Immersions, submersions and embeddings

The notions of immersion, submersion and embedding have generalizations to Banach manifolds. First, let $\mathcal{M}$ be a Banach-manifold, and $\mathcal{N} \subset \mathcal{M}$. We say that $\mathcal{N}$ is a *Banach-submanifold* if for every $m \in \mathcal{N}$, there is a chart $(U, \varphi, \mathbf{E})$ of $\mathcal{M}$, such that $m \in \mathcal{M}$, and a closed subspace $\mathbf{F} \subset \mathbf{E}$ such that

$$\varphi(U \cap \mathcal{N}) = \varphi(U) \cap \mathbf{F}.$$

Moreover, given a Banach-space $(\mathbf{E}, ||\cdot||_{\mathbf{E}})$, and a closed vector subspace $\mathbf{F} \subset \mathbf{E}$, we say that $\mathbf{F}$ *splits* in $\mathbf{E}$, if there is another closed vector subspace $\mathbf{G} \subset \mathbf{E}$ such that $\mathbf{E} = \mathbf{F} \oplus \mathbf{G}$. We say that $\mathbf{G}$ is a

topological complement of $\mathbf{F}$ in $\mathbf{E}$. Observe that every closed subspace of a Hilbert space splits (just take its orthogonal complement).

Definition C.6. Let $\mathcal{M}$ and $\mathcal{N}$ be two Banach-manifolds. Let $f\colon \mathcal{M} \to \mathcal{N}$ be smooth.

1. We say that f is a *(Banach)-immersion* if $\mathrm{d}_m f$ is injective and its image is a closed subspace of $T_{f(m)}\mathcal{N}$ that splits for each $m \in \mathcal{M}$.
2. We say that f is a *(Banach)-submersion* if $\mathrm{d}_m f$ is surjective and $\ker \mathrm{d}_m f$ splits for every $xm \in \mathcal{M}$. We say that x is a *regular value* of f.
3. We say that f is a *(Banach)-emebedding* if it is an immersion and a homeomorphism into its image.

If f is an embedding, then $f(\mathcal{M}) \subset \mathcal{N}$ is a submanifold (in the sense defined above). It is sometimes called an *embedded submanifold*. As we see, the main difference with respect to finite-dimensional manifolds is the split condition (which is always satisfied in finite-dimensional spaces). The split condition is needed to recover the local immersion and submersion theorems, that is, to see immersions as inclusions and submersions as projections in suitable coordinates.

C.1.3 Banach Lie groups

Unsurprisingly, a *Banach-Lie group* is defined as a group that has the structure of a Banach manifold, and such that the product and inversion maps are smooth, and a *Hilbert-Lie group* is defined analogously. In this setting, right and left-translations L_g and R_g are diffeomorphisms, just as in the finite-dimensional case. The Lie algebra is also defined quite similarly to the finite-dimensional case. Firstly, to vector fields $X \in \mathfrak{X}(\mathcal{M}), Y \in \mathfrak{X}(\mathcal{N})$ are *f-related* (where $f\colon \mathcal{M} \to \mathcal{N}$ is a smooth map) if $\mathrm{d}_m f(X_m) = Y_{f(m)}$ for all $m \in \mathcal{M}$. We thus say that a vector field $X \in \mathfrak{X}(G)$ is *left-invariant* if it is L_g-related to itself. The *Lie algebra* of G, $\mathrm{Lie}(G)$ is the set of all left invariant vector fields on G (we will keep the notation $\mathfrak{g}$ for the Lie algebra of finite-dimensional Lie groups).

Lemma C.7. Let G be a Banach-Lie group. Then, $\mathrm{Lie}(G)$ is a Lie subalgebra of $\mathfrak{X}(G)$.

Proof. It is obvious that $\mathrm{Lie}(G)$ is a vector space, so we need only check that it is closed under the Lie bracket operation, but this follows from the fact that if X_i and Y_i are f-related, $i = 1, 2$, then $[X_1, X_2]$ and $[Y_1, Y_2]$ are f-related as well (see [32, Page 117]). □

Observe that every left-invariant vector field is determined by its valued at the identity: $X_g = \mathrm{d}_e L_g(X_e)$, so we have a natural vector space isomorphism $T_eG \cong \mathrm{Lie}(G)$. As we do in the finite-dimensional case, we may use this isomorphism to endow T_eG with a Lie algebra structure (see [54, Sec. 1.1.2]).

Finally, Banach-Lie groups have a well-defined exponential map. This is not true for Lie groups modelled on arbitrary locally convex spaces. The definition is almost identical to the finite-dimensional one. If $\alpha\colon I \to M$ is a smooth curve, we define

$$\dot{\alpha}(t) := \mathrm{d}_t \alpha \left(\frac{\mathrm{d}}{\mathrm{d}t}\bigg|_t \right) \in T_{\alpha(t)}\mathcal{M}. \tag{C.2}$$

We say that a curve α is an *integral curve* of a vector field X if $X_{\alpha(t)} = \dot{\alpha}(t)$ for every $t \in I$. Given a Banach manifold, every vector field admits an integral curve defined on a maximal interval. The exponential map is just defined as

$$\begin{array}{rccc} \exp\colon & \mathrm{Lie}(G) & \longrightarrow & G \\ & X & \mapsto & \alpha(1) \end{array}$$

where α is the unique integral curve of X satisfying $\alpha(0) = e$. See [13] for more details. Finally, the adjoint and coadjoint actions of G on $\mathfrak{g}$ and $\mathfrak{g}^*$ admit direct generalizations to this context. To define Ad, simply observe that the map $I_g := L_{g^{-1}} \circ R_g$ is a diffeomorphism, hence we may just define $\mathrm{Ad}_g(X) := \mathrm{d}_e I_g(X)$ for $X \in \mathfrak{g}$ and $g \in G$. Similarly, we can just define Ad^* by setting

$$(\mathrm{Ad}^*_g \xi, X) := (\xi, \mathrm{Ad}_{g^{-1}}(X)).$$

C.1.4 Differential forms in Banach-manifolds.

Banach-manifolds are sufficiently well-behaved that we can still see differential forms as sections of adequate vector bundles (which is not the case for general infinite-dimensional manifolds, see for example [54, Appendix E] for details). Firstly, the definition of vector bundle can be adapted to a Banach-setting: A *Banach vector bundle* over a Banach manifold is a Banach manifold E, endowed with a smooth surjective map $\pi\colon E \to \mathcal{M}$ satisfying the following properties:

1. There is a Banach space $\mathbf{E}$ such that each point $x \in \mathcal{M}$ has an open neighborhood U, and a diffeomorphism $\Phi\colon \pi^{-1}(U) \to U_i \times \mathbf{E}$ that satisfies $\pi = \mathrm{pr}_U \circ \Phi$.

2. Each fiber $E_m := \pi^{-1}(m)$ is a vector space, and $\Phi_m := \Phi|_{E_m}\colon E_m \to \{m\} \times \mathbf{E}$ is a vector space isomorphism. The pairs (U, Φ) are called local trivializations.

3. For each pair of local trivializations (U, Φ), (V, Ψ), $\Phi_m \circ \Psi_m^{-1}\colon \mathbf{E} \to \mathbf{E}$ is a continuos vector space isomorphism whose inverse is continuous.

4. For each pair of local trivializations, (U, Φ), (V, Ψ), the map $U \cap V \to \mathrm{Aut}(\mathbf{E}), m \mapsto (\Phi \circ \Psi^{-1})_m$ is smooth.

For differential forms, we want to keep their essence and consider them as assignments of multilinear alternating maps from the tangent space to $\mathbb{R}$ to points of the manifold. Let $\mathrm{L}^r_{\mathrm{Alt}}(T\mathcal{M})$ be the vector bundle over $\mathcal{M}$ whose fibers are the Banach spaces of multilinear alternating maps from $T_m\mathcal{M}$ to $\mathbb{R}$. A differential form on $\mathcal{M}$ is defined as a section of this bundle. We keep the usual notation $\Omega^p_{\mathbb{R}}(\mathcal{M})$ for p-forms. As in the finite-dimensional case, $\Omega^p_{\mathbb{R}}(\mathcal{M})$ is a real vector space, and a $\mathscr{C}^\infty(\mathcal{M};\mathbb{R})$-module.

To define the exterior differential, one must observe that the usual definition in terms of local coordinates no longer makes sense in Banach manifolds. Instead, the exterior differential is defined using a coordinate-free relation that holds in the finite-dimensional setting as well.

Proposition C.8 (Exterior differential). Let $\mathcal{M}$ be a Banach manifold, and $\omega \in \Omega^p_{\mathbb{R}}(\mathcal{M})$. Then, there is a unique $p+1$-form on $\mathcal{M}$ such that for all smooth vector fields $X_1, \dots X_{p+1} \in \mathfrak{X}(\mathcal{M})$,

$$\begin{aligned} \mathrm{d}\omega(X_1, \dots X_{p+1}) = &\sum_{i=0}^{p} (-1)^i X_i\big(\omega(X_1, \dots \hat{X}_i, \dots, X_{p+1})\big) \\ &+ \sum_{i<j} (-1)^{i+j} \omega([X_i, X_j], X_1, \dots \hat{X}_i, \dots \hat{X}_j, \dots X_{p+1}). \end{aligned} \tag{C.3}$$

Proof. See [54, Prop. E.3] or [32, Prop. 3.2] □

Again, Banach manifolds are well-behaved enough so that most properties of the usual exterior differential are preserved. Namely, $\mathrm{d}^2 = 0$, and

$$\mathrm{d}(\omega \wedge \psi) = \mathrm{d}\omega \wedge \psi + (-1)^r \cdot \omega \wedge \mathrm{d}\psi$$

for $\omega \in \Omega^r_{\mathbb{R}}(\mathcal{M})$ and any differential form ψ. The exterior or wedge product is again defined analogously to the finite-dimensional case (see [32, Chapter V, section 3] for details).

C.2 Basics of locally convex spaces

We devote this section to introduce very briefly some concepts from the theory of locally convex spaces. We will need it to prove the Banach-version of the symplectic reduction theorem. Even though this is not why we introduce them, it is worth mentioning that locally convex spaces provide the local model for a generalization of Banach manifolds to locally convex manifolds. The manifolds so obtained, however, tend to behave very badly, as a lot of key theorems, namely, the implicit and inverse function theorems, do not hold.

First, recall that a *topological vector space*, TVS for short, is a vector space endowed with a topology such that addition and scalar products are continuous. Morphisms in the category **TVS** are continuous linear maps.

Definition C.9 (Seminorm). Let V be a TVS, a *seminorm* is a map $p\colon V \to [0,\infty)$ satisfying

1. $p(\lambda v) = |\lambda| \cdot p(v)$ for every $\lambda \in \mathbb{R}, v \in V$.
2. $p(v+w) \leq p(v) + p(w)$ for all $v, w \in V$.

We can now define locally convex spaces. Observe that a seminorm is just a norm for which the condition $p(v) = 0$ if and only if $v = 0$ is omitted. If we consider $p_i^{-1}(0)$, which is a vector subspace of E, then the quotient $E/p_i^{-1}(0)$ has a natural norm $||x + p_i^{-1}(0)|| := p(x)$ for any representative x of the equivalence class. Hence, the quotient has a natural topology as a normed vector space.

Definition C.10 (Locally convex TVS). Let V be a TVS, we say that is *locally convex* if there is a family of continuous seminorms $\{p_i\}_{i\in I}$ such that:

1. The topology of V is the initial topology with respect to the canonical projections $q_i\colon V \to V/p_i^{-1}(0)$.
2. If $v \in V$ and $p_i(v) = 0$ for all $i \in I$, then $v = 0$.

The second property of the definition implies that E is Hausdorff. [**54**, Prop. A.10] gives explicitly a subbase of the TVS topology on V and gives a proof of the Hausdorff condition. Observe that asking for V to be a TVS is superfluous. Indeed, if E is a vector space endowed with the initial topology with respect to the natural quotient maps then sum and product by scalars are continuous. Let us see it for sums. Let τ be initial topology on V, and τ_i be the metric topology on $V_i := V/p_i^{-1}(0)$ (which turns each E_i into a TVS). Consider the map $s\colon V \times V \to V, (x,y) \mapsto x+y$. By the universal property of the initial topology, s is continuous if and only if each $q_i \circ s$ is continuous. However, observe that the following diagram commutes

$$\begin{array}{ccc} (V,\tau) & \xrightarrow{\quad q_i \quad} & (V_i,\tau_i) \\ \uparrow{\scriptstyle s} & & \uparrow{\scriptstyle s_i} \\ (V\times V, \tau\times\tau) & \xrightarrow{q_i\times q_i} & (V_i\times V_i, \tau_\times\tau_i) \end{array}$$

where s_i is the sum in V_i (which is continuous). Hence, $q_i \circ s = s_i \circ (q_i \times q_i)$, the right-hand side being continuous. This means that s is continuous. The same reasoning works for scalar multiplication.

Locally convex TVS are important because they retain one of the most celebrated and useful results of functional analysis: the Hahn-Banach theorem.

Theorem C.11 (Hahn-Banach). Let V be a locally convex TVS. Then, for every pair of distinct points $u, v \in V$, there is a continuous linear map $\alpha\colon V \to \mathbb{R}$ such that $\alpha(x) \neq \alpha(y)$.

Proof. This is a standard result in functional analysis. See [**39**, Prop. 22.12] for a proof. □

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Notation index

i imaginary unit, $\sqrt{-1}$

A^* conjugate transpose of the matrix A

δ_{ij} Kronecker delta symbol

V^* dual of the vector space V

$\Omega^p_{\mathbb{R}}(M)$ set of p-forms on a smooth manifold M

$\Omega^p_{\mathbb{C}}(M)$ set of complex p-forms on an almost complex or complex manifold M

$\Omega^{(p,q)}_{\mathbb{C}}(M)$ set of complex (p,q)-forms on an almost complex or complex manifold M

$\Omega^{(p,q)}_{\mathbb{C}}(M;E)$, $\Omega^k_{\mathbb{C}}(M;E)$ set of bundle-valued complex forms

$\Gamma(M;E)$ set global sections for a smooth vector bundle $\pi\colon E \to M$

$\Gamma(U;M)$ set local sections for a smooth vector bundle $\pi\colon E \to M$

$\mathcal{O}(M;E)$ set global sections for a holomorphic vector bundle $\pi\colon E \to M$

$\mathcal{O}(U;E)$ set local sections for a holomorphic vector bundle $\pi\colon E \to M$

∇ connection on a vector bundle

$\nabla^{(1,0)}$ $\nabla^{(0,1)}$ componets of a connection

$\bar\partial_E$ Cauchy-Riemann operator

$\bar\partial_e$ integrable Dolbeault operator

$\Pi^{(0,1)}$, $\Pi^{(1,0)}$ projections onto the $(0,1)$ and $(1,0)$ components

F_∇ curvature of a connection on a vector bundle

$F^{(1,1)}_\nabla$, $F^{(0,2)}_\nabla$, $F^{(2,0)}_\nabla$ components of the curvature of a connection

d_∇ exterior covariant derivative associated to a connection

h hermitian fiber metric

$\mathcal{S}$ holomorphic or smooth map, manifold, bundle...

$(s_1, \dots s_r)$ local frame on a vector bundle

$(t_1, \dots t_r)$ local coframe on the dual of a vector bundle

$\mathscr{A}(E)$ set of unitary connections on a vector bundle

$\mathscr{A}^{(1,1)}(E)$ set of unitary connections of type $(1,1)$ on a vector bundle

$\mathscr{D}(E)$ set of Dolbeault operators on a vector bundle

$\mathscr{H}(E)$ set of integrable Dolbeault operators

$c_1(\nabla)$ first real Chern class

$c_1(E)$ Chern form

A connection or connection one-form on a principal bundle

$\mathrm{D}_A A$ curvature of a connection seen as an element of $\Omega^2_{\mathbb{R}}(P;\mathfrak{g})$

F_A curvature of a connection seen as an element of $\Omega^2_{\mathbb{R}}(M;\mathrm{ad}(P))$

$P(M,G)$, P_G principal bundle with structure group G over a manifold M

$P\times_G F$ associated bundle with typical fiber F

$H^A P$ horizontal distribution associated to a connection one-form A

$Y^{\mathbf{HL}}$ horizontal lift of a vector field Y with respect to a connection

$\mathscr{J}(P_G)$ set of almost complex G-structures on a principal G-bundle

$\bar{\mathscr{J}}(P_G)$ set of integrable almost complex G-structures on a principal G-bundle

$N\triangleleft G$ N is a normal subgroup of G

$Z(G)$ center of the Lie group G

$\mathfrak{z}(\mathfrak{g})$ center of the Lie algebra $\mathfrak{g}$

$\kappa_{\mathfrak{g}}$ Killing form of the Lie algebra $\mathfrak{g}$

e_G unit element of a group G

C_g map defined by $C_g\colon G\to G, h\mapsto ghg^{-1}$

$\mathrm{Aut}(G)$ group of isomorphisms of the Lie group G

$\mathrm{Aut}(Y)$ group of $\mathcal{S}$-isomorphisms of an $\mathcal{S}$-manifold Y

$\mathrm{Int}(G)$ $=\{C_g : g\in G\}$

$\mathrm{Out}(G)$ $=\mathrm{Aut}(G)/\mathrm{Int}(G)$

$\mathrm{M}_{n\times m}(\mathbb{K})$ set of $n\times m$ matrices with coefficients in the field $\mathbb{K}$

I_n unit matrix of order n

$\mathrm{O}(n,\mathbb{C})$ $=\{A\in \mathrm{M}_n(\mathbb{C}) : A\cdot A^t=\mathrm{I}_n\}$

$\mathrm{SO}(n,\mathbb{C})$ $=\{A\in \mathrm{M}_n(\mathbb{C}) : A\cdot A^t=\mathrm{I}_n$ and $\det A=1\}$

$\mathfrak{g}^{\mathbb{R}}$ underlying real Lie algebra to a complex Lie algebra

$L^p_2(M;E)$ Sobolev space of E valued sections

$\mathbf{\Omega}^p_2(M;\mathrm{ad}(P_G))$ Sobolev space of $\mathrm{ad}(P_G)$ valued p-forms

L_ω Lefschetz operator: $L_\omega\alpha=\omega\wedge\alpha$

Λ_ω Dual Lefschetz operator: $\langle L_\omega\alpha,\beta\rangle=\langle\alpha,\Lambda_\omega\beta\rangle$

$(\mathbf{E},||\cdot||_{\mathbf{E}})$ Banach space

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